Fundamentals *of* Algorithms

Data Assimilation

Methods, and Applications

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Data Assimilation

Fundamentals *of* Algorithms

Editor-in-Chief: Nicholas J. Higham, University of Manchester

The SIAM series on Fundamentals of Algorithms is a collection of short user-oriented books on state-of-the-art numerical methods. Written by experts, the books provide readers with sufficient knowledge to choose an appropriate method for an application and to understand the method's strengths and limitations. The books cover a range of topics drawn from numerical analysis and scientific computing. The intended audiences are researchers and practitioners using the methods and upper level undergraduates in mathematics, engineering, and computational science.

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Data Assimilation

Methods, Algorithms, and Applications

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Society for Industrial and Applied Mathematics
Philadelphia

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10 9 8 7 6 5 4 3 2 1

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Publisher	David Marshall
Acquisitions Editor	Elizabeth Greenspan
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Graphic Designer	Lois Sellers

Library of Congress Cataloging-in-Publication Data

Names: Asch, Mark. | Bocquet, Marc. | Nodet, Maëlle
Title: Data assimilation : methods, algorithms, and applications / Mark Asch, Université de Picardie Jules Verne, Amiens, France, Marc Bocquet, École des Ponts ParisTech/CEREA, Marne-la-Vallée, France, Maëlle Nodet, Université Grenoble Alpes, Grenoble, France.
Description: Philadelphia : Society for Industrial and Applied Mathematics, [2016] | Series: Fundamentals of algorithms ; 11 | Includes bibliographical references and index.
Identifiers: LCCN 2016035022 (print) | LCCN 2016039694 (ebook) | ISBN 9781611974539 | ISBN 9781611974546 (ebook) | ISBN 9781611974539 (print)
Subjects: LCSH: Inverse problems (Differential equations) | Numerical analysis. | Algorithms.
Classification: LCC QA378.5 .A83 2016 (print) | LCC QA378.5 (ebook) | DDC 515/.357--dc23
LC record available at https://lccn.loc.gov/2016035022

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Notation

$\mathbb{R}^n$ state space
$\mathbb{R}^p$ observation space
$\mathbb{R}^m$ ensemble space, $i = 1, \ldots, m$
t_k time, $k = 1, \ldots, K$
$\mathbf{I}$ identity matrix: $\mathbf{I}_n, \mathbf{I}_m, \mathbf{I}_p$
$\mathbf{x}$ vector
$\mathbf{x}^{\mathrm{t}}$ true state vector
$\mathbf{x}^{\mathrm{a}}$ analysis vector
$\mathbf{x}^{\mathrm{b}}$ background vector
$\mathbf{x}^{\mathrm{f}}$ forecast vector
$\mathbf{y}^{\mathrm{o}}$ observation vector
$\boldsymbol{\epsilon}^{\mathrm{a}}$ analysis error
$\boldsymbol{\epsilon}^{\mathrm{b}}$ background error
$\boldsymbol{\epsilon}^{\mathrm{f}}$ forecast error
$\boldsymbol{\epsilon}^{\mathrm{o}}$ observation error
$\boldsymbol{\epsilon}^{\mathrm{q}}$ model error
$\mathbf{M}_k$ linear model operator: $\mathbf{x}_{k+1} = \mathbf{M}_{k+1}\mathbf{x}_k$, with $\mathbf{M}_{k+1} = \mathbf{M}_{k+1:k}$ model from time step k to time step $k+1$; $\mathscr{M}$ nonlinear model operator
$\mathbf{X}_{\mathrm{a}}$ analysis perturbation matrix
$\mathbf{X}_{\mathrm{f}}$ forecast perturbation matrix
$\mathbf{P}^{\mathrm{f}}$ forecast error covariance matrix
$\mathbf{P}^{\mathrm{a}}$ analysis error covariance matrix
$\mathbf{K}$ Kalman gain matrix
$\mathbf{B}$ background error covariance matrix
$\mathbf{H}$ linearized observation operator; $\mathscr{H}$ nonlinear observation operator
$\mathbf{Q}$ model error covariance matrix
$\mathbf{R}$ observation error covariance matrix
$\mathbf{d}$ innovation vector
(j) iteration index of a variational assimilation (in parentheses).
$\mathbf{w}$ coefficients in ensemble space (ensemble transform)

Preface

This book places data assimilation (DA) into the broader context of inverse problems and the theory, methods, and algorithms that are used for their solution. It strives to provide a framework and new insight into the inverse problem nature of DA—the book emphasizes "why" and not just "how." We cover both statistical and variational approaches to DA (see Figure 1) and give an important place to the latest hybrid methods that combine the two. Since the methods and diagnostics are emphasized, readers will readily be able to apply them to their own, precise field of study. This will be greatly facilitated by numerous examples and diverse applications. The applications are taken from the following fields: geophysics and geophysical flows, environmental acoustics, medical imaging, mechanical and biomedical engineering, urban planning, economics, and finance.

In fact, this book is about *building bridges*—bridges between inverse problems and DA, bridges between variational and statistical approaches, bridges between statistics and inverse problems. These bridges will enable you to cross valleys and moats, thus avoiding the dangers that are most likely/possibly lurking down there. These bridges will allow you to fetch/go and get/retrieve different approaches and better understanding of the vast, and sometimes insular, domains of DA and inverse problems, stochastic and deterministic approaches, and direct and inverse problems. We claim that by assembling these, by reconciling these, we will be better armed to confront and tackle the grand societal challenges of today, broadly defined as "global change" issues—such as climate change, disaster prediction and mitigation, and nondestructive and noninvasive testing and imaging.

The aim of the book is thus to provide a comprehensive guide for advanced undergraduate and early graduate students and for practicing researchers and engineers engaged in (partial) differential equation–based DA, inverse problems, optimization, and optimal control—we will emphasize the close relationships among all of these. The reader will be presented with a statistical approach and a variational approach and will find pointers to all the numerical methods needed for either. Of course, the applications will furnish many case studies.

The book favours a continuous (infinite-dimensional) approach to the underlying inverse problems, and we do not make the distinction between continuous and discrete problems—every continuous problem, after discretization, yields a discrete (finite-dimensional) problem. Moreover, continuous problems admit a far richer and more extensive mathematical theory, and though DA (via the Kalman filter (KF)) is *in fine* a discrete approach, the variational analysis will be performed on the continuous model. Discrete inverse problems (finite dimensional) are very well presented in a number of excellent books, such as those of Lewis et al. [2006], Vogel [2002], and Hansen [2010], the latter of which has a strong emphasis on regularization methods.

Some advanced calculus and tools from linear algebra, real analysis, and numerical analysis are required in the presentation. We introduce and use Hadamard's well-posedness theory to explain and understand both *why things work* and *why they go wrong*. Throughout the book, we observe a maximum of mathematical rigor but with a minimum of formalism. This rigor is extremely important in practice, since it enables us to eliminate possible sources of error in the algorithmic and numerical implementations.

In summary, this is really a PDE-based book on inverse and DA modeling—readers interested in the specific application to meteorology or oceanography should additionally consult other sources, such as Lewis et al. [2006] and Evensen [2009]. Those who require a more mathematical approach to inverse problems are referred to Kirsch [1996] and Kaipio and Somersalo [2005], and for DA to the recent monographs of Law et al. [2015] and Reich and Cotter [2015].

Proposed pathways through the book are as follows (this depends on the level of the reader):

- The "debutant" reader is encouraged to study the first chapter in depth, since it will provide a basic understanding and the means to choose the most appropriate approach (variational or statistical).

- The experienced reader can jump directly to Chapter 2 or Chapter 3 according to the chosen or best-adapted approach.

- All readers are encouraged to initially skim through the examples and applications sections of Part III to be sure of the best match to their type of problem (by seeing what kind of problem is the closest to their own)—these can then be returned to later, after having mastered the basic methods and algorithms of Part I or eventually the advanced ones of Part II.

- For the most recent approaches, the reader or practitioner is referred to Part II and in particular to Chapters 4 and 7.

The authors would like to acknowledge their colleagues and students who accompanied, motivated, and inspired this book. MB thanks Alberto Carrassi, Jean-Matthieu Haussaire, Anthony Fillion, Victor Winiarek, Alban Farchi, and Sammy Metref. MN thanks Elise Arnaud, Arthur Vidard, Eric Blayo, and Claire Lauvernet. MA thanks in particular the CIMPA[1] and the Universidad Simon Bolivar in Caracas, Venezuela (where the idea for this book was born), for their hospitality. We thank the CIRM[2] for allowing us to spend two intensive weeks finalizing (in optimal conditions) the manuscript.

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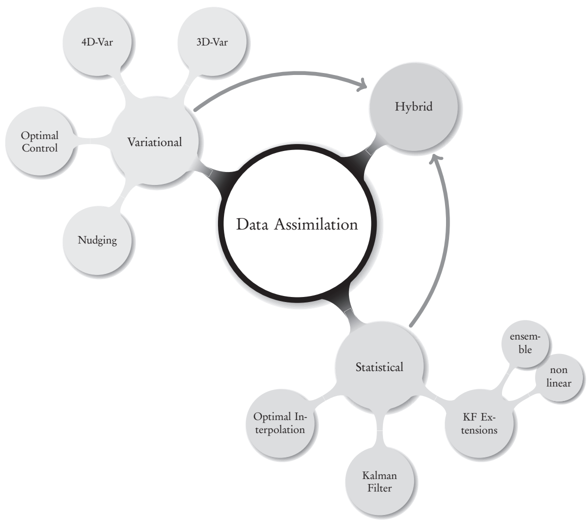


Figure 1. *The big picture for DA methods and algorithms.*

Part I

Basic methods and algorithms for data assimilation

After an introduction that sets up the general theoretical framework of the book and provides several simple (but important) examples, the two approaches for the solution of DA problems are presented: classical (variational) assimilation and statistical (sequential) assimilation. We begin with the variational approach. Here we take an *optimal control* viewpoint based on classical variational calculus and show its impressive power and generality. A sequence of carefully detailed inverse problems, ranging from an ODE-based to a nonlinear PDE-based case, are explained. This prepares the ground for the two major variational DA algorithms, 3D-Var and 4D-Var, that are currently used in most large-scale forecasting systems. For statistical DA, we employ a Bayesian approach, starting with optimal statistical estimation and showing how the standard KF is derived. This lays the foundation for various extensions, notably the ensemble Kalman filter (EnKF).

Chapter 1

Introduction to data assimilation and inverse problems

1.1 ▪ Introduction

What exactly is DA? The simplest view is that it is an approach/method for combining observations with model output with the objective of improving the latter. But do we really need DA? Why not just use the observations and average them or extrapolate them (as is done with regression techniques (McPherson, 2001), or just long-term averaging)? The answer is that we want to predict the state of a system, or its future, in the best possible way! For that we need to rely on models. But when models are not corrected periodically by reality, they can be of little value. Thus, we need to fit the model state in an optimal way to the observations, before an analysis or prediction is made. This fitting of a model to observations is a special case (but highly typical) of an *inverse problem*.

According to J. B. Keller [1966], two problems are inverse to each other if "the *formulation* of each involves all or part of the *solution* of the other." One of the two is named the direct problem, whereas the other is the inverse problem. The direct problem is usually the one that we can solve satisfactorily/easily. There is a back-and-forth transmission of information between the two. This is depicted in Figure 1.1, which represents a typical case: we replace the unknown (or partially known) medium by a model that depends on some unknown model parameters, $\boldsymbol{m}$. The inverse problem involves reversing the arrows—by comparing the simulations and the observations (at the array) to find the model parameters. In fact, the direct problem involves going from cause to effect, whereas the inverse problem attempts to go from the effects to the cause.

The comparison between model output and observations is performed by some form of optimization—recall that we seek an optimal match between simulations of the model and measurements taken of the system that we are trying to elucidate. This optimization takes two forms: classical and statistical. Let us explain. Classical optimization involves minimization of a positive, usually quadratic cost function that expresses the quantity that we seek to optimize. In most of the cases that we will deal with, this will be a function of the error between model and measurements—in this case we will speak of least-squares error minimization. The second form, statistical optimization, involves minimization of the variability or uncertainty of the model error and is based on statistical estimation theory.

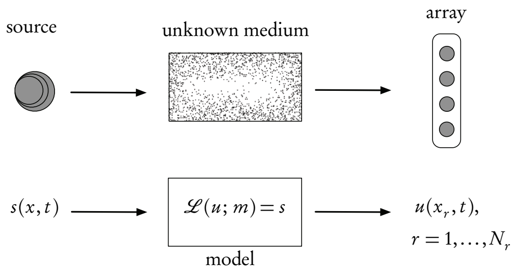


Figure 1.1. *Ingredients of an inverse problem: the physical reality (top) and the direct mathematical model (bottom). The inverse problem uses the difference between the model-predicted observations, u (calculated at the receiver array points, x_r), and the real observations measured on the array to find the unknown model parameters, m, or the source, s (or both).*

The main sources of inverse problems are science (social sciences included!) and engineering—in fact any process that we can model and measure satisfactorily. Often these problems concern the determination of the properties of some inaccessible region from observations on the boundary of the region, or at discrete instants over a given time interval. In other words, our information is *incomplete*. This incompleteness is the source of the major difficulties (and challenges…) that we will encounter in the solution of DA and inverse problems.

1.2 ▪ Uncertainty quantification and related concepts

Definition 1.1. *Uncertainty quantification (UQ) is the science of quantitative characterization and reduction of uncertainties in both computational and real-world applications. It tries to determine how likely certain outcomes are if some aspects of the system are not exactly known.*

The system-science paradigm, as expounded in Jordan [2015], exhibits the important place occupied by DA and inverse methods within the "deductive spiral"—see Figure 1.2. These methods furnish an essential link between the real world and the model of the system. They are intimately related to the concepts of *validation* and *verification*. Verification asks the question, "are we solving the equations correctly?"—this is an exercise in mathematics. Validation asks, "are we solving the correct equations?"—this is an exercise in physics. In geophysics, for example, the concept of validation is replaced by *evaluation*, since complete validation is not possible.

UQ is a basic component of model validation. In fact it is vital for characterizing our confidence in results coming out of modeling and simulation and provides a mathematically rigorous certification that is often needed in decision-making. In fact, it gives a precise notion of what constitutes a validated model by replacing the subjective concept of *confidence* by mathematically rigorous methods and measures.

There are two major categories of uncertainties: epistemic and aleatory. The first is considered to be reducible in that we can control it by improving our knowledge of

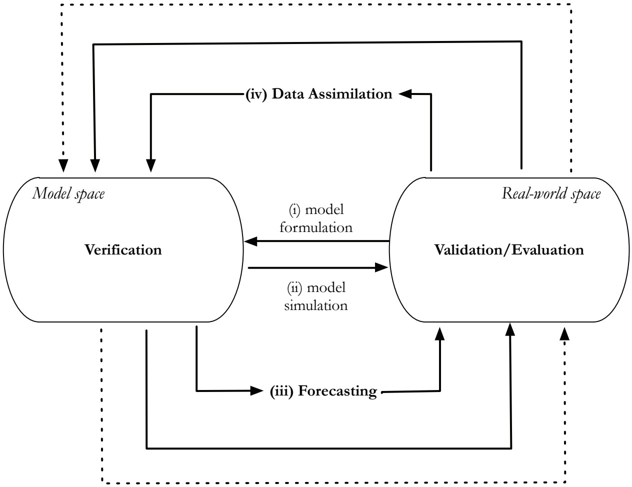


Figure 1.2. *The deductive spiral of system science (adapted from Jordan [2015]). The bottom half represents the direct problem (from model to reality); the top half represents the inverse problem (from reality to model). Starting from the center, with* (i) *model formulation and* (ii) *simulation, one works one's way around iteratively over* (iii) *forecasting and* (iv) *DA, while passing through the two phases of UQ (validation/evaluation and verification).*

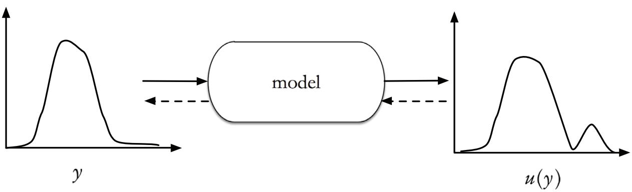


Figure 1.3. *UQ for a random quantity y: uncertainty propagation (left to right); uncertainty definition (right to left).*

the system. The second is assumed to be irreducible and has to do with the inherent noise in, or stochastic nature of, any natural system. Any computation performed under uncertainty will forcibly result in predictive simulations (see the introduction to Chapter 3 for more details on this point).

Uncertainty, in models of physical systems, is almost always represented as a probability density function (PDF) through samples, parameters, or kernels. The central objective of UQ is then to represent, propagate, and estimate this density—see Figure 1.3.

As a process, UQ can be decomposed into the following steps:

1. Define the system of interest, its response, and the desired performance measures.
2. Write a mathematical formulation of the system—governing equations, geometry, parameter values.
3. Formulate a discretized representation and the numerical methods and algorithms for its solution.
4. Perform the simulations and the analysis.
5. Loop back to step 1.

The numerical simulations themselves can be decomposed into three steps:

1. DA, whose objective is to compute the PDFs of the input quantities of interest. This is the major concern of this book and the methods described herein.
2. Uncertainty propagation, whose objective is to compute the PDFs of the output quantities of interest. This is usually the most complex and computationally intensive step and is generally based on Monte Carlo and stochastic Galerkin (finite element) methods—see Le Maitre and Knio [2010].
3. Certification, whose objective is to estimate the likelihood of specific outcomes and compare them with risk or operating margins.

For a complete, recent mathematical overview of UQ, the reader is referred to Owhadi et al. [2013]. There are a number of research groups dedicated to the subject—please consult the websites of UQ groups at Stanford University,[3] MIT,[4] and ETH Zurich[5] (for example).

1.3 ▪ Basic concepts for inverse problems: Well- and ill-posedness

There is a fundamental, mathematical distinction between the direct and the inverse problem: direct problems are (invariably) well-posed, whereas inverse problems are (notoriously) ill-posed. Hadamard [1923] defined the concept of a well-posed problem as opposed to an ill-posed one.

Definition 1.2. *A mathematical model for a physical problem is* well-posed *if it possesses the following three properties:*

WP1 *Existence of a solution.*

WP2 *Uniqueness of the solution.*

WP3 *Continuous dependence of the solution on the data.*

[3] `http://web.stanford.edu/group/uq/.`
[4] `http://uqgroup.mit.edu.`
[5] `http://www.sudret.ibk.ethz.ch.`

Note that existence and uniqueness together are also known as "identifiability," and the continuous dependence is related to the "stability" of the inverse problem. A more rigorous mathematical formulation is the following (see Kirsch [1996]).

Definition 1.3. *Let X and Y be two normed spaces, and let $K : X \to Y$ be a linear or nonlinear map between the two. The problem of finding x given y such that*

$$Kx = y$$

is well-posed if the following three properties hold:

WP1 *Existence—for every $y \in Y$ there is (at least) one solution $x \in X$ such that $Kx = y$.*

WP2 *Uniqueness—for every $y \in Y$ there is at most one $x \in X$ such that $Kx = y$.*

WP3 *Stability—the solution, x, depends continuously on the data, y, in that for every sequence $\{x_n\} \subset X$ with $Kx_n \to Kx$ as $n \to \infty$, we have that $x_n \to x$ as $n \to \infty$.*

This concept of ill-posedness will be the "red thread" running through the entire book. It will help us to understand and distinguish between direct and inverse models. It will provide us with basic comprehension of the methods and algorithms that will be used to solve inverse problems. Finally, it will assist us in the analysis of what went wrong in our attempt to solve the inverse problems.

1.4 ▪ Examples of direct and inverse problems

Take a parameter-dependent dynamical system,

$$\frac{dz}{dt} = g(t, z; \theta), \qquad z(t_0) = z_0,$$

with g known, z_0 an initial state, $\theta \in \Theta$ (a space, or set, of possible parameter values), and the state $z(t) \in \mathbb{R}^n$. We can now define the two classes of problems.

Direct Given parameters θ and initial state z_0, find $z(t)$ for $t \geq t_0$.

Inverse Given observations $z(t)$ for $t \geq t_0$, find $\theta \in \Theta$.

Since the observations are incompletely known (over space-time), they must be modeled by an observation equation,

$$f(t, \theta) = \mathcal{H} z(t, \theta),$$

where $\mathcal{H}$ is the observation operator (which could, in ideal situations, be the identity). Usually we have a finite number, p, of discrete (space-time) observations

$$\left\{ \tilde{y}_j \right\}_{j=1}^{p},$$

where

$$\tilde{y}_j \approx f(t_j, \theta)$$

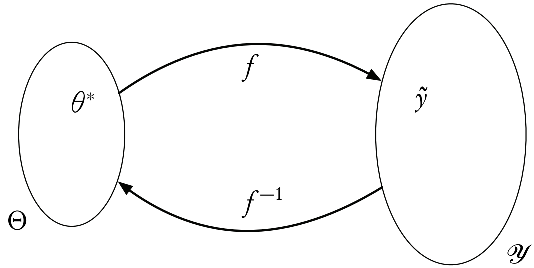
and the approximately equal sign denotes the possibility of measurement errors.

We now present a series of simple examples that clearly illustrate the three properties of well-posedness.

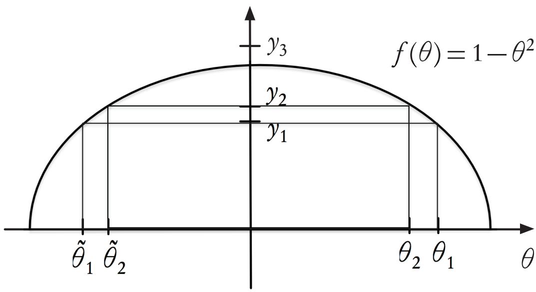
Example 1.4. (Simplest case—inspired by an oral presentation of H.T. Banks). Suppose that we have one observation, $\tilde{y}$, for $f(\theta)$ and we want to find the pre-image

$$\theta^* = f^{-1}(\tilde{y})$$

for a given $\tilde{y}$:



This problem can be severely ill-posed! Consider the following function:



Nonexistence There is no θ_3 such that $f(\theta_3) = y_3$.

Nonuniqueness $y_j = f(\theta_j) = f(\tilde{\theta}_j)$ for $j = 1, 2$.

Lack of continuity $|y_1 - y_2|$ small $\not\Rightarrow |f^{-1}(y_1) - f^{-1}(y_2)| = \left|\theta_1 - \tilde{\theta}_2\right|$ small.

Note that all three well-posedness properties, WP1, WP2, and WP3, are violated by this very basic case. Why is this so important? Couldn't we just apply a good least-squares algorithm (for example) to find the best possible solution? Let's try this. We define a mismatch-type cost function,

$$J(\theta) = |y_1 - f(\theta)|^2,$$

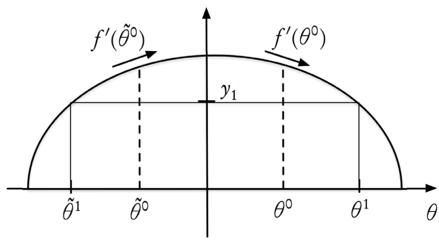
for a given y_1 and try to minimize this square error by applying a standard iterative scheme, such as direct search or gradient-based minimization [Quarteroni et al., 2007], to obtain a solution. For example, if we apply Newton's method, we obtain the following iteration:

$$\theta^{k+1} = \theta^k - \left[J'(\theta^k)\right]^{-1} J(\theta^k),$$

where

$$J'(\theta) = 2(y_1 - f(\theta))\big(-f'(\theta)\big).$$

Let us graphically perform a few iterations on the above function:



- $J'(\theta^0) = 2\overbrace{(y_1 - f(\theta^0))}^{(-)}\overbrace{(-f'(\theta^0))}^{(+)} < 0 \quad \Rightarrow \quad \theta^1 > \theta^0$, etc.,
- $J'(\tilde{\theta}^0) = 2\overbrace{(y_1 - f(\tilde{\theta}^0))}^{(-)}\overbrace{(-f'(\tilde{\theta}^0))}^{(-)} > 0 \quad \Rightarrow \quad \tilde{\theta}^1 < \tilde{\theta}^0$, etc.,

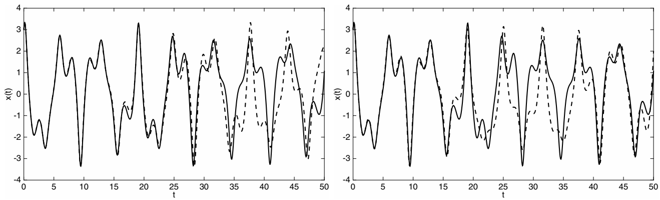
where in the last two formulas for J' we have indicated the sign (+/—) above each of the terms. We observe that in this simple case we have a highly unstable, oscillating behavior: at each step we move from positive to negative increments due to the changing sign of the gradient, and convergence is not possible. So what went wrong here? This behavior is not the fault of descent algorithms. It is a manifestation of the *inherent ill-posedness* of the problem. How to fix this problem has been the subject of much research over the past 50 years! Many remedies (fortunately) exist, such as explicit and implicit constrained optimizations, regularization, and penalization—these will be referred to, when necessary, in what follows. ∎

To further appreciate the complexity, let us briefly consider one of the remedies: Tykhonov regularization (TR)—see, for example, Engl et al. [1996] and Vogel [2002]. The idea here is to replace the ill-posed problem for $J(\theta) = |y_1 - f(\theta)|^2$ by a "nearby" problem for

$$J_\beta(\theta) = |y_1 - f(\theta)|^2 + \beta\,|\theta - \theta_0|^2,$$

where β is a suitably chosen regularization/penalization parameter. When it is done correctly, TR provides convexity and compactness,[6] thus ensuring the existence of a

[6]Convexity means that the function resembles a quadratic function, $f(x) = x^2$, with positive second derivative; compactness means that any infinite sequence of functions must get arbitrarily close to some function of the space.



(a) Initial error of 0.03% (b) Initial error of 0.06%

Figure 1.4. *Duffing's equation with small initial perturbations. Unperturbed (solid line) and perturbed (dashed line) trajectories.*

unique solution. However, even when done correctly, it *modifies the problem*, and new solutions may be far from the original ones. In addition, it is not trivial to regularize correctly or even to know if we have succeeded in finding a solution.

Example 1.5. The highly nonlinear Duffing's equation [Guckenheimer and Holmes, 1983],

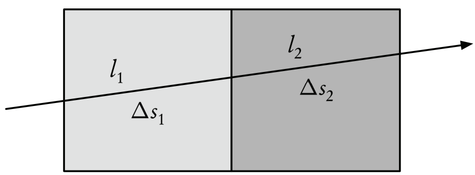
$$\ddot{x} + 0.05\dot{x} + x^3 = 7.5\cos t,$$

exhibits great *sensitivity to the initial conditions* (WP3). We will observe that two very closely spaced initial states can lead to a large discrepancy in the trajectories.

- Let $x(0) = 3$ and $\dot{x}(0) = 4$ be the true initial state.
- Introduce an error of 0.03% in the initial state—here we have an accurate forecast until $t = 35$ (see Figure 1.4(a)).
- Introduce an error of 0.06% in the initial state—here we only have an accurate forecast until $t = 20$ (see Figure 1.4(b)).

The initial perturbations are scarcely visible (and could result from measurement error), but the terminal states can differ considerably. ∎

Example 1.6. Seismic travel-time tomography provides an excellent example of non-uniqueness (WP2):



A signal seismic ray (or any other ray used in medical or other imaging) passes through a two-parameter block model.

- The *unknowns* are the two block slownesses (inverse of seismic velocity), $(\Delta s_1, \Delta s_2)$.
- The *data* consist of the observed travel time of the ray, Δt_1.
- The *model* is the linearized travel time equation

$$\Delta t_1 = l_1 \Delta s_1 + l_2 \Delta s_2,$$

  where l_j is the length of the ray in the jth block.

Clearly we have one equation for two unknowns, and hence there is *no unique solution*. In fact, for a given value of Δt_1, each time we fix Δs_1 we obtain a different Δs_2 (and vice versa). ∎

We hope the reader is convinced, based on these disarmingly simple examples, that inverse problems present a large number of potential pathologies. We can now proceed to examine the therapeutic tools that are at our disposal for attempting to "heal the patient."

1.5 ▪ DA methods

Definition 1.7. *DA is the approximation of the true state of some physical system at a given time by combining time-distributed observations with a dynamic model in an optimal way.*

DA can be classically approached in two ways: as variational DA and as statistical[7] DA—see Figure 1.5. They will be briefly presented here and then in far more detail in Chapters 2 and 3, respectively. Newer approaches are also becoming available: nudging methods, reduced methods, ensemble methods, and hybrid methods that combine variational and statistical[8] approaches. These are the subject of the chapters on advanced methods—see Part II.

In both we seek an optimal solution—statistically we will, for example, seek a solution with minimum variance, whereas variationally we will seek a solution that minimizes a suitable cost (or error) function. In fact, in certain special cases the two approaches are identical and provide exactly the same solution. However, the statistical approach, though often more complex and time-consuming, can provide a richer information structure: an average solution and some characteristics of its variability (probability distribution). Clearly a temperature forecast of "15°C for tomorrow" is much less informative than a forecast of "an *average* of 15°C with a *standard deviation* of 1.5°C," or "the *probability* of a temperature below 10°C is 0.125 for tomorrow," or, as is now quite common (on our smartphones), "there is a 60% chance of rain at 09h00 in New York." Ideally (and this will be our recommendation), one should attempt to combine the two into a single, "hybrid" approach. We can then take advantage of the relative rapidity and robustness of the variational approach, and at the same time obtain an information-rich solution thanks to the statistical/probabilistic approach. This is easily said but (as we will see) is not trivial to implement and will be highly problem dependent and probably computationally expensive. However, we can and

[7] Alternatives are "filtering" or "probabilistic."
[8] *ibid.*

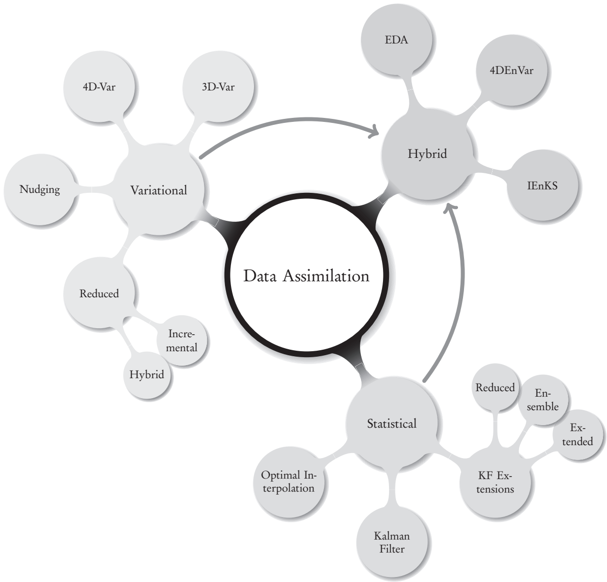


Figure 1.5. *DA methods: variational (Chapters* 2, 4, 5*); statistical (Chapters* 3, 5, 6*), hybrid (Chapter* 7*).*

will provide all the necessary tools (theoretical, algorithmic, numerical) and the indications for their implementation. It is worthwhile to point out that recently (since 2010) a number of the major weather-forecasting services across the world (Canada, France, United Kingdom, etc.) have started basing their operational forecasting systems on a new, hybrid approach, 4DEnVar (presented in Chapter 7), which combines 4D-Var (variational DA) with an ensemble, statistical approach.

To complete this introductory chapter, we will now briefly introduce and compare the two approaches of variational and statistical. Each one is subsequently treated, in far greater detail, in its own chapter—see Chapters 2 and 3, respectively.

1.5.1 ▪ Notation for DA and inverse problems

We begin by introducing the standard notation for DA problems as formalized by Ide et al. [1997]. We first consider a discrete model for the evolution of a physical (atmospheric, oceanic, mechanical, biological, etc.) system from time t_k to time t_{k+1}, described by a dynamic state equation

$$\mathbf{x}^{\mathrm{f}}(t_{k+1}) = \mathbf{M}_{k+1}\left[\mathbf{x}^{\mathrm{f}}(t_k)\right], \tag{1.1}$$

where $\mathbf{x}$ is the model's state vector of dimension n (see below for the definition of the superscripts) and $\mathbf{M}$ is the corresponding dynamics operator that can be time dependent. This operator usually results from a finite difference [Strikwerda, 2004] or finite element [Hughes, 1987] discretization of a (partial) differential equation (PDE). We associate an error covariance matrix $\mathbf{P}$ with the state $\mathbf{x}$ since the true state will differ from the simulated state (1.1) by random or systematic errors.

Observations, or measurements, at time t_k are defined by

$$\mathbf{y}^{\mathrm{o}} = \mathbf{H}_k\left[\mathbf{x}^t(t_k)\right] + \epsilon_k^{\mathrm{o}}, \tag{1.2}$$

where $\mathbf{H}$ is an *observation operator* that can be time dependent and ϵ^{o} is a *white noise process* with zero mean and associated covariance matrix $\mathbf{R}$ that describes instrument errors and representation errors due to the discretization. The observation vector, $\mathbf{y}_k^{\mathrm{o}} = \mathbf{y}^{\mathrm{o}}(t_k)$, has dimension p_k, which is usually much smaller than the state dimension, $p_k \ll n$.

Subscripts are used to denote the discrete time index, the corresponding spatial indices, or the vector with respect to which an error covariance matrix is defined; superscripts refer to the nature of the vectors/matrices in the DA process:

- "a" for *analysis*,
- "b" for *background* (or initial/first guess),
- "f" for *forecast*,
- "o" for *observation*, and
- "t" for the (unknown) *true* state.

Analysis is the process of approximating the true state of a physical system at a given time. Analysis is based on

- observational data,
- a model of the physical system, and
- background information on initial and boundary conditions.

An analysis that combines time-distributed observations and a dynamic model is called *data assimilation*.

Now let us introduce the continuous system. In fact, continuous time simplifies both the notation and the theoretical analysis of the problem. For a finite-dimensional system of ODEs, the equations (1.1)–(1.2) become

$$\dot{\mathbf{x}}^{\mathrm{f}} = \mathscr{M}(\mathbf{x}^{\mathrm{f}}, t),$$

and

$$\mathbf{y}^{\mathrm{o}}(t) = \mathscr{H}(\mathbf{x}^{\mathrm{t}}, t) + \epsilon,$$

where $\dot{()} = \mathrm{d}/\mathrm{d}t$ and $\mathscr{M}$ and $\mathscr{H}$ are nonlinear operators in continuous time for the model and the observation, respectively. This implies that $\mathbf{x}$, $\mathbf{y}$, and ϵ are also continuous-in-time functions. For PDEs, where there is in addition a dependence on space, attention must be paid to the function spaces, especially when performing variational analysis. Details will be provided in the next chapter. With a PDE model, the field (state) variable is commonly denoted by $\boldsymbol{u}(\mathbf{x}, t)$, where $\mathbf{x}$ represents the space variables (no longer the state variable as above!), and the model dynamics is now a nonlinear partial differential operator,

$$\mathcal{M} = \mathcal{M}\left[\partial_{\mathbf{x}}^{\alpha}, \boldsymbol{u}(\mathbf{x}, t), \mathbf{x}, t\right],$$

with $\partial_{\mathbf{x}}^{\alpha}$ denoting the partial derivatives with respect to the space variables of order up to $|\alpha| \leq m$, where m is usually equal to two and in general varies between one and four.

1.5.2 ▪ Statistical DA

Practical inverse problems and DA problems involve measured data. These data are inexact and are mixed with random noise. Only *statistical models* can provide rigorous, effective means for dealing with this measurement error. Let us begin with the following simple example.

1.5.2.1 ▪ A simple example

We want to estimate a *scalar* quantity, say the temperature or the ozone concentration at a fixed point in space. Suppose we have

- a model forecast, x^{b} (background, or a priori value), and
- a measured value, x^{o} (observation).

The simplest possible approach is to try a linear combination of the two,

$$x^{\mathrm{a}} = x^{\mathrm{b}} + w(x^{\mathrm{o}} - x^{\mathrm{b}}),$$

where x^{a} denotes the analysis that we seek and $0 \leq w \leq 1$ is a weight factor. We subtract the (always unknown) true state x^{t} from both sides,

$$x^{\mathrm{a}} - x^{\mathrm{t}} = x^{\mathrm{b}} - x^{\mathrm{t}} + w(x^{\mathrm{o}} - x^{\mathrm{t}} - x^{\mathrm{b}} + x^{\mathrm{t}}),$$

and, defining the three errors (analysis, background, observation) as

$$e^{\mathrm{a}} = x^{\mathrm{a}} - x^{\mathrm{t}}, \quad e^{\mathrm{b}} = x^{\mathrm{b}} - x^{\mathrm{t}}, \quad e^{\mathrm{o}} = x^{\mathrm{o}} - x^{\mathrm{t}},$$

we obtain

$$e^{\mathrm{a}} = e^{\mathrm{b}} + w(e^{\mathrm{o}} - e^{\mathrm{b}}) = w e^{\mathrm{o}} + (1 - w) e^{\mathrm{b}}.$$

If we have many realizations, we can take an ensemble average,[9] denoted by $\langle \cdot \rangle$:

$$\langle e^{\mathrm{a}} \rangle = \left\langle e^{\mathrm{b}} \right\rangle + w\left(\langle e^{\mathrm{o}} \rangle - \left\langle e^{\mathrm{b}} \right\rangle\right).$$

Now if these errors are centered (have zero mean, or the estimates of the true state are *unbiased*), then

$$\langle e^{\mathrm{a}} \rangle = 0$$

also. So we are logically led to look at the *variance* and demand that it be as small as possible. The variance is defined, using the above notation, as

$$\sigma^2 = \left\langle (e - \langle e \rangle)^2 \right\rangle.$$

[9] Please refer to a good textbook on probability and statistics for all the relevant definitions—e.g., DeGroot and Schervisch [2012].

So by taking variances of the error equation, and using the zero-mean property, we obtain

$$\sigma_{\mathrm{a}}^2 = \sigma_{\mathrm{b}}^2 + w^2 \left\langle \left(e^{\mathrm{o}} - e^{\mathrm{b}}\right)^2 \right\rangle + 2w \left\langle e^{\mathrm{b}} \left(e^{\mathrm{o}} - e^{\mathrm{b}}\right) \right\rangle.$$

This reduces to

$$\sigma_{\mathrm{a}}^2 = \sigma_{\mathrm{b}}^2 + w^2 \left(\sigma_{\mathrm{o}}^2 + \sigma_{\mathrm{b}}^2\right) - 2w\sigma_{\mathrm{b}}^2$$

if e^{o} and e^{b} are uncorrelated. Now, to compute a minimum, take the derivative of this last equation with respect to w and equate to zero,

$$0 = 2w \left(\sigma_{\mathrm{o}}^2 + \sigma_{\mathrm{b}}^2\right) - 2\sigma_{\mathrm{b}}^2,$$

where we have ignored all cross terms since the errors have been assumed to be independent. Finally, solving this last equation, we can write the optimal weight,

$$w_* = \frac{\sigma_{\mathrm{b}}^2}{\sigma_{\mathrm{o}}^2 + \sigma_{\mathrm{b}}^2} = \frac{1}{1 + \sigma_{\mathrm{o}}^2/\sigma_{\mathrm{b}}^2},$$

which, we notice, depends on the ratio of the observation and the background errors. Clearly $0 \le w_* \le 1$ and

- if the observation is perfect, $\sigma_{\mathrm{o}}^2 = 0$ and thus $w_* = 1$, the maximum weight;
- if the background is perfect, $\sigma_{\mathrm{b}}^2 = 0$ and $w_* = 0$, so the observation will not be taken into account.

We can now rewrite the analysis error variance as

$$\begin{aligned}
\sigma_{\mathrm{a}}^2 &= w_*^2 \sigma_{\mathrm{o}}^2 + (1 - w_*)^2 \sigma_{\mathrm{b}}^2 \\
&= \frac{\sigma_{\mathrm{b}}^2 \sigma_{\mathrm{o}}^2}{\sigma_{\mathrm{o}}^2 + \sigma_{\mathrm{b}}^2} \\
&= (1 - w_*)\sigma_{\mathrm{b}}^2 \\
&= \frac{1}{\sigma_{\mathrm{o}}^{-2} + \sigma_{\mathrm{b}}^{-2}},
\end{aligned}$$

where we suppose that $\sigma_{\mathrm{b}}^2,\ \sigma_{\mathrm{o}}^2 > 0$. In other words,

$$\frac{1}{\sigma_{\mathrm{a}}^2} = \frac{1}{\sigma_{\mathrm{o}}^2} + \frac{1}{\sigma_{\mathrm{b}}^2}.$$

Finally, the analysis equation becomes

$$x^{\mathrm{a}} = x^{\mathrm{b}} + \frac{1}{1 + \alpha}(x^{\mathrm{o}} - x^{\mathrm{b}}),$$

where $\alpha = \sigma_{\mathrm{o}}^2/\sigma_{\mathrm{b}}^2$. This is called the BLUE—best linear unbiased estimator—because it gives an unbiased, optimal weighting for a linear combination of two independent measurements.

We can isolate three special cases:

- If the observation is very accurate, $\sigma_{\mathrm{o}}^2 \ll \sigma_{\mathrm{b}}^2$, $\alpha \ll 1$, and thus $x^{\mathrm{a}} \approx x^{\mathrm{o}}$.

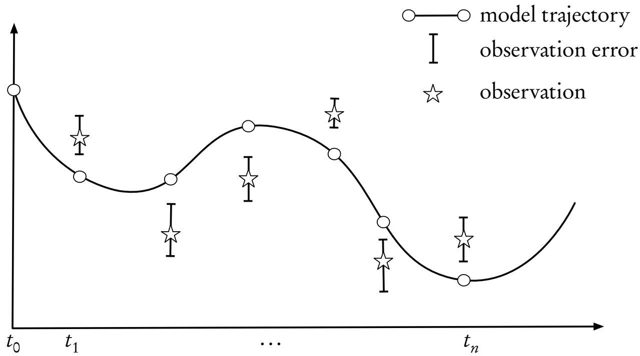


Figure 1.6. *Sequential assimilation. The x-axis denotes time; the y-axis is the assimilated variable.*

- If the background is accurate, $\alpha \gg 1$ and $x^{\mathrm{a}} \approx x^{\mathrm{b}}$.
- And, finally, if the observation and background variances are approximately equal, then $\alpha \approx 1$ and x^{a} is just the arithmetic average of x^{b} and x^{o}.

We can conclude that this simple, linear model does indeed capture the full range of possible solutions in a statistically rigorous manner, thus providing us with an "enriched" solution when compared with a nonprobabilistic, scalar response such as the arithmetic average of observation and background, which would correspond to only the last of the above three special cases.

1.5.2.2 ▪ The more general case: Introducing the Kalman filter

The above analysis of the temperature was based on a spatially dependent model. However, in general, the underlying process that we want to model will be time dependent. Within the significant toolbox of mathematical tools that can be used for statistical estimation from noisy sensor measurements, one of the most well-known and often-used tools is the Kalman filter (KF). The KF is named after Rudolph E. Kalman, who in 1960 published his famous paper describing a recursive solution to the *time-dependent* discrete-data linear filtering problem [Kalman, 1960].

We consider a dynamical system that evolves in time, and we seek to estimate a series of true states, $\mathbf{x}_k^{\mathrm{t}}$ (a sequence of random vectors), where discrete time is indexed by the letter k. These times are those when the observations or measurements are taken—see Figure 1.6. The assimilation starts with an unconstrained model trajectory from $t_0, t_1, \ldots, t_{k-1}, t_k, \ldots, t_n$ and aims to provide an optimal fit to the available observations/measurements given their uncertainties (error bars), depicted in the figure.

This situation is modeled by a stochastic system. We seek to estimate the state, $\mathbf{x} \in \mathbb{R}^n$, of a discrete-time dynamic process that is governed by the linear stochastic difference equation

$$\mathbf{x}_{k+1} = \mathbf{M}_{k+1}[\mathbf{x}_k] + \mathbf{w}_k,$$

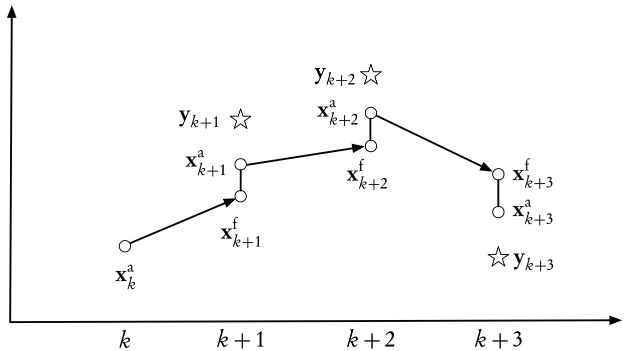


Figure 1.7. *Sequential assimilation scheme for the KF. The x-axis denotes time; the y-axis is the assimilated variable. We assume scalar variables.*

with a measurement/observation $\mathbf{y} \in \mathbb{R}^m$ defined by

$$\mathbf{y}_k = \mathbf{H}_k[\mathbf{x}_k] + \mathbf{v}_k.$$

The random vectors $\mathbf{w}_k$ and $\mathbf{v}_k$ represent the process/modeling and measurement/observation errors, respectively. They are assumed[10] to be independent, white, and with Gaussian/normal probability distributions,

$$\mathbf{w}_k \sim \mathcal{N}(0, \mathbf{Q}_k), \tag{1.3}$$
$$\mathbf{v}_k \sim \mathcal{N}(0, \mathbf{R}_k), \tag{1.4}$$

where $\mathbf{Q}$ and $\mathbf{R}$ are the covariance matrices (assumed known) and can in general be time dependent.

We can now set up a sequential DA scheme. The typical assimilation scheme is made up of two major steps: a *prediction/forecast* step and a *correction/analysis* step. At time t_k we have the result of a previous forecast, $\mathbf{x}^{\mathrm{f}}_k$ (the analogue of the background state $\mathbf{x}^{\mathrm{b}}_k$), and the result of an ensemble of observations in $\mathbf{y}_k$. Based on these two vectors, we perform an analysis that produces $\mathbf{x}^{\mathrm{a}}_k$. We then use the evolution model, which is usually (partial) differential equation–based, to obtain a prediction of the state at time t_{k+1}. The result of the forecast is denoted $\mathbf{x}^{\mathrm{f}}_{k+1}$ and becomes the background (or initial guess) for the next time step. This process is summarized in Figure 1.7.

We can now define forecast (a priori) and analysis (a posteriori) estimate errors in the same way as above for the scalar case, with their respective error covariance matrices, which generalize the variances used before, since we are now dealing with vector quantities. The goal of the KF is to compute an optimal a posteriori estimate, $\mathbf{x}^{\mathrm{a}}_k$, that is a linear combination of an a priori estimate, $\mathbf{x}^{\mathrm{f}}_k$, and a weighted difference between the actual measurement, $\mathbf{y}_k$, and the measurement prediction, $\mathbf{H}_k\left[\mathbf{x}^{\mathrm{f}}_k\right]$. This

[10]These assumptions are necessary in the KF framework. For real problems, they must often be relaxed.

is none other than the BLUE that we saw in the example above. The filter must be of the form

$$\mathbf{x}_k^{\mathrm{a}} = \mathbf{x}_k^{\mathrm{f}} + \mathbf{K}_k \left(\mathbf{y}_k - \mathbf{H}_k \mathbf{x}_k^{\mathrm{f}} \right), \tag{1.5}$$

where $\mathbf{K}$ is the Kalman gain. The difference $\left(\mathbf{y}_k - \mathbf{H}_k \mathbf{x}_k^{\mathrm{f}}\right)$ is called the *innovation* and reflects the discrepancy between the actual and the predicted measurements at time t_k. Note that for generality, the matrices are shown with a time dependence. Often this is not the case, and the subscripts k can then be dropped. The *Kalman gain* matrix, $\mathbf{K}$, is chosen to minimize the a posteriori error covariance equation. This is straightforward to compute: substitute (1.5) into the definition of the analysis error, then substitute in the error covariance equation, take the derivative of the trace of the result with respect to $\mathbf{K}$, set the result equal to zero, and, finally, solve for the optimal gain $\mathbf{K}$. The resulting optimal gain matrix is

$$\mathbf{K}_k = \mathbf{P}_k^{\mathrm{f}} \mathbf{H}^{\mathrm{T}} \left(\mathbf{H} \mathbf{P}_k^{\mathrm{f}} \mathbf{H}^{\mathrm{T}} + \mathbf{R} \right)^{-1},$$

where $\mathbf{P}_k^{\mathrm{f}}$ is the forecast error covariance matrix. Full details of this computation, as well as numerous examples, are provided in Chapter 3, where we will also generalize the approach to more realistic cases.

1.5.3 ▪ Variational DA

Unlike sequential/statistical assimilation (which emanates from estimation theory), variational assimilation is based on optimal control theory [Kwakernaak and Sivan, 1972; Friedland, 1986; Gelb, 1974; Tröltzsch, 2010], itself derived from the *calculus of variations*. The analyzed state is not defined as the one that maximizes a certain PDF, but as the one that *minimizes a cost function*. The minimization requires numerical optimization techniques. These techniques can rely on the *gradient* of the cost function, and this gradient will be obtained here with the aid of *adjoint methods*.

1.5.3.1 ▪ Adjoint methods: An introduction

All descent-based optimization methods require the computation of the gradient, ∇J, of a cost function, J. If the dependence of J on the control variables is complex or indirect, this computation can be very difficult. Numerically, we can always manage by computing finite increments, but this would have to be done in all possible perturbation directions. We thus need to find a less expensive way to compute the gradient. This will be provided by the *calculus of variations* and the *adjoint approach*.

A basic example: Let us consider a classical inverse problem known as a parameter identification problem, based on the ODE (of convection-diffusion type)

$$\begin{cases} -bu''(x) + cu'(x) = f(x), & 0 < x < 1, \\ u(0) = 0, \ u(1) = 0, \end{cases} \tag{1.6}$$

where $'$ depicts the derivative with respect to x, f is a given function, and b and c are *unknown (constant) parameters* that we seek to identify using observations of $u(x)$ on the interval $[0,1]$. The mismatch (or least-squares error) cost function is then

$$J(b,c) = \frac{1}{2} \int_0^1 \left(u(x) - u^{\mathrm{o}}(x)\right)^2 \mathrm{d}x,$$

where u^{o} is the observational data. The gradient of J can be calculated by introducing the *tangent linear model* (TLM). Perturbing the cost function by a small perturbation[11] in the direction α, with respect to the two parameters, b and c, gives

$$J(b+\alpha\delta b, c+\alpha\delta c) - J(b,c) = \frac{1}{2}\int_0^1 (\tilde{u}-u^{\mathrm{o}})^2 - (u-u^{\mathrm{o}})^2 \,\mathrm{d}x,$$

where $\tilde{u} = u_{b+\alpha\delta b, c+\alpha\delta c}$ is the perturbed solution and $u = u_{b,c}$ is the unperturbed one. Now we divide by α and pass to the limit $\alpha \to 0$ to obtain the directional derivative (with respect to the parameters, in the direction of the perturbations),

$$\hat{J}[b,c](\delta b, \delta c) = \int_0^1 (u-u^{\mathrm{o}})\,\hat{u}\,\mathrm{d}x, \tag{1.7}$$

where we have defined

$$\hat{u} = \lim_{\alpha\to 0} \frac{\tilde{u}-u}{\alpha}.$$

Then, passing to the limit in equation (1.6), we can define the TLM

$$\begin{cases} -b\hat{u}'' + c\hat{u}' = (\delta b)u'' - (\delta c)u', \\ \hat{u}(0) = 0, \ \hat{u}(1) = 0. \end{cases} \tag{1.8}$$

We would like to reformulate the directional derivative (1.7) to obtain a calculable expression for the gradient. For this we introduce the adjoint variable, p, satisfying *the adjoint model*

$$\begin{cases} -bp'' - cp' = (u-u^{\mathrm{o}}), \\ p(0) = 0, \ p(1) = 0. \end{cases} \tag{1.9}$$

Multiplying the TLM by this new variable, p, and integrating by parts enables us to finally write an explicit expression (see Chapter 2 for the complete derivation) for the gradient based on (1.7),

$$\nabla J(b,c) = \left(\int_0^1 pu''\,\mathrm{d}x, -\int_0^1 pu'\,\mathrm{d}x\right)^{\mathrm{T}},$$

or, separating the two components,

$$\nabla_b J(b,c) = \int_0^1 pu''\,\mathrm{d}x,$$

$$\nabla_c J(b,c) = -\int_0^1 pu'\,\mathrm{d}x.$$

Thus, for the additional cost of solving the adjoint model (1.9), we can compute the gradient of the cost function with respect to either one, or both, of the unknown parameters. It is now a relatively easy task to find (numerically) the optimal values of b and c that minimize J by using a suitable descent algorithm. This important example is fully developed in Section 2.3.2, where all the steps are explicitly justified.

Note that this method generalizes to (linear and nonlinear) time-dependent PDEs and to inverse problems where we seek to identify the *initial conditions*. This latter problem is exactly the 4D-Var problem of DA. All of this will be amply described in Chapter 2.

[11] The exact properties of the perturbation will be fully explained in Chapter 2.

Algorithm 1.1 Iterative 3D-Var (in its simplest form).

$k = 0$, $x = x_0$
while $||\nabla J|| > \epsilon$ or $j \leq j_{\max}$
  compute J
  compute ∇J
  gradient descent and update of x_{j+1}
  $j = j + 1$
end

1.5.3.2 ▪ 3D-Var

We have seen above that the BLUE requires the computation of an optimal gain matrix. We will show (in Chapters 2 and 3) that the optimal gain takes the form

$$\mathbf{K} = \mathbf{B}\mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{B}\mathbf{H}^{\mathrm{T}} + \mathbf{R})^{-1}$$

to obtain an analyzed state,

$$\mathbf{x}^{\mathrm{a}} = \mathbf{x}^{\mathrm{b}} + \mathbf{K}(\mathbf{y} - \mathbf{H}(\mathbf{x}^{\mathrm{b}})),$$

that minimizes what is known as the 3D-Var cost function,

$$J(x) = \frac{1}{2}\left(\mathbf{x} - \mathbf{x}^{\mathrm{b}}\right)^{\mathrm{T}} \mathbf{B}^{-1} \left(\mathbf{x} - \mathbf{x}^{\mathrm{b}}\right) + \frac{1}{2}(\mathbf{H}\mathbf{x} - \mathbf{y})^{\mathrm{T}} \mathbf{R}^{-1} (\mathbf{H}\mathbf{x} - \mathbf{y}), \tag{1.10}$$

where $\mathbf{R}$ and $\mathbf{B}$ (also denoted $\mathbf{P}^{\mathrm{f}}$) are the observation and background error covariance matrices, respectively. But the matrices involved in this calculation are often neither storable in memory nor manipulable because of their very large dimensions. The basic idea of variational methods is to overcome these difficulties by attempting to directly minimize the cost function, J. This minimization can be achieved, for inverse problems in general (and for DA in particular), by a combination of (1) an adjoint approach for the computation of the gradient of the cost function with (2) a descent algorithm in the direction of the gradient. For DA problems where there is no time dependence, the adjoint is not necessary and the approach is named 3D-Var, whereas for time-dependent problems we use the 4D-Var approach.

We recall that $\mathbf{R}$ and $\mathbf{B}$ are the observation and background error covariance matrices, respectively. When the observation operator $\mathbf{H}$ is linear, the gradient of J in (1.10) is given by

$$\nabla J = \mathbf{B}^{-1}\left(\mathbf{x} - \mathbf{x}^{\mathrm{b}}\right) - \mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}(\mathbf{y} - \mathbf{H}\mathbf{x}).$$

In the iterative 3D-Var Algorithm 1.1 we use as a stopping criterion the fact that ∇J is small or that the maximum number of iterations, $j_{\max}$, is reached.

1.5.3.3 ▪ A simple example of 3D-Var

We seek two temperatures, x_1 and x_2, in London and Paris. The climatologist gives us an initial guess (based on climate records) of $x^b = (10\ 5)^{\mathrm{T}}$, with background error covariance matrix

$$\mathbf{B} = \begin{pmatrix} 1 & 0.25 \\ 0.25 & 1 \end{pmatrix}.$$

Algorithm 1.2 4D-Var in its basic form

$j = 0$, $\mathbf{x} = \mathbf{x}_0$
while $||\nabla J|| > \epsilon$ **or** $j \le j_{\max}$
  (1) compute J with the direct model M and H
  (2) compute ∇J with adjoint model $\mathbf{M}^{\mathrm{T}}$ and $\mathbf{H}^{\mathrm{T}}$ (reverse mode)
  gradient descent and update of $\mathbf{x}_{j+1}$
  $j = j + 1$
end

We observe $\mathbf{y}^{\mathrm{o}} = 4$ in Paris, which implies that $\mathbf{H} = (0\ 1)$, with an observation error variance $\mathbf{R} = (0.25)$. We can now write the cost function (1.10) as

$$\begin{aligned}
J(\mathbf{x}) &= \begin{pmatrix} x_1 - 10 & x_2 - 5 \end{pmatrix} \begin{pmatrix} 1 & 0.25 \\ 0.25 & 1 \end{pmatrix}^{-1} \begin{pmatrix} x_1 - 10 \\ x_2 - 5 \end{pmatrix} + R^{-1}(x_2 - 4)^2 \\
&= \begin{pmatrix} x_1 - 10 & x_2 - 5 \end{pmatrix} \frac{16}{15} \begin{pmatrix} 1 & -0.25 \\ -0.25 & 1 \end{pmatrix} \begin{pmatrix} x_1 - 10 \\ x_2 - 5 \end{pmatrix} + 4(x_2 - 4)^2 \\
&= \frac{16}{15}\left((x_1 - 10)^2 + (x_2 - 5)^2 - 0.5(x_1 - 10)(x_2 - 5)\right) + 4(x_2 - 4)^2 \\
&= \frac{16}{15}\left(x_1^2 - 17.5x_1 + 100 + x_2^2 - 5x_2 - 0.5x_1x_2\right) + 4(x_2^2 - 8x + 16),
\end{aligned}$$

and its gradient can be easily seen to be

$$\nabla J(\mathbf{x}) = \frac{16}{15} \begin{pmatrix} 2x_1 - 0.5x_2 - 17.5 \\ 2x_2 - 5 - 0.5x_1 + \frac{15}{4}(2x_2 - 8) \end{pmatrix} = \frac{1}{15} \begin{pmatrix} 32x_1 - 8x_2 - 280 \\ -8x_1 + 152x_2 - 560 \end{pmatrix}.$$

The minimum is obtained for $\nabla J(\mathbf{x}) = 0$, which yields

$$x_1 = 9.8, \quad x_2 = 4.2.$$

This is an optimal estimate of the two temperatures, given the background and observation errors.

1.5.3.4 ▪ 4D-Var

In 4D-Var,[12] the cost function is still expressed in terms of the initial state, $\mathbf{x}_0$, but it will include the model because the observation $\mathbf{y}_i^{\mathrm{o}}$ at time i is compared to $\mathbf{H}_i(\mathbf{x}_i)$, where $\mathbf{x}_i$ is the state at time i initialized by $\mathbf{x}_0$ and the adjoint is not simply the transpose of a matrix but also the "transpose" of the model/operator dynamics. To compute this will require the use of a more general adjoint theory, which is introduced just after the following example and fully explained in Chapter 2.

In step (1) of Algorithm 1.2, we use the equations

$$\mathbf{d}_k = \mathbf{y}_k^{\mathrm{o}} - \mathbf{H}_k \mathbf{M}_k \mathbf{M}_{k-1} \ldots \mathbf{M}_2 \mathbf{M}_1 \mathbf{x}$$

and

$$J(\mathbf{x}) = \frac{1}{2}\left(\mathbf{x} - \mathbf{x}^{\mathrm{b}}\right)^{\mathrm{T}} \mathbf{B}^{-1} \left(\mathbf{x} - \mathbf{x}^{\mathrm{b}}\right) + \sum_{i=0}^{j} \mathbf{d}_i^{\mathrm{T}} \mathbf{R}_i^{-1} \mathbf{d}_i.$$

[12] The 4 refers to the additional time dimension.

In step (2), we use

$$\nabla J(\mathbf{x}) = \mathbf{B}^{-1}\left(\mathbf{x}-\mathbf{x}^{\mathrm{b}}\right) - \\ \left[\mathbf{H}_0^{\mathrm{T}}\mathbf{R}_0^{-1}\mathbf{d}_0 + \mathbf{M}_1^{\mathrm{T}}\left[\mathbf{H}_1^{\mathrm{T}}\mathbf{R}_1^{-1}\mathbf{d}_1 + \mathbf{M}_2^{\mathrm{T}}\left[\mathbf{H}_2^{\mathrm{T}}\mathbf{R}_2^{-1}\mathbf{d}_2 + \cdots + \mathbf{M}_j^{\mathrm{T}}\mathbf{H}_j^{\mathrm{T}}\mathbf{R}_j^{-1}\mathbf{d}_j\right]\right]\right],$$

where we have assumed that $\mathbf{H}$ and $\mathbf{M}$ are *linear*.

1.6 ▪ Some practical aspects of DA and inverse problems

In this brief section we point out some important practical considerations. It should now be clear that there are four basic ingredients in any inverse or DA problem:

1. observation or measured data;
2. a forward or direct model of the real-world context;
3. a backward or adjoint model in the variational case and a probabilistic framework in the statistical case; and
4. an optimization cycle.

But where does one start? The traditional approach, often employed in mathematical and numerical modeling, is to begin with some simplified, or at least well-known, situation. Once the above four items have been successfully implemented and tested on this instance, we then proceed to take into account more and more reality in the form of real data, more realistic models, more robust optimization procedures, etc. In other words, we introduce uncertainty, but into a system where we at least control some of the aspects.

1.6.1 ▪ Twin experiments

Twin experiments, or synthetic runs, are a basic and indispensable tool for all inverse problems. To evaluate the performance of a DA system we invariably begin with the following methodology:

1. Fix all parameters and unknowns and define a reference trajectory, obtained from a run of the direct model—call this the "truth."
2. Derive a set of (synthetic) measurements, or background data, from this "true" run.
3. Optionally, perturb these observations to generate a more realistic observed state.
4. Run the DA or inverse problem algorithm, starting from an initial guess (different from the "true" initial state used above), using the synthetic observations.
5. Evaluate the performance, modify the model/algorithm/observations, and cycle back to step 1.

Twin experiments thus provide a well-structured methodological framework. Within this framework we can perform different "stress tests" of our system. We can modify the observation network, increase or decrease (even switch off) the uncertainty, test the robustness of the optimization method, and even modify the model. In fact, these experiments can be performed on the full physical model or on some simpler (or reduced-order) model.

1.6.2 ▪ Toy models and other simplifications

Toy models are, by definition, simplified models that we can play with, yes, but these are of course "serious games." In certain complex physical contexts, of which meteorology is a famous example, we have well-established toy models, often of increasing complexity. These can be substituted for the real model, whose computational complexity is often too large, and provide a cheaper test-bed.

Some well-known examples of toy models are

- Lorenz models—see Lorenz [1963]—which are used as an avatar for weather simulations;
- various harmonic oscillators that are used to simulate dynamic systems; and
- famous examples such as the Ising model in physics, the Lotka–Volterra model in life sciences, and the Schelling model in social sciences;

See Marzuoli [2008] for a more general discussion.

1.7 ▪ To go further: Additional comments and references

- Examples of inverse problems: 11 examples can be found in Keller [1966] and 16 in Kirsch [1996].
- As the reader may have observed, the formulation and solution of DA and inverse problems require a wide range of tools and competencies in functional analysis, probability and statistics, variational calculus, numerical optimization, numerical approximation of (partial) differential equations, and stochastic simulation. This monograph will not provide all of this, so the reader must resort to other sources for the necessary background "tools." A few bibliographic recommendations are
  - DeGroot and Schervisch [2012] for probability and statistics;
  - Courant and Hilbert [1989a] for variational calculus;
  - Nocedal and Wright [2006] for numerical optimization;
  - Kreyszig [1978] and Reed and Simon [1980] for functional analysis;
  - Strikwerda [2004] for finite difference methods;
  - Hughes [1987] and Zienkiewicz and Taylor [2000] for finite element methods;
  - Press et al. [2007] for stochastic simulation;
  - Quarteroni et al. [2007] for basic numerical analysis (integration, solution of ODEs, etc.); and
  - Golub and van Loan [2013] for numerical linear algebra.

Chapter 2

Optimal control and variational data assimilation

2.1 ▪ Introduction

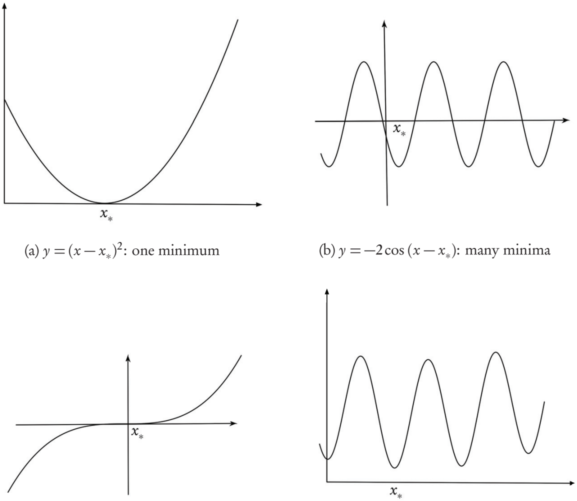
Unlike sequential assimilation (which emanates from statistical estimation theory and will be the subject of the next chapter), variational assimilation is based on optimal control theory.[13] The analyzed state is not defined as the one that maximizes a certain probability density function (PDF), but as the one that *minimizes a cost function.* The minimization requires numerical optimization techniques. These techniques can rely on the *gradient* of the cost function, and this gradient will be obtained here with the aid of *adjoint methods*.

The theory of adjoint operators, coming out of functional analysis, is presented in Kreyszig [1978] and Reed and Simon [1980]. A special case is that of matrix systems, which are simply the finite dimensional operator case. The necessary ingredients of optimization theory are described in Nocedal and Wright [2006].

In this chapter, we will show that the adjoint approach is an extremely versatile tool for solving a very wide range of inverse problems—DA problems included. This will be illustrated via a sequence of explicitly derived examples, from simple cases to quite complex nonlinear cases. We will show that once the theoretical adjoint technique is understood and mastered, almost any model equation can be treated and almost any inverse problem can be solved (at least theoretically). We will not neglect the practical implementation aspects that are vital for any real-world, concrete application. These will be treated in quite some detail since they are often the crux of the matter—that is, the crucial steps for succeeding in solving DA and inverse problems.

The chapter begins with a presentation of the *calculus of variations*. This theory, together with the concept of *ill-posedness*, is the veritable basis of inverse problems and DA, and its mastery is vital for formulating, understanding, and solving real-world problems. We then consider adjoint methods, starting from a general setting and moving on through a series of parameter identification problems—all of these in a differential equation (infinite-dimensional) setting. Thereafter, we study finite-dimensional cases, which lead naturally to the comparison of continuous and discrete adjoints. It is here that we will introduce *automatic differentiation*, which generalizes the calculation of the adjoint to intractable, complex cases. After all this preparation, we will be ready to study the two major variational DA approaches: 3D-Var and 4D-Var. Once

[13] This is basically true; however, 4D-Var applied to a chaotic model is in fact a sequential algorithm.



(a) $y = (x - x_*)^2$: one minimum

(b) $y = -2\cos(x - x_*)$: many minima

(c) $y = x^3$: saddle point

(d) $y = 0.015(x - x_*)^2 - 2\cos(x - x_*)$: one global minimum, many local minima

Figure 2.1. *A variety of local extrema, denoted by* x_*.

completed, we present a few numerical examples. We end the chapter with a brief description of a selection of advanced topics: preconditioning, reduced-order methods, and error covariance modeling. These will be expanded upon in Chapter 5.

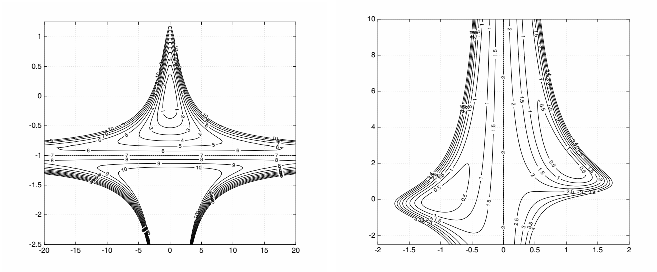
2.2 ▪ The calculus of variations

The calculus of variations is, to quote Courant and Hilbert [1989a], one of the "very central fields of analysis." It is also the central tool of variational optimization and DA, since it generalizes the theory of maximization and minimization. If we understand the workings of this theory well, we will be able to understand variational DA and inverse problems in a much deeper way and thus avoid falling into phenomenological traps. By this we mean that when, for a particular problem, things go wrong, we will have the theoretical and methodological distance/knowledge that is vital for finding a way around, over, or through the barrier (be it in the formulation or in the solution of the problem). Dear reader, please bear with us for a while, as we review together this very important theoretical tool.

To start out from a solid and well-understood setting, let us consider the basic theory of optimization[14] (maximization or minimization) of a continuous function

$$f(x, y, \ldots) \colon \mathbb{R}^d \to \mathbb{R}$$

[14] An excellent reference for a more complete treatment is the book of Nocedal and Wright [2006].



(a) $z = F_1(x,y)$.

(b) $z = F_2(x,y)$.

Figure 2.2. *Counterexamples for local extrema in* $\mathbb{R}^2$.

in a closed region Ω. We seek a point $\boldsymbol{x}_* = (x_*, y_*, \ldots) \in \mathbb{R}^d$ in Ω for which f has an extremum (maximum or minimum) in the vicinity of $\boldsymbol{x}_*$ (what is known as a *local* extremum—see Figure 2.1). A classical theorem of Weierstrass guarantees the existence of such an object.

Theorem 2.1. *Every continuous function in a bounded domain attains a maximal and a minimal value inside the domain or on its boundary.*

If f is differentiable in Ω and if $\boldsymbol{x}_*$ is an interior point, then the first derivatives of f, with respect to each of its variables, vanish at $\boldsymbol{x}_*$—we say that the gradient of f is equal to zero. However, this *necessary condition* is by no means sufficient because of the possible existence of saddle points. For example, see Figure 2.1c, where $f(x) = x^3$ at $\boldsymbol{x}_* = 0$. Moreover, as soon as we pass from $\mathbb{R}$ to even $\mathbb{R}^2$, we lose the simple Rolle's and intermediate-value theorem [Wikipedia, 2015c] results, and very counterintuitive things can happen. This is exhibited by the following two examples (see Figure 2.2):

- $F_1(x,y) = x^2(1+y)^3 + 7y^2$ has a single critical point—the gradient of F_1 vanishes at $(0,0)$—which is a local minimum, but not a global one. This cannot happen in one dimension because of Rolle's theorem. Note also that $(0,0)$ is *not* a saddle point.

- $F_2(x,y) = (x^2y - x - 1)^2 + (x^2 - 1)^2$ has exactly two critical points, at $(1,2)$ and $(-1,0)$, both of which are local minima—again, an impossibility in one dimension where we would have at least one additional critical point between the two.

A *sufficient condition* requires that the second derivative of the function exist and be positive. In this case the point $\boldsymbol{x}_*$ is indeed a local minimizer. The only case where things are simple is when we have smooth, convex functions [Wikipedia, 2015d]—in this case any local minimizer is a global minimizer, and, in fact, any stationary point is a global minimizer. However, in real problems, these conditions are (almost) never satisfied even though we will in some sense "convexify" our assimilation and inverse problems—see below.

Now, if the variables are subject to n constraints of the form $g_j(x,y,\ldots) = 0$ for $j = 1,\ldots,n$, then by introducing *Lagrange multipliers* we obtain the necessary conditions for an extremum. For this, we define an augmented function,

$$F = f + \sum_{j=1}^{n} \lambda_j g_j,$$

and write down the necessary conditions

$$\frac{\partial F}{\partial x} = 0, \quad \frac{\partial F}{\partial y} = 0, \quad \dots \quad (d \text{ equations}),$$

$$\frac{\partial F}{\partial \lambda_1} = g_1 = 0, \quad \dots, \quad \frac{\partial F}{\partial \lambda_n} = g_n = 0 \quad (n \text{ equations}),$$

which gives a system of equations (m equations in m unknowns, where $m = d + n$) that are then solved for $\boldsymbol{x}_* \in \mathbb{R}^d$ and λ_j, $j = 1, \dots, n$.

We will now generalize the above to finding the extrema of functionals.[15] The domain of definition becomes a *space of admissible functions*[16] in which we will seek the extremal member.

> The calculus of variations deals with the following problem: find the maximum or minimum of a functional, over the given domain of admissible functions, for which the functional attains the extremum with respect to all argument functions in a small neighborhood of the extremal argument function.

But we now need to generalize our definition of the vicinity (or neighborhood) of a function. Moreover, the problem may not have a solution because of the difficulty of choosing the set of admissible functions to be compact, which means that any infinite sequence of functions must get arbitrarily close to some function of the space—an *accumulation point*. However, if we restrict ourselves to necessary conditions (the vanishing of the first "derivative"), then the existence issue can be left open.

2.2.1 ▪ Necessary conditions for functional minimization

What follows dates from Euler and Lagrange in the 18th century. Their idea was to solve minimization problems by means of a general variational approach that reduces them to the solution of differential equations. Their approach is still completely valid today and provides us with a solid theoretical basis for the variational solution of inverse and DA problems.

We begin, as in Courant and Hilbert [1989a], with the "simplest problem" of the calculus of variations, where we seek a real-valued function $y(x)$, defined on the interval $[a, b]$, that minimizes the integral cost function

$$J[y] = \int_a^b F(x, y, y')\,\mathrm{d}x. \tag{2.1}$$

A classical example is to find the curve, $y(x)$, known as the *geodesic*, that minimizes the length between $x = a$ and $x = b$, in which case $F = \sqrt{1 + (y')^2}$. We will assume here

[15] These are functions of functions, rather than of scalar variables.

[16] An example of such a space is the space of continuous functions with continuous first derivatives, usually denoted $C^1(\Omega)$, where Ω is the domain of definition of each member function.

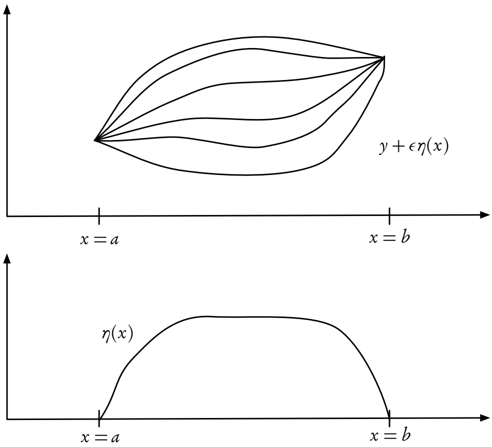


Figure 2.3. *Curve $\eta(x)$ and admissible functions $y+\epsilon\eta(x)$.*

that the functions F and y possess all the necessary smoothness (i.e., they are differentiable up to any order required). Suppose that y_* is the extremal function that gives the minimum value to J. This means that in a sufficiently small neighborhood (recall our difficulties with multiple extremal points—the same thing occurs for functions) of the function $y_*(x)$ the integral (2.1) is smallest when $y = y_*(x)$. To quantify this, we will define the *variation*, δy, of the function y. Let $\eta(x)$ be an arbitrary, smooth function defined on $[a,b]$ and vanishing at the endpoints, i.e., $\eta(a) = \eta(b) = 0$. We construct the new function $\widetilde{y} = y + \delta y$, with $\delta y = \epsilon\eta(x)$, where ϵ is an arbitrary (small) parameter—see Figure 2.3. This implies that all the functions $\widetilde{y}$ will lie in an arbitrarily small neighborhood of y_*, and thus the cost function $J[\tilde{y}]$, taken as a function of ϵ, must have its minimum at $\epsilon = 0$, and its "derivative" with respect to ϵ must vanish there. If we now take the integral

$$J(\epsilon) = \int_a^b F(x, y+\epsilon\eta, y' + \epsilon\eta')\,\mathrm{d}x$$

and differentiate it with respect to ϵ, we obtain at $\epsilon = 0$

$$J'(0) = \int_a^b \left(F_y\eta + F_{y'}\eta'\right)\mathrm{d}x = 0,$$

where the subscripts denote partial differentiation. Integrating the second term by parts and using the boundary values of η, we get

$$\int_a^b \eta\left(F_y - \frac{\mathrm{d}}{\mathrm{d}x}F_{y'}\right)\mathrm{d}x = 0,$$

and since this must hold for all functions η, we can invoke the following fundamental lemma.

Lemma 2.2. (Fundamental lemma of the calculus of variations). *If* $\int_{x_0}^{x_1} \eta(x)\phi(x)dx = 0$, *with* $\phi(x)$ *continuous, holds for all functions* $\eta(x)$ *vanishing on the boundary and continuous with two continuous derivatives, then* $\phi(x) = 0$ *identically.*

Using this lemma, we conclude that the necessary condition, known as the *Euler–Lagrange equation,* is

$$F_y - \frac{\mathrm{d}}{\mathrm{d}x} F_{y'} = 0. \tag{2.2}$$

We can expand the second term of this expression, obtaining (we have flipped the signs of the terms)

$$y'' F_{y'y'} + y' F_{y'y} + F_{y'x} - F_y = 0. \tag{2.3}$$

If we want to solve this equation for the highest-order derivative, we must require that

$$F_{y'y'} \neq 0,$$

which is known as the *Legendre condition* and plays the role of the second derivative in the optimization of scalar functions, by providing a *sufficient* condition for the existence of a maximum or minimum. We will invoke this later when we discuss the study of sensitivities with respect to individual parameters in multiparameter identification problems.

Example 2.3. Let us apply the Euler–Lagrange necessary condition to the geodesic problem with

$$J[y] = \int_a^b F(x, y, y')\,\mathrm{d}x = \int_a^b \sqrt{1 + (y')^2}\,\mathrm{d}x.$$

Note that F has no explicit dependence on the variables y and x. The partial derivatives of F are

$$F_y = 0, \quad F_{y'} = \frac{y'}{\sqrt{1+(y')^2}}, \quad F_{y'y} = 0, \quad F_{y'y'} = \left(1 + \left(y'\right)^2\right)^{-3/2}, \; F_{y'x} = 0.$$

Substituting in (2.3), we get $y'' = 0$, which implies that

$$y = cx + d,$$

and, unsurprisingly, we indeed find that a straight line is the shortest distance between two points in the Cartesian x-y plane. This result can be extended to finding the geodesic on a given surface by simply substituting parametric equations for x and y. ∎

2.2.2 ▪ Generalizations

The Euler–Lagrange equation (2.2) can be readily extended to functions of several variables and to higher derivatives. This will then lead to the generalization of the Euler–Lagrange equations that we will need in what follows.

In fact, we can consider a more general family of admissible functions, $y(x,\epsilon)$, with

$$\eta(x) = \frac{\partial}{\partial \epsilon} y(x,\epsilon)\bigg|_{\epsilon=0}.$$

Recall that we defined the variation of y ($\tilde{y} = y + \delta y$) as $\delta y = \epsilon\eta$. This leads to an analogous definition of the (first) *variation* of J,

$$\begin{aligned}
\delta J = \epsilon J'(0) &= \epsilon \int_{x_0}^{x_1} \left(F_y \eta + F_{y'}\eta'\right) \mathrm{d}x \\
&= \epsilon \int_{x_0}^{x_1} \left(F_y - \frac{\mathrm{d}}{\mathrm{d}x} F_{y'}\right) \eta \, \mathrm{d}x + \left[\epsilon F_{y'}\eta\right]_{x=x_0}^{x=x_1} \\
&= \int_{x_0}^{x_1} [F]_y \, \delta y \, \mathrm{d}x + \left[F_{y'}\delta y\right]_{x=x_0}^{x=x_1},
\end{aligned}$$

where

$$[F]_y \doteq \left(F_y - \frac{\mathrm{d}}{\mathrm{d}x} F_{y'}\right)$$

is the *variational derivative* of F with respect to y. We conclude that the *necessary condition for an extremum* is that the first variation of J be equal to zero for all admissible $y + \delta y$. The curves for which δJ vanishes are called *stationary functions*. Numerous examples can be found in Courant and Hilbert [1989a].

Let us now see how this fundamental result generalizes to other cases. If F depends on higher derivatives of y (say, up to order n), then

$$J[y] = \int_{x_0}^{x_1} F(x, y, y', \ldots, y^{(n)}) \, \mathrm{d}x,$$

and the Euler–Lagrange equation becomes

$$F_y - \frac{d}{dx} F_{y'} + \frac{d^2}{dx^2} F_{y''} - \cdots + (-1)^n \frac{d^n}{dx^n} F_{y^{(n)}} = 0.$$

If F consists of several scalar functions $(y_1, y_2, \ldots, y_n)$, then

$$J[y_1, y_2, \ldots, y_n] = \int_{x_0}^{x_1} F(x, y_1, \ldots, y_n, y_1', \ldots, y_n') \, \mathrm{d}x,$$

and the Euler–Lagrange equations are

$$F_{y_i} - \frac{d}{dx} F_{y_i'} = 0, \quad i = 1, \ldots, n.$$

If F depends on a single function of n variables and if Ω is a surface, then

$$J[y] = \int_{\Omega} F(x_1, \ldots, x_n, y, y_{x_1}, \ldots, y_{x_n}) \, \mathrm{d}x_1 \cdots \mathrm{d}x_n,$$

and the Euler–Lagrange equations are now PDEs:

$$F_y - \sum_{i=1}^{n} \frac{\partial}{\partial x_i} F_{y'_{x_i}} = 0, \quad i = 1, \ldots, n.$$

Finally, there is the case of several functions of several variables, which is just a combination of the above.

Example 2.4. We consider the case of finding an extremal function, u, of two variables, x and y, from the cost function

$$J[u] = \iint_{\Omega} F(x, y, u, u_x, u_y)\,\mathrm{d}x\,\mathrm{d}y \tag{2.4}$$

over the domain Ω. The necessary condition is

$$\delta J = \epsilon \left(\frac{d}{d\epsilon} J[u + \epsilon\eta] \right)_{\epsilon=0} = 0, \tag{2.5}$$

where $\eta(x,y)$ is a "nice" function satisfying zero boundary conditions on $\partial\Omega$, the boundary of Ω. Substituting (2.4) in (2.5), we obtain

$$\delta J = \epsilon \iint_{\Omega} \left(F_u \eta + F_{u_x} \eta_x + F_{u_y} \eta_y \right) \mathrm{d}x\,\mathrm{d}y = 0,$$

which we integrate by parts (by applying the Gauss divergence theorem), getting

$$\delta J = \epsilon \iint_{\Omega} \eta \left(F_u - \frac{\partial}{\partial x} F_{u_x} - \frac{\partial}{\partial y} F_{u_y} \right) \mathrm{d}x\,\mathrm{d}y = 0$$

(we have used the vanishing of η on the boundary), which yields the Euler–Lagrange equation

$$[F]_u = F_u - \frac{\partial}{\partial x} F_{u_x} - \frac{\partial}{\partial y} F_{u_y} = 0.$$

This can be expanded to

$$F_{u_x u_x} u_{xx} + 2F_{u_x u_y} u_{xy} + F_{u_y u_y} u_{yy} + F_{u_x u} u_x + F_{u_y u} u_y + F_{u_x x} + F_{u_y y} - F_u = 0.$$

We can now apply this result to the case where

$$F = \frac{1}{2}\left(u_x^2 + u_y^2 \right).$$

Clearly,

$$F_{u_x u_x} = F_{u_y u_y} = 1,$$

with all other terms equal to zero, and the Euler–Lagrange equation is precisely Laplace's equation,

$$\Delta u = u_{xx} + u_{yy} = 0,$$

which can be solved subject to the boundary conditions that must be imposed on u. ∎

2.2.3 ▪ Concluding remarks

The calculus of variations, via the Euler–Lagrange (partial) differential equations, provides a very general framework for minimizing functionals, taking into account both the functional to be minimized and the (partial) differential equations that describe the underlying physical problem. The calculus of variations also covers the minimization of more general integral equations, often encountered in imaging problems, but these will not be dealt with here. Entire books are dedicated to this subject—see, for example, Aster et al. [2012] and Colton and Kress [1998].

In what follows, for DA problems we will study a special case of calculus of variations and generalize it. Let us explain: the special case lies in the fact that our cost function will be a "mismatch" function, expressing the squared difference between measured values and simulated (predicted) values, integrated over a spatial (or space-time) domain. To this we will sometimes add "regularization" terms to ensure well-posedness. The generalization takes the form of the constraints that we add to the optimization problem: in the case of DA these constraints are (partial) differential equations that must be satisfied by the extremal function that we seek to compute.

2.3 ▪ Adjoint methods

Having, we hope by now, acquired an understanding of the calculus of variations, we will proceed to study the *adjoint approach* for solving (functional) optimization problems. We will emphasize the generality and the inherent power of this approach. Note that this approach is also used frequently in optimal control and optimal design problems—these are just special cases of what we will study here. We note that the Euler–Lagrange system of equations, amply seen in the previous section, will here be composed of the direct and adjoint equations for the system under consideration. A very instructive toy example of an ocean circulation problem can be found in Bennett [2004], where the Euler–Lagrange equations are carefully derived and their solution proposed using a special decomposition based on "representer functions."

In this section, we will start from a general setting for the adjoint method, and then we will back up and proceed progressively through a string of special cases, from a "simple" ODE-based inverse problem of parameter identification to a full-blown, nonlinear PDE-based problem. Even the impatient reader, who may be tempted to skip the general setting and go directly to the special case examples, is encouraged to study the general presentation, because a number of fundamental and very important points are dealt with here. After presenting the continuous case, the discrete (finite-dimensional) setting will be explained. This leads naturally to the important subject of *automatic differentiation*, which is often used today for automatically generating the adjoints of large production codes, though it can be very efficient for smaller codes too.

2.3.1 ▪ A general setting

We will now apply the calculus of variations to the solution of variational inverse problems. Let $\mathbf{u}$ be the state of a *dynamical system* whose behavior depends on model parameters $\mathbf{m}$ and is described by a differential operator equation

$$\mathbf{L}(\mathbf{u},\mathbf{m}) = \mathbf{f},$$

where $\mathbf{f}$ represents external forces. Define a *cost function*, $J(\mathbf{m})$, as an energy functional or, more commonly, as a misfit functional that quantifies the L^2-distance[17] between the observation and the model prediction $\mathbf{u}(\mathbf{x},t;\mathbf{m})$. For example,

$$J(\mathbf{m}) = \int_0^T \int_\Omega \left(\mathbf{u}(\mathbf{x},t;\mathbf{m}) - \mathbf{u}^{\mathrm{obs}}(\mathbf{x},t)\right)^2 \delta(\mathbf{x}-\mathbf{x}_r)\,\mathrm{d}\mathbf{x}\,\mathrm{d}t,$$

[17]The space L^2 is a Hilbert space of (measurable) functions that are square-integrable (in the Lebesgue sense). Readers unfamiliar with this should consult a text such as Kreyszig [1978] for this definition as well as all other (functional) analysis terms used in what follows.

where $x \in \Omega \subset \mathbb{R}^n$; $n = 2,3$; $0 \le t \le T$; δ is the Dirac delta function; and $\mathbf{x}_r$ are the observer positions. Our objective is to choose the model parameters, $\mathbf{m}$, as a function of the observed output, $\mathbf{u}^{\mathrm{obs}}$, such that the cost function, $J(\mathbf{m})$, is minimized.

We define the variation of $\mathbf{u}$ with respect to $\mathbf{m}$ in the direction $\delta\mathbf{m}$ (known as the Gâteaux differential, which is the directional derivative, but defined on more general spaces of functions) as

$$\delta\mathbf{u} \doteq \nabla_m \mathbf{u}\,\delta\mathbf{m},$$

where $\nabla_m(\cdot)$ is the gradient operator with respect to the model parameters (known, in the general case, as the Fréchet derivative). Then the corresponding directional derivative of J can be written as

$$\begin{aligned} \delta J &= \nabla_m J\,\delta\mathbf{m} \\ &= \nabla_u J\,\delta\mathbf{u} \\ &= \langle \nabla_u J_1\,\delta\mathbf{u}\rangle, \end{aligned} \tag{2.6}$$

where in the second line we have used the chain rule together with the definition of $\delta\mathbf{u}$, and in the third line $\langle\cdot\rangle$ denotes the space-time integral. Here we have passed the "derivative" under the integral sign, and J_1 is the integrand. There remains a major difficulty: the variation $\delta\mathbf{u}$ is impossible or unfeasible to compute numerically (for all directions $\delta\mathbf{m}$). To overcome this, we would like to eliminate $\delta\mathbf{u}$ from (2.6) by introducing an adjoint state (which can also be seen as a Lagrange multiplier).

To achieve this, we differentiate the state equation with respect to the model $\mathbf{m}$ and apply the necessary condition for optimality (disappearance of the variation) to obtain

$$\delta\mathbf{L} = \nabla_m\mathbf{L}\,\delta\mathbf{m} + \nabla_u\mathbf{L}\,\delta\mathbf{u} = 0.$$

Now we multiply this equation by an arbitrary test function $\mathbf{u}^\dagger$ (Lagrange multiplier) and integrate over space-time to obtain

$$\left\langle \mathbf{u}^\dagger \cdot \nabla_m\mathbf{L}\,\delta\mathbf{m}\right\rangle + \left\langle \mathbf{u}^\dagger \cdot \nabla_u\mathbf{L}\,\delta\mathbf{u}\right\rangle = 0.$$

Add this null expression to (2.6) and integrate by parts, regrouping terms in $\delta\mathbf{u}$:

$$\begin{aligned} \nabla_m J\,\delta\mathbf{m} &= \langle \nabla_u J_1\,\delta\mathbf{u}\rangle + \left\langle \mathbf{u}^\dagger \cdot \nabla_m\mathbf{L}\,\delta\mathbf{m}\right\rangle + \left\langle \mathbf{u}^\dagger \cdot \nabla_u\mathbf{L}\,\delta\mathbf{u}\right\rangle \\ &= \left\langle \delta\mathbf{u}\cdot\left(\nabla_u J_1^\dagger + \nabla_u\mathbf{L}^\dagger\mathbf{u}^\dagger\right)\right\rangle + \left\langle \mathbf{u}^\dagger\cdot\nabla_m\mathbf{L}\,\delta\mathbf{m}\right\rangle, \end{aligned}$$

where we have defined the adjoint operators $\nabla_u J_1^\dagger$ and $\nabla_u \mathbf{L}^\dagger$ via the appropriate inner products as

$$\langle \nabla_u J_1\,\delta\mathbf{u}\rangle = \left\langle \delta\mathbf{u}\cdot\nabla_u J_1^\dagger\right\rangle$$

and

$$\left\langle \mathbf{u}^\dagger\cdot\nabla_u\mathbf{L}\,\delta\mathbf{u}\right\rangle = \left\langle \delta\mathbf{u}\cdot\nabla_u\mathbf{L}^\dagger\mathbf{u}^\dagger\right\rangle.$$

Finally, to eliminate $\delta\mathbf{u}$, the adjoint state, $\mathbf{u}^\dagger$, should satisfy

$$\nabla_u\mathbf{L}^\dagger\mathbf{u}^\dagger = -\nabla_u J_1^\dagger,$$

which is known as the adjoint equation.

Once the adjoint solution, $\mathbf{u}^\dagger$, is found, the derivative/variation of the objective functional becomes

$$\nabla_m J\,\delta\mathbf{m} = \left\langle \mathbf{u}^\dagger\cdot\nabla_m\mathbf{L}\,\delta\mathbf{m}\right\rangle. \tag{2.7}$$

This key result enables us to compute the desired gradient, $\nabla_m J$, without the explicit knowledge of $\delta\mathbf{u}$. A number of *important remarks* are necessary here:

1. We obtain *explicit* formulas for the gradient with respect to each/any model parameter. Note that this has been done in a completely general setting, without any restrictions on the operator, $\mathbf{L}$, or on the model parameters, $\mathbf{m}$.

2. The *computational cost* is one solution of the adjoint equation, which is usually of the same order as (if not identical to) the direct equation,[18] but with a reversal of time.

3. The *variation* (Gâteaux derivative) of $\mathbf{L}$ with respect to the model parameters, $\mathbf{m}$, is, in general, straightforward to compute.

4. We have not considered boundary (or initial) conditions in the above general approach. In real cases, these are potential sources of difficulties for the use of the adjoint approach—see Section 2.3.9, where the *discrete adjoint* can provide a way to overcome this hurdle.

5. For complete mathematical rigor, the above development should be performed in an appropriate *Hilbert space* setting that guarantees the existence of all the inner products and adjoint operators—the interested reader could consult the excellent short course notes of Estep [2004] and references therein, or the monograph of Tröltzsch [2010].

6. In many real problems, the optimization of the misfit functional leads to *multiple local minima* and often to very "flat" cost functions—these are hard problems for gradient-based optimization methods. These difficulties can be (partially) overcome by a panoply of tools:

   (a) *Regularization* terms can alleviate the nonuniqueness problem—see Engl et al. [1996] and Vogel [2002].

   (b) *Rescaling* the parameters and/or variables in the equations can help with the "flatness"—this technique is often employed in numerical optimization—see Nocedal and Wright [2006].

   (c) *Hybrid* algorithms, which combine stochastic and deterministic optimization (e.g., simulated annealing), can be used to avoid local minima—see Press et al. [2007].

7. When measurement and modeling errors can be modeled by Gaussian distributions and a background (prior) solution exists, the objective function may be generalized by including suitable *covariance matrices*. This is the approach employed systematically in DA—see below for full details.

We will now present a series of examples where we apply the adjoint approach to increasingly complex cases. We will use two alternative methods for the derivation of the adjoint equation: a Lagrange multiplier approach and the *tangent linear model* (TLM) approach. After seeing the two in action, the reader can adopt the one that suits her/him best. Note that the Lagrangian approach supposes that we perturb the sought-for parameters (as seen above in Section 2.2) and is thus not applicable to inverting for constant-valued parameters, in which case we must resort to the TLM approach.

[18] Note that for nonlinear equations this may not be the case, and one may require four or five times the computational effort.

2.3.2 ▪ Parameter identification example

A basic example: Let us consider in more detail the parameter identification problem (already encountered in Chapter 1) based on the convection-diffusion equation (1.6),

$$\begin{cases} -bu''(x)+cu'(x)=f(x), \quad 0<x<1, \\ u(0)=0,\ u(1)=0, \end{cases} \tag{2.8}$$

where f is a given function in $L^2(0,1)$ and b and c are the unknown (constant) parameters that we seek to identify using observations of $u(x)$ on $[0,1]$. The least-squares error cost function is

$$J(b,c)=\frac{1}{2}\int_0^1 \left(u(x)-u^{\mathrm{obs}}(x)\right)^2 \mathrm{d}x.$$

Let us, once again, calculate its gradient by introducing the TLM. Perturbing the cost function by a small perturbation in the direction α with respect to the two parameters gives

$$J(b+\alpha\delta b, c+\alpha\delta c)-J(b,c)=\frac{1}{2}\int_0^1 \left(\tilde{u}-u^{\mathrm{obs}}\right)^2-\left(u-u^{\mathrm{obs}}\right)^2 \mathrm{d}x,$$

where $\tilde{u}=u_{b+\alpha\delta b,c+\alpha\delta c}$ is the perturbed solution and $u=u_{b,c}$ is the unperturbed one. Expanding and rearranging, we obtain

$$J(b+\alpha\delta b, c+\alpha\delta c)-J(b,c)=\frac{1}{2}\int_0^1 \left(\tilde{u}+u-2u^{\mathrm{obs}}\right)(\tilde{u}-u)\,\mathrm{d}x.$$

Now we divide by α on both sides of the equation and pass to the limit $\alpha\to 0$ to obtain the directional derivative (which is the derivative with respect to the parameters, in the direction of the perturbations),

$$\hat{J}[b,c](\delta b,\delta c)=\int_0^1 \left(u-u^{\mathrm{obs}}\right)\hat{u}\,\mathrm{d}x, \tag{2.9}$$

where we have defined

$$\hat{u}=\lim_{\alpha\to 0}\frac{\tilde{u}-u}{\alpha}, \quad \hat{J}[b,c](\delta b,\delta c)=\lim_{\alpha\to 0}\frac{J(b+\alpha\delta b,c+\alpha\delta c)-J(b,c)}{\alpha},$$

and we have moved the limit under the integral sign. Let us now use this definition to find the equation satisfied by $\hat{u}$. We have

$$\begin{cases} -(b+\alpha\delta b)\tilde{u}''+(c+\alpha\delta c)\tilde{u}'=f, \\ \tilde{u}(0)=0,\ \tilde{u}(1)=0, \end{cases}$$

and the given model (2.8),

$$\begin{cases} -bu''+cu'=f, \\ u(0)=0,\ u(1)=0. \end{cases}$$

Then, subtracting these two equations and passing to the limit (using the definition of $\hat{u}$), we obtain

$$\begin{cases} -b\hat{u}''-(\delta b)u''+c\hat{u}'+(\delta c)u'=0, \\ \hat{u}(0)=0,\ \hat{u}(1)=0. \end{cases}$$

We can now define the TLM

$$\begin{cases} -b\hat{u}'' + c\hat{u}' = (\delta b)u'' - (\delta c)u', \\ \hat{u}(0) = 0,\ \hat{u}(1) = 0. \end{cases} \tag{2.10}$$

We want to be able to reformulate the directional derivative (2.9) to obtain a calculable expression for the gradient. So we multiply the TLM (2.10) by a variable p and integrate twice by parts, transferring derivatives from $\hat{u}$ onto p:

$$-b\int_0^1 \hat{u}''p\,\mathrm{d}x + c\int_0^1 \hat{u}'p\,\mathrm{d}x = \int_0^1 \left((\delta b)u''\,\mathrm{d}x - (\delta c)u'\right)p\,\mathrm{d}x,$$

which gives (term by term)

$$\begin{aligned} \int_0^1 \hat{u}''p\,\mathrm{d}x &= \left[\hat{u}'p\right]_0^1 - \int_0^1 \hat{u}'p'\,\mathrm{d}x \\ &= \left[\hat{u}'p - \hat{u}p'\right]_0^1 + \int_0^1 \hat{u}p''\,\mathrm{d}x \\ &= \hat{u}'(1)p(1) - \hat{u}'(0)p(0) + \int_0^1 \hat{u}p''\,\mathrm{d}x \end{aligned}$$

and

$$\begin{aligned} \int_0^1 \hat{u}'p\,\mathrm{d}x &= [\hat{u}p]_0^1 - \int_0^1 \hat{u}p'\,\mathrm{d}x \\ &= -\int_0^1 \hat{u}p'\,\mathrm{d}x. \end{aligned}$$

Putting these results together, we have

$$-b\left(\hat{u}'(1)p(1) - \hat{u}'(0)p(0) + \int_0^1 \hat{u}p''\right) + c\left(-\int_0^1 \hat{u}p'\right) = \int_0^1 \left((\delta b)u'' - (\delta c)u'\right)p$$

or, grouping terms,

$$\int_0^1 \left(-bp'' - cp'\right)\hat{u} = b\hat{u}'(1)p(1) - b\hat{u}'(0)p(0) + \int_0^1 \left((\delta b)u'' - (\delta c)u'\right)p. \tag{2.11}$$

Now, to get rid of *all* the terms in $\hat{u}$ in this expression, we impose that p must satisfy *the adjoint model*

$$\begin{cases} -bp'' - cp' = (u - u^{\mathrm{obs}}), \\ p(0) = 0,\ p(1) = 0. \end{cases} \tag{2.12}$$

Integrating (2.12) and using the expression (2.11), we obtain

$$\int_0^1 (u - u^{\mathrm{obs}})\hat{u} = \int_0^1 \left(-bp'' - cp'\right)\hat{u} = (\delta b)\left(\int_0^1 pu''\right) + (\delta c)\left(-\int_0^1 pu'\right).$$

We recognize, in the last two terms, the L^2 inner product, which enables us, based on the key result (2.7), to finally write an explicit expression for the gradient, based on (2.9),

$$\nabla J(b,c) = \left(\int_0^1 pu''\mathrm{d}x, -\int_0^1 pu'\mathrm{d}x\right)^{\mathrm{T}}$$

or, separating the two components,

$$\nabla_b J(b,c) = \int_0^1 p u'' \mathrm{d}x, \tag{2.13}$$

$$\nabla_c J(b,c) = -\int_0^1 p u' \mathrm{d}x. \tag{2.14}$$

Thus, in this example, to compute the gradient of the least-squares error cost function, we must

- solve the direct equation (2.8) for u and derive u' and u'' from the solution, using some form of numerical differentiation (if we solved with finite differences), or differentiating the shape functions (if we solved with finite elements);
- solve the adjoint equation (2.12) for p (using the same solver[19] that we used for u);
- compute the two terms of the gradient, (2.13) and (2.14), using a suitable numerical integration scheme [Quarteroni et al., 2007].

Thus, for the additional cost of one solution of the adjoint model (2.12) plus a numerical integration, we can compute the gradient of the cost function with respect to either one, or both, of the unknown parameters. It is now a relatively easy task to find (numerically) the optimal values of b and c that minimize J by a suitable descent algorithm, for example, a quasi-Newton method [Nocedal and Wright, 2006; Quarteroni et al., 2007].

2.3.3 ▪ A simple ODE example: Lagrangian method

We now consider a variant of the convection-diffusion example, where the diffusion coefficient is spatially varying. This model is closer to many physical situations, where the medium is not homogeneous and we have zones with differing diffusive properties. The system is

$$\begin{cases} -\big(a(x)u'(x)\big)' - u'(x) = q(x), & 0 < x < 1, \\ u(0) = 0, \; u(1) = 0, \end{cases} \tag{2.15}$$

with the cost function

$$J[a] = \frac{1}{2}\int_0^1 \left(u(x) - u^{\mathrm{obs}}(x)\right)^2 dx,$$

where $u^{\mathrm{obs}}(x)$ denotes the observations on $[0,1]$. We now introduce an alternative approach for deriving the gradient, based on the Lagrangian (or variational formulation). Let the cost function be

$$J^*[a,p] = \frac{1}{2}\int_0^1 \left(u(x) - u^{\mathrm{obs}}(x)\right)^2 dx + \int_0^1 p\left(-(au')' - u' - q\right) dx,$$

[19] This is not true when we use a *discrete* adjoint approach—see Section 2.3.9.

noting that the second integral is zero when u is a solution of (2.15) and that the adjoint variable, p, can be considered here to be a Lagrange multiplier function. We begin by taking the variation of J^* with respect to its variables, a and p:

$$\delta J^* = \int_0^1 \left(u - u^{\text{obs}}\right)\delta u \, dx + \int_0^1 \delta p \overbrace{\left(-(a u')' - u' - q\right)}^{=0} dx + \int_0^1 p\left[\left(-\delta a\, u' - a\, \delta u'\right)'\right].$$

Now the strategy is to "kill terms" by imposing suitable, well-chosen conditions on p. This is achieved by integrating by parts and then defining the adjoint equation and boundary conditions on p as follows:

$$\begin{aligned}\delta J^* &= \int_0^1 \left[(u - u^{\text{obs}}) + p' - (a p')'\right]\delta u \, dx + \int_0^1 \delta a\, u' p' dx \\ &\quad \left[-p(\delta u + u'\delta a + a\delta u') + p' a \delta u\right]_0^1 \\ &= \int_0^1 \delta a\, u' p' dx,\end{aligned}$$

where we have used the zero boundary conditions on δu and assumed that the following adjoint system must be satisfied by p:

$$\begin{cases} -\left(a p'\right)' + p' = -(u - u^{\text{obs}}), & 0 < x < 1, \\ p(0) = 0, \ p(1) = 0. \end{cases} \tag{2.16}$$

And, as before, based on the key result (2.7), we are left with an explicit expression for the gradient,

$$\nabla_{a(x)} J^* = u' p'.$$

Thus, with one solution of the direct system (2.15) plus one solution of the adjoint system (2.16), we recover the gradient of the cost function with respect to the sought-for diffusion coefficient, $a(x)$.

2.3.4 ▪ Initial condition control

For DA problems in meteorology and oceanography, the objective is to reconstruct the *initial conditions* of the model. This is also the case in certain source identification problems for environmental pollution. We redo the above gradient calculations in this context. Let us consider the following system of (possibly nonlinear) ODEs:

$$\begin{cases} \dfrac{\mathrm{d}\mathbf{X}}{\mathrm{d}t} = \mathbf{M}(\mathbf{X}) & \text{in } \Omega \times [0, T], \\ \mathbf{X}(t = 0) = \mathbf{U}, \end{cases} \tag{2.17}$$

with the cost function

$$J(\mathbf{U}) = \frac{1}{2}\int_0^T \|\mathbf{H}\mathbf{X} - \mathbf{Y}^{\text{o}}\|^2 \, \mathrm{d}t,$$

where we have used the classical vector-matrix notation for systems of ODEs and $\|\cdot\|$ denotes the L^2-norm over the space variable. To compute the directional derivative, we perturb the initial condition $\mathbf{U}$ by a quantity α in the direction $\mathbf{u}$ and denote by $\tilde{\mathbf{X}}$ the corresponding trajectory, satisfying

$$\begin{cases} \dfrac{\mathrm{d}\tilde{\mathbf{X}}}{\mathrm{d}t} = \mathbf{M}(\tilde{\mathbf{X}}) \quad \text{in } \Omega \times [0,T], \\ \tilde{\mathbf{X}}(t=0) = \mathbf{U} + \alpha\mathbf{u}. \end{cases} \tag{2.18}$$

We then have

$$\begin{aligned} J(\mathbf{U}+\alpha\mathbf{u}) - J(\mathbf{U}) &= \frac{1}{2}\int_0^T \left\|\mathbf{H}\tilde{\mathbf{X}} - \mathbf{Y}^{\mathrm{o}}\right\|^2 - \|\mathbf{H}\mathbf{X} - \mathbf{Y}^{\mathrm{o}}\|^2 \,\mathrm{d}t \\ &= \frac{1}{2}\int_0^T \left(\mathbf{H}\tilde{\mathbf{X}} - \mathbf{Y}, \mathbf{H}\tilde{\mathbf{X}} - \mathbf{H}\mathbf{X} + \mathbf{H}\mathbf{X} - \mathbf{Y}\right) - (\mathbf{H}\mathbf{X} - \mathbf{Y}, \mathbf{H}\mathbf{X} - \mathbf{Y}) \\ &= \frac{1}{2}\int_0^T \left(\mathbf{H}\tilde{\mathbf{X}} - \mathbf{Y}, \mathbf{H}(\tilde{\mathbf{X}} - \mathbf{X})\right) - (\mathbf{H}\tilde{\mathbf{X}} - \mathbf{Y} - (\mathbf{H}\mathbf{X} - \mathbf{Y}), \mathbf{H}\mathbf{X} - \mathbf{Y}) \\ &= \frac{1}{2}\int_0^T \left(\mathbf{H}\tilde{\mathbf{X}} - \mathbf{Y}, \mathbf{H}(\tilde{\mathbf{X}} - \mathbf{X})\right) + (\mathbf{H}(\tilde{\mathbf{X}} - \mathbf{X}), \mathbf{H}\mathbf{X} - \mathbf{Y}). \end{aligned}$$

Now, we set

$$\hat{\mathbf{X}} = \lim_{\alpha \to 0} \frac{\tilde{\mathbf{X}} - \mathbf{X}}{\alpha},$$

and we compute the directional derivative,

$$\begin{aligned} \hat{J}[\mathbf{U}](u) &= \lim_{\alpha\to 0} \frac{J(\mathbf{U}+\alpha\mathbf{u}) - J(\mathbf{U})}{\alpha} \\ &= \frac{1}{2}\int_0^T \left(\mathbf{H}\mathbf{X} - \mathbf{Y}, \mathbf{H}\hat{\mathbf{X}}\right) + (\mathbf{H}\hat{\mathbf{X}}, \mathbf{H}\mathbf{X} - \mathbf{Y}) \\ &= \int_0^T (\mathbf{H}\hat{\mathbf{X}}, \mathbf{H}\mathbf{X} - \mathbf{Y}) \\ &= \int_0^T (\hat{\mathbf{X}}, \mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{X} - \mathbf{Y})). \end{aligned} \tag{2.19}$$

By subtracting the equations (2.18) and (2.17) satisfied by $\tilde{\mathbf{X}}$ and $\mathbf{X}$, we obtain

$$\begin{cases} \dfrac{\mathrm{d}(\tilde{\mathbf{X}} - \mathbf{X})}{\mathrm{d}t} = \mathbf{M}(\tilde{\mathbf{X}}) - \mathbf{M}\mathbf{X} = \left[\dfrac{\partial \mathbf{M}}{\partial \mathbf{X}}\right](\tilde{\mathbf{X}} - \mathbf{X}) + \dfrac{1}{2}(\tilde{\mathbf{X}} - \mathbf{X})^{\mathrm{T}}\left[\dfrac{\partial^2 \mathbf{M}}{\partial \mathbf{X}^2}\right](\tilde{\mathbf{X}} - \mathbf{X}) + \cdots, \\ (\tilde{\mathbf{X}} - \mathbf{X})(t=0) = \alpha\mathbf{u}. \end{cases}$$

Now we divide by α and pass to the limit $\alpha \to 0$ to obtain

$$\begin{cases} \dfrac{\mathrm{d}\hat{\mathbf{X}}}{\mathrm{d}t} = \left[\dfrac{\partial \mathbf{M}}{\partial \mathbf{X}}\right]\hat{\mathbf{X}}, \\ \hat{\mathbf{X}}(t=0) = \mathbf{u}. \end{cases} \tag{2.20}$$

These equations are the TLM.

We will now proceed to compute the adjoint model. As in the ODE example of Sections 1.5.3.1 and 2.3.2, we multiply the TLM (2.20) by $\mathbf{P}$ and integrate by parts on $[0,T]$. We find

$$\begin{aligned}
\int_0^T \left(\frac{\mathrm{d}\hat{\mathbf{X}}}{\mathrm{d}t}, \mathbf{P}\right) &= -\int_0^T \left(\hat{\mathbf{X}}, \frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t}\right) + \left[(\hat{\mathbf{X}}, \mathbf{P})\right]_0^T \\
&= -\int_0^T \left(\hat{\mathbf{X}}, \frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t}\right) + \left(\hat{\mathbf{X}}(T), \mathbf{P}(T)\right) - \left(\hat{\mathbf{X}}(0), \mathbf{P}(0)\right) \\
&= -\int_0^T \left(\hat{\mathbf{X}}, \frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t}\right) + \left(\hat{\mathbf{X}}(T), \mathbf{P}(T)\right) - (\mathbf{u}, \mathbf{P}(0))
\end{aligned}$$

and

$$\int_0^T \left(\left[\frac{\partial \mathbf{M}}{\partial \mathbf{X}}\right]\hat{\mathbf{X}}, \mathbf{P}\right) = \int_0^T \left(\hat{\mathbf{X}}, \left[\frac{\partial \mathbf{M}}{\partial \mathbf{X}}\right]^{\mathrm{T}} \mathbf{P}\right).$$

Thus, substituting in equation (2.20), we get

$$\int_0^T \left(\frac{\mathrm{d}\hat{\mathbf{X}}}{\mathrm{d}t} - \left[\frac{\partial \mathbf{M}}{\partial \mathbf{X}}\right]\hat{X}, \mathbf{P}\right) = 0 = \int_0^T \left(\hat{\mathbf{X}}, -\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t} - \left[\frac{\partial \mathbf{M}}{\partial \mathbf{X}}\right]^{\mathrm{T}} \mathbf{P}\right) + \left(\hat{\mathbf{X}}(T), \mathbf{P}(T)\right) - (\mathbf{u}, \mathbf{P}(0)).$$

Identifying with the directional derivative (2.19), we obtain the equations of the *adjoint model*

$$\begin{cases} \dfrac{\mathrm{d}\mathbf{P}}{\mathrm{d}t} + \left[\dfrac{\partial \mathbf{M}}{\partial \mathbf{X}}\right]^{\mathrm{T}} \mathbf{P} = \mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{X} - \mathbf{Y}), \\ \mathbf{P}(t = T) = 0, \end{cases} \tag{2.21}$$

which is a *backward* model, integrated from $t = T$ back down to $t = 0$.

We can now find the expression for the gradient. Using the adjoint model (2.21) in (2.19), we find

$$\begin{aligned}
\hat{J}[\mathbf{U}](\mathbf{u}) &= \int_0^T (\hat{\mathbf{X}}, \mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{X} - \mathbf{Y})) \\
&= \int_0^T \left(\hat{\mathbf{X}}, \frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t} + \left[\frac{\partial \mathbf{M}}{\partial \mathbf{X}}\right]^{\mathrm{T}} \mathbf{P}\right) \\
&= (-\mathbf{u}, \mathbf{P}(0)).
\end{aligned}$$

But, by definition,

$$\hat{J}[\mathbf{U}](\mathbf{u}) = (\nabla J_{\mathrm{U}}, \mathbf{u}),$$

and thus

$$\nabla J_{\mathrm{U}} = -\mathbf{P}(0).$$

Once again, with a single (backward) integration of the adjoint model, we obtain a particularly simple expression for the gradient of the cost function with respect to the control parameter.

2.3.5 ▪ Putting it all together: The case of a linear PDE

The natural extension of the ODEs seen above is the initial boundary value problem known as the diffusion equation:

$$\frac{\partial u}{\partial t} - \nabla\cdot(\nu\nabla u) = 0,\ x\in(0,L),\ t>0,$$
$$u(x,0)=u_0(x),\ u(0,t)=0,\ u(L,t)=\eta(t).$$

This equation has multiple origins emanating from different physical situations. The most common application is particle diffusion, where u is a concentration and ν is a diffusion coefficient. Then there is heat diffusion, for which u is temperature and ν is thermal conductivity. The equation is also found in finance, being closely related to the Black–Scholes model. Another important application is population dynamics. These diverse application fields, and hence the diffusion equation, give rise to a number of inverse and DA problems.

A variety of different controls can be applied to this system:

- *internal* control: $\nu(x)$—this is the parameter identification problem, also known as tomography;
- *initial* control: $\xi(x)=u_0(x)$—this is a source detection inverse or DA problem;
- *boundary* control: $\eta(t)=u(L,t)$—this is the “classical” boundary control problem, also a parameter identification inverse problem.

As above, we can define the cost function,

$$J[\nu,\xi,\eta] = \frac{1}{LT}\int_0^T\int_0^L (u-u^{\mathrm{o}})^2\,\mathrm{d}x\,\mathrm{d}t,$$

which is now a space-time multiple integral, and its related Lagrangian,

$$J^* = \frac{1}{LT}\int_0^T\int_0^L (u-u^{\mathrm{o}})^2\,\mathrm{d}x\,\mathrm{d}t + \frac{1}{LT}\int_0^T\int_0^L p\,[u_t-(\nu u_x)_x]\,\mathrm{d}x\,\mathrm{d}t.$$

Now take the variation of J^*,

$$\delta J^* = \frac{1}{LT}\int_0^T\int_0^L 2(u-u^{\mathrm{o}})\delta u\,\mathrm{d}x\,\mathrm{d}t + \frac{1}{LT}\int_0^T\int_0^L \delta p\,\overbrace{[u_t-(\nu u_x)_x]}^{=0}\,\mathrm{d}x\,\mathrm{d}t$$
$$+\frac{1}{LT}\int_0^T\int_0^L p\,[\delta u_t - (\delta\nu\, u_x + \nu\delta u_x)_x]\,\mathrm{d}x\,\mathrm{d}t,$$

and perform integration by parts to obtain

$$\delta J^* = \frac{1}{LT}\int_0^T\int_0^L \delta\nu\, u_x p_x\,\mathrm{d}x\,\mathrm{d}t - \frac{1}{LT}\int_0^L p\ \delta u|_{t=0}\,\mathrm{d}x + \frac{1}{LT}\int_0^T p\ \delta\eta|_{x=L}\,\mathrm{d}t, \quad (2.22)$$

where we have defined the adjoint equation as

$$\frac{\partial p}{\partial t} + \nabla\cdot(\nu\nabla u) = 2(u-u^{\mathrm{o}}),\quad x\in(0,L),\quad t>0,$$
$$p(0,t)=0,\quad p(L,t)=0,$$
$$p(x,T)=0.$$

As before, this equation is of the same type as the original diffusion equation but must be solved backward in time. Finally, from (2.22), we can pick off each of the three desired terms of the gradient:

$$
\begin{aligned}
\nabla_{\nu(x)} J^* &= \frac{1}{T}\int_0^T u_x p_x \,\mathrm{d}t, \\
\nabla_{u|_{t=0}} J^* &= -\, p|_{t=0}, \\
\nabla_{\eta|_{x=L}} J^* &= p|_{x=L}.
\end{aligned}
$$

Once again, at the expense of a single (backward) solution of the adjoint equation, we obtain explicit expressions for the gradient of the cost function with respect to each of the three control variables. This is quite remarkable and completely avoids “brute force” or exhaustive minimization, though, as mentioned earlier, we only have the guarantee of finding a local minimum. However, if we have a good starting guess, which is usually obtained from historical or other “physical” knowledge of the system, we are sure to arrive at a good (or, at least, better) minimum.

2.3.6 ▪ An adjoint “zoo”

As we have seen above, every (partial) differential operator has its very own adjoint form. We can thus derive, and categorize, adjoint equations for a whole variety of partial differential operators. Some common examples can be found in Table 2.1.

Table 2.1. *Adjoint forms for some common ordinary and partial differential operators.*

Operator	Adjoint
$\frac{du}{dx} - \gamma \frac{d^2u}{dx^2}$	$-\frac{dp}{dx} - \gamma \frac{d^2p}{dx^2}$
$\nabla \cdot (k\nabla u)$	$\nabla \cdot (k\nabla p)$
$\frac{\partial u}{\partial t} - c\frac{\partial^2 u}{\partial x^2}$	$-\frac{\partial p}{\partial t} - c\frac{\partial^2 p}{\partial x^2}$
$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x}$	$-\frac{\partial p}{\partial t} - c\frac{\partial p}{\partial x}$

The principle is simple: all second-order (or even) derivatives remain unchanged, whereas all first-order (or uneven) derivatives undergo a change of sign.

2.3.7 ▪ Application: Burgers’ equation (a nonlinear PDE)

We will now consider a more realistic application based on Burgers’ equation [Lax, 1973] with control of the initial condition and the boundary conditions. Burgers’ equation is a very good approximation to the Navier–Stokes equation in certain contexts where viscous effects dominate convective effects. The Navier–Stokes equation itself is the model equation used for all aerodynamic simulations and for many flow problems. In addition, it is the cornerstone of *numerical weather prediction* (NWP) codes.

The viscous Burgers' equation in the interval $x \in [0, L]$ is defined as

$$\begin{gathered}\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} - \nu\frac{\partial^2 u}{\partial x^2} = f,\\ u(0,t) = \psi_1(t), \quad u(L,t) = \psi_2(t),\\ u(x,0) = u_0(x).\end{gathered}$$

The control vector will be taken as a combination of the initial state and the two boundary conditions,

$$(u_0, \psi_1, \psi_2),$$

and the cost function is given by the usual mismatch,

$$J(u_0, \psi_1, \psi_2) = \frac{1}{2}\int_0^T \int_0^L \left(u - u^{\text{obs}}\right)^2 \,\mathrm{d}x\,\mathrm{d}t.$$

We know that the derivative of J in the direction[20] (h_u, h_1, h_2) is given (as above in (2.9)) by

$$\hat{J}[u_0, \psi_1, \psi_2](h_u, h_1, h_2) = \int_0^T \int_0^L \left(u - u^{\text{obs}}\right)\hat{u}\,\mathrm{d}x\,\mathrm{d}t,$$

where $\hat{u}$ is defined, as usual, by

$$\begin{aligned}\hat{u} &= \lim_{\alpha\to 0}\frac{\tilde{u} - u}{\alpha}\\ &= \lim_{\alpha\to 0}\frac{u(u_0 + \alpha h_u, \psi_1 + \alpha h_1, \psi_2 + \alpha h_2) - u(u_0, \psi_1, \psi_2)}{\alpha},\end{aligned}$$

which is the solution of the TLM

$$\begin{gathered}\frac{\partial \hat{u}}{\partial t} + \frac{\partial(u\hat{u})}{\partial x} - \nu\frac{\partial^2 \hat{u}}{\partial x^2} = 0,\\ \hat{u}(0,t) = h_1(t), \quad \hat{u}(L,t) = h_2(t),\\ \hat{u}(x,0) = h_u(x).\end{gathered}$$

We can now compute the equation of the adjoint model. As before, we multiply the TLM by p and integrate by parts on $[0, T]$. For clarity, we do this term by term:

$$\begin{aligned}\int_0^T \left(\frac{\partial \hat{u}}{\partial t}, p\right)\mathrm{d}t &= \int_0^T \int_0^L \frac{\partial \hat{u}}{\partial t} p\,\mathrm{d}x\,\mathrm{d}t\\ &= \int_0^L [\hat{u}\,p]_0^T\,\mathrm{d}x - \int_0^L \int_0^T \frac{\partial p}{\partial t}\hat{u}\,\mathrm{d}x\,\mathrm{d}t\\ &= \int_0^L (\hat{u}(T)p(x,T) - h_u\,p(x,0))\,\mathrm{d}x - \int_0^L \int_0^T \frac{\partial p}{\partial t}\hat{u}\,\mathrm{d}x\,\mathrm{d}t,\end{aligned}$$

[20] Instead of the δ notation, we have used another common form—the letter h—to denote the perturbation direction.

$$\begin{aligned}\int_0^T \left(\frac{\partial (u\hat{u})}{\partial x}, p\right) \mathrm{d}x &= \int_0^T \int_0^L \frac{\partial (u\hat{u})}{\partial x} p \,\mathrm{d}x\,\mathrm{d}t \\ &= \int_0^T [u\hat{u}p]_0^L \,\mathrm{d}t - \int_0^T \int_0^L u\hat{u}\frac{\partial p}{\partial x}\,\mathrm{d}x\,\mathrm{d}t \\ &= \int_0^T (\psi_2 h_2 p(L,t) - \psi_1 h_1 p(0,t))\,\mathrm{d}x - \int_0^T \int_0^L u\hat{u}\frac{\partial p}{\partial x}\,\mathrm{d}x\,\mathrm{d}t,\end{aligned}$$

$$\begin{aligned}&\int_0^T \left(\frac{\partial^2 \hat{u}}{\partial x^2}, p\right) \mathrm{d}t = \int_0^T \int_0^L \frac{\partial^2 \hat{u}}{\partial x^2} p \,\mathrm{d}x\,\mathrm{d}t \\ &= \int_0^T \left[p\frac{\partial \hat{u}}{\partial x}\right]_0^L \mathrm{d}t - \int_0^T \int_0^L \frac{\partial \hat{u}}{\partial x}\frac{\partial p}{\partial x}\,\mathrm{d}x\,\mathrm{d}t \\ &= \int_0^T \left[p\frac{\partial \hat{u}}{\partial x} - \hat{u}\frac{\partial p}{\partial x}\right]_0^L \mathrm{d}t + \int_0^T \int_0^L \hat{u}\frac{\partial^2 p}{\partial x^2}\,\mathrm{d}x\,\mathrm{d}t \\ &= \int_0^T \left(p(L,t)\frac{\partial \hat{u}}{\partial x}(L,t) - h_2\frac{\partial p}{\partial x}(L,t) - p(0,t)\frac{\partial \hat{u}}{\partial x}(0,t) + h_1\frac{\partial p}{\partial x}(0,t)\right)\mathrm{d}t \\ &\quad + \int_0^T \int_0^L \hat{u}\frac{\partial^2 p}{\partial x^2}\,\mathrm{d}x\,\mathrm{d}t.\end{aligned}$$

The natural initial[21] and boundary conditions for p are thus

$$p(x,T) = 0, \quad p(0,t) = p(L,t) = 0,$$

which give

$$\begin{aligned}0 &= \int_0^T \int_0^L \left(\frac{\partial \hat{u}}{\partial t} + \frac{\partial (u\hat{u})}{\partial x} - \nu\frac{\partial^2 \hat{u}}{\partial x^2}\right) p \;\mathrm{d}x\,\mathrm{d}t \\ &= \int_0^T \int_0^L \hat{u}\left(-\frac{\partial p}{\partial t} - u\frac{\partial p}{\partial x} - \nu\frac{\partial^2 p}{\partial x^2}\right)\mathrm{d}x\,\mathrm{d}t \\ &\quad + \int_0^L -h_u p(x,0)\,\mathrm{d}x + \int_0^T \nu h_2\frac{\partial p}{\partial x}(L,t) - \nu h_1\frac{\partial p}{\partial x}(0,t)\,\mathrm{d}t.\end{aligned}$$

In other words,

$$\begin{aligned}\int_0^T \int_0^L \hat{u}\left(-\frac{\partial p}{\partial t} - u\frac{\partial p}{\partial x} - \nu\frac{\partial^2 p}{\partial x^2}\right)\mathrm{d}x\,\mathrm{d}t = &-\int_0^L h_u p(x,0)\,\mathrm{d}x \\ &+ \int_0^T \nu h_2\frac{\partial p}{\partial x}(L,t) - \nu h_1\frac{\partial p}{\partial x}(0,t)\,\mathrm{d}t.\end{aligned}$$

We thus define the adjoint model as

$$\begin{gathered}\frac{\partial p}{\partial t} + u\frac{\partial p}{\partial x} - \nu\frac{\partial^2 p}{\partial x^2} = u - u^{\mathrm{obs}}, \\ p(0,t) = 0, \quad p(L,t) = 0, \\ p(x,T) = 0.\end{gathered}$$

[21] This is in fact a terminal condition, as we have encountered above.

Now we can rewrite the gradient of J in the form

$$\hat{J}[u_0,\psi_1,\psi_2](h_u,h_1,h_2) = -\int_0^L h_u p(x,t=0)\,\mathrm{d}x + \int_0^T \nu h_2 \frac{\partial p}{\partial x}(x=L,t) - \nu h_1 \frac{\partial p}{\partial x}(x=0,t)\,\mathrm{d}t,$$

which immediately yields

$$\nabla_{u_0} J = -p(x,t=0),$$
$$\nabla_{\psi_1} J = -\nu \frac{\partial p}{\partial x}(x=0,t),$$
$$\nabla_{\psi_2} J = \nu \frac{\partial p}{\partial x}(x=L,t).$$

These explicit gradients enable us to solve inverse problems for either (1) the initial condition, which is a data assimilation problem, or (2) the boundary conditions, which is an optimal boundary control problem, or (3) both. Another extension would be a parameter identification problem for ν. This would make an excellent project or advanced exercise.

2.3.8 ▪ Adjoint of finite-dimensional (matrix) operators

Suppose now that we have a solution vector, $\mathbf{x}$, of a *discretized* PDE, or of any other set of n equations. Assume that $\mathbf{x}$ depends as usual on a parameter vector, $\mathbf{m}$, made up of p components—these are sometimes called control variables, design parameters, or decision parameters. If we want to optimize these values for a given cost function, $J(\mathbf{x},\mathbf{m})$, we need to compute, as for the continuous case, the gradient, $\mathrm{d}J/\mathrm{d}\mathbf{m}$. As we have seen above, this should be possible with an adjoint method at a cost that is independent of p and comparable to the cost of a single solution for $\mathbf{x}$. In the finite-dimensional case, this implies the inversion of a linear system, usually $\mathcal{O}(n^3)$ operations. This efficiency, especially for large values of p, is what makes the solution of the inverse problem tractable—if it were not for this, many problems would be simply impossible to solve within reasonable resource limits.

We will first consider systems of linear algebraic equations, and then we can readily generalize to nonlinear systems of algebraic equations and to initial-value problems for linear systems of ODEs.

2.3.8.1 ▪ Linear systems

Let $\mathbf{x}$ be the solution of the $(n \times n)$ linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \tag{2.23}$$

and suppose that $\mathbf{x}$ depends on the parameters $\mathbf{m}$ through $\mathbf{A}(\mathbf{m})$ and $\mathbf{b}(\mathbf{m})$. Define a cost function, $J = J(\mathbf{x},\mathbf{m})$, that depends on $\mathbf{m}$ through $\mathbf{x}$. To evaluate the gradient of J with respect to $\mathbf{m}$ directly, we need to compute by the chain rule

$$\frac{\mathrm{d}J}{\mathrm{d}\mathbf{m}} = \frac{\partial J}{\partial \mathbf{m}} + \frac{\partial J}{\partial \mathbf{x}}\frac{\partial \mathbf{x}}{\partial \mathbf{m}} = J_{\mathbf{m}} + J_{\mathbf{x}}\mathbf{x}_{\mathbf{m}}, \tag{2.24}$$

where $J_{\mathbf{m}}$ is a $(p \times 1)$ column vector, $J_{\mathbf{x}}$ is a $(1 \times n)$ row vector, and $\mathbf{x}_{\mathbf{m}}$ is an $(n \times p)$ matrix. For a given function J the derivatives with respect to $\mathbf{x}$ and $\mathbf{m}$ are assumed to be easily computable. However, it is clearly much more difficult to differentiate $\mathbf{x}$ with respect to $\mathbf{m}$. Let us try and do this directly. We can differentiate, term by term, equation (2.23) with respect to the parameter m_i and solve for $\mathbf{x}_{m_i}$ from (applying the chain rule)

$$\mathbf{x}_{m_i} = \mathbf{A}^{-1}(\mathbf{b}_{m_i} - \mathbf{A}_{m_i}\mathbf{x}).$$

This must be done p times, rapidly becoming unfeasible for large n and p. Recall that p can be of the order of 10^6 in practical DA problems.

The adjoint method, which reduces this to a *single* solve, relies on the trick of adding zero in an astute way. We can do this, as was done above in the continuous case, by introducing a Lagrange multiplier. Since the residual vector $\mathbf{r}(\mathbf{x},\mathbf{m}) = \mathbf{A}\mathbf{x} - \mathbf{b}$ vanishes for the true solution $\mathbf{x}$, we can replace the function J by the augmented function

$$\hat{J} = J - \lambda^{\mathrm{T}}\mathbf{r}, \tag{2.25}$$

where we are free to choose λ at our convenience and we will use this liberty to make the difficult-to-compute term in (2.24), $\mathbf{x}_{\mathbf{m}}$, disappear. So let us take the expression for the gradient (2.24) and evaluate it at $\mathbf{r} = 0$,

$$\begin{aligned} \left.\frac{\mathrm{d}J}{\mathrm{d}\mathbf{m}}\right|_{\mathbf{r}=0} &= \left.\frac{\mathrm{d}\hat{J}}{\mathrm{d}\mathbf{m}}\right|_{\mathbf{r}=0} \\ &= J_{\mathbf{m}} - \lambda^{\mathrm{T}}\mathbf{r}_{\mathbf{m}} + \left(J_{\mathbf{x}} - \lambda^{\mathrm{T}}\mathbf{r}_{\mathbf{x}}\right)\mathbf{x}_{\mathbf{m}}. \end{aligned} \tag{2.26}$$

Then, to "kill" the troublesome $\mathbf{x}_{\mathbf{m}}$ term, we must require that $\left(J_{\mathbf{x}} - \lambda^{\mathrm{T}}\mathbf{r}_{\mathbf{x}}\right)$ vanish, which implies

$$\mathbf{r}_x^{\mathrm{T}}\lambda = J_{\mathbf{x}}^{\mathrm{T}}.$$

But $\mathbf{r}_{\mathbf{x}} = \mathbf{A}$, and hence λ must satisfy the *adjoint equation*

$$\mathbf{A}^{\mathrm{T}}\lambda = J_{\mathbf{x}}^{\mathrm{T}}, \tag{2.27}$$

which is a single $(n \times n)$ linear system. Equation (2.27) is of identical complexity as the original system (2.23), since the adjoint matrix $\mathbf{A}^{\mathrm{T}}$ has the same condition number, sparsity, and preconditioner as $\mathbf{A}$; i.e., if we have a numerical scheme (and hence a computer code) for solving the direct system, we will use precisely the same one for the adjoint.

With λ now known, we can compute the gradient of J from (2.26) as follows:

$$\begin{aligned} \left.\frac{\mathrm{d}J}{\mathrm{d}\mathbf{m}}\right|_{\mathbf{r}=0} &= J_{\mathbf{m}} - \lambda^{\mathrm{T}}\mathbf{r}_{\mathbf{m}} + 0 \\ &= J_{\mathbf{m}} - \lambda^{\mathrm{T}}(\mathbf{A}_{\mathbf{m}}\mathbf{x} - \mathbf{b}_{\mathbf{m}}). \end{aligned}$$

Once again, we assume that when $\mathbf{A}(\mathbf{m})$ and $\mathbf{b}(\mathbf{m})$ are explicitly known, this permits an easy calculation of the derivatives with respect to $\mathbf{m}$. If this is not the case, we must resort to automatic differentiation to compute these derivatives. The automatic differentiation approach will be presented below, after we have discussed nonlinear and initial-value problems.

2.3.8.2 ▪ Nonlinear systems

In general, the state vector $\mathbf{x}$ will satisfy a nonlinear functional equation of the general form

$$f(\mathbf{x},\mathbf{m}) = 0.$$

In this case the workflow is similar to the linear system. We start by solving for $\mathbf{x}$ with an iterative Newton-type algorithm, for example. Now define the augmented J as in (2.25), take the gradient as in (2.26), require that $\mathbf{r}_\mathbf{x}^\mathrm{T}\lambda = J_\mathbf{x}^\mathrm{T}$, and finally compute the gradient

$$\left.\frac{\mathrm{d}J}{\mathrm{d}\mathbf{m}}\right|_{\mathbf{r}=0} = J_\mathbf{m} - \lambda^\mathrm{T}\mathbf{r}_\mathbf{m}. \tag{2.28}$$

There is, of course, a slight modification needed: the adjoint equation is not simply the adjoint as in (2.27) but rather a tangent linear equation obtained by analytical (or automatic) differentiation of J with respect to $\mathbf{x}$.

2.3.8.3 ▪ Initial-value problems

We have, of course, seen this case in quite some detail above. Here we will reformulate it in matrix-vector form. We consider an initial-value problem for a linear, time-independent, homogeneous system of ODEs,

$$\dot{x} = \mathbf{B}x,$$

with $x(0) = b$. We know that the solution is given by

$$x(t) = \mathrm{e}^{\mathbf{B}t}b,$$

but this can be rewritten as a linear system, $\mathbf{A}x = b$, where $\mathbf{A} = \mathrm{e}^{-\mathbf{B}t}$. Now we can simply use our results from above. Suppose we want to minimize $J(\mathbf{x},\mathbf{m})$ based on the solution, x, at time, t. As before, we can compute the adjoint vector, λ, using (2.27),

$$\mathrm{e}^{-\mathbf{B}^\mathrm{T}t}\lambda = J_\mathbf{x}^\mathrm{T},$$

but this is equivalent to the adjoint ODE,

$$\dot{\lambda} = \mathbf{B}^\mathrm{T}\lambda,$$

with $\lambda(0) = J_\mathbf{x}^\mathrm{T}$. This is exactly what we would expect: solving for the adjoint state vector, λ, is a problem of the same complexity and type as that of finding the state vector, x. Clearly we are not obliged to use matrix exponentials for the solution, but we can choose among Runge–Kutta formulas, forward Euler, Crank–Nicolson, etc. [Quarteroni et al., 2007]. What about the important issue of stability? The eigenvalues of $\mathbf{B}$ and $\mathbf{B}^\mathrm{T}$ are complex conjugates and thus the stability of one (spectral radius less than one) implies the stability of the other. Finally, using (2.28), we obtain the gradient of the cost function in the time-dependent case,

$$\begin{aligned}\frac{\mathrm{d}J}{\mathrm{d}\mathbf{m}} &= J_\mathbf{m} - \lambda^\mathrm{T}(\mathbf{A}_\mathbf{m}\mathbf{x} - \mathbf{b}_\mathbf{m})\\ &= J_\mathbf{m} + \int_0^t \lambda^\mathrm{T}(t-t')\mathbf{B}_\mathbf{m}\mathbf{x}(t')\,\mathrm{d}t' + \lambda^\mathrm{T}\mathbf{b}_\mathbf{m},\end{aligned}$$

where we have differentiated the expression for $\mathbf{A}$. We observe that this computation of the gradient via the adjoint requires that we save in memory $\mathbf{x}(t')$ for all times $0 \le t' \le t$ to be able to compute the gradient. This is a well-known issue in adjoint approaches for time-dependent problems and can be dealt with in three ways (that are problem or, more precisely, dimension dependent):

1. Store everything in memory, if feasible.

2. If not, use some kind of checkpointing [Griewank and Walther, 2000], which means that we divide the time interval into a number of subintervals and store consecutively subinterval by subinterval.

3. Re-solve "simultaneously" forward and adjoint, and at the same time compute the integral; i.e., at each time step of the adjoint solution process, recompute the direct solution up to this time.

2.3.9 ▪ Continuous and discrete adjoints

In the previous section, we saw how to deal with finite-dimensional systems. This leads us naturally to the study of discrete adjoints, which can be computed by automatic differentiation, as opposed to analytical methods, where we took variations and used integration by parts. In the following discussion, the aim is not to show exactly how to write an adjoint code generator but to provide an understanding of the principles. Armed with this knowledge, the reader will be able to critically analyze (if needed) the eventual reasons for failure of the approach when applied to a real problem. An excellent reference is Hascoet [2012]—see also Griewank [2000].

To fix ideas, let us consider a second-order PDE of the general form (without loss of generality)

$$F(t, u, u_t, u_x, u_{xx}, \theta) = 0$$

and an objective function, $J(u, \theta)$, that depends on the unknown u and the parameters θ. As usual, we are interested in calculating the gradients of the cost function with respect to the parameters to find an optimal set of parameter values—usually one that attains the least-squares difference between simulated model predictions and real observations/measurements.

There are, in fact, two possible approaches for computing an adjoint state and the resulting gradients or sensitivities:

- discretization of the (analytical) adjoint, which we denote by **AtD** = Adjoint then Discretize (we have amply seen this above);

- adjoint of the discretization (the code), which we denote as **DtA** = Discretize then Adjoint.

The first is the *continuous* case, where we differentiate the PDE with respect to the parameters and then discretize the adjoint PDE to compute the approximate gradients. In the second, called the *discrete* approach, we first approximate the PDE by a discrete (linear or nonlinear) system and then differentiate the resulting discrete system with respect to the parameters. This is done by automatic differentiation of the code, which solves the PDE using tools such as TAPENADE, YAO, OpenAD, ADIFOR, ADMat, etc.—see `www.autodiff.org`. Note that numerical computation of gradients can be achieved by two other means: divided/finite differences or symbolic differentiation.[22] The first is notoriously unstable, and the latter cannot deal with complex functionals. For these reasons, the adjoint method is largely preferable.

In "simpler" problems, AtD is preferable,[23] but this assumes that we are able to calculate analytically the adjoint equation by integration by parts and that we can find compatible boundary conditions for the adjoint variable—see, for example, Bocquet [2012a]. This was largely developed above. In more realistic, complex cases, we must often resort to DtA, but then we may be confronted with serious difficulties each time the code is modified, since this implies the need to regenerate the adjoint. DtA is, however, well-suited for a nonexpert who does not need to have a profound understanding of the simulation codes to compute gradients. The DtA approach works for any cost functional, and no explicit boundary conditions are needed. However, DtA may turn out to be inconsistent with the adjoint PDE if a nonlinear, high-resolution scheme (such as upwinding) is used—a comparison of the two approaches can be found in Li and Petzold [2004], where the important question of consistency is studied and a simple example of the 1D heat equation is also presented.

2.4 ▪ Variational DA

2.4.1 ▪ Introduction

2.4.1.1 ▪ History

Variational DA was formally introduced by the meteorological community for solving the problem of numerical weather prediction (NWP).

In 1922, Lewis Fry Richardson published the first attempt at forecasting the weather numerically. But large errors were observed that were caused by inaccuracies in the fields used as the initial conditions in his analysis [Lynch, 2008], thus indicating the need for a DA scheme.

Originally, subjective analysis was used to correct the simulation results. In this approach, NWP forecasts were adjusted manually by meteorologists using their operational expertise and experience. Then objective analysis (e.g., Cressman's successive correction algorithm), which fitted data to grids, was introduced for automated DA. These objective methods used simple interpolation approaches (e.g., a quadratic polynomial interpolation scheme based on least-squares regression) and thus were 3D DA methods.

Later, 4D DA methods, called nudging, were developed. These are based on the simple idea of Newtonian relaxation and introduce into the right-hand side of the model dynamical equations a term that is proportional to the difference of the calculated meteorological variable and the observed value. This term has a negative sign and thus keeps the calculated state vector closer to the observations. Nudging can be interpreted as a variant of the Kalman filter (KF) with the gain matrix prescribed, rather than obtained from covariances. Various nudging algorithms are described in Chapter 4.

A major development was achieved by L. Gandin [1963], who introduced the statistical interpolation (or optimal interpolation) method, which developed earlier ideas of Kolmogorov. This is a 3D DA method and is a type of regression analysis that utilizes information about the spatial distributions of covariance functions of the errors

[22]By packages such as Maple, Mathematica, SAGE, etc.

[23]Though there are differences of opinion among practitioners who prefer the discrete adjoint for these cases as well. Thus, the final choice depends on one's personal experience and competence.

of the first guess field (previous forecast) and true field. The optimal interpolation algorithm is the reduced version of the KF algorithm in which the covariance matrices are not calculated from the dynamical equations but are predetermined. This is treated in Chapter 3.

Attempts to introduce KF algorithms as a 4D DA tool for NWP models came later. However, this was (and remains) a difficult task due to the very high dimensions of the computational grid and the underlying matrices. To overcome this difficulty, approximate or suboptimal KFs were developed. These include the ensemble Kalman filter (EnKF) and the reduced-rank Kalman filters (such as RRSQRT)—see Chapters 3, 5, and 6.

Another significant advance in the development of the 4D DA methods was the use of optimal control theory, also known as the variational approach. In the seminal work of Le Dimet and Talagrand [1986] based on earlier work of G. Marchuk, they were the first to apply the theory of Lions [1988] (see also Tröltzsch [2010]) to environmental modeling. The significant advantage of the variational approach is that the meteorological fields satisfy the dynamical equations of the NWP model, and at the same time they minimize the functional characterizing the difference between simulations and observations. Thus, a problem of constrained minimization is solved, as has been amply shown above in this chapter.

As has been shown by Lorenc [2003], Talagrand [2012], and others, all the abovementioned 4D DA methods are in some limit equivalent. Under certain assumptions they minimize the same cost function. However, in practical applications these assumptions are never fulfilled and the different methods perform differently. This raises the still disputed question: Which approach, Kalman filtering or variational assimilation, is better? Further fundamental questions arise in the application of advanced DA techniques. A major issue is that of the convergence of the numerical method to the global minimum of the functional to be minimized—please refer to the important discussions in the first two sections of this chapter.

The 4D DA method that is currently most successful is hybrid incremental 4D-Var (see below and Chapters 5 and 7), where an ensemble is used to augment the climatological background error covariances at the start of the DA time window, but the background error covariances are evolved during the time window by a simplified version of the NWP forecast model. This DA method is used operationally at major forecast centers, though there is currently a tendency to move toward the more efficient ensemble variational (EnVar) methods that will be described in Chapter 7.

2.4.1.2 ▪ Formulation

In variational DA we describe the state of the system by a state variable, $\mathbf{x}(t) \in \mathscr{X}$, a function of space and time that represents the physical variables of interest, such as current velocity (in oceanography), temperature, sea-surface height, salinity, biological species concentration, or chemical concentration. The evolution of the state is described by a system of (in general nonlinear) differential equations in a region Ω,

$$\begin{cases} \dfrac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathscr{M}(\mathbf{x}) \quad \text{in } \Omega \times [0, T], \\ \mathbf{x}(t=0) = \mathbf{x}_0, \end{cases} \tag{2.29}$$

where the initial condition is unknown (or inaccurately known). Suppose that we are in possession of observations $\mathbf{y}(t) \in \mathscr{O}$ and an observation operator $\mathscr{H}$ that describes the available observations. Then, to characterize the difference between the observations and the state, we define the objective (or cost) function

$$J(\mathbf{x}_0) = \frac{1}{2}\int_0^T \|\mathbf{y}(t) - \mathcal{H}(\mathbf{x}(\mathbf{x}_0, t))\|_{\mathcal{O}}^2 \, \mathrm{d}t + \frac{1}{2}\left\|\mathbf{x}_0 - \mathbf{x}^{\mathrm{b}}\right\|_{\mathcal{X}}^2, \tag{2.30}$$

where $\mathbf{x}^{\mathrm{b}}$ is the background (or first guess) and the second term plays the role of a regularization (in the sense of Tikhonov—see Vogel [2002] and Hansen [2010]). The two norms under the integral, in the finite-dimensional case, will be represented by the error covariance matrices **R** and **B**, respectively—see Chapter 1 and Section 2.4.3 below. Note that, for mathematical rigor, we have indicated the relevant functional spaces on which the norms are defined.

In the continuous context, the DA problem is formulated as follows: find the analyzed state, $\mathbf{x}_0^{\mathrm{a}}$, that minimizes J and satisfies

$$\mathbf{x}_0^{\mathrm{a}} = \operatorname{argmin} J(\mathbf{x}_0).$$

As seen above, the necessary condition for the existence of a (local) minimum is

$$\nabla J(\mathbf{x}_0^{\mathrm{a}}) = 0.$$

2.4.2 ▪ Adjoint methods in DA

To solve the above minimization problem for variational DA, we will use the adjoint approach. In summary, the *adjoint method* for DA is an iterative scheme that involves searching for the minimum of a scalar cost function with respect to a multidimensional initial state. The search algorithm is called a descent method and requires the derivative of the cost function with respect to arbitrary perturbations of the initial state. This derivative, or gradient, is obtained by running an adjoint model backward in time. Once the derivative is obtained, a direction that leads to lower cost has been identified, but the step size has not. Therefore, further calculations are needed to determine how far along this direction one needs to go to find a lower cost. Once this initial state is found, the next iteration is started. The algorithm proceeds until the minimum of the cost function is found. It should be noted that the adjoint method is used in 4D-Var to find the initial conditions that minimize a cost function. However, one could equally well have chosen to find the boundary conditions, or model parameters, as was done in the numerous examples presented in Section 2.3.

We point out that a truly unified derivation of variational DA should start from a probabilistic/statistical model. Then, as was mentioned above (see Section 1.5), we can obtain the 3D-Var model as a special case. We will return to this in Chapter 3. Here, as opposed to the presentation in Chapter 1, we will give a unified treatment of 3D- and 4D-Var that leads naturally to variants of the approach.

2.4.3 ▪ 3D-Var and 4D-Var: A unified framework

The 3D-Var and 4D-Var approaches were introduced in Chapter 1. Here we will recall the essential points of the formulation, present them in a unified fashion (after Talagrand [2012]), and expand on some concrete aspects and variants.

Unlike sequential/statistical assimilation (which emanates from estimation theory), we saw that variational assimilation is based on optimal control theory, itself derived from the *calculus of variations*. The analyzed state was defined as the one that *minimizes a cost function*. The minimization requires numerical optimization techniques. These techniques can rely on the *gradient* of the cost function, and this gradient will be obtained with the aid of *adjoint methods*, which we have amply discussed above. Note that variational DA is a particular usage of the adjoint approach.

Usually, 3D-Var and 4D-Var are introduced in a finite-dimensional or discrete context—this approach will be used in this section. For the infinite-dimensional or continuous case, we must use the calculus of variations and PDEs, as was done in the previous sections of this chapter.

We start out with the following cost function:

$$J(x) = \frac{1}{2}\left(\mathbf{x}-\mathbf{x}^{\mathrm{b}}\right)^{\mathrm{T}} \mathbf{B}^{-1}\left(\mathbf{x}-\mathbf{x}^{\mathrm{b}}\right) + \frac{1}{2}(\mathbf{H}\mathbf{x}-\mathbf{y})^{\mathrm{T}}\mathbf{R}^{-1}(\mathbf{H}\mathbf{x}-\mathbf{y}), \tag{2.31}$$

where, as was defined in the notation of Section 1.5.1, $\mathbf{x}$, $\mathbf{x}^{\mathrm{b}}$, and $\mathbf{y}$ are the state, the background state, and the measured state, respectively; $\mathbf{H}$ is the observation matrix (a linearization of the observation operator $\mathscr{H}$); and $\mathbf{R}$ and $\mathbf{B}$ are the observation and background error covariance matrices, respectively. This quadratic function attempts to strike a balance between some a priori knowledge about a background (or historical) state and the actual measured, or observed, state. It also assumes that we know and can invert the matrices $\mathbf{R}$ and $\mathbf{B}$—this, as will be pointed out below, is far from obvious. Furthermore, it represents the sum of the (weighted) background deviations and the (weighted) observation deviations.

2.4.3.1 ▪ The stationary case: 3D-Var

We note that when the background, $\mathbf{x}^{\mathrm{b}} = \mathbf{x}^{\mathrm{b}} + \epsilon^{\mathrm{b}}$, is available at some time t_k, together with observations of the form $\mathbf{y} = \mathbf{H}\mathbf{x}^{\mathrm{t}} + \epsilon^{\mathrm{o}}$ that have been acquired at the same time (or over a short enough interval of time when the dynamics can be considered stationary), then the minimization of (2.31) will produce an estimate of the system state at time t_k. In this case, the analysis is called three-dimensional variational analysis and is abbreviated as *3D-Var*.

We have seen above, in Section 1.5.2, that the best linear unbiased estimator (BLUE) requires the computation of an optimal gain matrix. We will show (in Chapter 3) that the optimal gain takes the form

$$\mathbf{K} = \mathbf{B}\mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{B}\mathbf{H}^{\mathrm{T}} + \mathbf{R})^{-1},$$

where $\mathbf{B}$ and $\mathbf{R}$ are the covariance matrices, to obtain an analyzed state,

$$\mathbf{x}^{\mathrm{a}} = \mathbf{x}^{\mathrm{b}} + \mathbf{K}(\mathbf{y} - \mathbf{H}(\mathbf{x}^{\mathrm{b}})).$$

But this is precisely the state that minimizes the 3D-Var cost function. This is quite easily verified by taking the gradient, term by term, of the cost function (2.31) and equating to zero,

$$\nabla J(\mathbf{x}^{\mathrm{a}}) = \mathbf{B}^{-1}\left(\mathbf{x}^{\mathrm{a}} - \mathbf{x}^{\mathrm{b}}\right) - \mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}(\mathbf{y} - \mathbf{H}\mathbf{x}^{\mathrm{a}}) = 0, \tag{2.32}$$

where

$$\mathbf{x}^{\mathrm{a}} = \operatorname{argmin} J(\mathbf{x}).$$

Solving the equation, we find

$$
\begin{aligned}
\mathbf{B}^{-1}\left(\mathbf{x}^{\mathrm{a}}-\mathbf{x}^{\mathrm{b}}\right) &= \mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\left(\mathbf{y}-\mathbf{H}\mathbf{x}^{\mathrm{a}}\right),\\
\left(\mathbf{B}^{-1}+\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)\mathbf{x}^{\mathrm{a}} &= \mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{y}+\mathbf{B}^{-1}\mathbf{x}^{\mathrm{b}},\\
\mathbf{x}^{\mathrm{a}} &= \left(\mathbf{B}^{-1}+\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{y}+\mathbf{B}^{-1}\mathbf{x}^{\mathrm{b}}\right)\\
&= \left(\mathbf{B}^{-1}+\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\left(\left(\mathbf{B}^{-1}+\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)\mathbf{x}^{\mathrm{b}}\right.\\
&\qquad \left.-\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\mathbf{x}^{\mathrm{b}}+\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{y}\right)\\
&= \mathbf{x}^{\mathrm{b}}+\left(\mathbf{B}^{-1}+\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\left(\mathbf{y}-\mathbf{H}\mathbf{x}^{\mathrm{b}}\right)\\
&= \mathbf{x}^{\mathrm{b}}+\mathbf{K}\left(\mathbf{y}-\mathbf{H}\mathbf{x}^{\mathrm{b}}\right), \qquad (2.33)
\end{aligned}
$$

where we have simply added and subtracted the term $\left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)\mathbf{x}^{\mathrm{b}}$ in the third-to-last line, and in the last line we have brought out what are known as the *innovation* term,

$$\mathbf{d}=\mathbf{y}-\mathbf{H}\mathbf{x}^{\mathrm{b}},$$

and the *gain matrix*,

$$\mathbf{K}=\left(\mathbf{B}^{-1}+\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}.$$

This matrix can be rewritten as

$$\mathbf{K}=\mathbf{B}\mathbf{H}^{\mathrm{T}}\left(\mathbf{R}+\mathbf{H}\mathbf{B}\mathbf{H}^{\mathrm{T}}\right)^{-1} \qquad (2.34)$$

using the well-known Sherman–Morrison–Woodbury formula of linear algebra [Golub and van Loan, 2013], which completely avoids the direct computation of the inverse of the matrix $\mathbf{B}$. The linear combination in (2.33) of a background term plus a multiple of the innovation is a classical result of linear-quadratic control theory [Friedland, 1986; Gelb, 1974; Kwakernaak and Sivan, 1972] and shows how nicely DA fits in with and corresponds to (optimal) control theory. The form of the gain matrix (2.34) can be explained quite simply. The term $\mathbf{H}\mathbf{B}\mathbf{H}^{\mathrm{T}}$ is the background covariance transformed to the observation space. The denominator term, $\mathbf{R}+\mathbf{H}\mathbf{B}\mathbf{H}^{\mathrm{T}}$, expresses the sum of observation and background covariances. The numerator term, $\mathbf{B}\mathbf{H}^{\mathrm{T}}$, takes the ratio of $\mathbf{B}$ and $\mathbf{R}+\mathbf{H}\mathbf{B}\mathbf{H}^{\mathrm{T}}$ back to the model space. This recalls (and is completely analogous to) the variance ratio,

$$\frac{\sigma_{\mathrm{b}}^2}{\sigma_b^2+\sigma_{\mathrm{o}}^2},$$

that appears in the optimal BLUE (see Chapter 1 and Chapter 3) solution. This is the case for a single observation, $\mathbf{y}$, of a quantity, $\mathbf{x}$,

$$
\begin{aligned}
x^{\mathrm{a}} &= x^{\mathrm{b}}+\frac{\sigma_{\mathrm{b}}^2}{\sigma_{\mathrm{b}}^2+\sigma_{\mathrm{o}}^2}(x^{\mathrm{o}}-x^{\mathrm{b}})\\
&= x^{\mathrm{b}}+\frac{1}{1+\alpha}(x^{\mathrm{o}}-x^{\mathrm{b}}),
\end{aligned}
$$

where

$$\alpha=\frac{\sigma_{\mathrm{o}}^2}{\sigma_{\mathrm{b}}^2}.$$

In other words, the best way to estimate the state is to take a weighted average of the background (or prior) and the observations of the state. And the best weight is the ratio of the mean squared errors (variances). This statistical viewpoint is thus perfectly reproduced in the 3D-Var framework.

2.4.3.2 ▪ The nonstationary case: 4D-Var

A more realistic, but complicated, situation arises when one wants to assimilate observations that are acquired over a time interval during which the system dynamics (flow, for example) cannot be neglected. Suppose that the measurements are available at a succession of instants, t_k, $k = 0, 1, \ldots, K$, and are of the form

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{x}_k + \epsilon_k^{\mathrm{o}}, \tag{2.35}$$

where $\mathbf{H}_k$ is a linear observation operator and ϵ_k^{o} is the observation error with covariance matrix $\mathbf{R}_k$, and suppose that these observation errors are uncorrelated in time. Now we add the dynamics described by the state equation,

$$\mathbf{x}_{k+1} = \mathbf{M}_{k+1}\mathbf{x}_k, \tag{2.36}$$

where we have neglected any model error.[24] We suppose also that at time index $k = 0$ we know the background state, $\mathbf{x}_0^{\mathrm{b}}$, and its error covariance matrix, $\mathbf{P}_0^{\mathrm{b}}$, and we suppose that the errors are uncorrelated with the observations in (2.35). Then a given initial condition, $\mathbf{x}_0$, defines a unique model solution, $\mathbf{x}_{k+1}$, according to (2.36). We can now generalize the objective function (2.31), which becomes

$$J(\mathbf{x}_0) = \frac{1}{2}\left(\mathbf{x}_0 - \mathbf{x}_0^{\mathrm{b}}\right)^T \left(\mathbf{P}_0^{\mathrm{b}}\right)^{-1} \left(\mathbf{x}_0 - \mathbf{x}_0^{\mathrm{b}}\right) + \frac{1}{2}\sum_{k=0}^{K} (\mathbf{H}_k \mathbf{x}_k - \mathbf{y}_k)^T \mathbf{R}_k^{-1} (\mathbf{H}_k \mathbf{x}_k - \mathbf{y}_k). \tag{2.37}$$

The minimization of $J(\mathbf{x}_0)$ will provide the initial condition of the model that fits the data most closely. This analysis is called *strong constraint four-dimensional variational assimilation*, abbreviated as *strong constraint 4D-Var*. The term *strong constraint* implies that the model found by the state equation (2.36) must be exactly satisfied by the sequence of estimated state vectors.

In the presence of model uncertainty, the state equation becomes

$$\mathbf{x}_{k+1}^{\mathrm{t}} = \mathbf{M}_{k+1}\mathbf{x}_k^{\mathrm{t}} + \eta_{k+1}, \tag{2.38}$$

where the model noise has covariance matrix $\mathbf{Q}_k$,which we suppose to be uncorrelated in time and uncorrelated with the background and observation errors. The objective function for the BLUE for the sequence of states

$$\{\mathbf{x}_k,\ k = 0, 1, \ldots, K\}$$

is of the form

$$\begin{aligned} J(\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_K) = {} & \frac{1}{2}\left(\mathbf{x}_0 - \mathbf{x}_0^{\mathrm{b}}\right)^{\mathrm{T}} \left(\mathbf{P}_0^{\mathrm{b}}\right)^{-1} \left(\mathbf{x}_0 - \mathbf{x}_0^{\mathrm{b}}\right) \\ & + \frac{1}{2}\sum_{k=0}^{K} (\mathbf{H}_k \mathbf{x}_k - \mathbf{y}_k)^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{H}_k \mathbf{x}_k - \mathbf{y}_k) \\ & + \frac{1}{2}\sum_{k=0}^{K-1} \left(\mathbf{x}_{k+1} - \mathbf{M}_{k+1}\mathbf{x}_k\right)^{\mathrm{T}} \mathbf{Q}_{k+1}^{-1} \left(\mathbf{x}_{k+1} - \mathbf{M}_{k+1}\mathbf{x}_k\right). \end{aligned} \tag{2.39}$$

[24] This will be taken into account in Section 2.4.7.5.

This objective function has become a function of the complete sequence of states

$$\{\mathbf{x}_k,\, k=0,1,\ldots,K\},$$

and its minimization is known as *weak constraint four-dimensional variational assimilation*, abbreviated as *weak constraint 4D-Var*. Equations (2.37) and (2.39), with an appropriate reformulation of the state and observation spaces, are special cases of the BLUE objective function—see Talagrand [2012].

All the above forms of variational assimilation, as defined by (2.31), (2.37), and (2.39), have been used for real-world DA, in particular in meteorology and oceanography. However, these methods are directly applicable to a vast array of other domains, among which we can cite geophysics and environmental sciences, seismology, atmospheric chemistry, and terrestrial magnetism. Examples of all these can be found in the applications chapters of Part III. We remark that in real-world practice, variational assimilation is performed on nonlinear models. If the extent of nonlinearity is sufficiently small (in some sense), then variational assimilation, even if it does not solve the correct estimation problem, will still produce useful results.

Some remarks concerning implementation: Now, our problem reduces to quantifying the covariance matrices and then, of course, computing the analyzed state. The quantification of the covariance matrices must result from extensive data studies (or the use of a KF approach—see Chapter 3). The computation of the analyzed state will be described in the next subsection—this will not be done directly, but rather by an adjoint approach for minimizing the cost functions. There is of course the inverse of $\mathbf{B}$ or $\mathbf{P}^{\mathrm{b}}$ to compute, but we remark that there appear only matrix-vector products of $\mathbf{B}^{-1}$ and $\left(\mathbf{P}^{\mathrm{b}}\right)^{-1}$, and we can thus define operators (or routines) that compute these efficiently without the need for large storage capacities.

2.4.3.3 ▪ The adjoint approach

We explain the adjoint approach in the case of strong constraint 4D-Var, taking into account a completely general nonlinear setting for the model and for the observation operators. Let $\mathbf{M}_k$ and $\mathbf{H}_k$ be the nonlinear model and observation operators, respectively. We reformulate (2.36) and (2.37) in terms of the nonlinear operators as

$$\begin{aligned} J(\mathbf{x}_0) = &\frac{1}{2}\left(\mathbf{x}_0-\mathbf{x}_0^{\mathrm{b}}\right)^{\mathrm{T}}\left(\mathbf{P}_0^{\mathrm{b}}\right)^{-1}\left(\mathbf{x}_0-\mathbf{x}_0^{\mathrm{b}}\right)\\ &+\frac{1}{2}\sum_{k=0}^{K}\left(\mathbf{H}_k(\mathbf{x}_k)-\mathbf{y}_k\right)^{\mathrm{T}}\mathbf{R}_k^{-1}\left(\mathbf{H}_k(\mathbf{x}_k)-\mathbf{y}_k\right), \end{aligned} \tag{2.40}$$

with the dynamics

$$\mathbf{x}_{k+1}=\mathbf{M}_{k+1}(\mathbf{x}_k), \quad k=0,1,\ldots,K-1. \tag{2.41}$$

The minimization problem requires that we now compute the gradient of J with respect to $\mathbf{x}_0$. The gradient is determined from the property that for a given perturbation $\delta\mathbf{x}_0$ of $\mathbf{x}_0$, the corresponding first-order variation of J is

$$\delta J=\left(\nabla_{\mathbf{x}_0}J\right)^{\mathrm{T}}\delta\mathbf{x}_0. \tag{2.42}$$

The perturbation is propagated by the tangent linear equation,

$$\delta\mathbf{x}_{k+1}=M_{k+1}\delta\mathbf{x}_k, \quad k=0,1,\ldots,K-1, \tag{2.43}$$

obtained by differentiation of the state equation (2.41), where M_{k+1} is the Jacobian matrix (of first-order partial derivatives) of $\mathbf{x}_{k+1}$ with respect to $\mathbf{x}_k$. The first-order variation of the cost function is obtained similarly by differentiation of (2.40),

$$\delta J = \left(\mathbf{x}_0 - \mathbf{x}_0^{\mathrm{b}}\right)^{\mathrm{T}} \left(\mathbf{P}_0^{\mathrm{b}}\right)^{-1} \delta \mathbf{x}_0 + \sum_{k=0}^{K} (\mathbf{H}_k(\mathbf{x}_k) - \mathbf{y}_k)^{\mathrm{T}} \mathbf{R}_k^{-1} \mathrm{H}_k \delta \mathbf{x}_k, \tag{2.44}$$

where H_k is the Jacobian of $\mathbf{H}_k$ and $\delta \mathbf{x}_k$ is defined by (2.43). This variation is a compound function of $\delta \mathbf{x}_0$ that depends on all the $\delta \mathbf{x}_k$'s. But if we can obtain a direct dependence on $\delta \mathbf{x}_0$ in the form of (2.42), eliminating the explicit dependence on $\delta \mathbf{x}_k$, then we will (as in the previous sections of this chapter) arrive at an explicit expression for the gradient, $\nabla_{\mathbf{x}_0} J$, of our cost function, J. This will be done, as we have done before, by introducing an adjoint state and requiring that it satisfy certain conditions—namely, the adjoint equation. Let us now proceed with this program.

We begin by defining, for $k = 0, 1, \ldots, K$, the adjoint state vectors $\mathbf{p}_k$ that belong to the dual of the state space. Now we take the null products (according to the tangent state equation (2.43)),

$$\mathbf{p}_k^{\mathrm{T}} \left(\delta\ x_k - M_k \delta \mathbf{x}_{k-1}\right),$$

and subtract them from the right-hand side of the cost function variation (2.44),

$$\begin{aligned} \delta J = {} & \left(\mathbf{x}_0 - \mathbf{x}_0^{\mathrm{b}}\right)^{T} \left(\mathbf{P}_0^{\mathrm{b}}\right)^{-1} \delta \mathbf{x}_0 + \sum_{k=0}^{K} (\mathbf{H}_k(\mathbf{x}_k) - \mathbf{y}_k)^{T} \mathbf{R}_k^{-1} \mathrm{H}_k \delta \mathbf{x}_k \\ & - \sum_{k=0}^{K} \mathbf{p}_k^{T} \left(\delta \mathbf{x}_k - M_k \delta \mathbf{x}_{k-1}\right). \end{aligned}$$

Rearranging the matrix products, using the symmetry of $\mathbf{R}_k$, and regrouping terms in $\delta \mathbf{x}$, we obtain

$$\begin{aligned} \delta J = {} & \left[\left(\mathbf{P}_0^{\mathrm{b}}\right)^{-1} \left(\mathbf{x}_0 - \mathbf{x}_0^{\mathrm{b}}\right) + \mathrm{H}_0^{\mathrm{T}} \mathbf{R}_0^{-1} (\mathbf{H}_0(\mathbf{x}_0) - \mathbf{y}_0) + M_0^{\mathrm{T}} \mathbf{p}_1 \right] \delta \mathbf{x}_0 \\ & + \left[\sum_{k=1}^{K-1} \mathrm{H}_k^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{H}_k(\mathbf{x}_k) - \mathbf{y}_k) - \mathbf{p}_k + M_k^{\mathrm{T}} \mathbf{p}_{k+1} \right] \delta \mathbf{x}_k \\ & + \left[\mathrm{H}_K^{\mathrm{T}} \mathbf{R}_K^{-1} (\mathbf{H}_K(\mathbf{x}_K) - \mathbf{y}_K) - \mathbf{p}_K \right] \delta \mathbf{x}_k. \end{aligned}$$

Notice that this expression is valid for any choice of the adjoint states, $\mathbf{p}_k$, and, in order to "kill" all $\delta \mathbf{x}_k$ terms, except $\delta \mathbf{x}_0$, we must simply impose that

$$\mathbf{p}_K = \mathrm{H}_K^{\mathrm{T}} \mathbf{R}_K^{-1} (\mathbf{H}_K(\mathbf{x}_K) - \mathbf{y}_K), \tag{2.45}$$

$$\mathbf{p}_k = \mathrm{H}_k^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{H}_k(\mathbf{x}_k) - \mathbf{y}_k) + M_k^{\mathrm{T}} \mathbf{p}_{k+1}, \quad k = K-1, \ldots, 1, \tag{2.46}$$

$$\mathbf{p}_0 = \left(\mathbf{P}_0^{\mathrm{b}}\right)^{-1} \left(\mathbf{x}_0 - \mathbf{x}_0^{\mathrm{b}}\right) + \mathrm{H}_0^{\mathrm{T}} \mathbf{R}_0^{-1} (\mathbf{H}_0(\mathbf{x}_0) - \mathbf{y}_0) + M_0^{\mathrm{T}} \mathbf{p}_1. \tag{2.47}$$

We recognize the backward adjoint equation for $\mathbf{p}_k$, and the only term remaining in the variation of J is then

$$\delta J = \mathbf{p}_0^{T} \delta \mathbf{x}_0,$$

so that $\mathbf{p}_0$ is the sought-for gradient, $\nabla_{\mathbf{x}_0} J$, of the objective function with respect to the initial condition, $\mathbf{x}_0$, according to (2.42). The system of equations (2.45)–(2.47) is

Algorithm 2.1 Iterative 3D-Var algorithm.

$k = 0$, $x = x_0$
while $||\nabla J|| > \epsilon$ **or** $k \leq k_{\max}$
  compute J with (2.31)
  compute ∇J with (2.32)
  gradient descent and update of x_{k+1}
  $k = k + 1$
end

the adjoint of the tangent linear equation (2.43). The term *adjoint* here corresponds to the transposes of the matrices $\mathrm{H}_k^{\mathrm{T}}$ and M_k^{T} that, as we have seen before, are the finite-dimensional analogues of an adjoint operator. We can now propose the "usual" algorithm for solving the optimization problem by the adjoint approach:

1. For a given initial condition, $\mathbf{x}_0$, integrate forward the (nonlinear) state equation (2.41) and store the solutions, $\mathbf{x}_k$ (or use some sort of checkpointing).

2. From the final condition, (2.45), integrate backward in time the adjoint equations (2.46).

3. Compute directly the required gradient (2.47).

4. Use this gradient in an iterative optimization algorithm to find a (local) minimum.

The above description for the solution of the 4D-Var DA problem clearly covers the case of 3D-Var, where we seek to minimize (2.31). In this case, we need only the transpose Jacobian H^T of the observation operator.

2.4.4 ▪ The 3D-Var algorithm

The matrices involved in the calculation of equation (2.33) are often neither storable in memory nor manipulable because of their very large dimensions, which can be as much as 10^6 or more. Thus, the direct calculation of the gain matrix, $\mathbf{K}$, is unfeasible. The 3D-Var variational method overcomes these difficulties by attempting to iteratively minimize the cost function, J. This minimization can be achieved, for inverse problems in general, by a combination of an adjoint approach for the computation of the gradient with a descent algorithm in the direction of the gradient. For DA problems where there is no time dependence, the adjoint operation requires only a matrix adjoint (and not the solution of an adjoint equation[25]), and the approach is called 3D-Var, whereas for time-dependent problems we will use the 4D-Var approach, which is presented in the next subsection.

The iterative 3D-Var Algorithm 2.1 is a classical case of an optimization algorithm [Nocedal and Wright, 2006] that uses as a stopping criterion the fact that ∇J is small or that the maximum number of iterations, $k_{\max}$, is reached. For the gradient descent, there is a wide choice of algorithmic approaches, but quasi-Newton methods [Nocedal and Wright, 2006; Quarteroni et al., 2007] are generally used and recommended.

[25] This may not be valid for complicated observation operators.

2.4.4.1 ▪ On the roles of R and B

The relative magnitudes of the errors due to measurement and background provide us with important information as to how much "weight" to give to the different information sources when solving the assimilation problem. For example, if background errors are larger than observation errors, then the analyzed state solution to the DA problem should be closer to the observations than to the background and vice versa.

The background error covariance matrix, $\mathbf{B}$, plays an important role in DA. This is illustrated by the following example.

Example 2.5. Effect of a single observation. Suppose that we have a single observation at a point corresponding to the jth element of the state vector. The observation operator is then

$$\mathbf{H} = \begin{pmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \end{pmatrix}.$$

The gradient of J is

$$\nabla J = \mathbf{B}^{-1}\left(\mathbf{x} - \mathbf{x}^{\mathrm{b}}\right) + \mathbf{H}^T\mathbf{R}^{-1}\left(\mathbf{H}\mathbf{x} - \mathbf{y}^{\mathrm{o}}\right).$$

Since it must be equal to zero at the minimum $\mathbf{x}^{\mathrm{a}}$,

$$\left(\mathbf{x}^{\mathrm{a}} - \mathbf{x}^{\mathrm{b}}\right) = \mathbf{B}\mathbf{H}^T\mathbf{R}^{-1}\left(\mathbf{y}^{\mathrm{o}} - \mathbf{H}\mathbf{x}^{\mathrm{a}}\right).$$

But $\mathbf{R} \doteq \sigma^2$; $\mathbf{H}\mathbf{x}^{\mathrm{a}} = x_j^{\mathrm{a}}$; and $\mathbf{B}\mathbf{H}^T$ is the jth column of $\mathbf{B}$, whose elements are denoted by $B_{i,j}$ with $i = 1, \ldots, n$. So we see that

$$\mathbf{x}^{\mathrm{a}} - \mathbf{x}^{\mathrm{b}} = \frac{y^{\mathrm{o}} - x_k^{\mathrm{a}}}{\sigma^2} \begin{pmatrix} B_{1,j} \\ B_{2,j} \\ \vdots \\ B_{n,j} \end{pmatrix}.$$

The increment is proportional to a column of $\mathbf{B}$. The choice of $\mathbf{B}$ is thus crucial and will determine how this observation provides information about what happens around the jth variable. ■

In the 4D-Var case, the increment at time t will be proportional to a single column of $\mathbf{M}\mathbf{B}\mathbf{M}^T$, which describes the error covariances of the background at the time, t, of the observation.

2.4.5 ▪ The 4D-Var algorithm

In this section, we reformulate the 4D-Var approach in a form that is better adapted to algorithmic implementation. As we have just seen, the 4D-Var method generalizes 3D-Var to the case where the observations are obtained at different times—this is depicted in Figure 2.4. As was already stated in Chapter 1, the difference between three-dimensional (3D-Var) and four-dimensional (4D-Var) DA is the use of a numerical forecast model in the latter. In 4D-Var, the cost function is still expressed in terms of the initial state, $\mathbf{x}_0$, but it includes the model because the observation $\mathbf{y}_k^{\mathrm{o}}$ at time k is compared to $\mathbf{H}_k(\mathbf{x}_k)$, where $\mathbf{x}_k$ is the state at time k initialized by $\mathbf{x}_0$ and the adjoint is not simply the transpose of a matrix, but the "transpose" of the model/operator dynamics.

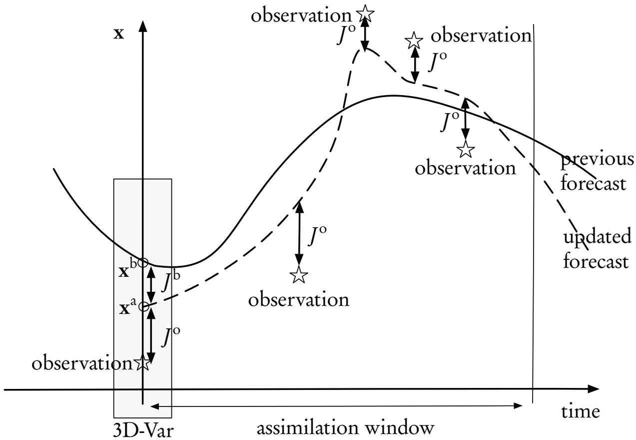


Figure 2.4. *3D- and 4D-Var.*

2.4.5.1 ▪ Cost function and gradient

The cost function (2.37) is still expressed in terms of the initial state, $\mathbf{x}$ (we have dropped the zero subscript, for simplicity), but it now includes the model because the observation $\mathbf{y}_k^\mathrm{o}$ at time k is compared to $\mathbf{H}_k(\mathbf{x}_k)$, where $\mathbf{x}_k$ is the state at time k initialized by $\mathbf{x}$. The cost function is the sum of the background and the observation errors,

$$J(\mathbf{x}) = J^\mathrm{b}(\mathbf{x}) + J^\mathrm{o}(\mathbf{x}),$$

where the background term is the same as above:

$$J^\mathrm{b}(\mathbf{x}) = \frac{1}{2}\left(\mathbf{x}-\mathbf{x}^\mathrm{b}\right)^T \mathbf{B}^{-1}\left(\mathbf{x}-\mathbf{x}^\mathrm{b}\right).$$

The background $\mathbf{x}^\mathrm{b}$, as with $\mathbf{x}$, is taken as a vector at the initial time, $k = 0$. The observation term is more complicated. We define

$$J^\mathrm{o}(\mathbf{x}) = \frac{1}{2}\sum_{k=0}^{K}\left(\mathbf{y}_k^\mathrm{o} - \mathbf{H}_k(\mathbf{x}_k)\right)^T \mathbf{R}_k^{-1}\left(\mathbf{y}_k^\mathrm{o} - \mathbf{H}_k(\mathbf{x}_k)\right),$$

where the state at time k is obtained by an iterated composition of the model matrix,

$$\begin{aligned}\mathbf{x}_k &= \mathbf{M}_{0\to k}(\mathbf{x})\\ &= \mathbf{M}_{k-1,k}\mathbf{M}_{k-2,k-1}\dots\mathbf{M}_{1,2}\mathbf{M}_{0,1}\mathbf{x}\\ &= \mathbf{M}_k\mathbf{M}_{k-1}\dots\mathbf{M}_2\mathbf{M}_1\mathbf{x}.\end{aligned}$$

This gives the final form of the observation term,

$$J^\mathrm{o}(\mathbf{x}) = \frac{1}{2}\sum_{k=0}^{K}\left(\mathbf{y}_k^\mathrm{o} - \mathbf{H}_k\mathbf{M}_k\mathbf{M}_{k-1}\dots\mathbf{M}_2\mathbf{M}_1\mathbf{x}\right)^T \mathbf{R}_k^{-1}\left(\mathbf{y}_k^\mathrm{o} - \mathbf{M}_k\mathbf{M}_{k-1}\dots\mathbf{M}_2\mathbf{M}_1\mathbf{x}\right).$$

Algorithm 2.2 4D-Var

$n=0,\ \mathbf{x}=\mathbf{x}_0$
while $\|\nabla J\| > \epsilon$ **or** $n \le n_{\max}$
  (1) compute J with the direct model $\mathbf{M}$ and $\mathbf{H}$
  (2) compute ∇J with adjoint model $\mathbf{M}^{\mathrm{T}}$ and $\mathbf{H}^{\mathrm{T}}$ (reverse mode)
  gradient descent and update of $\mathbf{x}_{n+1}$
  $n=n+1$
end

Now we can compute the gradient directly (whereas in the previous subsection we computed the variation, δJ):

$$\nabla J(\mathbf{x}) = \mathbf{B}^{-1}\left(\mathbf{x}-\mathbf{x}^{\mathrm{b}}\right) - \sum_{k=0}^{K} \mathbf{M}_1^T \mathbf{M}_2^T \ldots \mathbf{M}_{k-1}^T \mathbf{M}_k^T \mathbf{H}_k^T \mathbf{R}_k^{-1}\left(\mathbf{y}_k^{\mathrm{o}} - \mathbf{M}_k \mathbf{M}_{k-1} \ldots \mathbf{M}_2 \mathbf{M}_1 \mathbf{x}\right).$$

If we denote the innovation vector as

$$\mathbf{d}_k = \mathbf{y}_k^{\mathrm{o}} - \mathbf{H}_k \mathbf{M}_k \mathbf{M}_{k-1} \ldots \mathbf{M}_2 \mathbf{M}_1 \mathbf{x},$$

then we have

$$\begin{aligned}
-\nabla J^{\mathrm{o}}(\mathbf{x}) &= \sum_{k=0}^{K} \mathbf{M}_1^T \mathbf{M}_2^T \ldots \mathbf{M}_{k-1}^T \mathbf{M}_k^T \mathbf{H}_k^T \mathbf{R}_k^{-1} \mathbf{d}_k \\
&= \mathbf{H}_0^T \mathbf{R}_0^{-1} \mathbf{d}_0 + \mathbf{M}_1^T \mathbf{H}_1^T \mathbf{R}_1^{-1} \mathbf{d}_1 + \mathbf{M}_1^T \mathbf{M}_2^T \mathbf{H}_2^T \mathbf{R}_2^{-1} \mathbf{d}_2 + \cdots \\
&\quad + \mathbf{M}_1^T \ldots \mathbf{M}_{K-1}^T \mathbf{M}_K^T \mathbf{H}_K^T \mathbf{R}_K^{-1} \mathbf{d}_K \\
&= \mathbf{H}_0^T \mathbf{R}_0^{-1} \mathbf{d}_0 + \mathbf{M}_1^T \left[\mathbf{H}_1^T \mathbf{R}_1^{-1} \mathbf{d}_1 + \mathbf{M}_2^T \left[\mathbf{H}_2^T \mathbf{R}_2^{-1} \mathbf{d}_2 + \cdots + \mathbf{M}_K^T \mathbf{H}_K^T \mathbf{R}_K^{-1} \mathbf{d}_K\right]\right].
\end{aligned}$$

This factorization enables us to compute J^{o} followed by ∇J^{o} with one integration of the direct model followed by one integration of the adjoint model.

2.4.5.2 ▪ Algorithm

For Algorithm 2.2, in step (1) we use the equations

$$\mathbf{d}_k = \mathbf{y}_k^{\mathrm{o}} - \mathbf{H}_k \mathbf{M}_k \mathbf{M}_{k-1} \ldots \mathbf{M}_2 \mathbf{M}_1 \mathbf{x}$$

and

$$J(\mathbf{x}) = \frac{1}{2}\left(\mathbf{x}-\mathbf{x}^{\mathrm{b}}\right)^T \mathbf{B}^{-1}\left(\mathbf{x}-\mathbf{x}^{\mathrm{b}}\right) + \sum_{k=0}^{K} \mathbf{d}_k^T \mathbf{R}_k^{-1} \mathbf{d}_k.$$

In step (2), we use

$$\begin{aligned}
\nabla J(\mathbf{x}) = \mathbf{B}^{-1}\left(\mathbf{x}-\mathbf{x}^{\mathrm{b}}\right) - \big[&\mathbf{H}_0^T \mathbf{R}_0^{-1} \mathbf{d}_0 + \mathbf{M}_1^T \left[\mathbf{H}_1^T \mathbf{R}_1^{-1} \mathbf{d}_1 + \mathbf{M}_2^T \left[\mathbf{H}_2^T \mathbf{R}_2^{-1} \mathbf{d}_2 + \cdots \right.\right. \\
&\left.\left. + \mathbf{M}_K^T \mathbf{H}_K^T \mathbf{R}_K^{-1} \mathbf{d}_K\right]\right]\big].
\end{aligned}$$

2.4.5.3 ▪ A very simple scalar example

We consider an example with a single observation at time step 3 and a known background at time step 0. In this case, the 4D-Var cost function (2.37) for determining the initial state becomes scalar,

$$J(x_0) = \frac{1}{2}\frac{\left(x_0 - x_0^{\mathrm{b}}\right)^2}{\sigma_B^2} + \frac{1}{2}\sum_{k=1}^{K}\frac{\left(x_k - x_k^{\mathrm{o}}\right)^2}{\sigma_R^2},$$

where σ_B^2 and σ_R^2 are the (known) background and observation error variances, respectively. With a single observation at time step 3, the cost function is

$$J(x_0) = \frac{1}{2}\frac{\left(x_0 - x_0^{\mathrm{b}}\right)^2}{\sigma_B^2} + \frac{1}{2}\frac{\left(x_3 - x_3^{\mathrm{o}}\right)^2}{\sigma_R^2}.$$

The minimum is reached when the gradient of J disappears,

$$J'(x_0) = 0,$$

which can be computed as

$$\frac{\left(x_0 - x_0^{\mathrm{b}}\right)}{\sigma_B^2} + \frac{\left(x_3 - x_3^{\mathrm{o}}\right)}{\sigma_R^2}\frac{dx_3}{dx_2}\frac{dx_2}{dx_1}\frac{dx_1}{dx_0} = 0. \tag{2.48}$$

We now require a dynamic relation between the x_k's to compute the derivatives. To this end, let us take the most simple linear forecast model,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\alpha x,$$

with α a known positive constant. This is a typical model for describing decay, for example, of a chemical compound whose behavior over time is then given by

$$x(t) = x(0)\mathrm{e}^{-\alpha t}.$$

To obtain a discrete representation of the dynamics, we can use an upstream finite difference scheme [Strikwerda, 2004],

$$x(t_{k+1}) - x(t_k) = (t_{k+1} - t_k)\left[-\alpha x(t_{k+1})\right], \tag{2.49}$$

which can be rewritten in the explicit form

$$x(t + \Delta t) = \left(\frac{1}{1 + \alpha\Delta t}\right)x(t),$$

where we have assumed a fixed time step, $\Delta t = t_{k+1} - t_k$, for all k. We thus have the scalar relation

$$x_{k+1} = M(x_k) = \gamma x_k, \tag{2.50}$$

where the constant is

$$\gamma = \frac{1}{1 + \alpha\Delta t}.$$

The necessary condition (2.48) then becomes

$$\frac{\left(x_0 - x_0^{\mathrm{b}}\right)}{\sigma_B^2} + \frac{\left(x_3 - x_3^{\mathrm{o}}\right)}{\sigma_R^2}\gamma^3 = 0.$$

This can be solved for x_0 and then for x_3 to obtain the analyzed state

$$\begin{aligned} x_0 &= x_0^{\mathrm{b}} + \frac{\gamma^3\sigma_B^2}{\sigma_R^2}(x_3^{\mathrm{o}} - x_3) \\ &= x_0^{\mathrm{b}} + \frac{\gamma^3\sigma_B^2}{\sigma_R^2}\left(x_3^{\mathrm{o}} - \gamma^3 x_0^{\mathrm{b}}\right) \\ &= \frac{\sigma_R^2}{\sigma_R^2+\gamma^6\sigma_B^2}x_0^{\mathrm{b}} + \frac{\gamma^3\sigma_B^2}{\sigma_R^2+\gamma^6\sigma_B^2}\left(x_3^{\mathrm{o}} - \gamma^3 x_0^{\mathrm{b}}\right) \\ &= x_0^{\mathrm{b}} + \frac{\gamma^3\sigma_B^2}{\sigma_R^2+\gamma^6\sigma_B^2}\left[x_3^{\mathrm{o}} - \gamma^3 x_0^{\mathrm{b}}\right], \end{aligned}$$

where we have added and subtracted x_0^{b} to obtain the last line and used the system dynamics (2.50). Finally, by again using the dynamics, we find the 4D-Var solution

$$x_3 = \gamma^3 x_0^{\mathrm{b}} + \frac{\gamma^6\sigma_B^2}{\sigma_R^2+\gamma^6\sigma_B^2}\left[x_3^{\mathrm{o}} - \gamma^3 x_0^{\mathrm{b}}\right]. \tag{2.51}$$

Let us examine some asymptotic cases. If the parameter α tends to zero, then the dynamic gain, γ, tends to one and the model becomes stationary, with

$$x_{k+1} = x_k.$$

The solution then tends to the 3D-Var case, with

$$x_3 = x_0 = x_0^{\mathrm{b}} + \frac{\sigma_B^2}{\sigma_R^2+\sigma_B^2}\left[x_3^{\mathrm{o}} - x_0^{\mathrm{b}}\right]. \tag{2.52}$$

If the model is stationary, we can thus use all observations whenever they become available, exactly as in the 3D case.

The other asymptotic occurs when the step size tends to infinity and the dynamic gain goes to zero. The dynamic model becomes

$$x_{k+1} = 0,$$

with the initial condition $x_0 = x_0^{\mathrm{b}}$, and there is thus no connection between states at different time steps. Finally, if the observation is perfect, then $\sigma_R^2 = 0$ and

$$x_3 = x_3^{\mathrm{o}}.$$

But there is no link to x_0, and there is once again no dynamical connection between states at two different instants.

2.4.6 ▪ Practical variants of 3D-Var and 4D-Var

We have described above the simplest classical 3D-Var and 4D-Var algorithms. To overcome the numerous problems encountered in their implementation, there are several extensions and variants of these methods. We will describe two of the most important here. Further details can be found in Chapter 5.

2.4.6.1 ▪ Incremental 3D-Var and 4D-Var

We saw above that the adjoint of the complete model (2.40) is required for computing the gradient of the cost function. In NWP, the full nonlinear model is extremely complex [Kalnay, 2003]. To alleviate this, Courtier et al. [1994] proposed an incremental approach to variational assimilation, several variants of which now exist. Basically, the idea is to simplify the dynamical model (2.41) to obtain a formulation that is cheaper for the adjoint computation. To do this, we modify the tangent model (2.43), which becomes

$$\delta \mathbf{x}_{k+1} = \mathbf{L}_{k+1} \delta \mathbf{x}_k, \quad k = 0, 1, \ldots, K-1, \tag{2.53}$$

where $\mathbf{L}_k$ is an appropriately chosen simplified version of the Jacobian operator M_k. To preserve consistency, the basic model (2.41) must be appropriately modified so that the TLM corresponding to a known (e.g., from the background) reference solution, $\mathbf{x}_k^{(0)}$, is given by (2.53). This is easily done by letting the initial condition

$$\mathbf{x}_0 = \mathbf{x}_0^{(0)} + \delta \mathbf{x}_0$$

evolve according to (2.53) into

$$\mathbf{x}_k = \mathbf{x}_k^{(0)} + \delta \mathbf{x}_k.$$

The resulting dynamics are then linear.

Several possibilities exist for simplifying the objective function (2.40). One can linearize the observation operator H_k, as was done for the model M_k. We use the substitution

$$H_k(\mathbf{x}_k) \longmapsto H_k(\mathbf{x}_k^{(0)}) + \mathbf{N}_k \delta \mathbf{x}_k,$$

where $\mathbf{N}_k$ is some simplified linear approximation, which could be the Jacobian of H_k at $\mathbf{x}_k$. The objective function (2.40) then becomes

$$\begin{aligned} J_1(\delta \mathbf{x}_0) = & \frac{1}{2} \left(\delta \mathbf{x}_0 + \mathbf{x}_0^{(0)} - \mathbf{x}_0^{\mathrm{b}} \right)^T \left(\mathbf{P}_0^{\mathrm{b}} \right)^{-1} \left(\delta \mathbf{x}_0 + \mathbf{x}_0^{(0)} - \mathbf{x}_0^{\mathrm{b}} \right) \\ & + \frac{1}{2} \sum_{k=0}^{K} (\mathbf{N}_k \delta \mathbf{x}_k - \mathbf{d}_k)^T \mathbf{R}_k^{-1} (\mathbf{N}_k \delta \mathbf{x}_k - \mathbf{d}_k), \end{aligned} \tag{2.54}$$

where the $\delta \mathbf{x}_k$ satisfy (2.53) and the innovation at time k is $\mathbf{d}_k = \mathbf{y}_k - H_k(\mathbf{x}_k^{(0)})$. This objective function, J_1, is quadratic in the initial perturbation $\delta \mathbf{x}_0$, and the minimizer, $\delta \mathbf{x}_{0,m}$, defines an updated initial state

$$\mathbf{x}_0^{(1)} = \mathbf{x}_0^{(0)} + \delta \mathbf{x}_{0,m},$$

from which a new solution, $\mathbf{x}_k^{(1)}$, can be computed using the dynamics (2.41). Then we loop and repeat the whole process for $\mathbf{x}_k^{(1)}$. This defines a system of two-level nested loops (outer and inner) for minimizing the original cost function (2.40). The savings are thanks to the flexible choice that is possible for the simplified linearized operators $\mathbf{L}_k$ and $\mathbf{N}_k$. These can be chosen to ensure a reasonable trade-off between ease of implementation and physical fidelity. One can even modify the operator $\mathbf{L}_k$ in (2.53) during the minimization by gradually introducing more complex dynamics in the successive outer loops—this is the multi-incremental approach that is described in Section 5.4.1. Convergence issues are of course a major concern—see, for example, Tremolet [2007a].

These incremental methods together with the adjoint approach are what make variational assimilation computationally tractable. In fact, they have been used until now in most operational NWP systems that employ variational DA.

2.4.6.2 ▪ FGAT 3D-Var

This method, "first guess at appropriate time," (abbreviated FGAT) is best viewed as a special case of 4D-Var. It is in fact an extreme case of the incremental approach (2.53)–(2.54), in which the simplified linear operator $\mathbf{L}_k$ is set equal to the identity.

The process is 4D in the sense that the observations, distributed over the assimilation window, are compared with the computed values in the time integration of the assimilating model. But it is 3D because the minimization of the cost function (2.54) does not use the correct dynamics, i.e.,

$$\delta \mathbf{x}_{k+1} = \delta \mathbf{x}_k, \quad k = 0, 1, \ldots, K-1.$$

The FGAT 3D-Var approach, using a unique minimization loop (there is no nesting any more), has been shown to improve the accuracy of the assimilated variables. The reason for this is simple: FGAT uses a more precise innovation vector than standard 3D-Var, where all observations are compared with the same first-guess field.

2.4.7 ▪ Extensions and complements

2.4.7.1 ▪ Parameter estimation

If we want to optimize a set of parameters,

$$\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_p),$$

we need only include the control variables as terms in the cost function,

$$J(\mathbf{x}, \alpha) = J_1^{\mathrm{b}}(\mathbf{x}) + J_2^{\mathrm{b}}(\alpha) + J^{\mathrm{o}}(\mathbf{x}, \alpha).$$

The observation term includes a dependence on α, and it is often necessary to add a regularization term for α, such as

$$J_2^{\mathrm{b}}(\alpha) = \left\| \alpha - \alpha^{\mathrm{b}} \right\|^2, \text{ or } \quad J_2^{\mathrm{b}}(\alpha) = \left(\alpha - \alpha^{\mathrm{b}} \right) \mathbf{B}_\alpha^{-1} \left(\alpha - \alpha^{\mathrm{b}} \right),$$
$$\text{or} \quad J_2^{\mathrm{b}}(\alpha) = \| \nabla \alpha - \beta \|^2 .$$

2.4.7.2 ▪ Nonlinearities

When the nonlinearities in the model and/or the observation operator are weak, we can extend the 3D- and 4D-Var algorithms to take their effects into account. One can then define the *incremental 4D-Var algorithm*—see above.

2.4.7.3 ▪ Preconditioning

We recall that the condition number of a matrix $\mathbf{A}$ is the product $\|\mathbf{A}\| \|\mathbf{A}^{-1}\|$. In general, variational DA problems are badly conditioned. The rate of convergence of the minimization algorithms depends on the conditioning of the Hessian of the cost function: the closer it is to one, the better the convergence. For 4D-Var, the Hessian is equal to $\left(\mathbf{B}^{-1} + \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H}\right)$, and its condition number is usually very high.

Preconditioning [Golub and van Loan, 2013] is a technique for improving the condition number and thus accelerating the convergence of the optimization. We make a change of variable

$$\delta \mathbf{x} = \mathbf{x} - \mathbf{x}^{\mathrm{b}}$$

such that

$$\mathbf{w} = \mathbf{L}^{-1}\delta \mathbf{x}, \quad \mathbf{B}^{-1} = \mathbf{L}\mathbf{L}^{T},$$

where $\mathbf{L}$ is a given simple matrix. This is commonly used in meteorology and oceanography. The modified cost function is

$$\tilde{J}(\mathbf{w}) = \frac{1}{2}\mathbf{w}^{T}\mathbf{w} + \frac{1}{2}(\mathbf{H}\mathbf{L}\mathbf{w} - \mathbf{d})^{T}\mathbf{R}^{-1}(\mathbf{H}\mathbf{L}\mathbf{w} - \mathbf{d}),$$

and its Hessian is equal to

$$\tilde{J}'' = \mathbf{I} + \mathbf{L}^{T}\mathbf{H}^{T}\mathbf{R}^{-1}\mathbf{H}\mathbf{L}.$$

It is in general much better conditioned, and the resulting improvement in convergence can be spectacular.

2.4.7.4 ▪ Covariance matrix modeling

The modeling of the covariance matrices of the background error $\mathbf{B}$ and the observation error $\mathbf{R}$ is an important operational research subject. Reduced-cost models are particularly needed when the matrices are of high dimensions—in weather forecasting or turbulent flow control problems, for example, this can run into tens of millions. One may also be interested in having better-quality approximations of these matrices.

In background error covariance modeling [Fisher, 2003], compromises have to be made to produce a computationally viable model. Since we do not have access to the true background state, we must either separate out the information about the statistics of background error from the innovation statistics or derive statistics for a surrogate quantity. Both approaches require assumptions to be made, for example about the statistical properties of the observation error. The "separation" approach can be addressed by running an ensemble of randomly perturbed predictions, drawn from relevant distributions. This method of generating surrogate fields of background error is strongly related to the EnKF, which is fully described in Chapter 6—see also Evensen [2009].

Other approaches for modeling the $\mathbf{B}$ matrix by reduced bases, factorization, and spectral methods are fully described in Chapter 5.

2.4.7.5 ▪ Model error

In standard variational assimilation, we invert for the initial condition only. The underlying hypothesis that the model is perfectly known is not a realistic one. In fact, to take into account eventual model error, we should add an appropriate error term to the state equation and insert a cost term into the objective function. We thus arrive at a *parameter identification* inverse problem, similar to those already studied above in Section 2.3.

In the presence of model uncertainty, the state equation and objective functions become (see also the above equations (2.38) and (2.39))

$$\begin{cases} \dfrac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathcal{M}(\mathbf{x}) + \eta(t) \quad \text{in } \Omega \times [0, T], \\ \mathbf{x}(t = 0) = \mathbf{x}_0, \end{cases}$$

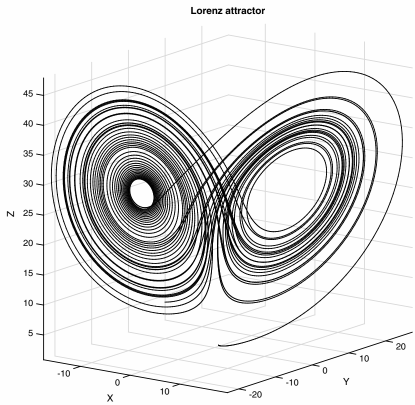


Figure 2.5. *Simulation of the chaotic Lorenz-63 system of three equations.*

where $\eta(t)$ is a suitably uncorrelated white noise. The new cost functional is

$$J(\mathbf{x}_0,\eta)=\frac{1}{2}\left\|\mathbf{x}_0-\mathbf{x}^{\mathrm{b}}\right\|_{\mathscr{X}}^2+\frac{1}{2}\int_0^T \|\mathbf{y}(t)-\mathscr{H}(\mathbf{x}(\mathbf{x}_0,t))\|_{\mathscr{O}}^2\,\mathrm{d}t+\frac{1}{2}\int_0^T \|\eta(t)\|_{\mathscr{E}}^2\,\mathrm{d}t,$$

where the model noise has covariance matrix $\mathbf{Q}$, which we suppose to be uncorrelated in time and uncorrelated with the background and observation errors. However, in cases with high dimensionality, this approach is not feasible, especially for practical problems. Numerous solutions have been proposed to overcome this problem—see Griffith and Nichols [2000], Tremolet [2007b], Vidard et al. [2004], and Tremolet [2007c].

2.5 ▪ Numerical examples

2.5.1 ▪ DA for the Lorenz equation

We study the nonlinear Lorenz system of equations [Lorenz, 1963],

$$\begin{aligned}\frac{dx}{dt}&=-\sigma(x-y),\\ \frac{dy}{dt}&=\rho x-y-xz,\\ \frac{dz}{dt}&=xy-\beta z,\end{aligned}$$

which exhibits chaotic behavior when we fix the parameter values $\sigma=10$, $\rho=28$, and $\beta=8/3$ (see Figure 2.5) This equation is a simplified model for atmospheric convection and is an excellent example of the lack of predictability. It is ill-posed in the sense of Hadamard. In fact, the solution switches between two stable orbits around the points

$$\left(\sqrt{\beta(\rho-1)}, \sqrt{\beta(\rho-1)}, \rho-1\right)$$

and

$$\left(-\sqrt{\beta(\rho-1)}, -\sqrt{\beta(\rho-1)}, \rho-1\right).$$

We now perform 4D-Var DA on this equation with only the observation term,

$$J^{\mathrm{o}}(\mathbf{x}) = \frac{1}{2}\sum_{i=0}^{n}\left(\mathbf{y}_k^{\mathrm{o}} - \mathbf{H}_k(\mathbf{x}_k)\right)^T \mathbf{R}_k^{-1}\left(\mathbf{y}_k^{\mathrm{o}} - \mathbf{H}_k(\mathbf{x}_k)\right).$$

This relatively simple model enables us to study a number of important effects and to answer the following practical questions:

- What is the influence of observation noise?
- What is the influence of the initial guess?
- What is the influence of the length of the assimilation window and the number of observations?

In addition, we can compare the performance of the standard 4D-Var with that of an incremental 4D-Var algorithm. All computations are based on the codes provided by A. Lawless of the DARC (Data Assimilation Research Centre) at Reading University [Lawless, 2002]. Readers are encouraged to obtain the software and experiment with it.

The assimilation results shown in Figures 2.6 and 2.7 were obtained from twin experiments with the following conditions:

- True initial condition is $(1.0, 1.0, 1.0)$.
- Initial guess is $(1.2, 1.2, 1.2)$.
- Time step is 0.05 seconds.
- Assimilation window is 2 seconds.
- Forecast window is 3 seconds.
- Observations are every 2 time steps.
- Number of outer loops (for incremental 4D-Var) is 4.

We remark that the incremental algorithm produces a more accurate forecast, over a longer period, in this case—see Figure 2.7.

2.5.2 ▪ Additional DA examples

Numerous examples of variational DA can be found in the advanced Chapters 4, 5, and 7, as well as in the applications sections—see Part III. Another rich source is the training material of the ECMWF [Bouttier and Courtier, 1997]—see `http://www.ecmwf.int/en/learning/education-material/lecture-notes`.

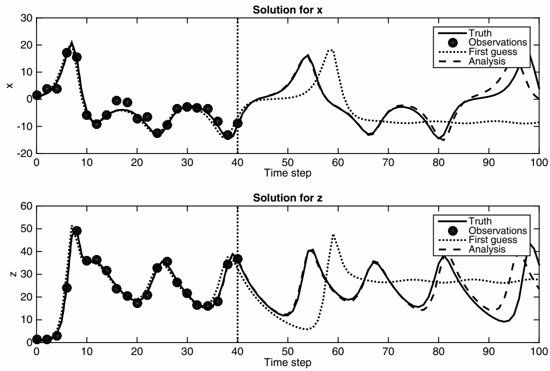


Figure 2.6. *Assimilation of the Lorenz-63 equations by standard 4D-Var, based on a twin experiment. The assimilation window is from step* 0 *to step* 40 *(2 seconds). The forecast window is from step* 41 *to* 100 *(3 seconds).*

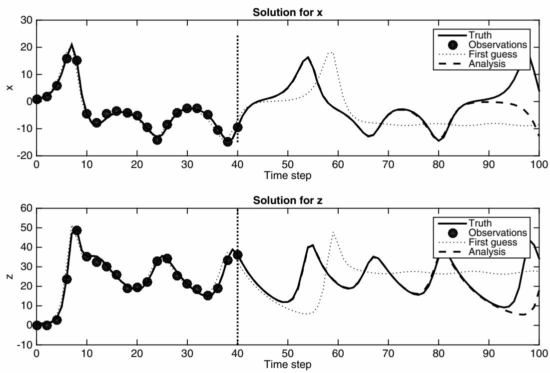


Figure 2.7. *Assimilation of the Lorenz-63 equations by incremental 4D-Var, based on a twin experiment. The assimilation window is from step* 0 *to step* 40 *(2 seconds). The forecast window is from step* 41 *to* 100 *(3 seconds).*

Chapter 3

Statistical estimation and sequential data assimilation

The road to wisdom?—Well, it's plain and simple to express:
Err
and err
and err again
but less
and less
and less.
—Piet Hein (1905–1996, Danish mathematician and inventor)

3.1 ▪ Introduction

In this chapter, we present the statistical approach to DA. This approach will be addressed from a Bayesian point of view. But before delving into the mathematical and algorithmic details, we will discuss some ideas about the history of weather forecasting and of the distinction between prediction and forecasting. For a broad, nontechnical treatment of prediction in a sociopolitical-economic context, the curious reader is referred to Silver [2012], where numerous empirical aspects of forecasting are also broached.

3.1.1 ▪ A long history of prediction

From Babylonian times, people have attempted to predict future events, for example in astronomy. Throughout the Renaissance and the Industrial Revolution there were vast debates on predictability.

In 1814, Pierre-Simon Laplace postulated that a perfect knowledge of the actual state of a system coupled with the equations that describe its evolution (natural laws) should provide perfect predictions! This touches on the far-reaching controversy between determinism and randomness/uncertainty … and if we go all the way down to the level of quantum mechanics, then due to Heisenberg's principle there cannot be a perfect prediction. However, weather (and many other physical phenomena) go no further than the molecular (not the atomic) level and as a result molecular chemistry and Newtonian physics are sufficient for weather forecasting. In fact, the (deterministic) PDEs that describe the large-scale circulation of air masses and oceans are remarkably precise and can reproduce an impressive range of meteorological conditions. This is equally true in a large number of other application domains, as described in Part III.

Weather forecasting is a success story: human and machine combining their efforts to understand and to anticipate a complex natural system. This is true for many other systems thanks to the broad applicability of DA and inverse problem methods and algorithms.

3.1.2 ▪ Stochastic versus deterministic

The simplest statistical approach to forecasting (rather like linear regression, but with a flat line) is to calculate the probability of an event (e.g., rain tomorrow) based on past knowledge and records—i.e., long-term averages. But these purely statistical predictions are of little value—they do not take into account the possibility and potential that we have of modeling the physics—this is where the progress (over the last 30 years) in numerical analysis and high-performance computing can come to the rescue. However, this is not a trivial pursuit, as we often notice when surprised by a rain shower, flood, stock market crash, or earthquake. So what goes wrong and impairs the accuracy/reliability of forecasts?

- The first thing that can go wrong is the resolution (spatial and temporal) of our numerical models … but this is an easier problem: just add more computing power, energy, and money!
- Second, and more important, is *chaos* (see Section 2.5.1), which applies to dynamic, nonlinear systems and is closely associated with the well-posedness issues of Chapter 1—note that this has nothing to do with randomness, but rather is related to the lack of predictability. In fact, in weather modeling, for example, after approximately one week only, chaos theory swamps the dynamic memory of the atmosphere (as "predicted" by the physics), and we are better off relying on climatological forecasts that are based on historical averaged data.
- Finally, there is our imprecise knowledge of the initial (and boundary) conditions for our physical model and hence our simulations—this loops back to the previous point and feeds the chaotic nature of the system. Our measurements are both incomplete and (slightly) inaccurate due to the physical limitations of the instruments themselves.

All of the above needs to be accounted for, as well as possible, in our numerical models and computational analysis. This can best be done with a probabilistic[26] approach.

3.1.3 ▪ Prediction versus forecast

The terms *prediction* and *forecast* are used interchangeably in most disciplines but deserve a more rigorous definition/distinction. Following the philosophy of Silver [2012], a prediction will be considered as a deterministic statement, whereas a forecast will be a probabilistic one. Here are two examples:

- "A major earthquake will strike Tokyo on May 28th" is a prediction, whereas "there is a 60% chance of a major earthquake striking Northern California over the next 25 years" is a forecast.

[26]Equivalently, a statistical or stochastic approach can be used.

- Extrapolation is another example of prediction and is in fact a very basic method that can be useful in some specific contexts but is generally too simplistic and can lead to very bad predictions and decisions.

We notice the UQ in the forecast statement. One way to implement UQ is through Bayesian reasoning—let us explain this now.

3.1.4 ▪ DA is fundamentally Bayesian

Thomas Bayes[27] believed in a rational world of Newtonian mechanics but insisted that by gathering evidence we can get closer and closer to the truth. In other words, rationality is probabilistic. Laplace, as we saw above, claimed that with perfect knowledge of the present and of the laws governing its evolution, we can attain perfect knowledge of the future. In fact it was Laplace who formulated what is known as Bayes' theorem. He considered probability to be "a waypoint between ignorance and knowledge." This is not bad ... it corresponds exactly to our endeavor and what we are trying to accomplish throughout this book: use models and simulations to reproduce and then predict (or, more precisely, forecast or better understand) the actual state and future evolutions of a complex system. For Laplace it was clear: we need a more thorough understanding of probability to make scientific progress!

Bayes' theorem is a very simple algebraic formula based on conditional probability (the probability of one event, A, occurring, knowing or given that another event, B, has occurred—see Section 3.2 below for the mathematical definitions):

$$p_{A|B} = \frac{p_{B|A}\, p_A}{p_B}.$$

It basically provides us with a reevaluated probability (posterior, $p_{A|B}$) based on the prior knowledge, $p_{B|A}\, p_A$, of the system that is normalized by the total knowledge that we have, p_B. To better understand and appreciate this result, let us consider a couple of simple examples that illustrate the importance of Bayesian reasoning.

Example 3.1. The famous example of breast cancer diagnosis from mammograms shows the importance and strength of priors. Based on epidemiological studies, the probability that a woman between the ages of 40 and 50 will be afflicted by a cancer of the breast is low, of the order of $p_A = 0.014$ or 1.4%. The question we want to answer is: If a woman in this age range has a positive mammogram (event B), what is the probability that she indeed has a cancer (event A)? Further studies have shown that the false-positive rate of mammograms is $p = 0.1$ or 10% of the time and that the correct diagnosis (true positive) has a rate of $p_{B|A} = 0.75$. So a positive mammogram, taken by itself, would seem to be serious news. However, if we do a Bayesian analysis that factors in the prior information, we get a different picture. Let us do this now. The posterior probability can be computed from Bayes' formula,

$$p_{A|B} = \frac{p_{B|A}\, p_A}{p_B} = \frac{0.75 \times 0.014}{0.75 \times 0.014 + 0.1 \times (1 - 0.014)} = 0.096,$$

and we conclude that the probability is only 10% in this case, which is far less worrisome than the overall 75% true-positive rate. So the false positives have dominated the result thanks to the fact that we have taken into account the prior information of

[27] English clergyman and statistician (1701–1761).

low cancer incidence in this age range. For this reason, there is a tendency in the medical profession today to recommend that women (without antecedents, which would increase the value of p_A) start having regular mammograms starting from age 50 only because, starting from this age, the prior probability is higher. ■

Example 3.2. Another good example comes from global warming, now called climate change, and we will see why it is so important to quantify uncertainty in the interest of scientific advancement and trust. The study of global warming started around the year 2001. At this time, it was commonly accepted, and scientifically justified, that CO_2 emissions caused and would continue to cause a rise in global temperatures. Thus, we could attribute a high prior probability, $p_A = 0.95$, to the hypothesis of global warming (event A). However, over the subsequent decade from 2001 to 2011, we have observed (event B) that global temperatures have *not* risen as expected—in fact they appeared to have decreased very slightly.[28] So, according to Bayesian reasoning, we should reconsider our estimation of the probability of global warming—the question is, to what extent? If we had a good estimate of the uncertainty in short-term patterns of temperature variations, then the downward revision of the prediction would not be drastic. By analyzing the historical data again, we find that there is a 15% chance that there is no net warming over a decade even if the global warming hypothesis holds—this is due to the inherent variability in the climate. On the other hand, if temperature variations were purely random, and hence unpredictable, then the chance of having a decade in which there is actually a cooling would be 50%. So let us compute the revised estimate for global warming with Bayes' formula. We find

$$p_{A|B} = \frac{p_{B|A} p_A}{p_B} = \frac{0.15 \times 0.95}{0.15 \times 0.95 + 0.5 \times (1 - 0.95)} = 0.851,$$

so we should revise our probability, in light of the last decade's evidence, from 95% to 85%. This is a truly honest approximation that takes into account the observations and revises the uncertainty. Of course, when we receive a new batch of measurements, we can recompute and obtain an update. This is precisely what DA seeks to achieve. The major difference resides in our possession (in the DA context) of a sophisticated model for actually computing the conditional probability, $p_{B|A}$, the probability of the data, or observations, given the parameters. ■

3.1.5 ▪ First steps toward a formal framework

Now let us begin to formalize. It can be claimed that a major part of scientific discovery and research deals with questions of this nature: what can be said about the value of an unknown, or inaccurately known, variable θ that represents the parameters of the system, if we have some measured data $\mathscr{D}$ and a model $\mathscr{M}$ of the underlying mechanism that generated the data? But this is precisely the Bayesian context,[29] where we seek a quantification of the uncertainty in our knowledge of the parameters that, according

[28] A recent paper, published in *Science*, has rectified this by taking into account the evolution of instrumentation since the start of the study. Indeed, it now appears that there has been a steady increase! Apparently, the "hiatus" was the result of a double observational artefact [see T.R. Karl et al., *Science Express*, 4 June 2015].

[29] See Barber [2012], where Bayesian reasoning is extensively developed in the context of machine learning.

to Bayes' theorem takes the form

$$p(\theta \mid \mathscr{D}) = \frac{p(\mathscr{D} \mid \theta)p(\theta)}{p(\mathscr{D})} = \frac{p(\mathscr{D} \mid \theta)p(\theta)}{\int_\theta p(\mathscr{D} \mid \theta)p(\theta)}.$$

Here, the physical model is represented by the conditional probability (also known as the *likelihood*) $p(\mathscr{D} \mid \theta)$, and the prior knowledge of the system by the term $p(\theta)$. The denominator is considered as a normalizing factor and represents the total probability of $\mathscr{D}$. From these we can then calculate the resulting posterior probability, $p(\theta \mid \mathscr{D})$.

The most probable estimator, called the maximum a posteriori (MAP) estimator, is the value that maximizes the posterior probability

$$\theta_* = \arg\max_\theta p(\theta \mid \mathscr{D}).$$

Note that for a flat, or uninformative, prior $p(\theta)$, the MAP is just the maximum likelihood, which is the value of θ that maximizes the likelihood $p(\mathscr{D} \mid \theta)$ of the model that generated the data, since in this case neither $p(\theta)$ nor the denominator plays a role in the optimization.

3.1.6 ▪ Concluding remarks (as an opening ...)

There are links between the above and the theories of state space, optimal control, and optimal filtering. We will study KFs, whose original theory was developed in this state space context, below—see Friedland [1986] and Kalman [1960].

The following was the theme of a recent Royal Meteorological Society meeting (Imperial College, London, April 2013): "Should weather and climate prediction models be deterministic or stochastic?"—this is a very important question that is relevant for other physical systems.

In this chapter, we will argue that uncertainty is an inherent characteristic of (weather and most other) predictions and thus that no forecast can claim to be complete without an accompanying estimation of its uncertainty—what we call uncertainty quantification (UQ).

3.2 ▪ Statistical estimation theory

In statistical modeling, the concepts of sample space, probability, and random variable play key roles. Readers who are already familiar with these concepts can skip this section. Those who require more background on probability and statistics should definitely consult a comprehensive treatment, such as DeGroot and Schervisch [2012] or the excellent texts of Feller [1968], Jaynes [2003], McPherson [2001], and Ross [1997].

A sample space, $\mathscr{S}$, is the set of all possible outcomes of a random, unpredictable experiment. Each outcome is a point (or an element) in the sample space. Probability provides a means for quantifying how likely it is for an outcome to take place. Random variables assign numerical values to outcomes in the sample space. Once this has been done, we can systematically work with notions such as average value, or mean, and variability.

It is customary in mathematical statistics to use capital letters to denote random variables (r.v.'s) and corresponding lowercase letters to denote values taken by the r.v. in its range. If $X : \mathscr{S} \to \mathbb{R}$ is an r.v., then for any $x \in \mathbb{R}$, by $\{X \leq x\}$ we mean $\{s \in \mathscr{S} \mid X(s) \leq x\}$.

Definition 3.3. *A* probability space $(\mathscr{S}, \mathscr{B}, \mathscr{P})$ *consists of a set* $\mathscr{S}$ *called the* sample space, *a collection* $\mathscr{B}$ *of (Borel) subsets of* $\mathscr{S}$, *and a* probability function $\mathscr{P} : \mathscr{B} \to \mathbb{R}_+$ *for which*

- $\mathscr{P}(\emptyset) = 0$,
- $\mathscr{P}(\mathscr{S}) = 1$, and
- $\mathscr{P}\left(\bigcup_i S_i\right) = \sum_i \mathscr{P}(S_i)$ for any disjoint, countable collection of sets $S_i \in \mathscr{B}$.

A *random variable* X is a measurable function $X : \mathscr{S} \to \mathbb{R}$. Associated with the r.v. X is its *distribution function*,

$$F_X(x) = \mathscr{P}\{X \leq x\}, \quad x \in \mathbb{R}.$$

The distribution function is nondecreasing and right continuous and satisfies

$$\lim_{x \to -\infty} F_X(x) = 0, \quad \lim_{x \to +\infty} F_X(x) = 1.$$

Definition 3.4. *A random variable X is called* discrete *if there exist countable sets* $\{x_i\} \subset \mathbb{R}$ *and* $\{p_i\} \subset \mathbb{R}_+$ *for which*

$$p_i = \mathscr{P}\{X = x_i\} > 0$$

for each i, and

$$\sum_i p_i = 1.$$

In this case, the PDF *for X is the real-valued function with discrete support*

$$p_X(x) = \begin{cases} p_i & \text{if } x = x_i, \quad i = 1, 2, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

The x_i's are the points of discontinuity of the distribution function,

$$F_X(x) = \sum_{\{i \mid x_i \leq x\}} p_X(x_i).$$

Definition 3.5. *A random variable X is called* continuous *if its distribution function, F_X, is absolutely continuous. In this case,*

$$F_X(x) = \int_{-\infty}^{x} p_X(u)\, \mathrm{d}u,$$

and there exists a derivative of F_X,

$$p_X(x) = \frac{\mathrm{d}F_X}{\mathrm{d}x},$$

that is called the probability density function (PDF) *for X.*

Definition 3.6. *The* mean, *or* expected value, *of an r.v. X is given by the integral*

$$\mathrm{E}(X) = \int_{-\infty}^{\infty} x\, \mathrm{d}F_X(x).$$

This is also known as the first moment of the random variable. If X is a continuous r.v., then

$$\mathrm{d}F_X(x) = p_X(x)\,\mathrm{d}x,$$

and, in the discrete case,

$$\mathrm{d}F_X(x) = p_X(x_i)\delta(x - x_i).$$

In the latter case,

$$\mathrm{E}(X) = \sum_i x_i\, p_X(x_i).$$

The expectation operator, E, *is a linear operator.*

Definition 3.7. *The* variance *of an r.v. X is given by*

$$\sigma^2 = \mathrm{E}\left[(X-\mu)^2\right] = \mathrm{E}(X^2) - (\mathrm{E}(X))^2,$$

where

$$\mu = \mathrm{E}(X).$$

Definition 3.8. *The* mode *is the value of x for which the PDF $p_X(x)$ attains its maximal value.*

Definition 3.9. *Two r.v.'s, X and Y, are* jointly distributed *if they are both defined on the same probability space, $(\mathcal{S}, \mathcal{B}, \mathcal{P})$.*

Definition 3.10. *A* random vector, $\mathbf{X} = (X_1, X_2, \ldots, X_n)$, *is a mapping from $\mathcal{S}$ into $\mathbb{R}^n$ for which all the components X_i are jointly distributed. The* joint distribution function *of $\mathbf{X}$ is given by*

$$F_{\mathbf{X}}(\mathbf{x}) = \mathcal{P}\left\{X_1 \le x_1, \ldots, X_n \le x_n\right\}, \quad \mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n.$$

The components X_i are independent *if the joint distribution function is the product of the distribution functions of the components,*

$$F_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^{n} F_{X_i}(x_i).$$

Definition 3.11. *A random vector $\mathbf{X}$ is* continuous *with* joint PDF $p_{\mathbf{X}}$ *if*

$$F_{\mathbf{X}}(\mathbf{x}) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} p_{\mathbf{X}}(\mathbf{u})\,\mathrm{d}u_1 \ldots \mathrm{d}u_n.$$

Definition 3.12. *The mean, or* expected value, *of a random vector $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ is the n-vector* $\mathrm{E}(\mathbf{X})$ *with components*

$$[\mathrm{E}(\mathbf{X})]_i = \mathrm{E}(X_i), \quad i = 1, \ldots, n.$$

The covariance *of $\mathbf{X}$ is the $n \times n$ matrix* $\mathrm{cov}(\mathbf{X})$ *with components*

$$[\mathrm{cov}(\mathbf{X})]_{ij} = \mathrm{E}\left[(X_i - \mu_i)(X_j - \mu_j)\right] = \sigma_{ij}^2, \quad 1 \le i, j \le n,$$

where

$$\mu_i = \mathrm{E}(X_i).$$

3.2.1 ▪ Gaussian distributions

A continuous random vector $\mathbf{X}$ has a Gaussian distribution if its joint PDF has the form

$$p_{\mathbf{X}}(\mathbf{x};\mu,\Sigma)=\frac{1}{\sqrt{(2\pi)^n\det(\Sigma)}}\exp\left(-\frac{1}{2}(\mathbf{x}-\mu)^T\Sigma^{-1}(\mathbf{x}-\mu)\right),$$

where $\mathbf{x},\mu\in\mathbb{R}^n$ and Σ is an $n\times n$ symmetric positive definite matrix. The mean is given by

$$\mathrm{E}(\mathbf{X})=\mu,$$

and the covariance matrix is

$$\mathrm{cov}(\mathbf{X})=\Sigma.$$

These two parameters completely characterize the distribution, and we indicate this situation by

$$\mathbf{X}\sim\mathcal{N}(\mu,\Sigma).$$

Note that in the *scalar* case we have the familiar *bell curve*,

$$p(x;\mu,\sigma)=\frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/(2\sigma^2)}.$$

3.2.2 ▪ Estimators and their properties

We now present some fundamental concepts of statistical estimation. A far more complete treatment can be found in Garthwaite et al. [2002], for example. Let us begin by defining estimation and estimators.

We suppose that we are in possession of a random sample $(x_1,x_2,\ldots,x_n)$ (of measurements, say) of the corresponding r.v.'s $X_1,X_2,\ldots,X_n$, whose PDF is $p_{\mathbf{X}}(\mathbf{x};\theta)$. We want to use the observed values $x_1,x_2,\ldots,x_n$ to estimate the parameter θ, which is either unknown or imprecisely known. We then calculate (see methods below) an estimate $\hat{\theta}$ of θ as a function of $(x_1,x_2,\ldots,x_n)$. The corresponding function $\hat{\theta}\,(X_1,X_2,\ldots,X_n)$, which is an r.v. itself, is an *estimator* for θ. In a given situation, there can exist a number of possible estimators (see example below), and thus the questions of how to choose the best one and what we mean by "best," have to be answered.

The first criterion, considered as indispensable in most circumstances, is that of *unbiasedness.*

Definition 3.13. *The estimator $\hat{\theta}$ for θ is an* unbiased *estimator if the expected value*

$$\mathrm{E}(\hat{\theta})=\theta.$$

The bias of $\hat{\theta}$ is the quantity $\mathrm{E}(\hat{\theta})-\theta$.

The notion of unbiasedness implies that the distribution of $\hat{\theta}$ (recall that $\hat{\theta}$ is an r.v.) is centered exactly at the value θ and that thus there is no tendency to either under- or overestimate this parameter.

Example 3.14. To estimate the mean of a (scalar-valued) normal distribution $\mathcal{N}(\mu,\sigma)$ from a sample of n values, the most evident estimator is the *sample mean*,

$$\hat{X}=\frac{1}{n}\sum_{i=1}^{n}X_i,$$

which is unbiased (this is also the case for other distributions, not only for Gaussians). However, there are numerous other unbiased estimators, for example, the median, the mid-range, and even X_1. ■

We conclude that unbiasedness is usually not enough for choosing an estimator and that we need some other criteria to settle this issue. The classical ones (in addition to unbiasedness) are known as consistency, efficiency, and sufficiency. We will not go into the details here but will concentrate on some optimality conditions.

3.2.3 ▪ Maximum likelihood estimation

Suppose a random vector $\mathbf{X}$ has a joint PDF $p_{\mathbf{X}}(\mathbf{x};\theta)$, where θ is an unknown parameter vector that we would like to estimate. Suppose also that we have a data vector $\mathbf{d}=(d_1,\ldots,d_n)$, a given realization of $\mathbf{X}$ (an outcome of a random experiment).

Definition 3.15. *A maximum likelihood estimator (MLE) for θ given $\mathbf{d}$ is a parameter vector $\hat{\theta}$ that maximizes the likelihood function*

$$L(\theta)=p_{\mathbf{X}}(\mathbf{d};\theta),$$

which is the joint PDF, considered as a function of θ. The MLE is also a maximizer of the log-likelihood function,

$$l(\theta)=\log p_{\mathbf{X}}(\mathbf{d};\theta).$$

3.2.4 ▪ Bayesian estimation

Now we can formalize the notions of Bayesian probability that were introduced in Sections 3.1.4 and 3.1.5. To this end, we must begin by discussing and defining conditional probability and conditional expectation.

Let $\mathbf{X}=(X_1,X_2,\ldots,X_n)$ and $\mathbf{Y}=(Y_1,Y_2,\ldots,Y_n)$ be jointly distributed discrete random vectors. Then $(\mathbf{X},\mathbf{Y})$ is also a discrete random vector.

Definition 3.16. *The joint PDF for $(\mathbf{X},\mathbf{Y})$ is given by*

$$p_{(\mathbf{X},\mathbf{Y})}(\mathbf{x},\mathbf{y})=\mathscr{P}\{\mathbf{X}=\mathbf{x},\quad \mathbf{Y}=\mathbf{y}\},\quad (\mathbf{x},\mathbf{y})\in\mathbb{R}^n\times\mathbb{R}^n.$$

The marginal PDF *of $\mathbf{X}$ is then defined as*

$$p_{\mathbf{X}}(\mathbf{x})=\sum_{\mathscr{P}\{\mathbf{Y}=\mathbf{y}\}>0} p_{(\mathbf{X},\mathbf{Y})}(\mathbf{x},\mathbf{y}),\quad \mathbf{x}\in\mathbb{R}^n. \tag{3.1}$$

The conditional PDF *for $\mathbf{Y}$ given $\mathbf{X}=\mathbf{x}$ is then defined as*

$$p_{(\mathbf{Y}|\mathbf{X})}(\mathbf{y}\mid\mathbf{x})=\frac{p_{(\mathbf{X},\mathbf{Y})}(\mathbf{x},\mathbf{y})}{p_{\mathbf{X}}(\mathbf{x})}, \tag{3.2}$$

where the denominator is nonzero.

So, the conditional probability $p(A|B)$ is the revised probability of an event A after learning that the event B has occurred.

Remark 3.17. *If* $\mathbf{X}$ *and* $\mathbf{Y}$ *are independent random vectors, then the conditional density function of* $\mathbf{Y}$ *given* $\mathbf{X}=\mathbf{x}$ *does not depend on* $\mathbf{x}$ *and satisfies*

$$p_{(\mathbf{Y}|\mathbf{X})}(\mathbf{y}\,|\,\mathbf{x}) = \frac{p_{\mathbf{X}}(\mathbf{x})p_{\mathbf{Y}}(\mathbf{y})}{p_{\mathbf{X}}(\mathbf{x})} = p_{\mathbf{Y}}(\mathbf{y}). \tag{3.3}$$

Definition 3.18. *Let* $\phi : \mathbb{R}^n \to \mathbb{R}^k$ *be a measurable mapping. The* conditional expectation *of* $\phi(\mathbf{Y})$ *given* $\mathbf{X}=\mathbf{x}$ *is*

$$\mathrm{E}(\phi(\mathbf{Y})\,|\,\mathbf{X}=\mathbf{x}) = \sum_{\mathscr{P}\{\mathbf{Y}=\mathbf{y}\}>0} \phi(\mathbf{y})p_{(\mathbf{Y}|\mathbf{X})}(\mathbf{y}\,|\,\mathbf{x}), \quad \mathbf{x}\in\mathbb{R}^n. \tag{3.4}$$

Remark 3.19. *For continuous random vectors* $\mathbf{X}$ *and* $\mathbf{Y}$*, we can define the analogous concepts by replacing the summations in* (3.1)–(3.4) *with appropriate integrals:*

$$p_{\mathbf{X}}(\mathbf{x}) = \int_{-\infty}^{\infty} p_{(\mathbf{X},\mathbf{Y})}(\mathbf{x},\mathbf{y})\,\mathrm{d}F_{\mathbf{Y}}(\mathbf{y}),$$

$$\mathrm{E}(\phi(\mathbf{Y})\,|\,\mathbf{X}=\mathbf{x}) = \int_{-\infty}^{\infty} \phi(\mathbf{y})p_{(\mathbf{Y}|\mathbf{X})}(\mathbf{y}\,|\,\mathbf{x})\,\mathrm{d}F_{\mathbf{Y}}(\mathbf{y}).$$

We are now ready to state Bayes' law, which relates the conditional random vector $\mathbf{X}|_{\mathbf{Y}=\mathbf{y}}$ to the inverse conditional random vector $\mathbf{Y}|_{\mathbf{X}=\mathbf{x}}$.

Theorem 3.20 (Bayes' law). *Let* $\mathbf{X}$ *and* $\mathbf{Y}$ *be jointly distributed random vectors. Then*

$$p_{(\mathbf{X}|\mathbf{Y})}(\mathbf{x}\,|\,\mathbf{y}) = \frac{p_{(\mathbf{Y}|\mathbf{X})}(\mathbf{y}\,|\,\mathbf{x})p_{\mathbf{X}}(\mathbf{x})}{p_{\mathbf{Y}}(\mathbf{y})}. \tag{3.5}$$

Proof. By the definition of conditional probability (3.2),

$$p_{(\mathbf{X}|\mathbf{Y})}(\mathbf{x}\,|\,\mathbf{y}) = \frac{p_{(\mathbf{X},\mathbf{Y})}(\mathbf{x},\mathbf{y})}{p_{\mathbf{Y}}(\mathbf{y})},$$

and the numerator is exactly equal to that of (3.5), once again by definition. ∎

Definition 3.21. *In the context of Bayes' law* (3.5)*, suppose that* $\mathbf{X}$ *represents the variable of interest and that* $\mathbf{Y}$ *represents an observable (measured) quantity that depends on* $\mathbf{X}$*. Then,*

- $p_{\mathbf{X}}(\mathbf{x})$ *is called the* a priori *PDF, or the* prior*;*
- $p_{(\mathbf{X}|\mathbf{Y})}(\mathbf{x}\,|\,\mathbf{y})$ *is called the* a posteriori *PDF, or the* posterior*;*
- $p_{(\mathbf{Y}|\mathbf{X})}(\mathbf{y}\,|\,\mathbf{x})$*, considered as a function of* $\mathbf{x}$*, is the* likelihood *function;*
- *the denominator, called the* evidence*,* $p_{\mathbf{Y}}(\mathbf{y})$*, can be considered as a normalization factor; and*
- *the posterior distribution is thus proportional to the product of the likelihood and the prior distribution or, in applied terms,*

$$p(\text{parameter}\,|\,\text{data}) \propto p(\text{data}\,|\,\text{parameter})p(\text{parameter}).$$

Remark 3.22. *A few fundamental remarks are in order here. First, Bayes' law plays a central role in probabilistic reasoning since it provides us with a method for inverting probabilities, going from $p(\mathbf{y} \mid \mathbf{x})$ to $p(\mathbf{x} \mid \mathbf{y})$. Second, conditional probability matches perfectly our intuitive notion of uncertainty. Finally, the laws of probability combined with Bayes' law constitute a complete reasoning system for which traditional deductive reasoning is a special case [Jaynes, 2003].*

3.2.5 ▪ Linear least-squares estimation: BLUE, minimum variance linear estimation

In this section we define the two estimators that form the basis of statistical DA. We show that these are optimal, which explains their widespread use.

Let $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ and $\mathbf{Z} = (Z_1, Z_2, \ldots, Z_m)$ be two jointly distributed, real-valued random vectors with finite expected squared components:

$$\mathrm{E}(X_i^2) < \infty, \quad i = 1, \ldots, n, \qquad \mathrm{E}(Z_j^2) < \infty, \quad j = 1, \ldots, m.$$

This is an existence condition and is necessary to have a rigorous, functional space setting for what follows.

Definition 3.23. *The* cross-correlation *matrix for* $\mathbf{X}$ *and* $\mathbf{Z}$ *is the* $n \times m$ *matrix* $\Gamma_{\mathbf{XZ}} = \mathrm{E}(\mathbf{X}\mathbf{Z}^T)$ *with entries*

$$[\Gamma_{\mathbf{XZ}}]_{ij} = \mathrm{E}(X_i Z_j), \quad i = 1, \ldots, n, \ j = 1, \ldots, m.$$

The autocorrelation *matrix for* $\mathbf{X}$ *is* $\Gamma_{\mathbf{XX}} = \mathrm{E}(\mathbf{X}\mathbf{X}^T)$, *with entries*

$$[\Gamma_{\mathbf{XX}}]_{ij} = \mathrm{E}(X_i X_j), \quad 1 \le i, j \le n.$$

Remark 3.24. *Note that* $\Gamma_{\mathbf{ZX}} = \Gamma_{\mathbf{XZ}}^T$ *and that* $\Gamma_{\mathbf{XX}}$ *is symmetric and positive semidefinite, i.e.,* $\forall \mathbf{x}$, $\mathbf{x}^{\mathrm{T}}\Gamma_{\mathbf{Xx}}\mathbf{x} \ge 0$. *Also, if* $\mathrm{E}(\mathbf{X}) = 0$, *then the autocorrelation reduces to the covariance,* $\Gamma_{\mathbf{XX}} = \mathrm{cov}(\mathbf{X})$.

We can relate the trace of the autocorrelation matrix to the second moment of the random vector $\mathbf{X}$.

Proposition 3.25. *If a random vector* $\mathbf{X}$ *has finite expected squared components, then*

$$\mathrm{E}\left(||\mathbf{X}||^2\right) = \mathrm{trace}(\Gamma_{\mathbf{XX}}).$$

We are now ready to formally define the BLUE. We consider a linear model,

$$\mathbf{z} = \mathbf{K}\mathbf{x} + \mathbf{N},$$

where $\mathbf{K}$ is an $m \times n$ matrix, $\mathbf{x} \in \mathbb{R}^n$ is deterministic, and $\mathbf{N}$ is a random (noise) n-vector with

$$\mathrm{E}(\mathbf{N}) = 0, \quad \mathbf{C}_{\mathbf{N}} = \mathrm{cov}(\mathbf{N}),$$

and $\mathbf{C}_{\mathbf{N}}$ is a known, nonsingular, $n \times n$ covariance matrix.

Definition 3.26. *The* best linear unbiased estimator (BLUE) *for* $\mathbf{x}$ *from the linear model* $\mathbf{z}$ *is the vector* $\hat{\mathbf{x}}_{\mathrm{BLUE}}$ *that minimizes the quadratic cost function*

$$J(\hat{\mathbf{x}}) = \mathrm{E}\left(\|\hat{\mathbf{x}} - \mathbf{x}\|^2\right)$$

subject to the constraints of linearity,

$$\hat{\mathbf{x}} = \mathbf{B}\mathbf{z}, \quad B \in \mathbb{R}^{n \times m},$$

and unbiasedness,

$$\mathrm{E}(\hat{\mathbf{x}}) = \mathbf{x}.$$

In the case of a full-rank matrix $\mathbf{K}$, the *Gauss–Markov theorem* [Sayed, 2003; Vogel, 2002] gives us an explicit form for the BLUE.

Theorem 3.27 (Gauss–Markov). *If* $\mathbf{K}$ *has full rank, then the BLUE is given by*

$$\hat{\mathbf{x}}_{\mathrm{BLUE}} = \hat{\mathbf{B}}\mathbf{z},$$

where

$$\hat{\mathbf{B}} = \left(\mathbf{K}^T \mathbf{C}_N^{-1} \mathbf{K}\right)^{-1} \mathbf{K}^T \mathbf{C}_N^{-1}.$$

Remark 3.28. *If the noise covariance matrix* $\mathbf{C}_N = \sigma^2 I$ *(white, uncorrelated noise), and* $\mathbf{K}$ *has full rank, then*

$$\hat{\mathbf{x}}_{\mathrm{BLUE}} = \left(\mathbf{K}^T \mathbf{K}\right)^{-1} \mathbf{K}^T \mathbf{z} = \mathbf{K}^\dagger \mathbf{z},$$

where $\mathbf{K}^\dagger$ *is called the pseudoinverse of* $\mathbf{K}$. *This corresponds, in the deterministic case, to the* least-squares problem

$$\min_{\mathbf{x}} \|\mathbf{K}\mathbf{x} - \mathbf{z}\|.$$

Due to the dependence of the BLUE on the inverse of the noise covariance matrix, it is unsuitable for the solution of noisy, ill-conditioned linear systems. To remedy this situation, we assume that $\mathbf{x}$ is a realization of a random vector $\mathbf{X}$, and we formulate a linear least-squares analogue of Bayesian estimation.

Definition 3.29. *Suppose that* $\mathbf{x}$ *and* $\mathbf{z}$ *are jointly distributed, random vectors with finite expected squares. The* minimum variance linear estimator (MVLE) *of* $\mathbf{x}$ *from* $\mathbf{z}$ *is given by*

$$\hat{\mathbf{x}}_{\mathrm{MVLE}} = \hat{\mathbf{B}}\mathbf{z},$$

where

$$\hat{\mathbf{B}} = \arg\min_{B \in \mathbb{R}^{n \times m}} \mathrm{E}\left(\|\mathbf{B}\mathbf{z} - \mathbf{x}\|^2\right).$$

Proposition 3.30. *If* $\Gamma_{\mathbf{ZZ}}$ *is nonsingular, then the MVLE of* $\mathbf{x}$ *from* $\mathbf{z}$ *is given by*

$$\hat{\mathbf{x}}_{\mathrm{MVLE}} = \left(\Gamma_{\mathbf{XZ}} \Gamma_{\mathbf{ZZ}}^{-1}\right) \mathbf{z}.$$

3.3 ▪ Examples of Bayesian estimation

In this section we provide some calculated examples of Bayesian estimation.

3.3.1 ▪ Scalar Gaussian distribution example

In this simple, but important, example we will derive in detail the parameters of the posterior distribution when the data and the prior are normally distributed. This will provide us with a richer understanding of DA.

We suppose that we are interested in forecasting the value of a *scalar* state variable, x, which could be a temperature, a wind velocity component, an ozone concentration, etc. We are in possession of a Gaussian prior distribution for x,

$$x \sim \mathcal{N}(\mu_X, \sigma_X^2),$$

with expectation μ and variance σ^2, which could come from a forecast model, for example. We are in possession of n independent, noisy observations,

$$\mathbf{y} = (y_1, y_2, \ldots, y_n),$$

each with conditional distribution

$$y_i \mid x \sim \mathcal{N}(x, \sigma^2),$$

that are conditioned on the true value of the parameter/process x. Thus, the conditional distribution of the data/observations is a product of Gaussian laws,

$$\begin{aligned} p(\mathbf{y} \mid x) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y_i - x)^2\right\} \\ &\propto \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x)^2\right\}. \end{aligned}$$

But from Bayes' law (3.5),

$$p(x \mid \mathbf{y}) \propto p(\mathbf{y} \mid x) p(x),$$

so using the data and the prior distributions/models, we have

$$\begin{aligned} p(x \mid \mathbf{y}) &\propto \exp\left\{-\frac{1}{2}\sum_{i=1}^{n}(y_i - x)^2/\sigma^2 + (x - \mu_X)^2/\sigma_X^2\right\} \\ &\propto \exp\left\{-\frac{1}{2}\left[x^2\left(\frac{n}{\sigma^2} + \frac{1}{\sigma_X^2}\right) - 2\left(\sum_{i=1}^{n}\frac{y_i}{\sigma^2} + \frac{\mu_X}{\sigma_X^2}\right)x\right]\right\}. \end{aligned}$$

Notice that this is the product of two Gaussians, which, by completing the square, can be show to be Gaussian itself. This produces the posterior distribution,

$$x \mid \mathbf{y} \sim \mathcal{N}\left(\mu_{x|\mathbf{y}}, \sigma_{x|\mathbf{y}}^2\right), \tag{3.6}$$

where

$$\mu_{x|\mathbf{y}} = \left(\frac{n}{\sigma^2} + \frac{1}{\sigma_X^2}\right)^{-1} \left(\sum_{i=1}^{n} \frac{y_i}{\sigma^2} + \frac{\mu_X}{\sigma_X^2}\right)$$

and

$$\sigma_{x|\mathbf{y}}^2 = \left(\frac{n}{\sigma^2} + \frac{1}{\sigma_X^2}\right)^{-1}.$$

Let us now study more closely these two parameters of the posterior law. We first remark that the inverse of the posterior variance, called the posterior *precision*, is equal to the sum of the prior precision, $1/\sigma_X^2$, and the data precision, n/σ^2. Second, the posterior mean, or conditional expectation, can also be written as a sum of two terms:

$$\begin{aligned}
\mathrm{E}(x \mid \mathbf{y}) &= \frac{\sigma^2 \sigma_X^2}{\sigma^2 + n\sigma_X^2} \left(\frac{n}{\sigma^2}\bar{y} + \frac{\mu_X}{\sigma_X^2}\right) \\
&= w_y \bar{y} + w_{\mu_X} \mu_X,
\end{aligned}$$

where the sample mean,

$$\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i,$$

and the two weights,

$$w_y = \frac{n\sigma_X^2}{\sigma^2 + n\sigma_X^2}, \quad w_{\mu_X} = \frac{\sigma^2}{\sigma^2 + n\sigma_X^2},$$

add up to

$$w_y + w_{\mu_X} = 1.$$

We observe immediately that the posterior mean is the weighted sum/average of the data mean ($\bar{y}$) and the prior mean (μ_X). Now let us examine the weights themselves. If there is a large uncertainty in the prior, then $\sigma_X^2 \to \infty$ and hence $w_y \to 1$, $w_{\mu_X} \to 0$ and the likelihood dominates the prior, leading to what is known as the sampling distribution for the posterior:

$$p(x \mid \mathbf{y}) \to \mathcal{N}(\bar{y}, \sigma^2/n).$$

If we have a large number of observations, then $n \to \infty$ and the posterior now tends to the sample mean, whereas if we have few observations, then $n \to 0$ and the posterior

$$p(x \mid \mathbf{y}) \to \mathcal{N}(\mu_X, \sigma_X^2)$$

tends to the prior. In the case of equal uncertainties between data and prior, $\sigma^2 = \sigma_X^2$, and the prior mean has the weight of a single additional observation. Finally, if the uncertainties are small, either the prior is infinitely more precise than the data ($\sigma_X^2 \to 0$) or the data are perfectly precise ($\sigma^2 \to 0$).

We end this example by rewriting the posterior mean and variance in a special form. Let us start with the mean:

$$\begin{aligned}
\mathrm{E}(x \mid \mathbf{y}) &= \mu_X + \frac{n\sigma_X^2}{\sigma^2 + n\sigma_X^2}(\bar{y} - \mu_X) \\
&= \mu_X + G(\bar{y} - \mu_X).
\end{aligned} \tag{3.7}$$

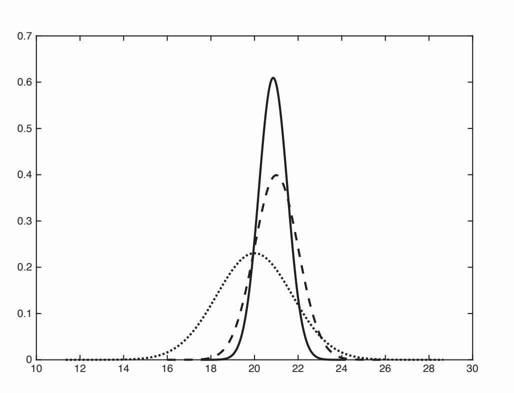


Figure 3.1. *Scalar Gaussian distribution example. Prior* $\mathcal{N}(20, 3)$ *(dotted), instrument* $\mathcal{N}(x, 1)$ *(dashed), and posterior* $\mathcal{N}(20.86, 0.43)$ *(solid) distributions.*

We conclude that the prior mean μ_X is adjusted toward the sample mean $\bar{y}$ by a gain (or amplification factor) of $G = 1/(1+\sigma^2/n\sigma_X^2)$, multiplied by the innovation $\bar{y}-\mu_X$, and we observe that the variance ratio, between data and prior, plays an essential role. In the same way, the posterior variance can be reformulated as

$$\sigma_{x|\mathbf{y}}^2 = (1-G)\sigma_X^2, \tag{3.8}$$

and the posterior variance is thus updated from the prior variance according to the same gain G. These last two equations, (3.7) and (3.8), are fundamental for a good understanding of DA, since they clearly express the interplay between prior and data and the effect that each has on the posterior.

Let us illustrate this with two initial numerical examples. Suppose we have a prior distribution $x \sim \mathcal{N}(\mu_X, \sigma_X^2)$ with mean 20 and variance 3. Suppose that our data model has the conditional law $y_i \mid x \sim \mathcal{N}(x, \sigma^2)$ with variance 1. Here the data are relatively precise compared to the prior. Say we have acquired two observations, $\mathbf{y} = (19, 23)'$. Now we can use (3.7) and (3.8) to compute the posterior distribution:

$$\mathrm{E}(x \mid \mathbf{y}) = 20 + \frac{6}{1+6}(21-20) = 20.86,$$
$$\sigma_{x|\mathbf{y}}^2 = \left(1 - \frac{6}{7}\right)3 = 0.43,$$

thus yielding the posterior distribution

$$y_i \mid x \sim \mathcal{N}(20.86, 0.43),$$

which represents the update of the prior according to the observations and takes into account all the uncertainties available—see Figure 3.1. In other words, we have obtained a complete forecast at a given point in time.

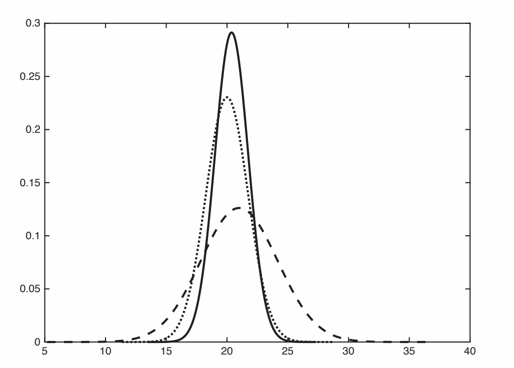


Figure 3.2. *Scalar Gaussian distribution example. Prior* $\mathcal{N}(20, 3)$ *(dotted), instrument* $\mathcal{N}(x, 10)$ *(dashed), and posterior* $\mathcal{N}(20.375, 1.875)$ *(solid) distributions.*

Now consider the same prior, $x \sim \mathcal{N}(20, 3)$, but with a relatively uncertain/imprecise observation model, $y_i \mid x \sim \mathcal{N}(x, 10)$, and the same two measurements, $\mathbf{y} = (19, 23)'$. Redoing the above calculations, we now find

$$\mathrm{E}(x \mid \mathbf{y}) = 20 + \frac{6}{16}(21 - 20) = 20.375,$$
$$\sigma^2_{x|\mathbf{y}} = \left(1 - \frac{6}{16}\right)3 = 1.875,$$

thus yielding the new posterior distribution,

$$y_i \mid x \sim \mathcal{N}(20.375, 1.875),$$

which has virtually the same mean but a much larger variance—see Figure 3.2, where the scales on both axes have changed.

3.3.2 ▪ Estimating a temperature

Suppose that the outside temperature measurement gives 2°C and the instrument has an error distribution that is Gaussian with mean $\mu = 2$ and variance $\sigma^2 = 0.64$—see the dashed curve in Figure 3.3. This is the model/data distribution. We also suppose that we have a prior distribution that estimates the temperature, with mean $\mu = 0$ and variance $\sigma^2 = 1.21$. The prior comes from either other observations, a previous model forecast, or physical and climatological constraints—see the dotted curve in Figure 3.3. By combining these two, using Bayes' formula, we readily compute the posterior distribution of the temperature given the observations, which has mean $\mu = 1.31$ and variance $\sigma^2 = 0.42$. This is the update or the analysis—see the solid curve in Figure 3.3.

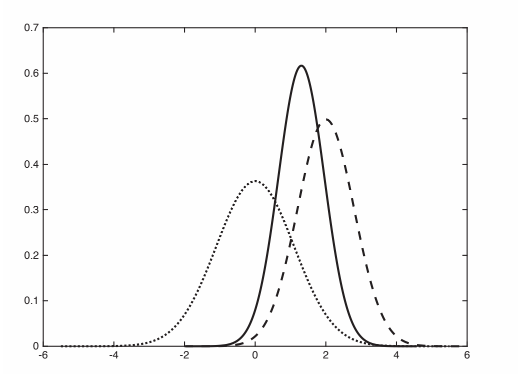


Figure 3.3. *A Gaussian product example for forecasting temperature: prior (dotted), instrument (dashed), and posterior (solid) distributions.*

The code for this calculation can be found in `gaussian_product.m` [DART toolbox, 2013].

3.3.3 ▪ Estimating the parameters of a pendulum

We present an example of a simple mechanical system and seek an estimation of its parameters from noisy measurements. Consider a model for the angular displacement, x_t, of an ideal pendulum (no friction, no drag),

$$x_t = \sin(\theta t) + \epsilon_t,$$

where ϵ_t is a Gaussian noise with zero mean and variance σ^2, the pendulum parameter is denoted by θ, and t is time. From these noisy measurements (suppose that the instrument is not very accurate) of x_t we want to estimate θ, which represents the physical properties of the pendulum—in fact $\theta = \sqrt{g/L}$, where g is the gravitational constant and L is the pendulum's length. Using this physical model, can we estimate (or infer) the unknown physical parameters of the pendulum?

If the measurements are independent, then the likelihood of a set of T observations $x_1, \ldots, x_T$ is given by the product

$$p(x_1, \ldots, x_T | \theta) = \prod_{t=1}^{T} p(x_t | \theta).$$

In addition, suppose that we have some prior estimation (before obtaining the measurements) of the probabilities of a set of possible values of θ. Then the posterior distribution of θ, given the measurements, can be calculated from Bayes' law, as seen above,

$$p(\theta | x_1, \ldots, x_T) \propto p(\theta) \prod_{t=1}^{T} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_t - \sin(\theta t))^2},$$

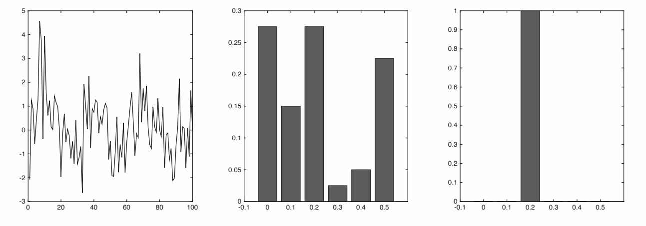


Figure 3.4. *Bayesian estimation of noisy pendulum parameter, $\theta = 0.2$. Observations of 100 noisy positions (left). Prior distribution of parameter values (center). Posterior distribution for θ (right).*

where we have omitted the denominator. We are given the following table of priors:

$[\theta_{\min}, \theta_{\max}]$	$p(\theta_{\min} < \theta < \theta_{\max})$
$[0, 0.05]$	0.275
$[0.05, 0.15]$	0.15
$[0.15, 0.25]$	0.275
$[0.25, 0.35]$	0.025
$[0.35, 0.45]$	0.05
$[0.45, 0.55]$	0.225

After performing numerical simulations, we observe (see Figure 3.4) that the posterior for θ develops a prominent peak for a large number ($T = 100$) of measurements, centered around the real value $\theta = 0.2$ (which was used to generate the time series, x_t).

3.3.4 ▪ Vector/multivariate Gaussian distribution example

As a final case that will lead us naturally to the following section, let us consider the vector/multivariate extension of the example in Section 3.3.1. We will now study a vector process, $\mathbf{x}$, with n components and a prior distribution

$$\mathbf{x} \sim \mathcal{N}(\mu, \mathbf{B}),$$

where the mean vector μ and the covariance matrix $\mathbf{B}$ are assumed to be known (as usual from historical data, model forecasts, etc.). The observation now takes the form of a data vector, $\mathbf{y}$, of dimension p and has the conditional distribution/model:

$$\mathbf{y} \mid \mathbf{x} \sim \mathcal{N}(\mathbf{H}\mathbf{x}, \mathbf{R}),$$

where the $(p \times n)$ observation matrix $\mathbf{H}$ maps the process to the measurements and the error covariance matrix $\mathbf{R}$ is known. These are exactly the same matrices that we have already encountered in the variational approach—see Chapters 1 and 2. The difference is that now our modeling is placed in a richer, Bayesian framework.

As before, we would like to calculate the posterior conditional distribution of $\mathbf{x} \mid \mathbf{y}$, given by

$$p(\mathbf{x} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{x}) p(\mathbf{x}).$$

Just as with the scalar/univariate case, the product of two Gaussians is Gaussian, and the posterior law is the multidimensional analogue of (3.6) and can be shown to take the form

$$\mathbf{x} \mid \mathbf{y} \sim \mathcal{N}\left(\mu_{x|\mathbf{y}}, \Sigma_{x|\mathbf{y}}\right),$$

where

$$\mu_{x|\mathbf{y}} = \left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H} + \mathbf{B}^{-1}\right)^{-1}\left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}y + \mathbf{B}^{-1}\mu\right)$$

and

$$\Sigma_{x|\mathbf{y}} = \left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H} + \mathbf{B}^{-1}\right)^{-1}.$$

As above, we will now rewrite the posterior mean and variance in a special form. The posterior conditional mean becomes

$$\begin{aligned} \mathrm{E}(\mathbf{x} \mid \mathbf{y}) &= \left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H} + \mathbf{B}^{-1}\right)^{-1}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}y + \left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H} + \mathbf{B}^{-1}\right)^{-1}\mathbf{B}^{-1}\mu \\ &= \mu + \mathbf{K}(y - \mathbf{H}\mu), \end{aligned} \tag{3.9}$$

where the gain matrix is now

$$\mathbf{K} = \mathbf{B}\mathbf{H}^{\mathrm{T}}(\mathbf{R} + \mathbf{H}\mathbf{B}\mathbf{H}^{\mathrm{T}})^{-1}.$$

In the same manner, the posterior conditional covariance matrix can be reformulated as

$$\begin{aligned} \Sigma(\mathbf{x} \mid \mathbf{y}) &= \left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H} + \mathbf{B}^{-1}\right)^{-1} \\ &= (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{B}, \end{aligned} \tag{3.10}$$

with the same gain matrix $\mathbf{K}$ as for the posterior mean. As before, these last two equations, (3.9) and (3.10), are fundamental for a good understanding of DA, since they clearly express the interplay between prior and data and the effect that each has on the posterior. They are, in fact, the veritable foundation of DA.

3.3.5 ▪ Connections with variational and sequential approaches

As was already indicated in the first two chapters of this book, the link between variational approaches and optimal BLUE is well established. The BLUE approach is also known as kriging in spatial geostatistics, or optimal interpolation (OI) in oceanography and atmospheric science. In the special, but quite widespread, case of a multivariate Gaussian model (for data and priors), the posterior mode (which is equivalent to the mean in this case) can equally be obtained by minimizing the quadratic objective function (2.31),

$$J(\mathbf{x}) = \frac{1}{2}\left(\mathbf{x} - \mathbf{x}^{\mathrm{b}}\right)^{\mathrm{T}}\mathbf{B}^{-1}\left(\mathbf{x} - \mathbf{x}^{\mathrm{b}}\right) + \frac{1}{2}(\mathbf{H}\mathbf{x} - \mathbf{y})^{\mathrm{T}}\mathbf{R}^{-1}(\mathbf{H}\mathbf{x} - \mathbf{y}).$$

This is the fundamental link between variational and statistical approaches. Though strictly equivalent to the Bayes formulation, the variational approach has, until now, been privileged for operational, high-dimensional DA problems—though this is changing with the arrival of new hardware and software capabilities for treating "big data and extreme computing" challenges [Reed and Dongarra, 2015].

Since many physical systems are dynamic and evolve in time, we could improve our estimations considerably if, as new measurements became available, we could simply update the previous optimal estimate of the state process without having to redo all computations. The perfect framework for this sequential updating is the KF, which we will now present in detail.

3.4 ▪ Sequential DA and Kalman filters

We have seen that DA, from the statistical/Bayesian point of view, strives to have as complete a knowledge as possible of the a posteriori probability law, that is, the conditional law of the state given the observations. But it is virtually impossible to determine the complete distribution, so we seek instead an estimate of its statistical parameters, such as its mean and/or its variance. Numerous proven statistical methods can lead to best or optimal estimates [Anderson and Moore, 1979; Garthwaite et al., 2002; Hogg et al., 2013; Ross, 2014]; for example, the minimum variance (MV) estimator is the conditional mean of the state given the observations, and the maximum a posteriori (MAP) estimator produces the mode of the conditional distribution. As seen above, assuming Gaussian distributions for the measurements and the process, we can determine the complete a posteriori law, and, in this case, it is clear that the MV and MAP estimators coincide. In fact, the MV estimator produces the optimal interpolation (OI) or kriging equations, whereas the MAP estimator leads to 3D-Var. In conclusion, for the case of a linear observation operator together with Gaussian error statistics, 3D-Var and OI are strictly equivalent.

So far we have been looking at the spatial estimation problem, where all observations are distributed in space but at a single instant in time. For stationary stochastic processes, the mean and covariance are constant in time [Parzen, 1999; Ross, 1997], so such a DA scheme could be used at different times based on the invariant statistics. This is not so rare: in practice, for global NWP, the errors have been considered stationary over a one-month time scale.[30] However, for general environmental applications, the governing equations vary with time and we must take into account nonstationary processes.

Within the significant box of mathematical tools that can be used for statistical estimation from noisy sensor measurements over time, one of the most well known and often used tools is the Kalman filter (KF). The KF is named after Rudolph E. Kalman, who in 1960 published his famous paper describing a recursive solution to the discrete data linear filtering problem [Kalman, 1960]. There exits a vast literature on the KF, and a very "friendly" introduction to the general idea of the KF can be found in Chapter 1 of Maybeck [1979]. As just stated above, it would be ideal and very efficient if, as new data or measurements became available, we could easily update the previous optimal estimates without having to recompute everything. The KF provides exactly this solution.

To this end, we will now consider a dynamical system that evolves in time, and we will seek to estimate a series of *true* states, $\mathbf{x}_k^{\mathrm{t}}$ (a sequence of random vectors), where discrete time is indexed by the letter k. These times are those when the observations or measurements are taken, as shown in Figure 3.5. The assimilation starts with an unconstrained model trajectory from $t_0, t_1, \ldots, t_{k-1}, t_k, \ldots, t_n$ and aims to provide an optimal fit to the available observations/measurements given their uncertainties (error bars). For example, in current synoptic scale weather forecasts, $t_k - t_{k-1} = 6$ hours; the time step is less for the convective scale.

3.4.1 ▪ Bayesian modeling

Let us recall the principles of Bayesian modeling from Section 3.2 on statistical estimation and rewrite them in the terminology of the DA problem. We have a vector, $\mathbf{x}$, of (unknown) unobserved quantities of interest (temperature, pressure, wind, etc.) and

[30] This assumption is no longer employed at MétéoFrance or ECWMF, for example.

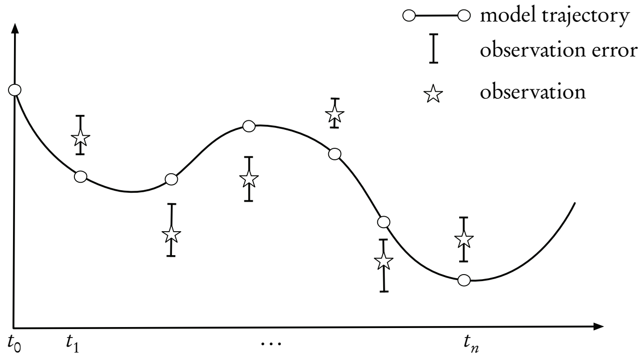


Figure 3.5. *Sequential assimilation: a computed model trajectory, observations, and their error bars.*

a vector, $\mathbf{y}$, of (known) observed data (at various locations, and at various times). The full joint probability model can always be factored into two components,

$$\begin{aligned} p(\mathbf{x},\mathbf{y}) &= p(\mathbf{y}\,|\,\mathbf{x})p(\mathbf{x}) \\ &= p(\mathbf{x}\,|\,\mathbf{y})p(\mathbf{y}), \end{aligned}$$

and thus

$$p(\mathbf{x}\,|\,\mathbf{y}) = \frac{p(\mathbf{y}\,|\,\mathbf{x})p(\mathbf{x})}{p(\mathbf{y})},$$

provided that $p(\mathbf{y}) \neq 0$.

The KF can be rigorously derived from this Bayesian perspective following the presentation above in Section 3.3.4.

3.4.2 ▪ Stochastic model of the system

We seek to estimate the state $\mathbf{x} \in \mathbb{R}^n$ of a discrete-time dynamic process that is governed by the linear stochastic difference equation

$$\mathbf{x}_{k+1} = \mathbf{M}_{k+1}\mathbf{x}_k + \mathbf{w}_k, \tag{3.11}$$

with a measurement/observation $\mathbf{y} \in \mathbb{R}^m$:

$$\mathbf{y}_k = \mathbf{H}_k\mathbf{x}_k + \mathbf{v}_k. \tag{3.12}$$

Note that $\mathbf{M}_{k+1}$ and $\mathbf{H}_k$ are considered linear here. The random vectors $\mathbf{w}_k$ and $\mathbf{v}_k$ represent the process/modeling and measurement/observation errors, respectively. They are assumed to be independent, white-noise processes with Gaussian/normal probability distributions

$$\begin{aligned} \mathbf{w}_k &\sim \mathcal{N}(0,\mathbf{Q}_k), \\ \mathbf{v}_k &\sim \mathcal{N}(0,\mathbf{R}_k), \end{aligned}$$

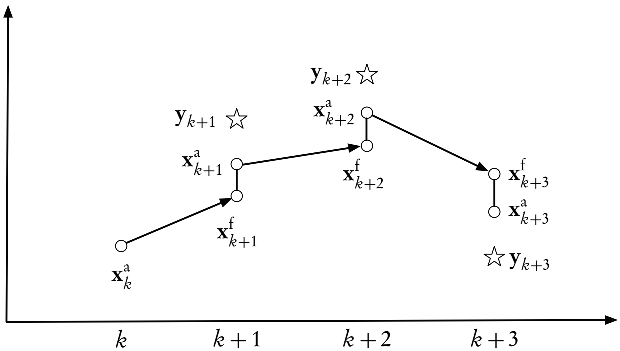


Figure 3.6. *Sequential assimilation scheme for the KF. The x-axis denotes time; the y-axis denotes the values of the state and observation vectors.*

where $\mathbf{Q}$ and $\mathbf{R}$ are the covariance matrices (assumed known) of the modeling and observation errors, respectively. All these assumptions about unbiased and uncorrelated errors (in time and between each other) are not limiting, since extensions of the standard KF can be developed should any of these not be valid—see below and Chapters 5, 6, and 7.

We note that for a broader mathematical view on the above system, we could formulate everything in terms of stochastic differential equations (SDEs). Then the theory of Itô can provide a detailed solution of the problem of optimal filtering as well as existence and uniqueness results—see Oksendal [2003], where one can find such a precise mathematical formulation.

3.4.3 ▪ Sequential assimilation scheme

The typical assimilation scheme is made up of two major steps: a prediction/forecast step and a correction/analysis step. At time t_k we have the result of a previous forecast, $\mathbf{x}^{\mathrm{f}}_k$ (the analogue of the background state $\mathbf{x}^{\mathrm{b}}_k$), and the result of an ensemble of observations in $\mathbf{y}_k$. Based on these two vectors, we perform an analysis that produces $\mathbf{x}^{\mathrm{a}}_k$. We then use the evolution model to obtain a prediction of the state at time t_{k+1}. The result of the forecast is denoted $\mathbf{x}^{\mathrm{f}}_{k+1}$ and becomes the background, or initial guess, for the next time step. This process is summarized in Figure 3.6. The KF problem can be summarized as follows: given a prior/background estimate, $\mathbf{x}^{\mathrm{f}}$, of the system state at time t_k, what is the best update/analysis, $\mathbf{x}^{\mathrm{a}}_k$, based on the currently available measurements, $\mathbf{y}_k$?

We can now define forecast (a priori) and analysis (a posteriori) estimate errors as

$$\mathbf{e}^{\mathrm{f}}_k = \mathbf{x}^{\mathrm{f}}_k - \mathbf{x}^{\mathrm{t}}_k,$$
$$\mathbf{e}^{\mathrm{a}}_k = \mathbf{x}^{\mathrm{a}}_k - \mathbf{x}^{\mathrm{t}}_k,$$

where $\mathbf{x}_k^{\mathrm{t}}$ is the (unknown) true state. Their respective error covariance matrices are

$$\begin{aligned}
\mathbf{P}_k^{\mathrm{f}} &= \operatorname{cov}(\mathbf{e}_k^{\mathrm{f}}) = E\left[\mathbf{e}_k^{\mathrm{f}}(\mathbf{e}_k^{\mathrm{f}})^{\mathrm{T}}\right],\\
\mathbf{P}_k^{\mathrm{a}} &= \operatorname{cov}(\mathbf{e}_k^{\mathrm{a}}) = E\left[\mathbf{e}_k^{\mathrm{a}}(\mathbf{e}_k^{\mathrm{a}})^{\mathrm{T}}\right].
\end{aligned} \tag{3.13}$$

The goal of the KF is to compute an optimal a posteriori estimate, $\mathbf{x}_k^{\mathrm{a}}$, that is a linear combination of an a priori estimate, $\mathbf{x}_k^{\mathrm{f}}$, and a weighted difference between the actual measurement, $\mathbf{y}_k$, and the measurement prediction, $\mathbf{H}_k\mathbf{x}_k^{\mathrm{f}}$. This is none other than the BLUE that we have seen above. The filter is thus of the linear, recursive form

$$\mathbf{x}_k^{\mathrm{a}} = \mathbf{x}_k^{\mathrm{f}} + \mathbf{K}_k\left(\mathbf{y}_k - \mathbf{H}_k\mathbf{x}_k^{\mathrm{f}}\right). \tag{3.14}$$

The difference $\mathbf{d}_k = \mathbf{y}_k - \mathbf{H}_k\mathbf{x}_k^{\mathrm{f}}$ is called the *innovation* and reflects the discrepancy between the actual and the predicted measurements at time t_k. Note that for generality, the matrices are shown with a time dependence. When this is not the case, the subscripts k can be dropped. The *Kalman gain* matrix, $\mathbf{K}$, is chosen to minimize the a posteriori error covariance equation (3.13).

To compute this *optimal gain* requires a careful derivation. Begin by substituting the observation equation (3.12) into the linear filter equation (3.14):

$$\begin{aligned}
\mathbf{x}_k^{\mathrm{a}} &= \mathbf{x}_k^{\mathrm{f}} + \mathbf{K}_k\left(\mathbf{H}_k\mathbf{x}_k^{\mathrm{t}} + \mathbf{v}_k - \mathbf{H}_k\mathbf{x}_k^{\mathrm{f}}\right)\\
&= \mathbf{x}_k^{\mathrm{f}} + \mathbf{K}_k\left(\mathbf{H}_k\left(\mathbf{x}_k^{\mathrm{t}} - \mathbf{x}_k^{\mathrm{f}}\right) + \mathbf{v}_k\right).
\end{aligned}$$

Now place this last expression into the definition of $\mathbf{e}_k^{\mathrm{a}}$:

$$\begin{aligned}
\mathbf{e}_k^{\mathrm{a}} &= \mathbf{x}_k^{\mathrm{a}} - \mathbf{x}_k^{\mathrm{t}}\\
&= \mathbf{x}_k^{\mathrm{f}} + \mathbf{K}_k\left(\mathbf{H}_k\left(\mathbf{x}_k^{\mathrm{t}} - \mathbf{x}_k^{\mathrm{f}}\right) + \mathbf{v}_k\right) - \mathbf{x}_k^{\mathrm{t}}\\
&= \mathbf{K}_k\left(-\mathbf{H}_k\left(\mathbf{x}_k^{\mathrm{f}} - \mathbf{x}_k^{\mathrm{t}}\right) + \mathbf{v}_k\right) + \left(\mathbf{x}_k^{\mathrm{f}} - \mathbf{x}_k^{\mathrm{t}}\right).
\end{aligned}$$

Then substitute in the error covariance equation (3.13):

$$\begin{aligned}
\mathbf{P}_k^{\mathrm{a}} &= \mathrm{E}\left[\mathbf{e}_k^{\mathrm{a}}(\mathbf{e}_k^{\mathrm{a}})^{\mathrm{T}}\right]\\
&= \mathrm{E}\left[\left(\mathbf{K}_k\left(\mathbf{v}_k - \mathbf{H}_k\left(\mathbf{x}_k^{\mathrm{f}} - \mathbf{x}_k^{\mathrm{t}}\right)\right) + \left(\mathbf{x}_k^{\mathrm{f}} - \mathbf{x}_k^{\mathrm{t}}\right)\right)\left(\mathbf{K}_k\left(\mathbf{v}_k - \mathbf{H}_k\left(\mathbf{x}_k^{\mathrm{f}} - \mathbf{x}_k^{\mathrm{t}}\right)\right) + \left(\mathbf{x}_k^{\mathrm{f}} - \mathbf{x}_k^{\mathrm{t}}\right)\right)^{\mathrm{T}}\right].
\end{aligned}$$

Now perform the indicated expectations over the r.v.'s, noting that $\left(\mathbf{x}_k^{\mathrm{f}} - \mathbf{x}_k^{\mathrm{t}}\right) = \mathbf{e}_k^{\mathrm{f}}$ is the a priori estimation error, that this error is uncorrelated with the observation error $\mathbf{v}_k$, that by definition $\mathbf{P}_k^{\mathrm{f}} = \mathrm{E}\left[\mathbf{e}_k^{\mathrm{f}}(\mathbf{e}_k^{\mathrm{f}})^T\right]$ and that $\mathbf{R}_k = \mathrm{E}\left[\mathbf{v}_k\mathbf{v}_k^{\mathrm{T}}\right]$. We thus get

$$\begin{aligned}
\mathbf{P}_k^{\mathrm{a}} &= \mathrm{E}\left[\left(\mathbf{K}_k\left(\mathbf{v}_k - \mathbf{H}_k\mathbf{e}_k^{\mathrm{f}}\right) + \mathbf{e}_k^{\mathrm{f}}\right)\left(\mathbf{K}_k\left(\mathbf{v}_k - \mathbf{H}_k\mathbf{e}_k^{\mathrm{f}}\right) + \mathbf{e}_k^{\mathrm{f}}\right)^{\mathrm{T}}\right]\\
&= \mathrm{E}\left[\left(\mathbf{e}_k^{\mathrm{f}}\right)\left(\mathbf{e}_k^{\mathrm{f}}\right)^{\mathrm{T}} - \left(\mathbf{K}_k\mathbf{H}_k\mathbf{e}_k^{\mathrm{f}}\right)\left(\mathbf{K}_k\mathbf{H}_k\mathbf{e}_k^{\mathrm{f}}\right)^{\mathrm{T}} + (\mathbf{K}_k\mathbf{v}_k)(\mathbf{K}_k\mathbf{v}_k)^{\mathrm{T}}\right]\\
&= (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^{\mathrm{f}}(\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)^{\mathrm{T}} + \mathbf{K}_k\mathbf{R}_k\mathbf{K}_k^{\mathrm{T}}.
\end{aligned} \tag{3.15}$$

Note that this is a completely general formula for the updated covariance matrix and that it is valid for any gain $\mathbf{K}_k$, not necessarily optimal.

Now we still need to compute the optimal gain that minimizes the matrix entries along the principal diagonal of $\mathbf{P}_k^{\mathrm{a}}$, since these terms are the ones that represent the estimation error variances for the entries of the state vector itself. We will use the classical approach of variational calculus, by taking the derivative of the trace of the result with respect to $\mathbf{K}$ and then setting the resulting derivative expression equal to zero. But for this, we require two results from matrix differential calculus [Petersen and Pedersen, 2012]. These are

$$\frac{\mathrm{d}}{\mathrm{d}\mathbf{A}}\operatorname{Tr}(\mathbf{AB}) = \mathbf{B}^{\mathrm{T}},$$
$$\frac{\mathrm{d}}{\mathrm{d}\mathbf{A}}\operatorname{Tr}\left(\mathbf{ACA}^{\mathrm{T}}\right) = 2\mathbf{AC},$$

where Tr denotes the matrix trace operator and we assume that $\mathbf{AB}$ is square and that $\mathbf{C}$ is a symmetric matrix. The derivative of a scalar quantity with respect to a matrix is defined as the matrix of derivatives of the scalar with respect to each element of the matrix. Before differentiating, we expand (3.15) to obtain

$$\mathbf{P}_k^{\mathrm{a}} = \mathbf{P}_k^{\mathrm{f}} - \mathbf{K}_k\mathbf{H}_k\mathbf{P}_k^{\mathrm{f}} - \mathbf{P}_k^{\mathrm{f}}\mathbf{H}_k^{\mathrm{T}}\mathbf{K}_k^{\mathrm{T}} + \mathbf{K}_k\left(\mathbf{H}_k\mathbf{K}_k\mathbf{H}_k^{\mathrm{T}} + \mathbf{R}_k\right)\mathbf{K}_k^{\mathrm{T}}.$$

There are two linear terms and one quadratic term in $\mathbf{K}_k$. To minimize the trace of $\mathbf{P}_k^{\mathrm{a}}$, we can now apply the above matrix differentiation formulas (supposing that the individual squared errors are also minimized when their sum is minimized) to obtain

$$\frac{\mathrm{d}}{\mathrm{d}\mathbf{K}_k}\operatorname{Tr}\mathbf{P}_k^{\mathrm{a}} = -2\left(\mathbf{H}_k\mathbf{P}_k^{\mathrm{f}}\right)^{\mathrm{T}} + 2\mathbf{K}_k\left(\mathbf{H}_k\mathbf{K}_k\mathbf{H}_k^{\mathrm{T}} + \mathbf{R}_k\right).$$

Setting this last result equal to zero, we can finally solve for the optimal gain. The resulting $\mathbf{K}$ that minimizes equation (3.13) is given by

$$\mathbf{K}_k = \mathbf{P}_k^{\mathrm{f}}\mathbf{H}_k^{\mathrm{T}}\left(\mathbf{H}_k\mathbf{P}_k^{\mathrm{f}}\mathbf{H}_k^{\mathrm{T}} + \mathbf{R}_k\right)^{-1}, \tag{3.16}$$

where we remark that $\mathbf{H}\mathbf{P}_k^{\mathrm{f}}\mathbf{H}_k^{\mathrm{T}} + \mathbf{R}_k = \mathrm{E}\left[\mathbf{d}_k\mathbf{d}_k^{\mathrm{T}}\right]$ is the covariance of the innovation. Looking at this expression for $\mathbf{K}_k$, we see that when the measurement error covariance, $\mathbf{R}_k$, approaches zero, the gain, $\mathbf{K}_k$, weights the innovation more heavily, since

$$\lim_{\mathbf{R}\to 0}\mathbf{K}_k = \mathbf{H}_k^{-1}.$$

On the other hand, as the a priori error estimate covariance, $\mathbf{P}_k^{\mathrm{f}}$, approaches zero, the gain, $\mathbf{K}_k$, weights the innovation less heavily, and

$$\lim_{\mathbf{P}_k^{\mathrm{f}}\to 0}\mathbf{K}_k = 0.$$

Another way of thinking about the weighting of $\mathbf{K}$ is that as the measurement error covariance, $\mathbf{R}$, approaches zero, the actual measurement, $\mathbf{y}_k$, is "trusted" more and more, while the predicted measurement, $\mathbf{H}_k\mathbf{x}_k^{\mathrm{f}}$, is trusted less and less. On the other hand, as the a priori error estimate covariance, $\mathbf{P}_k^{\mathrm{f}}$, approaches zero, the actual measurement, $\mathbf{y}_k$, is trusted less and less, while the predicted measurement, $\mathbf{H}_k\mathbf{x}_k^{\mathrm{f}}$, is trusted more and more—see the computational example below.

The covariance matrix associated with the optimal gain can now be computed from (3.15). We already have

$$\begin{aligned}\mathbf{P}_k^{\mathrm{a}} &= (\mathbf{I}-\mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^{\mathrm{f}}(\mathbf{I}-\mathbf{K}_k\mathbf{H}_k)^{\mathrm{T}}+\mathbf{K}_k\mathbf{R}_k\mathbf{K}_k^{\mathrm{T}}\\ &= \mathbf{P}_k^{\mathrm{f}}-\mathbf{K}_k\mathbf{H}_k\mathbf{P}_k^{\mathrm{f}}-\mathbf{P}_k^{\mathrm{f}}\mathbf{H}_k^{\mathrm{T}}\mathbf{K}_k^{\mathrm{T}}+\mathbf{K}_k\left(\mathbf{H}_k\mathbf{K}_k\mathbf{H}_k^{\mathrm{T}}+\mathbf{R}_k\right)\mathbf{K}_k^{\mathrm{T}},\end{aligned}$$

and, substituting the optimal gain (3.16), we can derive three more alternative expressions:

$$\mathbf{P}_k^{\mathrm{a}} = \mathbf{P}_k^{\mathrm{f}}-\mathbf{P}_k^{\mathrm{f}}\mathbf{H}_k^{\mathrm{T}}\left(\mathbf{H}_k\mathbf{P}_k^{\mathrm{f}}\mathbf{H}_k^{\mathrm{T}}+\mathbf{R}_k\right)^{-1}\mathbf{H}_k\mathbf{P}_k^{\mathrm{f}},$$

$$\mathbf{P}_k^{\mathrm{a}} = \mathbf{P}_k^{\mathrm{f}}-\mathbf{K}_k\left(\mathbf{H}_k\mathbf{P}_k^{\mathrm{f}}\mathbf{H}_k^{\mathrm{T}}+\mathbf{R}_k\right)\mathbf{K}_k^{\mathrm{T}},$$

and

$$\mathbf{P}_k^{\mathrm{a}} = (\mathbf{I}-\mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^{\mathrm{f}}. \tag{3.17}$$

Each of these four expressions for $\mathbf{P}_k^{\mathrm{a}}$ would give the same results with perfectly precise arithmetic, but in real-world applications some may perform better numerically. In what follows, we will use the simplest form (3.17), but this is by no means restrictive, and any one of the others could be substituted.

3.4.3.1 ▪ Predictor/forecast step

We start from a previous analyzed state, $\mathbf{x}_k^{\mathrm{a}}$, or from the initial state if $k=0$, characterized by the Gaussian PDF $p(\mathbf{x}_k^{\mathrm{a}} \mid \mathbf{y}_{1:k}^{\mathrm{o}})$ of mean $\mathbf{x}_k^{\mathrm{a}}$ and covariance matrix $\mathbf{P}_k^{\mathrm{a}}$. We use here the classical notation $\mathbf{y}_{i:j}=(\mathbf{y}_i,\mathbf{y}_{i+1},\ldots,\mathbf{y}_j)$ for $i \le j$ that denotes conditioning on all the observations in the interval. An estimate of $\mathbf{x}_{k+1}^{\mathrm{t}}$ is given by the dynamical model, which defines the forecast as

$$\mathbf{x}_{k+1}^{\mathrm{f}} = \mathbf{M}_{k+1}\mathbf{x}_k^{\mathrm{a}}, \tag{3.18}$$

$$\mathbf{P}_{k+1}^{\mathrm{f}} = \mathbf{M}_{k+1}\mathbf{P}_k^{\mathrm{a}}\mathbf{M}_{k+1}^{T}+\mathbf{Q}_{k+1}, \tag{3.19}$$

where the expression for $\mathbf{P}_{k+1}^{\mathrm{f}}$ is obtained from the dynamics equation and the definition of the model noise covariance, $\mathbf{Q}$.

3.4.3.2 ▪ Corrector/analysis step

At time t_{k+1}, the PDF $p(\mathbf{x}_{k+1}^{\mathrm{f}} \mid \mathbf{y}_{1:k}^{\mathrm{o}})$ is known, thanks to the mean, $\mathbf{x}_{k+1}^{\mathrm{f}}$, and covariance matrix, $\mathbf{P}_{k+1}^{\mathrm{f}}$, just calculated, as well as the assumption of a Gaussian distribution. The analysis step then consists of correcting this PDF using the observation available at time t_{k+1} to compute $p(\mathbf{x}_{k+1}^{\mathrm{a}} \mid \mathbf{y}_{k+1:1}^{\mathrm{o}})$. This comes from the BLUE in the dynamical context and gives

$$\mathbf{K}_{k+1} = \mathbf{P}_{k+1}^{\mathrm{f}}\mathbf{H}^{T}\left(\mathbf{H}\mathbf{P}_{k+1}^{\mathrm{f}}\mathbf{H}^{T}+\mathbf{R}_{k+1}\right)^{-1}, \tag{3.20}$$

$$\mathbf{x}_{k+1}^{\mathrm{a}} = \mathbf{x}_{k+1}^{\mathrm{f}}+\mathbf{K}_{k+1}\left(\mathbf{y}_{k+1}-\mathbf{H}\mathbf{x}_{k+1}^{\mathrm{f}}\right), \tag{3.21}$$

$$\mathbf{P}_{k+1}^{\mathrm{a}} = \left(\mathbf{I}-\mathbf{K}_{k+1}\mathbf{H}\right)\mathbf{P}_{k+1}^{\mathrm{f}}. \tag{3.22}$$

The predictor–corrector loop is illustrated in Figure 3.7 and can be immediately transposed into an operational algorithm.

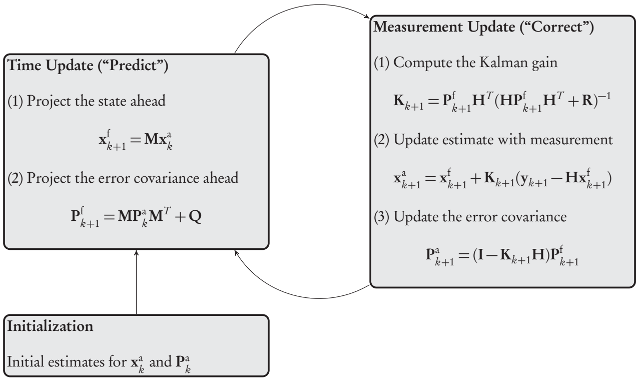


Figure 3.7. *Kalman filter loop.*

3.4.4 ▪ Note on the relation between Bayes and BLUE

If we know that the a priori and the observation data are both Gaussian, Bayes' rule can be applied to compute the a posteriori PDF. The a posteriori PDF is then Gaussian, and its parameters are given by the BLUE equations. Hence, with Gaussian PDFs and a linear observation operator, there is no need to use Bayes' rule. The BLUE equations can be used instead to compute the parameters of the resulting PDF. Since the BLUE provides the same result as Bayes' rule, it is the best estimator of all.

In addition (see the previous chapter), one can recognize the 3D-Var cost function. By minimizing this cost function, 3D-Var finds the MAP estimate of the Gaussian PDF, which is equivalent to the MV estimate found by the BLUE.

3.5 ▪ Implementation of the Kalman Filter

We now describe some important implementation issues and discuss ways to overcome the difficulties that they give rise to.

3.5.1 ▪ Stability and convergence

Stability is a concern for any dynamic system. The KF will be uniformly asymptotically stable if the system model itself is controllable and observable. The reader is referred to Friedland [1986], Gelb [1974], and Maybeck [1979] for detailed explanations of these concepts.

If the model is linear and time invariant (i.e., system matrices do not vary with time), the autocovariances will converge toward steady-state values. Consequently, the KF gain will converge toward a steady-state KF gain value that can be precalculated by solving an algebraic Ricatti equation. It is quite common to use only the steady-state gain in applications. For a nonlinear system, the gain may vary with the operating point (if the system matrix of the linearized model varies with the operating point).

In practical applications, the gain may be recalculated as the operating point changes.

In practical situations, there are a vast number of different sources for nonconvergence. In Grewal and Andrews [2001], the reader can find a very well explained presentation of all these (and many more). In particular, as we will point out, there are various remedies for

- convergence, divergence, and failure to converge;
- testing for unpredictable behavior;
- effects due to incorrect modeling;
- reduced-order and suboptimal filtering (see Chapter 5);
- reduction of round-off errors and computational expenses;
- analysis and repair of covariance matrices (see next subsection).

3.5.2 ▪ Filter divergence and covariance matrices

If the a priori statistical information is not well specified, the filter might underestimate the variances of the state errors, $\mathbf{e}_k^{\mathrm{a}}$. Too much confidence is put in the state estimation and too little confidence is put in the information contained in the observations. The effect of the analysis is minimized, and the gain becomes too small. In the most extreme case, observations are simply rejected. This is known as *filter divergence*, where the filter seems to behave well, with low predicted analysis error variance, but where the analysis is in fact drifting away from the reality.

Very often filter divergence is easy to diagnose:

- state error variances are small,
- the time sequence of innovations is biased, and
- the Kalman gains tend to zero as time increases.

It is thus important to monitor the innovation sequence and check that it is "white," i.e., unbiased and normally distributed. If this is not the case, then some of your assumptions are not valid.

There are a few rules to follow to avoid divergence:

- Do not underestimate model errors; rather, overestimate them.
- If possible, it is better to use an adaptive scheme to tune model errors by estimating them on the fly using the innovations.
- Give more weight to recent data, thus reducing the filter's memory of older data and forcing the data into the KF.
- Place some empirical, relevant lower bound on the Kalman gains.

3.5.3 ▪ Problem size and optimal interpolation

The straightforward application of the KF implies the "propagation" of an $n \times n$ covariance matrix at each time step. This can result in a very large problem in terms of computations and storage. If the state has a spatial dimension of 10^7 (which is not uncommon in large-scale geophysical and other simulations), then the covariance matrices will be of order 10^{14}, which will exceed the resources of most available computer installations. To overcome this, we must resort to various suboptimal schemes (an example of which is detailed below) or switch to ensemble approaches (see Chapters 6 and 7).

If the computational cost of propagating $\mathbf{P}^{\mathrm{a}}_{k+1}$ is an issue, we can use a *frozen covariance matrix*,

$$\mathbf{P}^{\mathrm{a}}_{k} = \mathbf{P}^{\mathrm{b}}, \quad k = 1, \ldots, n.$$

This defines the OI class of methods. Under this simplifying hypothesis, the two-step assimilation cycle defined above becomes the following:

1. **Forecast:**

$$\begin{aligned}
\mathbf{x}^{\mathrm{f}}_{k+1} &= \mathbf{M}_{k+1}\mathbf{x}^{\mathrm{a}}_{k},\\
\mathbf{P}^{\mathrm{f}}_{k+1} &= \mathbf{P}^{\mathrm{b}}.
\end{aligned}$$

2. **Analysis:**

$$\begin{aligned}
\mathbf{K}_{k+1} &= \mathbf{P}^{\mathrm{b}}\mathbf{H}^{T}\left(\mathbf{H}\mathbf{P}^{\mathrm{b}}\mathbf{H}^{T} + \mathbf{R}_{k+1}\right)^{-1},\\
\mathbf{x}^{\mathrm{a}}_{k+1} &= \mathbf{x}^{\mathrm{f}}_{k+1} + \mathbf{K}_{k+1}\left(\mathbf{y}_{k+1} - \mathbf{H}\mathbf{x}^{\mathrm{f}}_{k+1}\right),\\
\mathbf{P}^{\mathrm{a}}_{k+1} &= \mathbf{P}^{\mathrm{b}}.
\end{aligned}$$

There are at least two ways to compute the static covariance matrix $\mathbf{P}^{\mathrm{b}}$. The first is an analytical formulation,

$$\mathbf{P}^{\mathrm{b}} = \mathbf{D}^{1/2}\mathbf{C}\mathbf{D}^{1/2},$$

where $\mathbf{D}$ is a diagonal matrix of variances and $\mathbf{C}$ is a correlation matrix that can be defined, for example, as

$$C_{ij} = \left(1 + ah + \frac{1}{3}a^2h^2\right)e^{-ah},$$

where a is a tuneable parameter, h is the grid size, and the exponential function provides a local spatial dependence effect that often corresponds well to the physics. The second approach uses an ensemble of N_{e} snapshots of the state vector taken from a model free run, from which we compute the first and second statistical moments as follows:

$$\mathbf{x}^{\mathrm{b}} = \frac{1}{N_{\mathrm{e}}}\sum_{l=1}^{N_{\mathrm{e}}}\mathbf{x}_l,$$

$$\mathbf{P}^{\mathrm{b}} = \frac{1}{N_{\mathrm{e}}-1}\sum_{l=1}^{N_{\mathrm{e}}}\left(\mathbf{x}_l - \mathbf{x}^{\mathrm{b}}\right)\left(\mathbf{x}_l - \mathbf{x}^{\mathrm{b}}\right)^{\mathrm{T}}.$$

The static approach is more suited to successive assimilation cycles that are separated by a long enough time delay so that the corresponding dynamical states are sufficiently decorrelated.

Other methods are detailed in the sections on reduced methods—see Chapter 5.

3.5.4 ▪ Evolution of the state error covariance matrix

In principle, equation (3.19) generates a symmetric matrix. In practice, this may not be the case, and numerical truncation errors may lead to an asymmetric covariance matrix and a subsequent collapse of the filter. A remedy is to add an extra step to enforce symmetry, such as

$$\mathbf{P}^{\mathrm{f}}_{k+1} = \frac{1}{2}\left(\mathbf{P}^{\mathrm{f}}_{k+1} + (\mathbf{P}^{\mathrm{f}}_{k+1})^{\mathrm{T}}\right),$$

or a square root decomposition—see Chapter 5.

3.6 ▪ Nonlinearities and extensions of the KF

In real-life problems, we are most often confronted with a nonlinear process and/or a nonlinear measurement operator. Our dynamic system now takes the more general form

$$\begin{aligned}\mathbf{x}_{k+1} &= M_{k+1}(\mathbf{x}_k) + \mathbf{w}_k,\\ \mathbf{y}_k &= H_k(\mathbf{x}_k) + \mathbf{v}_k,\end{aligned}$$

where M_k now represents a nonlinear function of the state at time step k and H_k represents the nonlinear observation operator.

To deal with these nonlinearities, one approach is to linearize about the current mean and covariance, which is called the *extended Kalman filter* (EKF). This approach and its variants are presented in Chapter 6.

As previously mentioned, the KF is only optimal in the case of Gaussian statistics and linear operators, in which case the first two moments (the mean and the covariances) suffice to describe the PDF entering the estimation problem. Practitioners report that the linearized extension to nonlinear problems, the EKF, only works for moderate deviations from linearity and Gaussianity. The ensemble Kalman filter (EnKF) [Evensen, 2009] is a method that has been designed to deal with nonlinearities and non-Gaussian statistics, whereby the PDF is described by an ensemble of N_{e} time-dependent states $\mathbf{x}_{k,e}$. This method is presented in detail in Chapter 6. The appeal of this approach is its conceptual simplicity, the fact that it does not require any TLM or adjoint model (see Chapter 2), and the fact that it is extremely well suited to parallel programming paradigms, such as MPI [Gropp et al., 2014].

What happens if both the models are nonlinear and the PDFs are non-Gaussian? The KF and its extensions are no longer optimal and, more important, can easily fail the estimation process. Another approach must be used. A promising candidate is the *particle filter,* which is described below. The particle filter (see [Doucet and Johansen, 2011] and references therein) works sequentially in the spirit of the KF, but unlike the latter, it handles an ensemble of states (the particles) whose distribution approximates the PDF of the true state. Bayes' rule (3.5) and the marginalization formula (3.1) are explicitly used in the estimation process. The linear and Gaussian hypotheses can then be ruled out, in theory. In practice, though, the particle filter cannot yet be applied to very high dimensional systems (this is often referred to as "the curse of dimensionality").

Finally, there is a new class of hybrid methods, called *ensemble variational methods,* that attempt to combine variational and ensemble approaches—see Chapter 7 for a detailed presentation. The aim is to seek compromises to exploit the best aspects of (4D) variational and ensemble DA algorithms.

For further details of all these extensions, the reader should consult the advanced methods section (Part II) and the above references.

3.7 ▪ Particle filters for geophysical applications

Can we actually design a filtering numerical algorithm that converges to the Bayesian solution? Such a numerical approach would typically belong to the class of *sequential Monte Carlo* methods. That is to say, a PDF is represented by a discrete sample of the targeted PDF. Rather than trying to compute the exact solution of the Bayesian filtering equations, the transformations of such filtering (Bayes' rule for the analysis; model propagation for the forecast) are applied to the members of the sample. The statistical properties of the sample, such as the moments, are meant to be those of the targeted PDF. Obviously this sampling strategy can only be exact in the asymptotic limit, that is, in the limit where the number of members (or particles) goes to infinity.

This is the focus of a large body of applied mathematics that led to the design of many very successful Monte Carlo type methods [see, for instance, Doucet et al., 2001]. However, they have mostly been applied to very low dimensional systems (only a few dimensions). Their efficiency for high-dimensional models has been studied more recently, in particular thanks to a strong interest in these approaches in the geosciences. In the following, we give a brief biased overview of the subject as seen by the geosciences DA community.

3.7.1 ▪ Sequential Monte Carlo

The most popular and simple algorithm of Monte Carlo type that solves the Bayesian filtering equations is called the *bootstrap particle filter* [Gordon et al., 1993]. Its description follows.

3.7.1.1 ▪ Sampling

Let us consider a sample of particles $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m\}$. The related PDF at time t_k is $p_k(\mathbf{x})$, where $p_k(\mathbf{x}) \simeq \sum_{i=1}^m \omega_i^k \delta(\mathbf{x}-\mathbf{x}_k^i)$, δ is the Dirac mass, and the sum is meant to be an approximation of the exact density that the sample emulates. A positive scalar, ω_k^i, weights the importance of particle i within the ensemble. At this stage, we assume that the weights, ω_i^k, are uniform and $\omega_i^k = 1/m$.

3.7.1.2 ▪ Forecast

At the forecast step, the particles are propagated by the model without approximation, $p_{k+1}(\mathbf{x}) \simeq \sum_{i=1}^m \omega_k^i \delta(\mathbf{x} - \mathbf{x}_{k+1}^i)$, with $\mathbf{x}_{k+1}^i = \mathcal{M}_{k+1}(\mathbf{x}_k)$. A stochastic noise can be optionally added to the dynamics of each particle (see below).

3.7.1.3 ▪ Analysis

The analysis step of the particle filter is extremely simple and elegant. The rigorous implementation of Bayes' rule ascribes to each particle a statistical weight that corresponds to the likelihood of the particle given the data. The weight of each particle is updated according to (see Figure 3.8)

$$\omega_{k+1}^{\mathrm{a},i} \propto \omega_{k+1}^{\mathrm{f},i} p(\mathbf{y}_{k+1}|\mathbf{x}_{k+1}^i). \tag{3.23}$$

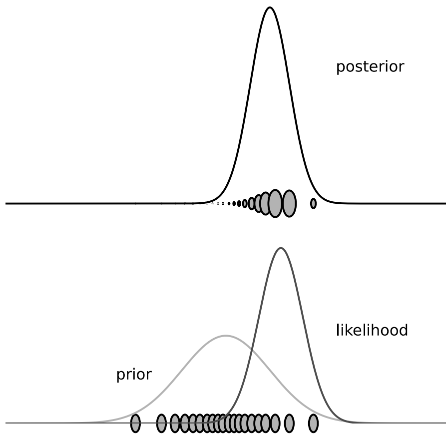


Figure 3.8. *Analysis of the particle filter. The initial ensemble of particles is sampled from a normal prior, with equal weights (bottom). Given an observation with Gaussian noise and the relative state likelihood (bottom), the particle filter analysis ascribes a weight to each particle, which is proportional to the likelihood of the particle given the observation (top). The major axis of the ellipses, representing the particles, is proportional to the particle weight.*

It is remarkable that the analysis is carried out with only a few multiplications. It does not involve inverting any system or matrix, as opposed, for instance, to the KF.

3.7.1.4 ▪ Resampling

Unfortunately, these normalized statistical weights have a potentially large amplitude of fluctuation. Even worse, as sequential filtering progresses, one particle (one trajectory of the model) will stand out from the others. Its weight will largely dominate the others ($\omega_i \lesssim 1$), while the other weights will vanish. Then the particle filter becomes very inefficient as an estimating tool since it has lost its variability. This phenomenon is called *degeneracy* of the particle filter [Kong et al., 1994]. An example of such degeneracy is given in Figure 3.9, where the statistical properties of the biggest weight are studied on an extensive meteorological toy model of 40 and 80 variables. In a degenerate case, the maximum weight will often reach 1 or close to 1, whereas in a balanced case, values very close to 1 will be less frequent.

One way to mitigate this phenomenon is to resample the particles by redrawing a sample with uniform weights from the degenerate distribution. After resampling, all particles have the same weight: $\omega_k^i = 1/m$.

The particle filter is very efficient for highly nonlinear models but with low dimensionality. Unfortunately, it is not suited for DA systems with models of high dimension, as soon as the dimension exceeds, say, about 10. Avoiding degeneracy requires a great number of particles. This number typically increases exponentially with

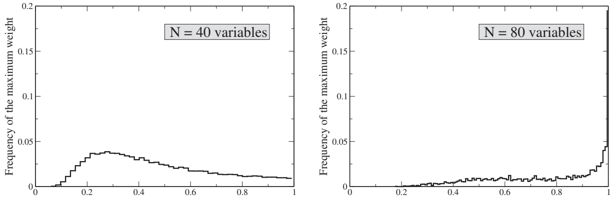


Figure 3.9. *On the left: statistical distribution of the maximal weight of the particle bootstrap filter in a balanced case. The physical system is a Lorenz-*95 *model with* 40 *variables [Lorenz and Emmanuel,* 1998*]. On the right: the same particle filter is applied to a Lorenz-*95 *low-order model, but with* 80 *variables. The maximal weight clearly degenerates with a peak close to* 1.

the system state space dimension. This is because the support of the prior PDF overlaps exponentially less with the support of the likelihood as the dimension of the state space of the systems increases. This is known as the *curse of dimensionality*.

For the forecast step, it could also be crucial to introduce stochastic perturbations of the states. Indeed, the ensemble will become impoverished with the many resamplings that it has to undergo. To enrich the sample, it is necessary to stochastically perturb the states of the system.

3.7.2 ▪ Application in the geosciences

The applicability of particle filters to high-dimensional models has been investigated in the geosciences [van Leeuwen, 2009; Bocquet et al., 2010]. The impact of the curse of dimensionality has been quantitatively studied in Snyder et al. [2008]. It has been shown, on a heuristic basis, that the number of particles m required to efficiently track the system must scale like the variance of the log-likelihood,

$$\ln(m) \propto \mathrm{Var}[\ln(p(\mathbf{y}|\mathbf{x}))], \tag{3.24}$$

which usually scales like the size of the system for typical geophysical problems. It is known [see, for instance, MacKay, 2003] that using an importance proposal to guide the particles toward regions of high probability will not change this trend, albeit with a smaller proportionality factor in (3.24). Snyder et al. [2015] confirmed this and gave bounds to the optimal proposal for particle filters that use an importance proposal leading to a minimal variance in the weights. They conclude again on the exponential dependence of the effective ensemble size with the problem dimension.

When smoothing is combined with a particle filter (which becomes a particle smoother) over a DA window, alternative and more efficient particle filters can be designed, such as the implicit particle filter [Morzfeld et al., 2012].

Particle filters can nevertheless be useful for high-dimensional models if the significant degree of nonlinearity is confined to a small subspace of the state space. For instance, in Lagrangian DA, the errors on the location of moving observation platforms have significantly non-Gaussian statistics. In this case, these degrees of freedom can be addressed with a particle filter, while the rest is controlled by an EnKF, which is practical for high-dimensional models [Slivinski et al., 2015].

If we drop the assumption that a particle filter should have the proper Bayesian asymptotic limit, it becomes possible to design nonlinear filters for DA with high-dimensional models such as the equal-weight particle filter (see [Ades and van Leeuwen, 2015] and references therein).

Finally, if the system cannot be split, then a solution to implement a particle filter in high dimension could come from localization, just as with the EnKF (Chapter 6). This was proven to be more difficult because locally updated particles cannot easily be glued together into global particles. However, an ensemble transform representation that has been built for the EnKF [Bishop et al., 2001] is better suited to ensure a smoother gluing of the local updates [Reich, 2013]. An astute merging of the particles has been shown to yield a local particle filter that could outperform the EnKF in specific regimes with a moderate number of particles [Poterjoy, 2016].

3.8 ▪ Examples

In this section we present a number of examples of special cases of the KF—both analytical and numerical. Though they may seem overly simple, the intention is that you, the user, gain the best possible feeling and intuition regarding the actual operation of the filter. This understanding is essential for more complex cases, such as those presented in the advanced methods and applications chapters.

Example 3.31. *Case without observations*. Here, the observation matrix $\mathbf{H}_k = 0$ and thus $\mathbf{K}_k = 0$ as well. Hence the KF equations (3.18)–(3.22) reduce to

$$\begin{aligned} \mathbf{x}^{\mathrm{f}}_{k+1} &= \mathbf{M}_{k+1}\mathbf{x}^{\mathrm{a}}_{k}, \\ \mathbf{P}^{\mathrm{f}}_{k+1} &= \mathbf{M}_{k+1}\mathbf{P}^{\mathrm{a}}_{k}\mathbf{M}^{T}_{k+1} + \mathbf{Q}_{k+1}, \end{aligned}$$

and

$$\begin{aligned} \mathbf{K}_{k+1} &= 0, \\ \mathbf{x}^{\mathrm{a}}_{k+1} &= \mathbf{x}^{\mathrm{f}}_{k+1}, \\ \mathbf{P}^{\mathrm{a}}_{k+1} &= \mathbf{P}^{\mathrm{f}}_{k+1}. \end{aligned}$$

Thus, we can completely eliminate the analysis stage of the algorithm to obtain

$$\begin{aligned} \mathbf{x}^{\mathrm{f}}_{k+1} &= \mathbf{M}_{k+1}\mathbf{x}^{\mathrm{f}}_{k}, \\ \mathbf{P}^{\mathrm{f}}_{k+1} &= \mathbf{M}_{k+1}\mathbf{P}^{\mathrm{f}}_{k}\mathbf{M}^{T}_{k+1} + \mathbf{Q}_{k+1}, \end{aligned}$$

initialized by

$$\begin{aligned} \mathbf{x}^{\mathrm{f}}_{0} &= \mathbf{x}_{0}, \\ \mathbf{P}^{\mathrm{f}}_{0} &= \mathbf{P}_{0}. \end{aligned}$$

The model then runs without any input of data, and if the dynamics are neutral or unstable, the forecast error will grow without limit. For example, in a typical NWP assimilation cycle, where observations are obtained every 6 hours, the model runs for 6 hours without data. During this period, the forecast error grows and is damped only when the data arrives, thus giving rise to the characteristic "sawtooth" pattern of error variance evolution. ∎

Example 3.32. *Perfect observations at all grid points*. In the case of perfect observations, the observation error covariance matrix $\mathbf{R}_k = 0$ and the observation operator $\mathbf{H}$ is the identity. Hence the KF equations (3.18)–(3.22) reduce to

$$\mathbf{x}^{\mathrm{f}}_{k+1} = \mathbf{M}_{k+1}\mathbf{x}^{\mathrm{a}}_{k},$$
$$\mathbf{P}^{\mathrm{f}}_{k+1} = \mathbf{M}_{k+1}\mathbf{P}^{\mathrm{a}}_{k}\mathbf{M}^{T}_{k+1} + \mathbf{Q}_{k+1},$$

and

$$\mathbf{K}_{k+1} = \mathbf{P}^{\mathrm{f}}_{k+1}\mathbf{H}^{T}\left(\mathbf{H}\mathbf{P}^{\mathrm{f}}_{k+1}\mathbf{H}^{T}\right)^{-1} = \mathbf{I},$$
$$\mathbf{x}^{\mathrm{a}}_{k+1} = \mathbf{x}^{\mathrm{f}}_{k+1} + \left(\mathbf{y}_{k+1} - \mathbf{x}^{\mathrm{f}}_{k+1}\right),$$
$$\mathbf{P}^{\mathrm{a}}_{k+1} = \left(\mathbf{I} - \mathbf{K}_{k+1}\mathbf{H}\right)\mathbf{P}^{\mathrm{f}}_{k+1} = 0.$$

This is obviously another case of ideal observations, and we can once again completely eliminate the analysis stage to obtain

$$\mathbf{x}^{\mathrm{f}}_{k+1} = \mathbf{M}_{k+1}\mathbf{x}^{\mathrm{f}}_{k},$$
$$\mathbf{P}^{\mathrm{f}}_{k+1} = \mathbf{Q}_{k+1},$$

with initial conditions

$$\mathbf{x}^{\mathrm{f}}_{0} = \mathbf{y}_{0},$$
$$\mathbf{P}^{\mathrm{f}}_{0} = 0.$$

Since $\mathbf{R}$ is in fact the sum of measurement and representation errors, $\mathbf{R} = 0$ implies that the only scales that are observed are those resolved by the model. The forecast is thus an integration of the observed state, and the forecast error reduces to the model error. ■

Example 3.33. *Scalar case.* As in Section 2.4.5, let us consider the same scalar example, but this time apply the KF to it. We take the simplest linear forecast model,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\alpha x,$$

with α a known positive constant. We assume the same discrete dynamics considered in (2.49) with a single observation at time step 3.

The stochastic system (3.11)–(3.12) is

$$x^{\mathrm{t}}_{k+1} = M(x^{\mathrm{t}}_{k}) + w_k,$$
$$y_{k+1} = x^{\mathrm{t}}_{k} + v_k,$$

where $w_k \sim \mathcal{N}(0, \sigma_Q^2)$, $v_k \sim \mathcal{N}(0, \sigma_R^2)$, and $x^{\mathrm{t}}_0 - x^{\mathrm{b}}_0 \sim \mathcal{N}(0, \sigma_B^2)$. The KF steps are as follows:

Forecast:

$$x^{\mathrm{f}}_{k+1} = M(x^{\mathrm{a}}_{k}) = \gamma x_k,$$
$$P^{\mathrm{f}}_{k+1} = \gamma^2 P^{\mathrm{a}}_{k} + \sigma_Q^2.$$

Analysis:

$$
\begin{aligned}
K_{k+1} &= P^{\mathrm{f}}_{k+1} H \left(H^2 P^{\mathrm{f}}_{k+1} + \sigma_R^2 \right)^{-1}, \\
x^{\mathrm{a}}_{k+1} &= x^{\mathrm{f}}_{k+1} + K_{k+1}(x^{\mathrm{obs}}_{k+1} - H x^{\mathrm{f}}_{k+1}), \\
P^{\mathrm{a}}_{k+1} &= (1 - K_{k+1} H) P^{\mathrm{f}}_{k+1} = \left(\frac{1}{P^{\mathrm{f}}_{k+1}} + \frac{1}{\sigma_R^2} \right)^{-1}, \quad H = 1.
\end{aligned}
$$

Initialization:

$$
\begin{aligned}
x^{\mathrm{a}}_0 &= x^{\mathrm{b}}_0, \\
P^{\mathrm{a}}_0 &= \sigma_B^2.
\end{aligned}
$$

We start with the initial state, at time step $k = 0$. The initial conditions are as above. The forecast is

$$
\begin{aligned}
x^{\mathrm{f}}_1 &= M(x^{\mathrm{a}}_0) = \gamma x^{\mathrm{b}}_0, \\
P^{\mathrm{f}}_1 &= \gamma^2 \sigma_B^2 + \sigma_Q^2.
\end{aligned}
$$

Since there is no observation available, $H = 0$, and the analysis gives

$$
\begin{aligned}
K_1 &= 0, \\
x^{\mathrm{a}}_1 &= x^{\mathrm{f}}_1 = \gamma x^{\mathrm{b}}_0, \\
P^{\mathrm{a}}_1 &= P^{\mathrm{f}}_1 = \gamma^2 \sigma_B^2 + \sigma_Q^2.
\end{aligned}
$$

At the next time step, $k = 1$, and the forecast gives

$$
\begin{aligned}
x^{\mathrm{f}}_2 &= M(x^{\mathrm{a}}_1) = \gamma^2 x^{\mathrm{b}}_0, \\
P^{\mathrm{f}}_2 &= \gamma^2 P^{\mathrm{a}}_1 + \sigma_Q^2 = \gamma^4 \sigma_B^2 + (\gamma^2 + 1)\sigma_Q^2.
\end{aligned}
$$

Once again there is no observation available, $H = 0$, and the analysis yields

$$
\begin{aligned}
K_2 &= 0, \\
x^{\mathrm{a}}_2 &= x^{\mathrm{f}}_2 = \gamma^2 x^{\mathrm{b}}_0, \\
P^{\mathrm{a}}_2 &= P^{\mathrm{f}}_2 = \gamma^4 \sigma_B^2 + (\gamma^2 + 1)\sigma_Q^2.
\end{aligned}
$$

Moving on to $k = 2$, we have the new forecast:

$$
\begin{aligned}
x^{\mathrm{f}}_3 &= M(x^{\mathrm{a}}_2) = \gamma^3 x^{\mathrm{b}}_0, \\
P^{\mathrm{f}}_3 &= \gamma^2 P^{\mathrm{a}}_2 + \sigma_Q^2 = \gamma^6 \sigma_B^2 + (\gamma^4 + \gamma^2 + 1)\sigma_Q^2.
\end{aligned}
$$

Now there is an observation, x^{o}_3, available, so $H = 1$, and the analysis is

$$
\begin{aligned}
K_3 &= P^{\mathrm{f}}_3 \left(P^{\mathrm{f}}_3 + \sigma_R^2 \right)^{-1}, \\
x^{\mathrm{a}}_3 &= x^{\mathrm{f}}_3 + K_3 (x^{\mathrm{o}}_3 - x^{\mathrm{f}}_3), \\
P^{\mathrm{a}}_3 &= (1 - K_3) P^{\mathrm{f}}_3.
\end{aligned}
$$

Substituting and simplifying, we find

$$x_3^{\mathrm{a}} = \gamma^3 x_0^{\mathrm{b}} + \frac{\gamma^6\sigma_B^2 + (\gamma^4+\gamma^2+1)\sigma_Q^2}{\sigma_R^2 + \gamma^6\sigma_B^2 + (\gamma^4+\gamma^2+1)\sigma_Q^2}\left(x_3^{\mathrm{o}} - \gamma^3 x_0^{\mathrm{b}}\right). \tag{3.25}$$

Case 1: Assume we have a perfect model. Then $\sigma_Q^2 = 0$ and the KF state (3.25) becomes

$$x_3^{\mathrm{a}} = \gamma^3 x_0^{\mathrm{b}} + \frac{\gamma^6\sigma_B^2}{\sigma_R^2 + \gamma^6\sigma_B^2}\left(x_3^{\mathrm{o}} - \gamma^3 x_0^{\mathrm{b}}\right),$$

which is precisely the 4D-Var expression (2.51) obtained before.

Case 2: When the parameter α tends to zero, then γ tends to one, the model is stationary, and the KF state (3.25) becomes

$$x_3^{\mathrm{a}} = x_0^{\mathrm{b}} + \frac{\sigma_B^2 + 3\sigma_Q^2}{\sigma_R^2 + \sigma_B^2 + 3\sigma_Q^2}\left(x_3^{\mathrm{o}} - x_0^{\mathrm{b}}\right),$$

which, when $\sigma_Q^2 = 0$, reduces to the 3D-Var solution,

$$x_3^{\mathrm{a}} = x_0^{\mathrm{b}} + \frac{\sigma_B^2}{\sigma_R^2 + \sigma_B^2}\left(x_3^{\mathrm{o}} - x_0^{\mathrm{b}}\right),$$

that was obtained before in (2.52).

Case 3: When α tends to infinity, then γ goes to zero, and we are in the case where there is no longer any memory with

$$x_3^{\mathrm{a}} = \frac{\sigma_Q^2}{\sigma_R^2 + \sigma_Q^2} x_3^{\mathrm{o}}.$$

Then, if the model is perfect, $\sigma_Q^2 = 0$ and $x_3^{\mathrm{a}} = 0$. If the observation is perfect, $\sigma_R^2 = 0$ and $x_3^{\mathrm{a}} = x_3^{\mathrm{o}}$.

This example shows the complete chain, from the KF solution through the 4D-Var and finally reaching the 3D-Var solution. We hope that this clarifies the relationship between the three and demonstrates why the KF provides the most general solution possible. ■

Example 3.34. *Brownian motion.* Here we compute a numerical application of the scalar case seen above in Example 3.33. We have the following state and measurement equations:

$$\begin{aligned} x_{k+1} &= x_k + w_k, \\ y_{k+1} &= x_k + v_k, \end{aligned}$$

where the dynamic transition matrix $M_k = 1$ and the observation operator $H = 1$. Let us suppose constant error variances of $Q_k = 1$ and $R_k = 0.25$ for the process and measurement errors, respectively. Here the KF equations (3.18)–(3.22) reduce to

$$\begin{aligned} x_{k+1}^{\mathrm{f}} &= x_k^{\mathrm{a}}, \\ P_{k+1}^{\mathrm{f}} &= P_k^{\mathrm{a}} + 1, \end{aligned}$$

and

$$
\begin{aligned}
K_{k+1} &= P^{\mathrm{f}}_{k+1}\left(P^{\mathrm{f}}_{k+1}+0.25\right)^{-1},\\
x^{\mathrm{a}}_{k+1} &= x^{\mathrm{f}}_{k+1}+K_{k+1}\left(y_{k+1}-x^{\mathrm{f}}_{k+1}\right),\\
P^{\mathrm{a}}_{k+1} &= \left(I-K_{k+1}\right)P^{\mathrm{f}}_{k+1}.
\end{aligned}
$$

By substituting for P^{f}_{k+1} from the forecast equation, we can rewrite the Kalman gain in terms of P^{a}_k as

$$
K_{k+1} = \frac{P^{\mathrm{a}}_k+1}{P^{\mathrm{a}}_k+1.25},
$$

and we obtain the update for the error variance:

$$
P^{\mathrm{a}}_{k+1} = \frac{P^{\mathrm{a}}_k+1}{4P^{\mathrm{a}}_k+5}.
$$

Plugging into the analysis equation, we now have the complete update:

$$
\begin{aligned}
x^{\mathrm{a}}_{k+1} &= x^{\mathrm{a}}_k+\frac{P^{\mathrm{a}}_k+1}{P^{\mathrm{a}}_k+1.25}\left(y_{k+1}-x^{\mathrm{a}}_k\right),\\
P^{\mathrm{a}}_{k+1} &= \frac{P^{\mathrm{a}}_k+1}{4P^{\mathrm{a}}_k+5}.
\end{aligned}
$$

Let us now, manually, perform a couple of iterations. Taking as initial conditions

$$
x^{\mathrm{a}}_0 = 0, \quad P^{\mathrm{a}}_0 = 0,
$$

we readily compute, for $k=0$,

$$
\begin{aligned}
K_1 &= \frac{1}{1.25} = 0.8,\\
x^{\mathrm{a}}_1 &= 0+K_1(y_1-0) = 0.8y_1,\\
P^{\mathrm{a}}_1 &= \frac{1}{5} = 0.2.
\end{aligned}
$$

Then for $k=1$,

$$
\begin{aligned}
K_2 &= \frac{0.2+1}{0.2+1.25} \approx 0.8276,\\
x^{\mathrm{a}}_2 &= 0.8y_1+K_2(y_2-0.8y_1) \approx 0.138y_1+0.828y_2,\\
P^{\mathrm{a}}_2 &= \frac{0.2+1}{0.8+5} = \frac{6}{29} \approx 0.207.
\end{aligned}
$$

One more step for $k=2$ gives

$$
\begin{aligned}
K_3 &= \frac{6/29+1}{6/29+1.25} \approx 0.8284,\\
x^{\mathrm{a}}_3 &= 0.138y_1+0.828y_2+K_3(y_3-0.138y_1-0.828y_2) \approx 0.024y_1+0.142y_2+0.828y_3,\\
P^{\mathrm{a}}_3 &= \frac{6/29+1}{24/29+5} \approx 0.207.
\end{aligned}
$$

Let us see what happens in the limit, $k \to \infty$. We observe that $P_{k+1} \approx P_k$; thus

$$P^{\mathrm{a}}_{\infty} = \frac{P^{\mathrm{a}}_{\infty} + 1}{4P^{\mathrm{a}}_{\infty} + 5},$$

which is a quadratic equation for P^{a}_{∞}, whose solutions are

$$P^{\mathrm{a}}_{\infty} = \frac{1}{2}\left(-1 \pm \sqrt{2}\right).$$

The positive definite solution is

$$P^{\mathrm{a}}_{\infty} = \frac{1}{2}\left(-1 + \sqrt{2}\right) \approx 0.2071,$$

and hence

$$K_{\infty} = \frac{2 + 2\sqrt{2}}{3 + 2\sqrt{2}} \approx 0.8284.$$

We observe in this case that the KF tends toward a steady-state filter after only two steps. The reasons for this rapid convergence are that the dynamics are neutral and that the observation error covariance is relatively small when compared to the process error, $R \ll Q$, which means that the observations are relatively precise compared to the model error. In addition, the state (being scalar) is completely observed whenever an observation is available.

In conclusion, the combination of dense, precise observations with steady, linear dynamics will always lead to a stable filter. ■

Example 3.35. *Estimation of a random constant.* In this simple numerical example, let us attempt to estimate a scalar random constant, for example, a voltage. Let us assume that we have the ability to take measurements of the constant, but that the measurements are corrupted by a 0.1 volt root mean square (rms) white measurement noise (e.g., our analog-to-digital converter is not very accurate). In this example, our process is governed by the state equation

$$x_k = M x_{k-1} + w_k = x_{k-1} + w_k$$

and the measurement equation

$$y_k = H x_k + v_k = x_k + v_k.$$

The state, being constant, does not change from step to step, so $M = I$. Our noisy measurement is of the state directly, so $H = 1$. We are in fact in the same Brownian motion context as the previous example.

The time update (forecast) equations are

$$\begin{aligned} x^{\mathrm{f}}_{k+1} &= x^{\mathrm{a}}_{k}, \\ P^{\mathrm{f}}_{k+1} &= P^{\mathrm{a}}_{k} + Q, \end{aligned}$$

and the measurement update (analysis) equations are

$$\begin{aligned} K_{k+1} &= P^{\mathrm{f}}_{k+1}(P^{\mathrm{f}}_{k+1} + R)^{-1}, \\ x^{\mathrm{a}}_{k+1} &= x^{\mathrm{f}}_{k+1} + K_{k+1}(y_{k+1} - x^{\mathrm{f}}_{k+1}), \\ P^{\mathrm{a}}_{k+1} &= (1 - K_{k+1})P^{\mathrm{f}}_{k+1}. \end{aligned}$$

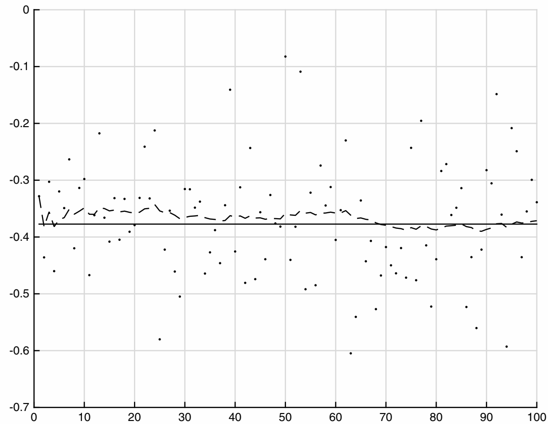


Figure 3.10. *Estimating a constant—simulation with $R = 0.01$. True value (solid), measurements (dots), KF estimation (dashed). The x-axis denotes time; the y-axis denotes the state variable.*

Initialization. Presuming a very small process variance, we let $Q = 1.\mathrm{e}-5$. We could certainly let $Q = 0$, but assuming a small but nonzero value gives us more flexibility in "tuning" the filter, as we will demonstrate below. Let's assume that from experience we know that the true value of the random constant has a standard Gaussian probability distribution, so we will "seed" our filter with the guess that the constant is 0. In other words, before starting, we let $x_0 = 0$. Similarly, we need to choose an initial value for P_k^a; call it P_0. If we were absolutely certain that our initial state estimate was correct, we would let $P_0 = 0$. However, given the uncertainty in our initial estimate, x_0, choosing $P_0 = 0$ would cause the filter to initially and always believe that $x_k^a = 0$. As it turns out, the alternative choice is not critical. We could choose almost any $P_0 \neq 0$ and the filter would eventually converge. We will start our filter with $P_0 = 1$.

Simulations. To begin with, we randomly chose a scalar constant $y = -0.37727$. We then simulated 100 distinct measurements that had an error normally distributed around zero with a standard deviation of 0.1 (remember we presumed that the measurements are corrupted by a 0.1 volt rms white measurement noise).

In the first simulation we fixed the measurement variance at $R = (0.1)^2 = 0.01$. Because this is the "true" measurement error variance, we would expect the "best" performance in terms of balancing responsiveness and estimate variance. This will become more evident in the second and third simulations. Figure 3.10 depicts the results of this first simulation. The true value of the random constant, $x = -0.37727$, is given by the solid line, the noisy measurements by the dots, and the filter estimate by the remaining dashed curve.

In Figures 3.11 and 3.12 we can see what happens when the measurement error variance, R, is increased or decreased by a factor of 100. In Figure 3.11, the filter was told

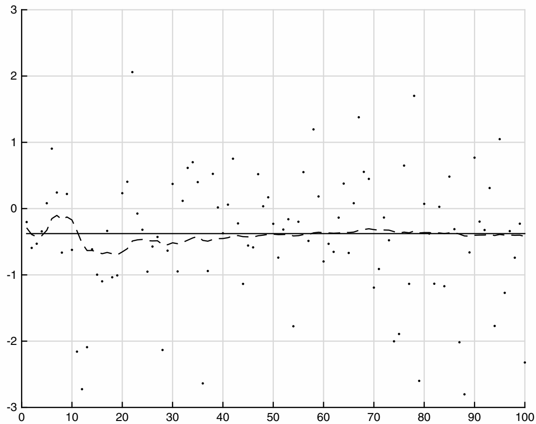


Figure 3.11. *Estimating a constant—simulation with $R = 1$. True value (solid), measurements (dots), KF estimation (dashed). The x-axis denotes time; the y-axis denotes the state variable.*

that the measurement variance was 100 times as great (i.e., $R = 1$), so it was "slower" to believe the measurements. In Figure 3.12, the filter was told that the measurement variance was 1/100th the size (i.e., $R = 0.0001$), so it was very "quick" to believe the noisy measurements.

While the estimation of a constant is relatively straightforward, this example clearly demonstrates the workings of the KF. In Figure 3.11 in particular the Kalman "filtering" is evident, as the estimate appears considerably smoother than the noisy measurements. We observe the speed of convergence of the variance in Figure 3.13.

Here is the MATLAB code used to perform the simulations.

```
% SCALAR EXAMPLE (estimate a constant):
%
% Define the system as a constant of  -0.37727 volts:
clear s
s.x =  -0.37727;
s.A = 1;
% Define a process noise:
s.Q = 0.00001; % variance
% Define the voltmeter to measure the voltage itself:
s.H = 1;
% Define a measurement error:
s.R   = 0.01^2; % variance, hence stdev^2
Rstd  = sqrt(s.R); % random measurement noise stdev
% Specify an initial state:
s.x =  -0.37727;
s.P = 1;
% Generate random voltages and perform the filter operation.
tru=[]; % true voltage
for t=1:100
```

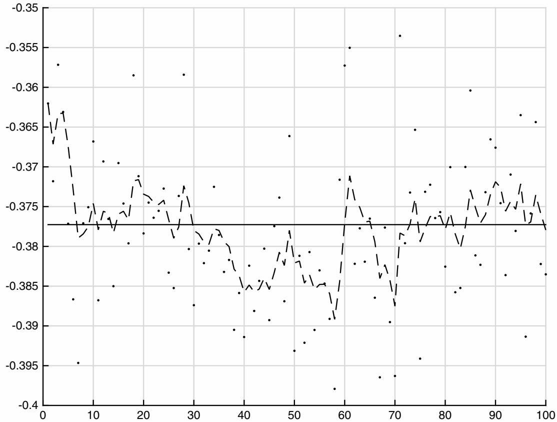


Figure 3.12. *Estimating a constant—simulation with $R = 0.0001$. True value (solid), measurements (dots), KF estimation (dashed). The x-axis denotes time; the y-axis denotes the state variable.*

```
    tru(end+1) =  -0.37727;
    s(end).y = tru(end) + Rstd*randn; % create a measurement
    s(end+1)=kalmanf(s(end)); % perform a Kalman filter iteration
end
% plot measurement data:
figure, hold on, grid on
hy=plot([s(1:end-1).y],'r.');
% plot a-posteriori state estimates:
hk=plot([s(2:end).x],'b--');
% plot true data
ht=plot(tru,'g-');
%legend([hy hk ht],'observations','Kalman output','true voltage',0)
%title('Estimating a constant')
hold off
% KALMANF - updates a system state vector estimate based upon an
%           observation, using a discrete Kalman filter.
%
% Version 1.1, August 13, 2015
%
% This function is based on the original of Michael C. Kleder
%
% INTRODUCTION
%
% Applying the filter to a basic linear system is actually very
% easy.
% This MATLAB file demonstrates that.
%
% An excellent paper on Kalman filtering at the introductory level,
% without detailing the mathematical underpinnings, is
% "An Introduction to the Kalman Filter"
```

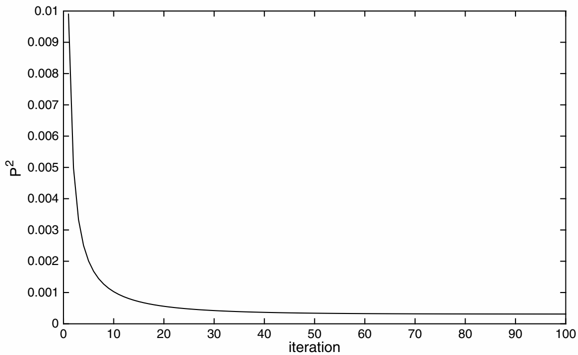


Figure 3.13. *Estimating a constant—convergence of the variance with* $R = 0.01$.

```
% Greg Welch and Gary Bishop, University of North Carolina
% http://www.cs.unc.edu/~welch/kalman/kalmanIntro.html
%
% PURPOSE:
%
% The purpose of each iteration of a Kalman filter is to update
% the estimate of the state vector of a system (and the covariance
% of that vector) based upon the information in a new observation.
% The version of the Kalman filter in this function assumes that
% observations occur at fixed discrete time intervals. Also, this
% function assumes a linear system, meaning that the time evolution
% of the state vector can be calculated by means of a state
% transition matrix.
%
% USAGE:
%
% s = kalmanf(s)
%
% "s" is a "system" struct containing various fields used as input
% and output. The state estimate "x" and its covariance "P" are
% updated by the function. The other fields describe the mechanics
% of the system and are left unchanged. A calling routine may change
% these other fields as needed if state dynamics are time dependent;
% otherwise, they should be left alone after initial values are set.
% The exceptions are the observation vector "y"
%
% SYSTEM DYNAMICS:
%
% The system evolves according to the following difference
% equations, where quantities are further defined below:
%
% x = Ax + w        meaning the state vector x evolves during one
%                   time step by premultiplying by the "state
%                   transition matrix" A. There is also Gaussian
%                   process noise w.
% y = Hx + v        meaning the observation vector y is a linear
```

```
%                    function of the state vector, and this linear
%                    relationship is represented by premultiplication
%                    by "observation matrix" H. There is also
%                    Gaussian measurement noise v.
% where w ~ N(0,Q) meaning w is Gaussian noise with covariance Q
%       v ~ N(0,R) meaning v is Gaussian noise with covariance R
%
% VECTOR VARIABLES:
%
% s.x = state vector estimate. In the input struct, this is the
%       "a priori" state estimate (prior to the addition of the
%       information from the new observation). In the output struct,
%       this is the "a posteriori" state estimate (after the new
%       measurement information is included).
% s.y = observation vector
%
% MATRIX VARIABLES:
%
% s.A = state transition matrix (defaults to identity).
% s.P = covariance of the state vector estimate. In the input
%       struct, this is "a priori," and in the output it is
%       "a posteriori" (required unless autoinitializing as
%       described below).
% s.Q = process noise covariance (defaults to zero).
% s.R = measurement noise covariance (required).
% s.H = observation matrix (defaults to identity).
%
% NORMAL OPERATION:
%
% (1) define all state definition fields: A,H,Q,R
% (2) define initial state estimate: x,P
% (3) obtain observation vector: y
% (4) call the filter to obtain updated state estimate: x,P
% (5) return to step (3) and repeat
%
% INITIALIZATION:
%
% If an initial state estimate is unavailable, it can be obtained
% from the first observation as follows, provided that there are
% the same number of observable variables as state variables.
% This "auto-initialization" is done automatically if s.x is
% absent or NaN.
%
% x = inv(H)yz
% P = inv(H)*R*inv(H')
%
% This is mathematically equivalent to setting the initial state
% estimate covariance to infinity.
%

function s = kalmanf(s)

% set defaults for absent fields:
if ~isfield(s,'y'); error('Observation␣vector␣missing'); end
if ~isfield(s,'x'); s.x=nan*y; end
if ~isfield(s,'P'); s.P=nan; end
if ~isfield(s,'A'); s.A=eye(length(x)); end
if ~isfield(s,'Q'); s.Q=zeros(length(x)); end
if ~isfield(s,'R'); error('Observation␣covariance␣missing'); end
if ~isfield(s,'H'); s.H=eye(length(x)); end

if isnan(s.x)
   % initialize state estimate from first observation
```

```
    if diff(size(s.H))
        error('Observation␣matrix␣must␣be␣square␣and␣invertible␣' ...
                'for␣state␣auto-initialization')
    end
    s.x = inv(s.H)*s.y;
    s.P = inv(s.H)*s.R*inv(s.H');
else

    % This is the code which implements the discrete Kalman filter:

    % Prediction for state vector and covariance:
    s.x = s.A*s.x;
    s.P = s.A * s.P * s.A' + s.Q;

    % Compute Kalman gain factor:
    K = (s.P)*(s.H')*inv(s.H*s.P*s.H'+s.R);

    % Correction based on observation:
    s.x = s.x + K*(s.y-s.H*s.x);
    s.P = s.P - K*s.H*s.P;

    % Note that the desired result, which is an improved estimate
    % of the system state vector x and its covariance P, was obtained
    % in only five lines of code, once the system was defined. (That's
    % how simple the discrete Kalman filter is to use.)
end
return
```

∎

Example 3.36. *Estimation of a linear trajectory*. We now go one step further and introduce some simple linear dynamics into the problem. We consider a train moving at an unknown, constant velocity, and we would like to predict both its position and its velocity from noisy observations of the position only. Note that this example has many concrete applications in automatic pilots, robotics, and other inertial navigation instruments, where KFs are extensively employed. Let us write down the equations.

The equation of motion for the actual position, x, is

$$x(t) = x_0 + st,$$

where x_0 is the initial position and s is the constant speed of the train. For state space notation, we introduce the state vector,

$$\mathbf{x} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix},$$

where x is the position and $\dot{x}$ is the velocity. The state equation can then be written as

$$\mathbf{x}_{k+1} = \mathbf{M}\mathbf{x}_k + \mathbf{w}_k,$$

where we have assumed that the system is perturbed by a white Gaussian noise, $\mathbf{w}$, with known covariance. From the dynamics, we deduce the discrete form of the matrix,

$$\mathbf{M} = \begin{bmatrix} 1 & \mathrm{d}t \\ 0 & 1 \end{bmatrix},$$

where $\mathrm{d}t$ is the time step increment and corresponds to the instants when measurements are taken. The observation is scalar,

$$y_k = \mathbf{H}\mathbf{x}_k + v_k,$$

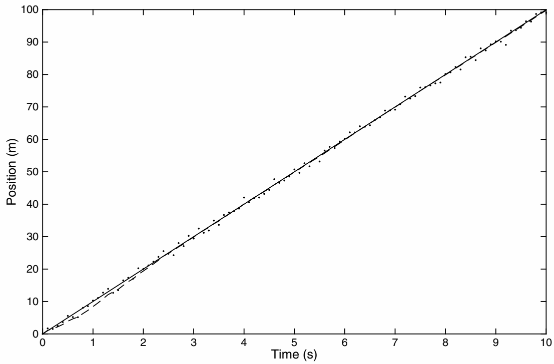


Figure 3.14. *Position estimation for constant-velocity dynamics. True value (solid); measurements (dots); KF estimation (dashed).*

with

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

We initialize our problem with $x_0 = 0$ and $\dot{x}_0 = 0.5\dot{x}^{\mathrm{t}}$, where the (unknown) true velocity $\dot{x}^{\mathrm{t}} = 10\ \mathrm{m\,s^{-1}}$. We assume that measurements are taken every $\mathrm{d}t = 0.1$ s from $t = 0$ until $t = 10$ and that they are subject to a noise with standard deviation of $1\ \mathrm{ms^{-1}}$. We will suppose that the initial error covariance matrix $\mathbf{P}_0 = \mathbf{I}$ and that the process noise is small ($R = 0.0001$). We want to predict the train's position 2 seconds ahead, that is, at $t = 12$ s. We will use the KF, as we need an accurate and smooth estimate for the velocity to predict the train's position in the future.

The numerical computation gives the results and predictions shown in Figures 3.14, 3.15, 3.16, 3.17, and 3.18. The filter does a good job and "steers" a smoother path among the noisy measurements. When compared to an extrapolation, based on a running average for the velocity, the KF clearly outperforms other methods when it comes to the prediction of the position in the future. The convergence of the filter parameters to zero is a sign that the model is adequate and that the data and model are consistent and in good agreement.

The reader is strongly encouraged to modify the code of the previous example to reproduce these results and then to adjust the problem's parameters and observe the effects—as was done in the previous example. ∎

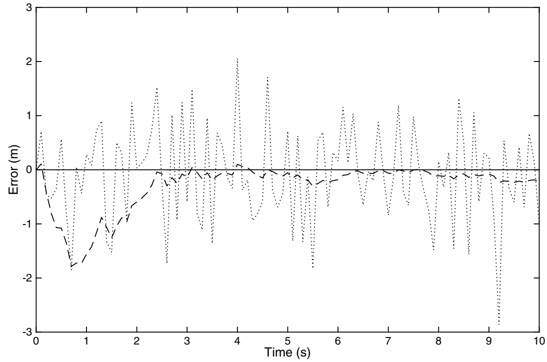


Figure 3.15. *Position estimation errors for constant-velocity dynamics. True value (solid); measurements (dotted); KF estimation (dashed).*

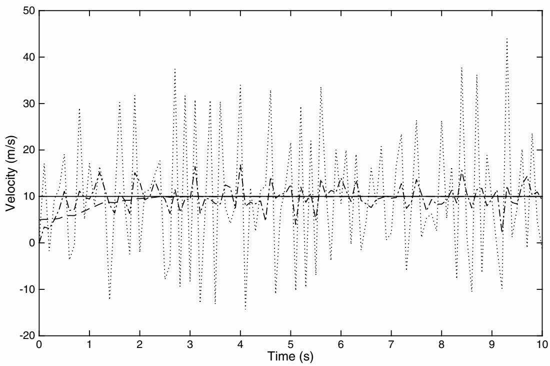


Figure 3.16. *Velocity estimation results for constant-velocity dynamics. True value (solid); from raw measurements (dotted); from running average of raw measurements (dotted-dashed); KF estimation (dash).*

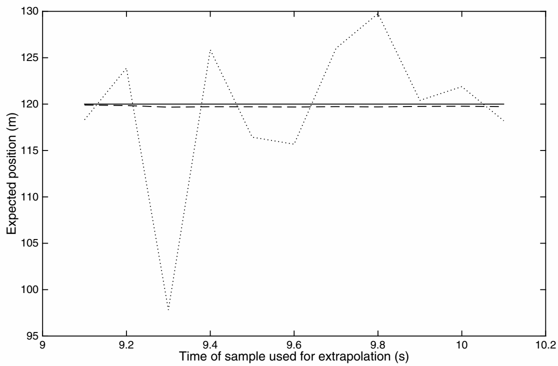


Figure 3.17. *Extrapolation of position 2 seconds ahead for constant-velocity dynamics. True value (solid); from running average of raw measurements (dotted); KF estimation (dashed).*

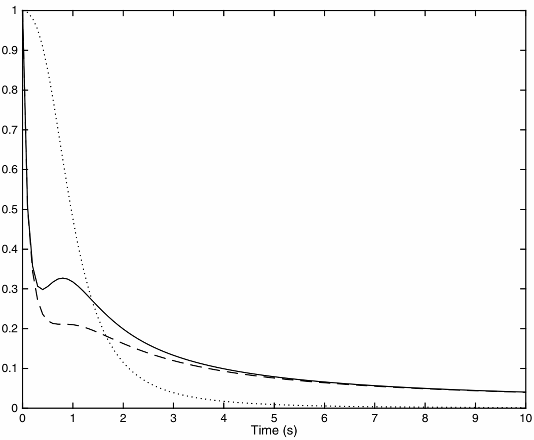


Figure 3.18. *Convergence of the KF parameters for constant-velocity dynamics. K (solid); P_{11} (dashed); P_{22} (dotted).*

Part II

Advanced methods and algorithms for data assimilation

DA has undergone many recent developments. These aim at overcoming some of the operational challenges that confront the users of these methods. In particular, they generalize the approaches to nonlinear and non-Gaussian contexts.

Chapter 4

Nudging methods

Nudging, or Newtonian relaxation, is a simple yet dynamic method that aims to dynamically adjust the model toward the observations. The idea is simply to insert a feedback term into the model equation that is proportional to the observation–model misfit and *nudges* the model state toward the observations, as shown in Figure 4.1.

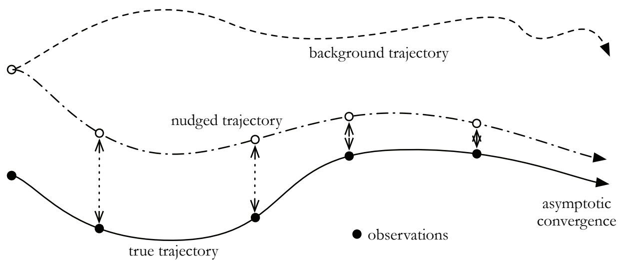


Figure 4.1. *Schematic representation of the nudging method.*

In geosciences nudging was first called the "dynamic initialization technique" and was introduced in the 1970s by Anthes [1974] for hurricane forecasting and by Hoke and Anthes [1976] for weather forecasting. Since then it has been used in a wide range of applications, as a first step toward more sophisticated DA methods. Its main advantage is its simplicity of implementation since there is no need for adjoint development, nor for a complex analysis/best linear unbiased estimator (BLUE) computation step. It is sufficient to implement an observation operator and to slightly modify the model to add a feedback term.

It is still used nowadays when more complex methods need to be avoided. However, more recent implementations are preferable, e.g., spectral nudging (see Section 4.1.5) or back and forth nudging (BFN; see Section 4.2).

4.1 ▪ Nudging

4.1.1 ▪ Principle

The nudging method can be explained with a very simple ODE. Let us assume that the model state variable $\mathbf{x}$ is a vector-valued real function of time $t \in \mathbb{R}, \mathbf{x}(t) \in \mathbb{R}^n$, and a solution of the differential model equation

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathscr{M}(\mathbf{x}(t)), \quad \mathbf{x}(t=0) = \mathbf{x}_0,$$

where $\mathscr{M} : \mathbb{R}^n \to \mathbb{R}^n$ is a given function, possibly nonlinear. We then assume that we have p observations of the system $\mathbf{y}^{\mathrm{o}}(t) \in \mathbb{R}^p$ distributed over time, and that we have an observation operator $\mathscr{H}$, possibly nonlinear, that maps, at each observation time, the state $\mathbf{x} \in \mathbb{R}^n$ into the observation space $\mathbb{R}^p$, so that $\mathscr{H}(\mathbf{x}(t)) \in \mathbb{R}^p$ and $\mathbf{y}^{\mathrm{o}}(t)$ are comparable objects. We call the vector

$$\boldsymbol{\epsilon}^{\mathrm{o}}(t) = \mathscr{H}(\mathbf{x}(t)) - \mathbf{y}^{\mathrm{o}}(t) \in \mathbb{R}^p$$

the observation error. The nudging method, also called Newtonian relaxation, consists of looking for a solution, $\mathbf{x}(t)$, of the equation

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathscr{M}(\mathbf{x}(t)) - \mathbf{K}\{\mathscr{H}(\mathbf{x}(t)) - \mathbf{y}^{\mathrm{o}}(t)\}, \quad \mathbf{x}(t=0) = \mathbf{x}_0,$$

where $\mathbf{K}$ is a matrix with n rows and p columns. The idea is simply to relax, or nudge, the model state vector $\mathbf{x}$ toward the observations. In the limit where $\mathbf{K}$ is very small, $\mathbf{x}$ only solves the model equation; if $\mathbf{K}$ is very large, then $\mathscr{H}(\mathbf{x}(t))$ is close in some sense to $\mathbf{y}^{\mathrm{o}}$. An ideal choice of $\mathbf{K}$ is then a compromise between being close to the observations and satisfying the model equation, which is a classical motto in DA. A simplified linear case will allow us to see this more clearly. So we now assume that $\mathscr{M} = \mathbf{M}$ is a matrix, $p = n$, and $\mathscr{H} = \mathbf{I}$ is the identity. In this case, the nudging equation becomes

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{M}\mathbf{x} - \mathbf{K}(\mathbf{x} - \mathbf{y}^{\mathrm{o}}), \quad \mathbf{x}(t=0) = \mathbf{x}_0,$$

and its solution is

$$\mathbf{x}(t) = e^{-(\mathbf{K}-\mathbf{M})t} \int_0^t e^{(\mathbf{K}-\mathbf{M})s} \mathbf{K}\mathbf{y}^{\mathrm{o}} \, \mathrm{d}s + e^{-(\mathbf{K}-\mathbf{M})t} \mathbf{x}_0.$$

If $\mathbf{K}$ is symmetric positive definite, then we have (see, e.g., Auroux and Blum [2005])

$$\mathbf{x}(t) \underset{\mathbf{K}\to\infty}{\longrightarrow} \mathbf{y}^{\mathrm{o}}(t) \quad \forall t > 0.$$

Let us mention that the nudging method has been well known since the 1960s in automatics and control theory, in the linear case where the model, $\mathbf{M}$, and observation operator, $\mathbf{H}$, are both matrices. In these domains it is called the Luenberger observer [Luenberger, 1966] or asymptotic observer. Indeed, under some hypotheses, $\mathbf{K}$ can be chosen so that the error between the true model state and the nudging equation solution tends to zero when t goes to infinity, thus the name *asymptotic* observer.

4.1.2 ▪ Link with the KF

In a discrete-time framework, one time step of the nudging equation can be written as

$$\mathbf{x}_{k+1} = \mathbf{M}_{k+1:k}\mathbf{x}_k - \mathbf{K}\left\{\mathbf{H}_{k+1}(\mathbf{M}_{k+1:k}(\mathbf{x}_k)) - \mathbf{y}^{\mathrm{o}}_{k+1}\right\}.$$

If we call the vector $\mathbf{M}_{k+1:k}(\mathbf{x}_k)$ the forecast, then this equation can be seen as a predictor-corrector scheme that is very similar to the Kalman filter (KF) equations in the "no model error" case,

$$\mathbf{x}^{\mathrm{a}}_{k+1} = \mathbf{x}^{\mathrm{f}}_{k+1} + \mathbf{K}\left\{\mathbf{y}^{\mathrm{o}}_{k+1} - \mathbf{H}_{k+1}(\mathbf{x}^{\mathrm{f}}_{k+1})\right\}, \qquad \mathbf{x}^{\mathrm{f}}_{k+1} = \mathbf{M}_{k+1:k}(\mathbf{x}^{\mathrm{a}}_k).$$

The similarity between these two equations is obvious; they only differ by the choice of the $\mathbf{K}$ matrix. As we know that the KF is optimal in the linear Gaussian case, then this is the same for the nudging, provided the $\mathbf{K}$ matrix is exactly the associated Kalman gain matrix. Of course, in real nudging applications, the Kalman gain matrix is out of reach, so that the nudging method may be seen as a degraded yet simple KF.

4.1.3 ▪ Advantages and drawbacks

The nudging method is very simple, which sums up both its advantages and drawbacks: no complex implementation, but suboptimal results. Compared to variational assimilation, nudging does not require any adjoint model, whose development is a time-consuming task. Compared to Kalman filtering, it does not require complex implementation or the initialization of covariance matrices. Its implementation consists of, essentially

1. coding the observation operator as in 4D-Var and Kalman filtering;
2. tuning the $\mathbf{K}$ matrix, which can be chosen diagonal or even scalar; and
3. slightly modifying the model equations to incorporate the additional term $\mathbf{K}\{\mathscr{H}(\mathbf{x}(t)) - \mathbf{y}^{\mathrm{o}}(t)\}$.

It can be a first step toward a more elaborate DA method, which will require the same observation operator anyway, and it can give some hints about the potential of the observations to correct toward the model state. In some cases, such as small observation error, decorrelated in space and constant over time, it can be sufficient to achieve the desired results.

However, the nudging method with a diagonal $\mathbf{K}$ can produce poor results if, for example, the observation errors are correlated in space, or if the errors vary much in time—see Wunsch [2006] for more details. In this kind of framework, the specification of $\mathbf{K}$ is crucial, but its tuning is not automated and relies on numerical experimentation as well as the skill of the practitioner. Of course the tuning of $\mathbf{K}$ could be automated as well as optimized—see above the relationship with the KF gain matrix and below the optimal nudging scheme. However, in this case the ease of implementation and very likely the low computational cost would be lost.

4.1.4 ▪ Optimal nudging

As we have seen before, the nudging scheme can be seen, in the linear case, as a degraded, suboptimal form of the KF. It is therefore very tempting to improve the quality of the nudging matrix $\mathbf{K}$ to be closer to the KF optimality. However, the Kalman gain matrix can be very costly to implement and compute, so it may seem preferable to find a way to improve $\mathbf{K}$ at a lower cost. As we have seen before, if the nudging coefficient is too small, then the model equation dominates and the observation has little effect on the variable. On the contrary, if it is too large, then the observation dominates, as well as the observation error if it is substantial. We want to find the optimal compromise between these two effects.

To do so, an optimal nudging procedure has been proposed by Zou et al. [1992]. They apply variational parameter estimation to the determination of the $\mathbf{K}$ matrix coefficients, in the case where $\mathbf{K}$ is chosen diagonal—but not space dependent, i.e., one coefficient is required for each physical variable. To do so, they form the cost function

$$J(\mathbf{K}) = J^{\mathrm{o}}(\mathbf{x}(\mathbf{K})) + J^{\mathrm{b}}(\mathbf{K}),$$

where

- $\mathbf{x}$ follows the nudging equation

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathscr{M}(x) - \mathbf{K}(\mathbf{x} - \mathbf{x}^{\mathrm{o}}), \quad \mathbf{x}(t=0) = \mathbf{x}_0;$$

- J^{o} measures the observation misfit, $\mathbf{x} - \mathbf{x}^{\mathrm{o}}$; and
- J^{b} is a regularization term for $\mathbf{K}$, i.e., a norm of $\mathbf{K} - \mathbf{K}^{\mathrm{b}}$, where $\mathbf{K}^{\mathrm{b}}$ is a first guess for $\mathbf{K}$.

The estimation of $\mathbf{K}$ is then performed as in regular variational assimilation (see Chapter 2): implementation of the adjoint model, computation of the gradient of the cost function with respect to $\mathbf{K}$, and iterative optimization of $\mathbf{K}$ using a gradient descent type minimization algorithm.

This method retains the advantage of simplicity, because the optimization concerns a small number of coefficients, as $\mathbf{K}$ is kept diagonal and the same nudging coefficient is used for a given variable at all grid points. Therefore, the optimization takes place in a low-dimensional space and can be very efficiently implemented at a small additional cost. The gain in using optimized coefficients is, however, great, as demonstrated by the numerical experiments of Zou et al. [1992]. A parallel study by Stauffer and Bao [1993] also highlighted the interest of this method while pointing out some difficulties, namely the sensitivity of the optimized nudging coefficients to their first guess and to the choice of the regularization term $J^{\mathrm{b}}(\mathbf{K})$, which could lead to poor results if badly specified.

A more recent study [Vidard et al., 2003] on the subject of optimal nudging proposed to use it in the framework of model error control in variational assimilation. While being similar to Zou et al. [1992], the approach is formulated slightly differently and allows us to take into account model error in some sense, which is quite a difficult problem in variational assimilation. The model equation is the same as in Zou et al. [1992],

$$\mathbf{x}_{k+1} = \mathscr{M}_{k+1:k}(\mathbf{x}_k) - \mathbf{K}_{k+1}\left\{\mathscr{H}_{k+1}(\mathscr{M}_{k+1:k}(\mathbf{x}_k)) - \mathbf{y}^{\mathrm{o}}_{k+1}\right\},$$

except that the nudging term is more general because it includes an observation operator, and also because it is seen as a model error term. Then the control vector is not only the initial condition, as in traditional 4D-Var, but augmented by the coefficients of the matrix $\mathbf{K}$. The cost function now has three terms,

$$J(\mathbf{x}_0, \mathbf{K}) = J^{\mathrm{o}}(\mathbf{x}) + J^{\mathrm{b}}(\mathbf{x}_0) + J^{\mathrm{nudg}}(\mathbf{x}, \mathbf{K}),$$

where

- $\mathbf{x}$ follows the nudging equation above;
- J^{o} measures the observation misfit $\mathscr{H}(\mathbf{x})-\mathbf{x}^{\mathrm{o}}$;
- J^{b} is the background cost, i.e., a norm of $\mathbf{x}_0-\mathbf{x}^{\mathrm{b}}$, where $\mathbf{x}^{\mathrm{b}}$ is the background;
- J^{nudg} is the nudging cost, formulated in terms of model error control (see Section 2.4.7.5 for the control of model error in a variational framework),

  $$J^{\mathrm{nudg}}=\sum_{k,\mathrm{time}}\left\{\mathbf{K}_k(\mathscr{H}_k(\mathscr{M}_k(\mathbf{x}_{k-1}))-\mathbf{y}_k^{\mathrm{o}})\right\}^{\mathrm{T}}\mathbf{Q}_k^{-1}\left\{\mathbf{K}_k(\mathscr{H}_k(\mathscr{M}_k(\mathbf{x}_{k-1}))-\mathbf{y}_k^{\mathrm{o}})\right\},$$

  with $\mathbf{Q}_k$ the model error covariance matrix; and
- $\mathbf{K}_k$ may vary over the time steps k, and $\mathbf{K}_k$ may not be diagonal.

As we can see, the nudging here has two roles: first, as expected, it nudges the model toward the observation; second, it allows some kind of model error control. If $\mathbf{K}$ is not assumed to be diagonal, then this 4D optimal nudging scheme (4D-ON) provides better results than 4D-Var with an imperfect model. The authors also showed that in some cases it is better to choose $\mathbf{K}$ diagonal, and 4D-ON still provides improved results compared to 4D-Var and allows for numerical simplifications that greatly reduce the computational cost.

4.1.5 ▪ Spectral nudging

In this subsection, the equations are presented in the *continuous* case, because it simplifies the spectral formulation, using Fourier decomposition.

Global atmosphere or ocean forecasting is often limited to coarse-scale predictions because of computing power limitations. Regional forecasting allows us to use a higher-resolution model and to predict finer-scale phenomena. However, when using a regional model, it is crucial to correctly specify the boundary conditions. Nudging was used to provide such specifications in Davies [1976]. Let us denote by x_R the regional model state and assume that the model evolution equation is

$$\frac{\mathrm{d}x_R}{\mathrm{d}t}=\mathscr{M}(x_R).$$

Then the equation is modified by nudging using the global model state x_G,

$$\frac{\mathrm{d}x_R}{\mathrm{d}t}=\mathscr{M}(x_R)-K(x_R-x_G), \tag{4.1}$$

where the nudging parameter $K\in\mathbb{R}$ is space dependent: K is positive around the boundary of the regional domain and decreases to zero in the interior of the domain, its decrease being a function of the distance to the boundary.

This method allows us to take into account the global model information in a smooth way, but only near the boundary. It was later modified to take into account the global model everywhere inside the regional model (see Waldron et al. [1996] and more recent applications in von Storch and Langenberg [2000] and Miguez-Macho [2004]). The idea is that the large-scale trend of the regional model, being a refinement of the global model, should coincide with the global one. Therefore, the spectral nudging method performs nudging of the regional model toward the global one, as in equation (4.1), but only for large scales. Thus,

$$\frac{\mathrm{d}x_R}{\mathrm{d}t} = \mathscr{M}(x_R) - \sum_{|p|<P} \sum_{|q|<Q} K_{p,q}(x_{p,q,R} - x_{p,q,G}) e^{ix\frac{2\pi p}{D_x}} e^{iy\frac{2\pi p}{D_y}},$$

where the regional domain is a rectangle of size D_x in longitude x and D_y in longitude y, and $x_{p,q,R}$ (resp. $x_{p,q,G}$) are the regional (resp. global) model Fourier coefficients. We can see in this equation that the nudging coefficient K depends on the scale, and that it only happens for large scales, in other words, for Fourier frequencies smaller than P and Q. For small scales there is no nudging and the regional model alone drives the dynamics.

4.2 ▪ Back-and-forth nudging

4.2.1 ▪ Backward nudging

We have seen that the standard nudging method, under the hypotheses of Section 4.1.1, has asymptotic properties: the model state converges toward the observations when the time tends to infinity. In DA, time windows are always finite so that the nudging is suboptimal. The best fit to the observations is generally obtained at the end of the time window, as the state is nudged over time to the observations. However, in DA one is often interested in the initial state of the system, not the final state. This remark is the starting point of the backward nudging idea [Auroux, 2003]: "reverse" the model and perform nudging backward in time. To do so we first write formally, in the ODE framework, the backward model equation with a final condition,

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathscr{M}(\mathbf{x}(t)), \quad \mathbf{x}(t=T) = \mathbf{x}_T. \tag{4.2}$$

A change of variable, $t' = T - t$, leads to

$$-\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t'} = \mathscr{M}(\mathbf{x}(t')), \quad \mathbf{x}(t'=0) = \mathbf{x}_T,$$

which is a forward equation in t' with an initial condition and can be rewritten as

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t'} = -\mathscr{M}(\mathbf{x}(t')), \quad \mathbf{x}(t'=0) = \mathbf{x}_T. \tag{4.3}$$

The backward nudging then consists of performing nudging on this last equation,

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t'} = -\mathscr{M}(\mathbf{x}(t')) - \mathbf{K}\left\{\mathscr{H}(\mathbf{x}(t')) - \mathbf{y}^{\mathrm{o}}(t')\right\}, \quad \mathbf{x}(t'=0) = \mathbf{x}_T. \tag{4.4}$$

The formal backward equation (4.2) and its forward equivalent (4.3) are generally ill-posed (e.g., when the forward model contains some kind of diffusive process) and may explode numerically. The presence of the nudging term in (4.4) then plays a double role: first it acts as a stabilizer in the numerical equation (putting the eigenvalues of the system back in the negative real plane), and second it nudges the system to the observations, providing a best fit at the initial time.

Auroux [2003] provides applications of backward nudging to the Lorenz model, as well as a barotropic quasi-geostrophic ocean model, and shows improved results compared to direct nudging, with an error one third to one fourth the size on future forecasts. Figure 4.2 presents the rms errors obtained after assimilation for this latter model, where the improvement associated with backward nudging is highlighted.

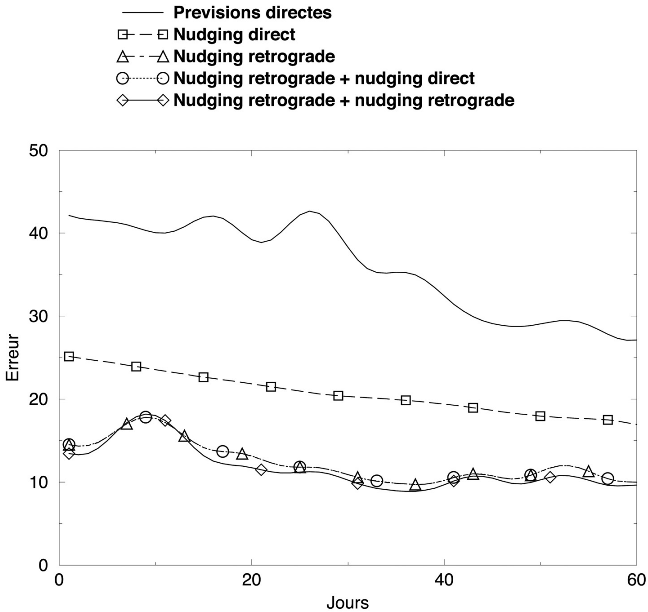


Figure 4.2. *Rms error for DA on the quasi-geostrophic ocean model for various nudging methods. Reprinted with permission from D. Auroux [2003].*

4.2.2 ▪ Back-and-forth nudging algorithm

Backward nudging improves forward nudging, but it still has the same major drawback, namely acting on a finite time window so that time stops before convergence happens. The BFN algorithm [Auroux and Blum, 2005] addresses this problem by alternating iteratively forward and backward nudging. The idea is that one iteration of forward and backward nudging over a duration T could replace forward nudging during a double duration $2T$. Of course it is not strictly equivalent, but in essence this is the basis of the BFN algorithm: by iterating back and forth, we attain an asymptotic behavior, as if time were going to infinity.

Algorithm 4.1 can be implemented as follows (see also Figure 4.3):

1. Initialization of $\mathbf{x}_0$.
2. Forward nudging starting from $\mathbf{x}_0$. At the end of the time window it has reached the state $\mathbf{x}(T)$.

Algorithm 4.1 BFN algorithm.

INITIALIZATION:

$$\tilde{\mathbf{x}}_{-1}(0) = \mathbf{x}_0$$

BFN LOOP OVER $j \geq 0$:
Forward nudging:

$$(F)\begin{cases} \dfrac{\mathrm{d}\mathbf{x}_j}{\mathrm{d}t} = \mathscr{M}(\mathbf{x}_j) - \mathbf{K}(\mathscr{H}(\mathbf{x}_j) - \mathbf{y}^{\mathrm{o}}), \\ \mathbf{x}_j(0) = \tilde{\mathbf{x}}_{j-1}(0). \end{cases}$$

Backward nudging:

$$(B)\begin{cases} \dfrac{\mathrm{d}\tilde{\mathbf{x}}_j}{\mathrm{d}t} = \mathscr{M}(\tilde{\mathbf{x}}_j) + \mathbf{K}'(\mathscr{H}(\tilde{\mathbf{x}}_j) - \mathbf{y}^{\mathrm{o}}), \\ \tilde{\mathbf{x}}_j(T) = \mathbf{x}_j(T). \end{cases}$$

3. Backward nudging starting at time T from $\mathbf{x}(T)$. At the beginning of the time window it will have reached the state $\tilde{\mathbf{x}}(0)$.

4. Back to step 2: forward nudging with $\mathbf{x}(0)$ initialized with $\tilde{\mathbf{x}}(0)$.

5. Back to step 3, and repeat backward and forward nudging integrations until convergence.

We note that the forward nudging matrix $\mathbf{K}$ may differ from the backward nudging matrix $\mathbf{K}'$.

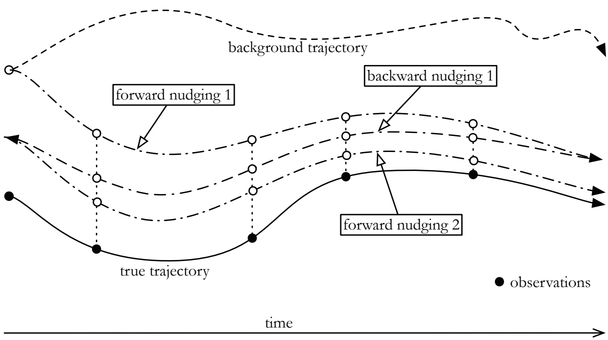


Figure 4.3. *Schematic representation of the BFN method.*

Auroux and Blum [2005] and Auroux and Blum [2008] detail some properties of the BFN algorithm, which we will recall in Section 4.2.3.

4.2.3 ▪ Algorithm properties

4.2.3.1 ▪ Convergence

Auroux and Blum [2005] prove the convergence in a very elementary case, where the model $\mathbf{M}$ is linear and $\mathbf{H}$ is the identity operator, so that the state vector is fully and perfectly observed. In this case, we can explicitly compute the state vectors $\mathbf{x}_j(t)$ and $\tilde{\mathbf{x}}_j(t)$ and prove that if $\mathbf{K}$ and $\mathbf{K}'$ are large enough, both converge as the iteration index j goes to infinity toward limit trajectories $\mathbf{x}_\infty(t)$ and $\tilde{\mathbf{x}}_\infty(t)$. We can see in this special case that the convergence can be achieved in the finite time interval $[0,T]$, contrary to forward nudging, which requires $t \to \infty$, using the BFN trick of iterating nudging steps.

Moreover, if $\mathbf{K}=\mathbf{K}'$ and matrices $\mathbf{K}$ and $\mathbf{M}$ commute, then

$$\mathbf{x}_\infty(t) \underset{\mathbf{K}\to\infty}{\longrightarrow} \mathbf{y}^{\mathrm{o}}(t), \quad \tilde{\mathbf{x}}_\infty(t) \underset{\mathbf{K}\to\infty}{\longrightarrow} \mathbf{y}^{\mathrm{o}}(t) \quad \forall t \in [0,T].$$

Still, in the linear case, if $\mathbf{K}=\mathbf{K}'$ and if the limit trajectories are equal, $\mathbf{x}_\infty(t) = \tilde{\mathbf{x}}_\infty(t)$, then the limit satisfies both of the following equations:

$$\begin{aligned} &(\mathrm{F}) \quad \frac{d\mathbf{x}_\infty}{dt} = \mathbf{M}\mathbf{x}_\infty - \mathbf{K}(\mathbf{H}\mathbf{x}_\infty - \mathbf{y}^{\mathrm{o}}),\\ &(\mathrm{B}) \quad \frac{d\mathbf{x}_\infty}{dt} = \mathbf{M}\mathbf{x}_\infty + \mathbf{K}(\mathbf{H}\mathbf{x}_\infty - \mathbf{y}^{\mathrm{o}}). \end{aligned}$$

Subtracting the equations, we obtain the following property for $\mathbf{x}_\infty$:

$$\mathbf{K}(\mathbf{H}\mathbf{x}_\infty(t) - \mathbf{y}^{\mathrm{o}}(t)) = 0.$$

4.2.3.2 ▪ Variational interpretation

In Auroux and Blum [2008], the following variational interpretation of nudging is proposed in the simplified case where

- $\mathcal{H} = \mathbf{H}$ is linear;
- $\mathcal{M} = \mathbf{M}$ is linear and symmetric; and
- the nudging matrix is $\mathbf{K} = \mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}$, where $\mathbf{R}$ is the observation error covariance matrix.

We denote by k the time-stepping index. The nudging equation is discretized in time as

$$\frac{\mathbf{x}_{k+1} - \mathbf{x}_k}{\Delta t} = \mathbf{M}\mathbf{x}_{k+1} - \mathbf{K}(\mathbf{H}\mathbf{x}_{k+1} - \mathbf{y}^{\mathrm{o}}),$$

so that $\mathbf{x}_{k+1}$ is the solution of the optimization problem

$$\min_{\mathbf{x}} \left[\frac{1}{2}(\mathbf{x}-\mathbf{x}_k)^{\mathrm{T}}(\mathbf{x}-\mathbf{x}_k) - \frac{\Delta t}{2}\mathbf{x}^{\mathrm{T}}\mathbf{M}\mathbf{x} + \frac{\Delta t}{2}(\mathbf{H}\mathbf{x}-\mathbf{y}^{\mathrm{o}})^{\mathrm{T}}\mathbf{R}^{-1}(\mathbf{H}\mathbf{x}-\mathbf{y}^{\mathrm{o}}) \right].$$

The first two terms are the energy of the discrete model and the last one is the observation misfit. As in variational assimilation, the solution is therefore at each time step a compromise between minimizing the energy of the system and being close to the observations. This interpretation also provides a nice way to specify the nudging matrix, $\mathbf{K}$, while still retaining some error covariance information.

4.2.4 ▪ Well-posedness of the BFN: Theoretical studies

Theoretical results about one-dimensional transport equations have been given by Auroux and Nodet [2012]. The well-posedness of the BFN equations and/or the convergence of the algorithm were studied for various cases. In this framework, x is the space variable, the model state is called u, and the model is continuous in space. A generic BFN step was then written as

$$\begin{array}{ll} \text{(F)} & \partial_t u = M_1(u) + M_2(u) - K(t,x)(u - u^{\mathrm{o}}), \quad u(0) = u_0, \\ \text{(B)} & \partial_t \tilde{u} = M_1(\tilde{u}) + M_2(\tilde{u}) + K'(t,x)(\tilde{u} - u^{\mathrm{o}}), \quad \tilde{u}(T) = u(T), \end{array}$$

where u^{o} are observations of the system satisfying the forward equation without nudging. Let us remark that there is no observation operator $\mathscr{H}$, but K depends on t and x, so that it contains the information about where/when the observations are available or not. More precisely, if $K = 0$ at some point in space and time, then it means there is no observation.

Four types of equations were considered over the space interval $x \in [0,1]$:

- one-dimensional linear transport: $M_1(u) = -a(x)\partial_x u$, where $a(x)$ is a smooth given function;
- Burgers' equation: $M_1(u) = -u\partial_x u$;
- with a viscous term: $M_2(u) = \nu\partial_{xx} u$, where $\nu > 0$ is a given constant;
- without viscosity: $M_2(u) = 0$.

And various cases were considered for $K(t,x)$, with $K'(t,x) = \text{constant} \times K(t,x)$:

1. $K(t,x) = K \in \mathbb{R}$, so that u is observed everywhere and for all time.
2. $K(t,x) = K1_{[t_1,t_2]}(t)$ with $K \in \mathbb{R}$ and $0 < t_1 < t_2 < T$, so that u is observed everywhere during a given time interval.
3. $K(t,x) = K(x)$ with $\text{Support}(K) \subset [a,b]$ and $0 < a < b < 1$, so that u is unobserved on an open space with nonempty interior.

The results are summed up in Table 4.1. In the realistic case where the model state is not observed everywhere and there is viscosity, we can see that the BFN equations are not well-posed, even for a linear transport equation. Intuitively this is quite natural, because the backward equation contains a viscous term with the wrong sign, which is of course ill-posed. This is confirmed numerically, as the backward nudging constant K' must be chosen very large for the backward equation to be stable.

4.2.5 ▪ BFN algorithm improvement

As has been studied by Auroux and Nodet [2012] and recalled above, the BFN algorithm is ill-posed in the presence of viscous terms and where the observations are sparse, because of the backward equation. Of course, numerically this is not a deadlock, because it is sufficient to set the backward nudging constant K' to a very large number to counteract the largest eigenvalue of the viscous term. However, this can be an issue because the backward equation may force the model state toward the observations too brutally and one may wish to perform the backward nudging more subtly.

Table 4.1. *Summary of the theoretical results about BFN equations.*

	Linear transport	Burgers'
Inviscid	• (B) and (F) equations well-posed. • BFN algorithm converges. • Rate of convergence is a function of $K(t,x)$.	• Results similar to the linear case.
Viscous	• Case (1) and (2) (observations everywhere): (B) and (F) equations well-posed; BFN converges. • Case (3) (no observation on an open subset with nonempty interior): **Ill-posed**; no solution backward.	• **Ill-posed** for every case (even (1)); no regularity of the solution.

This is why the algorithm was modified by Auroux et al. [2013] to account for viscous equations. In this framework, it is recommended to change the sign of the viscosity in the backward equation so that it becomes well-posed and the nudging can be done with a reasonably small nudging parameter. More precisely, the model is separated into two parts, $\mathcal{M}$ for the nonviscous terms and D for the viscous terms:

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathcal{M}(\mathbf{x}) + D(\mathbf{x}), \quad \mathbf{x}(t=0) = \mathbf{x}_0.$$

The diffusive-BFN algorithm can be written as

$$\begin{aligned}
&\text{(F)} \quad \frac{\mathrm{d}\mathbf{x}_j}{\mathrm{d}t} = \mathcal{M}(\mathbf{x}_j) + D(\mathbf{x}_j) - \mathbf{K}(\mathcal{H}(\mathbf{x}_j) - \mathbf{y}^{\mathrm{o}}), \quad \mathbf{x}_j(0) = \tilde{\mathbf{x}}_{j-1}(0),\\
&\text{(B)} \quad \frac{\mathrm{d}\tilde{\mathbf{x}}_j}{\mathrm{d}t} = \mathcal{M}(\tilde{\mathbf{x}}_j) - D(\tilde{\mathbf{x}}_j) + \mathbf{K}'(\mathcal{H}(\tilde{\mathbf{x}}_j) - \mathbf{y}^{\mathrm{o}}), \quad \tilde{\mathbf{x}}_j(T) = \mathbf{x}_j(T).
\end{aligned}$$

The motivation for doing so is twofold. First, it solves the problem of having to use a huge backward $\mathbf{K}'$. Second, in geoscience applications, the viscous term, D, is often an artificial parameterization of subscale phenomena and used to dissipate energy, so that it is "natural" to reverse it in the backward equation to maintain a physically realistic energy dissipation.

This improved version of the algorithm allowed the authors to obtain convergence where the classic BFN was unable to do so, with the presence of viscous terms, as well as sparse and noisy observations.

4.2.6 ▪ Why or when to choose the BFN algorithm?

The BFN algorithm is a cheap DA method, which has two major advantages. First, its implementation is extremely simple. Second, it converges fast. Therefore, it can be an ideal choice for a first dive into DA, with a minimal methodological and numerical investment.

Let us emphasize that the BFN is always preferable to the simple forward nudging method for three reasons. First, its implementation is almost equivalent and very simple either way. Second, it overcomes the finite-time window problem of forward nudging, for which only asymptotic convergence is assured. Last, it provides fast convergence in a few iterations.

However, it can be too coarse for realistic applications. Indeed, although interesting results have been obtained so far with realistic oceanic or atmospheric models (see, e.g., Boilley and Mahfouf [2012] and Ruggiero et al. [2015]), it has not been tested operationally with real data. In the latter case, it could be recommended to use a few BFN iterations as a preprocessing tool to accelerate a more classic 4D-Var or KF algorithm (see, e.g., an academic illustration in Auroux [2009]).

Chapter 5

Reduced methods

5.1 ▪ Overview of reduction methods

Reduction methods aim at reducing the dimension of the workspace and therefore the computational cost of the problem. As DA often deals with large-scale problems, the tools offered by reduction can be very useful. This chapter gives an overview of the methods, as well as their applications within the DA framework.

The two following Chapters 6 and 7 focus more on specific widely used DA methods that employ such reductions.

5.1.1 ▪ What is problem reduction?

The general aim of reduced methods, or problem reduction, or dimension reduction, is to approximate a full-scale high-dimensional problem by a new, much smaller dimensional problem. This idea is widespread in applied mathematics. For example, in image compression, high-dimensional pixel representations of images are replaced by a reduced thresholded wavelet decomposition. Also, in signal processing, truncated Fourier series can be very good approximations of a given signal. Many other examples could be given, and the final idea would be similar: reduce the dimension of the problem to increase the computing efficiency and/or to allow larger problems to be considered.

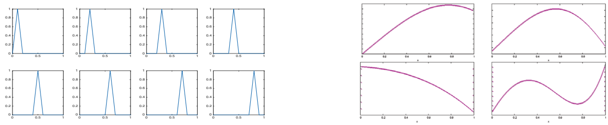
The principle of reduced methods is to change variables from the vector space X of the system state into a reduced-dimensional state $\tilde{X} \subset X$. In general, the full solution (in X) is replaced by one of its projections on the reduced space $\tilde{X}$, and the reduced algorithms provide ways of computing this projection. The change of variable is usually performed via a change of basis, from a large-dimensional generic basis to a problem-adapted smaller basis, see, e.g., illustration in Figure 5.1.

The question is then how to choose an adequate basis, adapted to the problem, so that the subspace it spans is a good approximation of the full space. Before we explain how to choose a new basis, we review the various forms of reduction in DA below.

5.1.2 ▪ Some reduced bases used in DA

Various families of vectors can be used as reduced bases in DA: polynomial families; Lyapunov-type vectors; eigen/singular vectors; Fourier/sine/cosine families;

$$\mathbf{x} = \sum_{j=1}^{n} x_j \boldsymbol{\phi}_j \in X \quad \longrightarrow \quad \mathbf{x} \simeq \tilde{\mathbf{x}} = \sum_{i=1}^{r} v_i \mathbf{e}_i \in \tilde{X} \subset X$$



$(\boldsymbol{\phi}_j)_{j=1\ldots n}$ generic basis
n large

$(\mathbf{e}_i)_{i=1\ldots r}$ adapted basis
reduced rank $r \ll n$

Figure 5.1. *Example of dimension reduction.*

and X-lets such as wavelets, curvelets, etc. We will give some details about each type of basis below.

5.1.2.1 ▪ Singular vectors and related bases

A widely used class of bases is related to eigen/singular value decompositions:

- forward/backward singular vectors,
- Lyapunov vectors,
- bred vectors,
- nonlinear singular vectors,
- principal component analysis (PCA), also called proper orthogonal decomposition (POD), empirical orthogonal functions (EOFs), etc.

Before we introduce forward/backward singular vectors, let us briefly recall the closely related linear algebra singular value decomposition (SVD). If $\mathbf{A}$ is an $n \times m$ real matrix, then it has a factorization, or SVD [Golub and van Loan, 2013], of the form

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^{\mathrm{T}},$$

where Σ is an $n \times m$ rectangular diagonal matrix with nonnegative real numbers on the diagonal, called the singular values, and $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices of sizes $n \times n$ and $m \times m$ containing the left and right singular vectors, respectively. The nonzero singular values are the square roots of the nonzero eigenvalues of $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ and $\mathbf{A}^{\mathrm{T}}\mathbf{A}$, and $\mathbf{U}$ contains the eigenvectors of $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ and $\mathbf{V}$ the eigenvectors of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$. Let us note here that both matrices $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ and $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ are symmetric positive semidefinite so that their eigenvalue decompositions exist and their eigenvalues are positive. Traditionally, the singular values are listed in decreasing order in the matrix Σ.

As we will see later, SVD provides a nice way to approximate a matrix $\mathbf{A}$ while reducing its computing storage requirements. First, choose a truncation index r so that we consider the first r singular values to be important and the others not. Then define $\mathbf{U}_r$ (respectively $\mathbf{V}_r$) as an $n \times r$ (resp. $m \times r$) matrix containing the first r left (resp. right) singular vectors, and Σ_r as an $r \times r$ diagonal matrix containing the r

largest singular values on the diagonal. Then the matrix $\mathbf{A}_r$ provides an approximation of $\mathbf{A}$:

$$\mathbf{A} \simeq \mathbf{A}_r = \mathbf{U}_r \Sigma_r \mathbf{V}_r.$$

This approximation gives the best approximation of $\mathbf{A}$ by a matrix of rank r,

$$||\mathbf{A} - \mathbf{A}_r||_{\mathrm{F}} = \min_{\mathrm{rank}(B) \leq r} ||\mathbf{A} - B||_{\mathrm{F}},$$

where $||\cdot||_{\mathrm{F}}$ is the Frobenius matrix norm. The rate of decrease of the singular values also gives information about the approximation error, $\mathbf{A} - \mathbf{A}_r$, but we will not go into details here. We refer to the reference book by Golub and van Loan [2013] for more details.

We can now define the classical bases used in DA relying on eigen/singular value decomposition. Let us denote by $\mathscr{M}(t_1, t_2)$ a nonlinear evolution model from time t_1 to t_2. If $\mathbf{x}(t_1)$ denotes the state of the system at time t_1, then this state evolves to $\mathbf{x}(t_2)$ at time t_2 according to the formula $\mathbf{x}(t_2) = \mathscr{M}(t_1, t_2)\mathbf{x}(t_1)$.

Let us now denote its tangent operator by $\mathbf{M}(t_1, t_2)$. This operator is a matrix, and if $\mathbf{z}(t_1)$ denotes a perturbation of the system at time t_1, then we have

$$\mathscr{M}(t_1, t_2)(\mathbf{x}(t_1) + \mathbf{z}(t_1)) = \mathscr{M}(t_1, t_2)(\mathbf{x}(t_1)) + \mathbf{M}(t_1, t_2)\mathbf{z}(t_1) + o(\mathbf{z}(t_1)). \tag{5.1}$$

We are interested in vectors for which the amplification of an initial perturbation is maximal. We can define the ratio

$$\rho(\mathbf{z}(t_1)) = \frac{||\mathscr{M}(t_1, t_2)(\mathbf{x}(t_1) + \mathbf{z}(t_1)) - \mathscr{M}(t_1, t_2)(\mathbf{x}(t_1))||}{||\mathbf{z}(t_1)||},$$

as well as its approximation when using the first-order Taylor formula (5.1),

$$\zeta(\mathbf{z}(t_1)) = \frac{||\mathbf{M}(t_1, t_2)\mathbf{z}(t_1)||}{||\mathbf{z}(t_1)||}.$$

Using the adjoint operator of $\mathbf{M}(t_1, t_2)$ and the scalar product $\langle \cdot, \cdot \rangle$ associated with the norm $||\cdot||$, we can rewrite ζ as

$$\zeta(\mathbf{z}(t_1)) = \frac{\sqrt{\langle \mathbf{z}(t_1), \mathbf{M}(t_1, t_2)^{\mathrm{T}} \mathbf{M}(t_1, t_2)\mathbf{z}(t_1) \rangle}}{||\mathbf{z}(t_1)||}. \tag{5.2}$$

An interesting type of vector is the kind that amplifies these ratios the most. Indeed, for numerical weather prediction (NWP), for example, a good and reliable forecast needs to be stable around the "correct" atmosphere trajectory. The most unstable/-variable vectors are the ones that need to be controlled and most precisely determined. From these ratios we obtain the following families of vectors:

Forward/backward singular vectors. We assume that t_1 and t_2 are both finite and the tangent linear hypothesis is valid.[31] We look for the maximizer of ζ, which is denoted by $\mathbf{z}_1^*(t_1)$ and called the first forward singular vector,

$$\zeta(\mathbf{z}_1^*(t_1)) = \max_{\mathbf{z}(t_1)} \zeta(\mathbf{z}(t_1)).$$

[31]This means that arbitrary perturbations to the initial state evolve approximately linearly in time.

By induction we define the following most variable vectors, called the forward singular vectors,

$$\zeta(\mathbf{z}_i^*(t_1)) = \max_{\mathbf{z}(t_1)\perp \mathrm{Span}(\mathbf{z}_1^*(t_1),\dots,\mathbf{z}_{i-1}^*(t_1))} \zeta(\mathbf{z}(t_1)), \quad i \geq 2.$$

The definition (5.2) allows us to say that the forward singular vectors are in fact the singular vectors of the matrix $\mathbf{M}(t_1, t_2)$, in other words, the eigenvectors associated with the largest eigenvalues of the positive operator $\mathbf{M}(t_1, t_2)^{\mathrm{T}}\mathbf{M}(t_1, t_2)$. Similarly, we can wonder which are the vectors at time t_2 that have been most amplified from time t_1, and we get the backward singular vectors, which are the eigenvectors associated with the largest eigenvalues of $\mathbf{M}(t_1, t_2)\mathbf{M}(t_1, t_2)^{\mathrm{T}}$.

Lyapunov vectors. The forward/backward singular vectors are associated with a finite time interval $[t_1, t_2]$. We can wonder what are the asymptotics of the eigenvectors of $\mathbf{M}^{\mathrm{T}}\mathbf{M}$ and $\mathbf{M}\mathbf{M}^{\mathrm{T}}$ when one of the bounds tends to infinity. The forward and backward Lyapunov vectors are defined as the limit of the singular vectors when $t_2 \to \infty$ (forward) or $t_1 \to -\infty$ (backward) [Legras and Vautard, 1996].

Nonlinear singular vectors. If we do not assume the tangent linear hypothesis, we can define the vectors that amplify the ratio ρ the most:

$$\rho(\mathbf{z}_1^*(t_1)) = \max_{\mathbf{z}(t_1)} \rho(\mathbf{z}(t_1)),$$

$$\rho(\mathbf{z}_i^*(t_1)) = \max_{\mathbf{z}(t_1)\perp \mathrm{Span}(\mathbf{z}_1^*(t_1),\dots,\mathbf{z}_{i-1}^*(t_1))} \rho(\mathbf{z}(t_1)), \quad i \geq 2.$$

These vectors are called the nonlinear singular vectors. Contrary to the singular vectors, we have few theoretical results about them [Mu, 2000], and they cannot be computed as eigenvectors. The forward nonlinear singular vectors can be computed fairly easily using constrained minimization algorithms, but the backward vectors are out of reach.

Bred modes, or breeding method. These vectors correspond to the backward Lyapunov vectors, but without the tangent linear hypothesis. They were introduced by Toth and Kalnay [1995] in a purely algorithmic DA framework for weather forecasting, without any mathematical theory.

Another family of vectors is related to the mathematical notion of eigenvectors/singular vectors, called principal components in the statistics community, empirical orthogonal functions (EOFs) in oceanography and DA, and proper orthogonal decomposition (POD) in automatics and engineering.

Principal components, or POD, EOF, or Karhunen–Loeve decomposition. We assume that we have a sample $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m)$ of system states, i.e., m vectors of dimension n. Principal component analysis (PCA) is a statistical method that consists of finding the largest inertial axes of the corresponding point cloud. To do so we consider the orthogonal projection of the sample $\{\mathbf{x}_i\}_{i=1,\dots,m}$ on a vector $\mathbf{u}_1$ and we look for $\mathbf{u}_1$ such that the variance of the projected sample is maximal. Then $\mathbf{u}_1$ is called the first principal component. The other principal components are found similarly and form an orthogonal family. More precisely, let us denote by $\mathbf{u}$ a unitary vector and by $\mathbf{p}_i$ the orthogonal projection of $\mathbf{x}_i$ on the line directed by $\mathbf{u}$. The inertia of the cloud with respect to $\mathbf{u}$ is given by

$$I_{\mathbf{u}}(\mathbf{x}_1,\ldots,\mathbf{x}_m)=\sum_{i=1}^{m}||\mathbf{x}_i-\mathbf{p}_i||^2.$$

This can be seen physically as the residual inertia around $\mathbf{u}$, and this quantity must therefore be minimal. Algebraic manipulations give

$$I_{\mathbf{u}}(\mathbf{x}_1,\ldots,\mathbf{x}_m)=m||\overline{\mathbf{x}}-\overline{\mathbf{p}}||^2+\sum_{i=1}^{m}||\mathbf{x}_i-\overline{\mathbf{x}}||^2-\sum_{i=1}^{m}||\mathbf{p}_i-\overline{\mathbf{p}}||^2,$$

with $\overline{\mathbf{x}}=\frac{1}{m}\sum_{i=1}^{m}\mathbf{x}_i$ and $\overline{\mathbf{p}}=\frac{1}{m}\sum_{i=1}^{m}\mathbf{p}_i$. As $\sum_{i=1}^{m}||\mathbf{x}_i-\overline{\mathbf{x}}||^2$ does not depend on $\mathbf{u}$, it is equivalent to minimizing

$$m||\overline{\mathbf{x}}-\overline{\mathbf{p}}||^2-\sum_{i=1}^{m}||\mathbf{p}_i-\overline{\mathbf{p}}||^2.$$

In other words, $\overline{\mathbf{p}}$ must be equal to $\overline{\mathbf{x}}$ and $\sum_{i=1}^{m}||\mathbf{p}_i-\overline{\mathbf{p}}||^2$ must be maximized. As $\mathbf{p}_i-\overline{\mathbf{p}}$ is the projection of $\mathbf{x}_i-\overline{\mathbf{x}}$, its norm is equal to $|\langle\mathbf{x}_i-\overline{\mathbf{x}},\mathbf{u}\rangle|$, where $\langle\cdot\rangle$ is the inner product. Let us denote by $\mathbf{X}$ the matrix whose ith line contains the vector $\mathbf{x}_i-\overline{\mathbf{x}}$. We thus have to maximize the quantity

$$||\mathbf{X}\mathbf{u}||^2=\mathbf{u}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}\mathbf{u}=\frac{\mathbf{u}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}\mathbf{u}}{\mathbf{u}^{\mathrm{T}}\mathbf{u}},$$

which is maximal when $\mathbf{u}=\mathbf{u}_1$ is the eigenvector of $\mathbf{X}^{\mathrm{T}}\mathbf{X}$ associated with the largest eigenvalue, μ_1. In other words, $\mathbf{u}_1$ is the first right singular vector of $\mathbf{X}$. Similarly, the other principal components are the eigenvectors associated with the next largest eigenvalues. The larger the eigenvalue, the larger the part of the system variability represented by the associated principal component. Also, the variability explained by the first r principal components is given by

$$\frac{\sum_{i=1}^{r}\mu_i}{\sum_{i=1}^{m}\mu_i}. \tag{5.3}$$

In geosciences these are called EOFs, and the interested reader can consult, e.g., Hannachi et al. [2007] for a extensive review about EOFs and their variants, as well as their use in atmospheric sciences.

As we will see later, these bases are useful in DA, both for variational and filtering methods.

5.1.2.2 ▪ Multiscale analysis

Another class of families comes from multiscale analysis: Fourier series, spectral decomposition, wavelets, etc. The idea is to change variables into the space of the multiscale decomposition and to truncate the decomposition to reduce the working space dimension. In other words, instead of representing a state vector in the state space, we consider its multiscale series decomposition and we assume that a few modes are enough to get an accurate approximation,

$$\mathbf{x}\in X=\sum_{i=1}^{\infty}x_i\mathbf{e}_i\simeq\sum_{i=1}^{r}x_i\mathbf{e}_i\in\tilde{X},$$

where $\mathbf{e}_i$ are the multiscale basis vectors, such as Fourier modes, spectral modes, wavelets, etc.; x_i are the coefficients of $\mathbf{x}$ in this basis; and r is the reduced dimension. Fast algorithms, e.g., fast Fourier transform (FFT) or, similarly fast wavelet transform, are available to compute the coefficients x_i and to recompose the signal from the coefficient sequence.

Once again, these families are useful in DA, both in filtering and variational. As more details are not required for the remainder of this chapter, we simply refer the reader to classical works on this subject, e.g., by Mallat [1989, 2008], and to the vast online tutorial material at, e.g., `http://www.wavelet.org`.

5.1.2.3 ▪ Polynomials

Polynomial families, such as Taylor, Lagrange, and Hermite, have not been used in DA yet, but were used for a similar problem: the optimal control of PDEs, namely the Navier–Stokes equation, by Ito and Ravindran [1998]. In this paper the authors project the solution into a subspace spanned by various polynomial families and perform the optimization in the reduced space, thus gaining a lot of computing time without losing too much in terms of accuracy. This idea is similar to the reduced 4D-Var approach using singular vectors, EOFs, or bred modes, which will be detailed later on.

5.1.3 ▪ Reduced methods in DA

Different approaches to using reduction methods in DA can be mentioned: the first class of approaches reduces the model or the covariance matrices, but not the DA scheme per se, while the second class of approaches reduces the DA method, i.e., reduces the working dimension of the algorithm cores.

Reduction of the model or the covariance matrices. We can mention two approaches in this class. In the first approach we reduce the model, not the DA method. In this case the DA method remains identical, but the model computing time is much smaller, thus allowing time saving. However, the reduced dimension of the model might make it cruder and more prone to biases or errors.

In the second approach we use matrix rank reduction, for error covariance matrix modeling in both variational and filtering methods. In this case the full model is retained and the algorithm does not change dimension. In other words, if n is the dimension of the state vector, the minimization (for 4D-Var) and the inversion required in the Kalman gain matrix computation (for Kalman filter [KF]) are both performed in dimension n. What changes is essentially the rank of the covariance matrix ($\mathbf{B}$ or $\mathbf{P}^{\mathrm{f}}$), which is smaller.

Reduction of the DA method. In the filtering framework, reduction means covariance matrix rank reduction for the KF and rewriting of the algorithm so that the matrix inversion required to compute the gain, $\mathbf{K}$, is done in a reduced dimension.

In the variational framework, we reduce the control space/minimization space dimension to speed up the 4D-Var DA algorithm. In classical 4D-Var, the minimization space is equal to the parameter space, usually the model state space for the initial condition plus additional dimensions for the other poorly known parameters—boundary conditions or physical/numerical parameters. With reduction, the minimization space is a subspace of the full parameter space: the optimization problem is projected into a reduced subspace.

All of these approaches will be detailed in the following sections of this chapter. They will also, to a large extent, be the focus of the subsequent chapters.

5.2 ▪ Model reduction

The very first approach using model reduction techniques for DA consists of simply reducing the model. In other words, all the machinery of model reduction methods, algorithms, and libraries can be used as is. From the DA method point of view, nothing changes: regular algorithms are used with a different (reduced) model. Therefore, we will not go into too many details here, as model reduction per se goes well beyond the scope of this book, and we will briefly present examples of the various ways to use reduced models within DA schemes.

5.2.1 ▪ Reduction of the full model

5.2.1.1 ▪ Simple models

Model simplification through reduction has been used widely in the past, when DA methods were an emerging hot topic, and while computing power was not large. For example, Miller and Cane [1989] use a simple model of sea-level anomalies in the Pacific tropical ocean, essentially a superposition of linear wave equations, with a reduced number of waves, to demonstrate the use of a KF with a parametric modeling of the model error. At that time, the aim was to demonstrate the capability of the assimilation method to adequately make use of the observations, so that linear assumptions, allowing the use of the superposition principle and simplified models, sometimes oversimplified, were commonplace.

5.2.1.2 ▪ Proper orthogonal decomposition

Later on, the computing power increased as well as the complexity of the state-of-the-art models, and the efficiency of DA methods was well established. The focus was then on reducing the cost of the model to be able to handle either large-dimensional problems, where the size of the control vector is 10^7–10^9, or real-time DA applications such as, e.g., flow control applications. For example, in Cao et al. [2006b] POD was developed to build a reduced model of the Pacific tropical ocean, and then 4D-Var assimilation was performed with this reduced model and its adjoint. On the Kalman side, Farrell and Ioannou [2001] built a reduced model with balanced truncation (variant of the POD method using POD modes and a second kind of optimal vectors) and used it within a KF algorithm. On the real-time side, we can cite Artana et al. [2012], in which the authors combined POD and 4D-Var assimilation for fluid flow control applications. As in Cao et al. [2006a], they designed reduced direct and adjoint models to be used within the 4D-Var minimization algorithm.

5.2.1.3 ▪ Wavelets

Wavelets also provide a convenient way to reduce the dimension of the model. The idea is to retain only a small fraction of coefficients in the wavelet decomposition through a thresholding procedure that retains only the largest ones. One of the advantages of wavelets over the other reduction techniques is that they are local; in other words, the support of each individual wavelet is local in space. This is not the case for SVD, POD/EOF, Fourier, or polynomial bases, for which every basis member has a global support. Wavelets can therefore be very good for local refinements or capturing fine local scales, while being cheap and sparse. This is why Tangborn and Zhang [2000] chose to build a reduced model using wavelets. They then proceeded to use the KF with the reduced model, similarly to what has been done for POD.

5.2.2 ▪ Reduction focused on covariance matrices

One of the major drawbacks of the DA algorithms in large dimensions is the prohibitive cost of the covariance matrices: they are too large to be fully estimated or even stored in memory, and too large to be used directly for numerical computations. Most of the solutions to the curse of dimensionality for these matrices rely on this simple principle: find a reduced and thus cheaper way to compute and represent or evolve the covariance matrices. In this subsection we will focus on methods dealing with covariance matrices only, which do not change the core dimension of the DA algorithm, i.e., the dimension of the space in which the inversion (KF)/optimization (Var) takes place.

5.2.2.1 ▪ B matrix modeling by factorization and spectral methods

The question of the initialization of the error covariance matrix ($\mathbf{B}$ or $\mathbf{P}_0^{\mathrm{f}}$) is crucial for every DA method, sequential as well as variational. As the $\mathbf{B}$ matrix is generally huge in geosciences applications (up to $10^9 \times 10^9$), it is impossible to store it in memory as is. In other words, every modeling attempt aims at reducing its dimension. As the square root of the $\mathbf{B}$ matrix is often required in variational assimilation, factorizations of $\mathbf{B}^{1/2}$ using sparse matrices have been proposed, relying also on the symmetry property,

$$\mathbf{B} = \mathbf{B}^{1/2}\mathbf{B}^{\mathrm{T}/2}.$$

Because every operational center for geosciences forecasting has its own factorization, the literature on this subject is extremely abundant, but we can cite Bannister [2008] for a review. Such factorizations include the spectral methods, which aim at reducing the covariances using independent spectral modes, e.g., Fourier on the sphere for atmospheric sciences. The idea is to decompose $\mathbf{B}^{1/2}$ into a product of sparse operators, e.g.,

$$\mathbf{B}^{1/2} = \mathbf{L}\boldsymbol{\Sigma}\mathbf{C}^{1/2},$$

where $\mathbf{L}$ is a physical balance operator; $\boldsymbol{\Sigma}$ a spectral transform, e.g., Fourier series, spherical harmonics, etc.; and $\mathbf{C}$ a diagonal or block diagonal matrix containing the correlations in spectral space. As the spectral modes are global, there are also variants of this method using wavelets instead of Fourier modes to provide local inhomogeneity, as presented, e.g., in the review by Fisher [2003]. Let us also mention the diffusion equation method by Weaver and Courtier [2001], which, while not being a reduced method per se, models the $\mathbf{B}$ matrix thanks to a simple equation resolution.

As every operational center uses its own particular modeling for the $\mathbf{B}$ matrix, more details will not be given here, but the interested reader can refer to Fisher [2003], Fischer et al. [2005], Brousseau et al. [2011], and Rawlins et al. [2007], and references therein.

5.2.2.2 ▪ Covariance propagation using SVD

One of the most expensive steps of the (extended) KF is the error covariance matrix propagation (3.19),

$$\mathbf{P}^{\mathrm{f}}_{k+1} = \mathbf{M}_{k+1}\mathbf{P}^{\mathrm{a}}_{k}\mathbf{M}^{\mathrm{T}}_{k+1} + \mathbf{Q}_k, \tag{5.4}$$

where $\mathbf{M}_{k+1}$ is the linear model step from time k to time $k+1$, or its linearized version for the extended Kalman filter (EKF). Model reduction has been applied by Cohn and Todling [1996] to tackle this issue and render it computationally affordable. The idea was to perform an SVD of the matrix $\mathbf{M}_{k+1}$, to truncate the SVD to keep a small number, r, of significant singular values, while retaining most of the model dynamics. This allows us to simplify the matrix products of equation (5.4) and considerably reduce the operations count, and therefore the computational cost of the algorithm. However, this has to be done at every step of the model, thus adding a nonnegligible burden to the algorithm cost. This is why this idea is not sufficient in itself and has to be extended to the reduction of the full filter algorithm. We will see, e.g., the SEEK filter in Section 5.3.1, which provides an efficient way of doing so.

5.2.2.3 ▪ Ensemble/EOF estimations of the B matrix for variational assimilation

As we have said before, the question of the initialization of the error covariance matrix is crucial for every DA method. This is especially true for three-dimensional variational assimilation, where the background error covariance matrix **B** is not propagated in time and must therefore be correctly specified from the start. Indeed, the **B** matrix has multiple roles in the 3D-Var framework:

- information spreading: the **B** matrix spreads information from an observed geographical point to an unobserved one;
- information smoothing in space;
- balance properties: the **B** matrix spreads information from an observed variable to an unobserved one and helps preserve physical properties (if correctly specified, e.g., in the ocean geostrophy or temperature/salinity equilibrium);
- preconditioning of the assimilation: as the cost function is generally ill conditioned, a change of variable involving $\mathbf{B}^{1/2}$ usually helps reduce the condition number of the problem.

This paragraph deals with estimations of the **B** matrix using only a small number of well-chosen vectors, EOFs, or ensemble members.

EOFs have been used to reduce the rank of the **B** matrix and provide a low-cost approximation in a variational context [Robert et al., 2005]. Using the notation of Section 5.1.2.1, the authors computed the EOFs of their ocean model using a sequence $(\mathbf{x}_i)_i$ of model outputs computed with various perturbations of the input parameters. They assume that the covariance matrix of the centered samples is a good approximation of **B**:

$$\mathbf{B} \simeq \frac{1}{m}\mathbf{X}\mathbf{X}^{\mathrm{T}}.$$

Then, keeping only the r leading EOFs, they provide a cheap way to approximate $\frac{1}{m}\mathbf{X}^{\mathrm{T}}\mathbf{X}$ and **B** using the truncated SVD of $\frac{1}{\sqrt{m}}\mathbf{X}$,

$$\mathbf{B} \simeq \frac{1}{m}\mathbf{X}\mathbf{X}^{\mathrm{T}} \simeq \mathbf{U}_r\Sigma_r^2\mathbf{U}^{\mathrm{T}}.$$

Recently, an ensemble method has been applied in oceanography to estimate the **B** matrix in a variational framework. Daget et al. [2009] create an ensemble by applying perturbations to the forcing parameters, here surface fluxes such as wind stress, and use this ensemble to compute, using a Monte Carlo estimator, an estimation of the background error covariance matrix **B**. With a very small number of members, they still get encouraging results.

This method is also used at the European Center for Medium-range Weather Forecasts (ECMWF) to diagnose the background error covariances (see Fisher [2003] for a review). Contrary to the previous study, they do not produce a **B** matrix to be used in the DA algorithm, but they do provide a diagnosis. They also build an ensemble to produce an estimation of the **B** matrix, using forecasts produced by older and perturbed analyses.

5.3 ▪ Filtering algorithm reduction

This section deals with methods aiming to reduce the core dimension of the KF, without necessarily reducing the whole direct model. The dimensions involved are:

- n, the dimension of the system space (size of the model state vector);
- p, the number of observations (size of the observation vector); and
- r, a reduced dimension, usually such that $r \ll n$ and $r \ll p$.

The core dimension of the KF can be found in the Kalman gain matrix equation (3.20),

$$\mathbf{K} = \mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}} + \mathbf{R})^{-1}.$$

First attempts at reducing this equation were done by Cohn and Todling [1996]. In the framework of square root filters (see below and Chapter 6), they proposed to use a truncated eigenvalue decomposition of $\mathbf{M}\mathbf{P}^{\mathrm{a}}\mathbf{M}^{\mathrm{T}}$ to speed up the computation of **K**. However, there was no propagation of the eigenvectors in their filter, so the eigenvalue decomposition had to be performed at each time step, implying a noticeable computational burden.

A similar idea was proposed later on by Verlaan and Heemink [1997]. The forecast error covariance matrix $\mathbf{P}^{\mathrm{f}}$ was replaced by its truncated SVD. As the SVD was a costly step of the algorithm, it was performed only once, in the framework of a steady-state filter.

Either way, these approaches were a first step in the direction of filter reduction but not completely satisfactory because of the computational cost of the added SVD/EVD. The SEEK filter, in the next subsection, overcomes this difficulty by avoiding these computations and always providing a low-rank approximation of the covariance matrices.

5.3.1 ▪ The Singular Evolutive Extended Kalman Filter

The singular evolutive extended Kalman (SEEK) filter algorithm, which was first introduced by Pham et al. [1998], is a square-root type KF (see Chapter 6 for more details about these filters). Recent reviews about the SEEK filter can also be found in Brasseur and Verron [2006] and Rozier et al. [2007]. The idea is to replace the analysis error covariance matrix $\mathbf{P}^{\mathrm{a}}$ by a lower-rank SVD-induced approximation. The propagation of the analysis covariance error of the KF is governed by (3.19)

$$\mathbf{P}^{\mathrm{f}}_{k+1} = \mathbf{M}_{k+1}\mathbf{P}^{\mathrm{a}}_{k}\mathbf{M}^{\mathrm{T}}_{k+1} + \mathbf{Q}_k, \tag{5.5}$$

but, as said before, the size of the involved matrices is prohibitive. The first idea of the SEEK filter is then to replace this costly step by a low-cost one.

The initialization of the filter is given by

$$\mathbf{x}^{\mathrm{f}}_0 = \mathbf{x}^{\mathrm{b}}, \quad \mathbf{P}^{\mathrm{f}}_0 = \mathbf{B} = \mathbf{S}^{\mathrm{f}}_0(\mathbf{S}^{\mathrm{f}}_0)^{\mathrm{T}},$$

where the matrix $\mathbf{S}^{\mathrm{f}}_0$ (size $n \times r$) contains the first r EOFs of the system. We also refer to page 144 for more details about the SEEK filter initialization.

We will now assume by induction that the matrix $\mathbf{P}^{\mathrm{a}}_{k+1}$ can be approximated by a low-rank square root factorization,

$$\mathbf{P}^{\mathrm{a}}_{k+1} \simeq \mathbf{S}^{\mathrm{a}}_{k+1}(\mathbf{S}^{\mathrm{a}}_{k+1})^{\mathrm{T}},$$

where $\mathbf{S}^{\mathrm{a}}_{k+1}$ (size $n \times r$) contains the r analyzed EOFs at time i. The propagation of the analysis of covariance step of the KF (5.5) is replaced by

$$\mathbf{P}^{\mathrm{f}}_{k+1} = \mathbf{M}_{k+1}\mathbf{S}^{\mathrm{a}}_{k}(\mathbf{S}^{\mathrm{a}}_{k})^{\mathrm{T}}\mathbf{M}^{\mathrm{T}}_{k+1} + \mathbf{Q}_k$$

or, in other words,

$$\mathbf{P}^{\mathrm{f}}_{k+1} = \tilde{\mathbf{S}}^{\mathrm{f}}_{k+1}(\tilde{\mathbf{S}}^{\mathrm{f}}_{k+1})^{\mathrm{T}} + \mathbf{Q}_k, \text{ with } \tilde{\mathbf{S}}^{\mathrm{f}}_{k+1} = \mathbf{M}_{k+1}\mathbf{S}^{\mathrm{a}}_{k}. \tag{5.6}$$

Let us remark here that the previous equations assumed the model $\mathbf{M}$ was linear, but the filter can also be written when it is not linear; see Verron et al. [1999]. In this case, $\tilde{\mathbf{S}}^{\mathrm{f}}_{k+1}$ is computed column by column, as follows. First, we propagate the analysis vector to get the forecast

$$\mathbf{x}^{\mathrm{f}}_{k+1} = \mathscr{M}_{k+1:k}(\mathbf{x}^{\mathrm{a}}_{k}).$$

Then we propagate the r columns of $\tilde{\mathbf{S}}^{\mathrm{f}}_{k+1}$ using the nonlinear model $\mathscr{M}_{k+1:k}$,

$$(\tilde{\mathbf{S}}^{\mathrm{f}}_{k+1})_j = \mathscr{M}_{k+1:k}(\mathbf{x}^{\mathrm{a}}_{k} + (\mathbf{S}^{\mathrm{a}}_{k})_j) - \mathscr{M}_{k+1:k}(\mathbf{x}^{\mathrm{a}}_{k}), \quad j = 1, \ldots, r,$$

where $(\tilde{\mathbf{S}}^{\mathrm{f}}_{k+1})_j$ is the jth column vector of the reduced matrix $\tilde{\mathbf{S}}^{\mathrm{f}}_{k+1}$. Of course, if the model is linear or if the tangent linear hypothesis is valid, then this equation is equivalent to (5.6). This can be seen as an evolution equation, namely the forecast step, for the jth EOF. In this respect, the SEEK filter is very similar to the ensemble Kalman filter (EnKF; see Chapter 6) in the sense that the computation of the forecast error covariance matrix is replaced by the propagation of a reduced number of state vectors: EOFs here, ensemble members for the EnKF.

To avoid multiple SVD computations, because they take place at each filter step, the SEEK filter requires $\mathbf{P}^{\mathrm{f}}$ and $\mathbf{P}^{\mathrm{a}}$ to keep the same rank, r, so this calls for some adjustment, essentially hypotheses about the $\mathbf{Q}$ matrix, so that $\mathbf{P}^{\mathrm{f}}$ and $\tilde{\mathbf{S}}^{\mathrm{f}}$ still have the same rank. More details about this point can be found in Brasseur and Verron [2006]. We assume that $\mathbf{P}^{\mathrm{f}}$ can therefore be written as

$$\mathbf{P}^{\mathrm{f}}_{k+1} = \mathbf{S}^{\mathrm{f}}_{k+1}(\mathbf{S}^{\mathrm{f}}_{k+1})^{\mathrm{T}},$$

where $\mathbf{S}^{\mathrm{f}}$ and $\tilde{\mathbf{S}}^{\mathrm{f}}$ have the same reduced rank, r.

Algorithm 5.1 SEEK filter equations.

Analysis step:

1. $\mathbf{K}_k = \mathbf{S}_k^{\mathrm{f}} \left[\mathbf{I}_r + (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})\right]^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1}$,
2. $\mathbf{x}_k^{\mathrm{a}} = \mathbf{x}_k^{\mathrm{f}} + \mathbf{S}_k^{\mathrm{f}} \left[\mathbf{I}_r + (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})\right]^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{y}_k^{\mathrm{o}} - \mathbf{H}_k \mathbf{x}_k^{\mathrm{f}})$,
3. $\mathbf{P}^{\mathrm{a}} = \mathbf{S}_k^{\mathrm{a}} (\mathbf{S}_k^{\mathrm{a}})^{\mathrm{T}}$, with $\mathbf{S}_k^{\mathrm{a}} = \mathbf{S}_k^{\mathrm{f}} \left[\mathbf{I}_r + (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})\right]^{-1/2}$.

Forecast step:

4. $\mathbf{x}_{k+1}^{\mathrm{f}} = \mathscr{M}_{k+1:k} \mathbf{x}_k^{\mathrm{a}}$,
5. $(\tilde{\mathbf{S}}_{k+1}^{\mathrm{f}})_j = \mathscr{M}_{k+1:k}(\mathbf{x}_k^{\mathrm{a}} + (\mathbf{S}_k^{\mathrm{a}})_j) - \mathscr{M}_{k+1:k}(\mathbf{x}_k^{\mathrm{a}}), \quad j = 1, \dots, r$,
6. $\mathbf{P}_{k+1}^{\mathrm{f}} = \tilde{\mathbf{S}}_{k+1}^{\mathrm{f}} (\tilde{\mathbf{S}}_{k+1}^{\mathrm{f}})^{\mathrm{T}} + \mathbf{Q}_k$.

Similarly, the analysis step of the KF can be then replaced by a low-cost one, using reduced-rank matrices

$$\mathbf{K}_k = \mathbf{S}_k^{\mathrm{f}} \left[\mathbf{I}_r + (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})\right]^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1}.$$

Let us remark here that, as $\mathbf{H}$ has p rows and n columns, $\mathbf{S}^{\mathrm{f}}$ has n rows and r columns, and $\mathbf{R}$ has p rows and p columns, the matrix to be inverted has r rows and r columns. This is to be compared to the full Kalman gain formula, where the inversion takes place in the observation space (size $p \times p$). The update of the analysis vector is then

$$\mathbf{x}_k^{\mathrm{a}} = \mathbf{x}_k^{\mathrm{f}} + \mathbf{S}_k^{\mathrm{f}} \left[\mathbf{I}_r + (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})\right]^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{y}_k^{\mathrm{o}} - \mathbf{H}_k \mathbf{x}_k^{\mathrm{f}}).$$

We can see that the analysis increment, $\mathbf{x}_k^{\mathrm{a}} - \mathbf{x}_k^{\mathrm{f}}$, is a linear combination of the columns of $\mathbf{S}_k^{\mathrm{f}}$. In other words, it is a linear combination of the EOFs at time k. Finally, the updated analysis covariance matrix becomes

$$\mathbf{P}^{\mathrm{a}} = \mathbf{S}_k^{\mathrm{a}} (\mathbf{S}_k^{\mathrm{a}})^{\mathrm{T}}, \text{ with } \mathbf{S}_k^{\mathrm{a}} = \mathbf{S}_k^{\mathrm{f}} \left[\mathbf{I}_r + (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})^{\mathrm{T}} \mathbf{R}_k^{-1} (\mathbf{H}_k \mathbf{S}_k^{\mathrm{f}})\right]^{-1/2},$$

where the inverse of the square root also takes place in dimension r. This equation can be seen as the analysis step for the EOFs' update.

To recapitulate, Algorithm 5.1 summarizes the SEEK filter steps.

SEEK filter initialization. A few words should be said here about the initialization of the filter, which requires a background and a background error covariance matrix,

$$\mathbf{x}_0^{\mathrm{f}} = \mathbf{x}^{\mathrm{b}}, \quad \mathbf{P}_0^{\mathrm{f}} = \mathbf{B}.$$

The choice of $\mathbf{x}^{\mathrm{b}}$ is a classical problem, not specific to the SEEK filter. The $\mathbf{B}$ matrix, however, can also be chosen through reduced-order modeling. Indeed, as we have seen before, the first r EOFs span a subspace for which the projected variance of the sample is maximal, that is to say the error is minimal. Therefore, the first r EOFs provide an optimal low-rank approximation of the covariance matrix of a given sample of system state vectors. The initialization process of the SEEK filter is implemented as follows:

1. Design an experiment to compute a sample of N system states, using the direct model $\mathscr{M}$, $(\mathbf{x}_j)_{j=1..N}$. Form the matrix $\mathbf{X}$ that has $\mathbf{x}_j - \frac{1}{N}\sum_{l=1}^{N}\mathbf{x}_l$ as columns. This step should be done carefully, as the variability contained in the $\mathbf{x}_j$'s determines the variability of the EOFs and the $\mathbf{B}$ matrix. The sensitive parameters of the model, as well as a "correct" number of model runs, N, should be sampled and tuned carefully.

2. Perform the SVD of matrix $\mathbf{X}$.

3. Choose an adequate number, r, for the reduced rank, i.e., the total number of EOFs. This step is both crucial and complex, as r will be a compromise between the final accuracy of the filter and its computing requirements. The original work by Pham et al. [1998] states that the singular value associated with the $(r+1)$th EOF should be smaller than one, as well as those of all the EOFs not chosen, to assure the filter's stability. However, this criterion is difficult to measure in practice. Instead, the behavior of the ratio (5.3) giving the percentage of variability explained by the first r EOFs, as a function of r, can be a good indicator to find a suitable r. Usually, in large-scale oceanography, satisfactory results are obtained with r equal to a few dozens to a few hundreds.

4. Form the matrix $\mathbf{S}_0^{\mathrm{f}}$ (size $n \times r$) containing the first r EOFs in columns.

5. Finally, set
$$\mathbf{x}_0^{\mathrm{f}} = \mathbf{x}^{\mathrm{b}}, \quad \mathbf{P}_0^{\mathrm{f}} = \mathbf{B} = \mathbf{S}_0^{\mathrm{f}}(\mathbf{S}_0^{\mathrm{f}})^{\mathrm{T}}.$$

This filter has known some evolutions and generalizations in the past, e.g., the interpolated version [Pham et al., 1998] to replace linearization by interpolation and the associated smoother [Cosme et al., 2010]. We refer to the chapter about ensemble filters, Chapter 6, for more details about reduced ensemble smoothers.

The SEEK filter is an efficient example of reduced-rank square root filters (RRSQRT) [Verlaan and Heemink, 1997], which make the most of the square root factorization of the covariance matrices and reformulate the KF steps using mostly the square roots instead of the full matrices. The Singular Evolutive Interpolated Kalman (SEIK) filter [Pham, 2001] is an interpolated variant of the SEEK filter that can also be interpreted as an ensemble filter with a reduced ensemble. This general class of filters will be detailed in Chapter 6.

5.3.2 ▪ Wavelet rank reduction for Kalman filtering

As we have seen in Section 5.2.1.3, wavelets have been used to produce reduced models. Tangborn and Zhang [2000] performed wavelet model reduction for the forecast propagation but changed variables back into physical space for the analysis, so that the Kalman gain was still very costly to compute. Tangborn [2004] and Auger and Tangborn [2004] upgraded the algorithm to do all computations, analysis included, in the possibly truncated wavelet space, thus reducing the computational cost of the KF.

We can also cite Chin and Mariano [1999] and Chui and Chen [1999], who propose a reduction of the error space by projection onto wavelet space. This is quite similar to the SEEK, which projects onto a space spanned by EOFs instead. As before, the authors argue that using wavelets allows local inhomogeneity, contrary to global functions such as EOFs or Fourier modes.

5.3.3 ▪ The EnKF

The ensemble Kalman filter (EnKF), which will be described in more detail in Chapter 6, is also a reduced method in the sense that it requires the propagation and analysis of a small number of ensemble members, instead of full-rank covariance matrices. The idea behind the EnKF is simply a Monte Carlo estimator of the covariance matrix, which we recall here briefly.

Given a sample of independent observations, $\epsilon_1, \ldots, \epsilon_p$, of the n-dimensional vector ε corresponding, e.g., to the background error, then we can compute with $\epsilon_1, \ldots, \epsilon_p$ an unbiased estimation of the covariance matrix of ε,

$$\mathrm{Cov}(\varepsilon) = \mathrm{E}\Big[(\varepsilon - \mathrm{E}(\varepsilon))(\varepsilon - \mathrm{E}(\varepsilon))^{\mathrm{T}}\Big] \simeq \frac{1}{p-1}\sum_{k=1}^{p}(\epsilon_k - \overline{\epsilon})(\epsilon_k - \overline{\epsilon})^{\mathrm{T}},$$

where $\overline{\epsilon}$ is the mean of the sample $\{\epsilon_k\}_{k=1,\ldots,p}$, defined by

$$\overline{\epsilon} = \frac{1}{p}\sum_{k=1}^{p}\epsilon_k.$$

As for the SEEK filter, but with ensemble members instead of EOFs, the idea is then to change the filter equations so that the covariance matrices do not have to be explicitly computed, but so that the forecast and analysis can be done solely on the sample.

5.4 ▪ Reduced methods for variational assimilation

This section deals with reduced methods within variational assimilation, mostly four-dimensional. Contrary to the simple use of reduced models, in this section we will see methods for which at least two direct models coexist: the "full" model and a reduced version with which the optimization is performed.

5.4.1 ▪ Multigrid and (multi)incremental approaches

5.4.1.1 ▪ Incremental 4D-Var

Incremental 4D-Var is a variation of 4D-Var that takes into account small nonlinearities while still using quadratic optimization. The hypothesis "small nonlinearities" can be stated as

$$\mathscr{M}_{k+1:k}\ldots\mathscr{M}_{1:0}(\mathbf{x}) - \mathscr{M}_{k+1:k}\ldots\mathscr{M}_{1:0}(\mathbf{x}^{\mathrm{b}}) \simeq \mathbf{M}_{k+1}\ldots\mathbf{M}_1(\mathbf{x}-\mathbf{x}^{\mathrm{b}}),$$

where $\mathscr{M}_{k+1:k}$ represents the full nonlinear model time step and $\mathbf{M}_{k+1}$ is a linear approximation of $\mathscr{M}_{k+1:k}$. Similarly, one asks for

$$\begin{aligned}&\mathscr{H}_{k+1}\mathscr{M}_{k+1:k}\ldots\mathscr{M}_{1:0}(\mathbf{x}) - \mathscr{H}_{k+1}\mathscr{M}_{k+1:k}\ldots\mathscr{M}_{1:0}(\mathbf{x}^{\mathrm{b}})\\ \simeq\ &\mathbf{H}_{k+1}\big(\mathscr{M}_{k+1:k}\ldots\mathscr{M}_{1:0}(\mathbf{x}) - \mathscr{M}_{k+1:k}\ldots\mathscr{M}_{1:0}(\mathbf{x}^{\mathrm{b}})\big),\end{aligned}$$

where $\mathbf{H}_{k+1}$ is required to be a good linear approximation of the possibly nonlinear observation operator $\mathscr{H}_{k+1}$. Then we define the *increment* $\boldsymbol{\delta}\mathbf{x}$ as

$$\boldsymbol{\delta}\mathbf{x} = \mathbf{x} - \mathbf{x}^{\mathrm{b}},$$

and the incremental 4D-Var cost function as a function of $\boldsymbol{\delta}\mathbf{x}$ is

$$\tilde{J}(\boldsymbol{\delta}\mathbf{x}) = \tilde{J}^{\mathrm{b}}(\boldsymbol{\delta}\mathbf{x}) + \tilde{J}^{\mathrm{o}}(\boldsymbol{\delta}\mathbf{x}),$$

$$\tilde{J}^{\mathrm{b}}(\boldsymbol{\delta}\mathbf{x}) = \boldsymbol{\delta}\mathbf{x}^{\mathrm{T}}\mathbf{B}^{-1}\boldsymbol{\delta}\mathbf{x},$$

$$\tilde{J}^{\mathrm{o}}(\boldsymbol{\delta}\mathbf{x}) = \sum_{k=1}^{K}(\mathbf{d}_k - \mathbf{H}_k\mathbf{M}_k \ldots \mathbf{M}_1\boldsymbol{\delta}\mathbf{x})^{\mathrm{T}}\mathbf{R}_k^{-1}(\mathbf{d}_k - \mathbf{H}_k\mathbf{M}_k \ldots \mathbf{M}_1\boldsymbol{\delta}\mathbf{x}),$$

where $\mathbf{d}_k$ is the *innovation vector*

$$\mathbf{d}_k = \mathbf{y}_k^{\mathrm{o}} - \mathscr{H}_k\mathscr{M}_k \ldots \mathscr{M}_1(\mathbf{x}^{\mathrm{b}}).$$

Thus, the incremental cost function, $\tilde{J}(\boldsymbol{\delta}\mathbf{x})$, is an approximation of the full 4D-Var cost function, and it is quadratic, thus easier to minimize. The incremental 4D-Var algorithm allows us to take into account small nonlinearities as it periodically updates the linearized operators $\mathbf{M}_k$ and $\mathbf{H}_k$. Algorithm 5.2 sums up the procedure, and Figure 5.2 provides a synthetic sketch.

Incremental 4D-Var is a widely used method in variational assimilation for two main reasons. First, it requires the minimization of only quadratic functionals that are not nonconvex. Second, it allows the use of a simplified model in the inner loop.

The second point is relevant to reduced methods, because it is often a reduced, cheaper, version of the model that is used in the inner loop, as illustrated by Figure 5.2. The variety of such applications is huge, as numerous teams did and still do incremental 3D-/4D-Var. The cheaper, reduced, version of the tangent linear model (TLM) can be obtained in various ways. First, it can be done by simplification of the physics, in particular of the nondifferentiable processes. Second, it can be achieved using a coarser resolution, i.e., fewer physical points on the grid. Finally, the tangent model can be approximated by an actual reduced model, e.g., using truncated spherical harmonics or POD/EOFs.

Regarding this last item, we can mention Lawless et al. [2008], who apply the balanced truncation method, which is a variation of the POD method, within an incremental 4D-Var. They compare the balanced truncation with an equivalently costly reduced model consisting of halving the grid-point spacing and show the superiority of the former.

5.4.1.2 ▪ Multi-incremental 4D-Var

The incremental 4D-Var usually carries two models: the full nonlinear model for the outer loop and a reduced linear version for the inner loop. The multi-incremental version of the algorithm is a refinement consisting of using various reduced models for the various inner loops [Veersé and Thepaut, 1998]. This method is used operationally in Europe (Météo-France, ECMWF, UK MetOffice) with various ways of reducing the model, e.g., varying resolution, truncated harmonics or Fourier series, simplified physics, etc. During the first outer loop iteration, the inner loop is performed with a coarse, simplified, cheap model, so that an approximate minimum is reached quickly. Then, during the second outer iteration, a more accurate but still reduced model is

Algorithm 5.2 Incremental 4D-Var.

Initialize: $\mathbf{x}^{\mathrm{r}} = \mathbf{x}^{\mathrm{g}}$
($\mathbf{x}^{\mathrm{r}}$ is called the *reference state*; $\mathbf{x}^{\mathrm{g}}$ is the first guess.)

OUTER LOOP BEGINS

1. Integrate the full nonlinear model: $\mathbf{x}_k^{\mathrm{r}} = \mathscr{M}_{k:0}[\mathbf{x}^{\mathrm{r}}]$.
2. Compute the innovation vector $\mathbf{d}_k$ thanks to the nonlinear observation operator.

   INNER LOOP BEGINS

   (a) Compute the incremental cost function, $\tilde{J}(\delta\mathbf{x})$, using the linear approximations $\mathbf{M}$ and $\mathbf{H}$, approximations performed around the reference trajectory $\mathbf{x}^{\mathrm{r}}$.
   (b) Compute the gradient, $\nabla\tilde{J}(\delta\mathbf{x})$, thanks to the adjoint operators $\mathbf{M}^{\mathrm{T}}$ and $\mathbf{H}^{\mathrm{T}}$.
   (c) Minimize $\tilde{J}$, e.g., with a gradient descent method, such as conjugate gradient, quasi-Newton, etc.

   INNER LOOP ENDS

3. Update the analysis increment $\delta\mathbf{x}^{\mathrm{a}} = \delta\mathbf{x}$.
4. Update the reference trajectory $\mathbf{x}^{\mathrm{r}} = \mathbf{x}^{\mathrm{r}} + \delta\mathbf{x}^{\mathrm{a}}$.

OUTER LOOP ENDS

Compute the analysis: $\mathbf{x}^{\mathrm{a}} = \mathbf{x}^{\mathrm{r}}$, $\mathbf{x}_k^{\mathrm{a}} = \mathscr{M}_{k:0}[\mathbf{x}^{\mathrm{a}}]$.

---

used, and so on. This approach allows the use of very coarse approximations for the first iterations, thus reducing the computational cost of the overall algorithm.

However, contrary to incremental 4D-Var, there is no convergence theorem, and divergence is well known after a few outer loops [Tremolet, 2007a]. In a simplified framework, it was even shown by Neveu [2011] that this approach first converges quickly to the solution but then quickly diverges, and that the optimal number of iterations was difficult to grasp. It is nevertheless used operationally because convergence is very fast during the first iterations and in practice few of them are performed anyway, because of the numerical cost of the optimization.

5.4.1.3 ▪ Multigrid methods

Multigrid methods have been developed for decades in linear algebra, and extended to nonlinear systems. Let us recall briefly the principle, and consider for simplicity a linear equation,

$$\mathbf{A}\mathbf{x} = \mathbf{b},$$

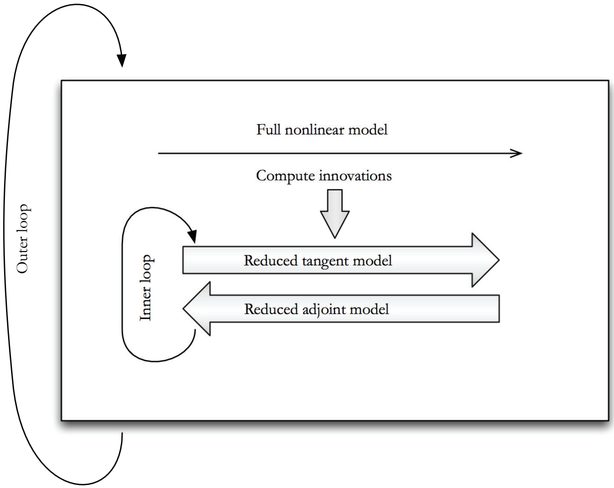


Figure 5.2. *Incremental 4D-Var with reduced models.*

where $\mathbf{A}$ is a given square matrix, $\mathbf{b}$ a vector, and $\mathbf{x}$ the unknown. This equation may come, e.g., from the discretization of a PDE, so that the vector $\mathbf{x}$ is a model state on a given grid in space and/or time. This system can be solved using iterative methods, such as Jacobi or Gauss–Seidel (see, e.g., Golub and van Loan [2013]). It has been shown that these methods have good rates of convergence for small scales/high frequencies but quite poor convergence for large scales/low frequencies. The idea of the multigrid method is to provide corrections of this problem by coarsening the grids: the residual error is projected onto a coarser resolution grid so that low frequencies on the fine original grid become high frequencies on the coarse grid, and a correction is provided from the coarse grid. This can be illustrated as follows with two grids:

1. Perform a few iterations to solve $\mathbf{A}_f\mathbf{x} = \mathbf{b}_f$ on the fine original grid; obtain a first guess $\mathbf{x}_f$ of the solution on the fine grid.

2. Compute the fine-grid residual error: $\mathbf{r}_f = \mathbf{A}_f\mathbf{x}_f - \mathbf{b}_f$.

3. Project the error, $\mathbf{r}_f$, onto the coarse grid to obtain $\mathbf{r}_c$. As the error is essentially located on the large scales, one should have $\mathbf{r}_c \simeq \mathbf{r}_f$.

4. Perform a few iterations with the coarse grid solver, $\mathbf{A}_c\mathbf{y} = \mathbf{r}_c$, to obtain a coarse grid correction $\mathbf{y}_c$.

5. Update $\mathbf{x} = \mathbf{x}_f + \mathbf{y}_c$, and perform a few more iterations on the fine grid.

Of course the multigrid method can be extended to more than two grids, and multiple cycles can be done by alternating resolutions on the various grids. This has also been adapted to the Newton and Gauss–Newton methods to solve a DA-type equation, $\nabla J(\mathbf{x}) = 0$. More details can be found in Debreu et al. [2015] and Neveu [2011]. This approach is quite promising. However, it needs to be implemented on more realistic cases. Moreover, it still requires a few costly iterations on the fine grid, so that for now it is not competitive compared to the multi-incremental approach, which has an excellent convergence rate.

5.4.2 ▪ Reduced 4D-Var

Like reduced Kalman filtering (see Section 5.3), reduced 4D-Var [Durbiano, 2001] aims at reducing the dimension in the core part of the algorithm, namely the minimization. The control variable $\mathbf{x} \in \mathbb{R}^n$, which can contain as always the initial state of the system, numerical parameters, boundary conditions, forcing parameters, etc., is projected onto a lower-dimension space around the background

$$\mathbf{x} = \mathbf{x}^{\mathrm{b}} + \sum_{i=1}^{r} c_i \mathbf{l}_i,$$

where r is the reduced dimension, $\{c_1, \ldots, c_r\}$ are real coefficients, and $\{\mathbf{l}_1, \ldots, \mathbf{l}_r\}$ is a suitable reduced basis of vectors in $\mathbb{R}^n$. In the framework of incremental 4D-Var, the control variable would be simply

$$\delta\mathbf{x} = \sum_{i=1}^{r} c_i \mathbf{l}_i,$$

so that now $\delta\mathbf{x} \in \mathrm{Span}(\mathbf{l}_1, \ldots, \mathbf{l}_r)$. The principle of reduced 4D-Var is then to minimize the reduced incremental cost function

$$\tilde{J}_r(c_1, \ldots, c_r) = \tilde{J}_r^{\mathrm{b}}(c_1, \ldots, c_r) + \tilde{J}_r^{\mathrm{o}}(c_1, \ldots, c_r),$$

$$\tilde{J}_r^{\mathrm{b}}(c_1, \ldots, c_r) = (\textstyle\sum_{i=1}^{r} c_i \mathbf{l}_i)^{\mathrm{T}} \mathbf{B}_r^{-1} (\sum_{i=1}^{r} c_i \mathbf{l}_i),$$

$$\tilde{J}_r^{\mathrm{o}}(c_1, \ldots, c_r) = \sum_{k=1}^{K} \left[\mathbf{d}_k - \mathbf{H}_k \mathbf{M}_k \ldots \mathbf{M}_1 (\textstyle\sum_{i=1}^{r} c_i \mathbf{l}_i)\right]^{\mathrm{T}} \mathbf{R}_k^{-1} \left[\mathbf{d}_k - \mathbf{H}_k \mathbf{M}_k \ldots \mathbf{M}_1 (\sum_{i=1}^{r} c_i \mathbf{l}_i)\right],$$

where $\mathbf{d}_k$ is the innovation vector, as in incremental 4D-Var (see Algorithm 5.2). The background error covariance matrix $\mathbf{B}_r$ is also reduced using the EOFs,

$$\mathbf{B}_r = \mathbf{L} \boldsymbol{\Delta}_r \mathbf{L}^{\mathrm{T}},$$

where $\mathbf{L}$ is the matrix containing the $\mathbf{l}_i$'s in columns and $\boldsymbol{\Delta}_r$ is a diagonal matrix.

The reduced cost function $\tilde{J}_r$ is now a function from $\mathbb{R}^r$ to $\mathbb{R}$. In other words, the minimization is now done in a space of dimension r, usually between a few tens and a few hundreds, instead of n, usually around millions or billions, so that it is much cheaper computationally.

This algorithm was introduced by Durbiano [2001], and various bases were considered for $\mathbf{l}_1, \ldots, \mathbf{l}_r$: EOFs, forward/backward singular vectors, Lyapunov vectors, bred vectors (see Section 5.1.2.1 for more details about the bases).

The algorithm was implemented for a 2D shallow-water model, and the study concluded the superiority of the EOFs. A later article [Robert et al., 2005] implemented with success the method in a more realistic primitive equations ocean model, but still in the idealized twin experiment framework, with simulated data instead of real observations.

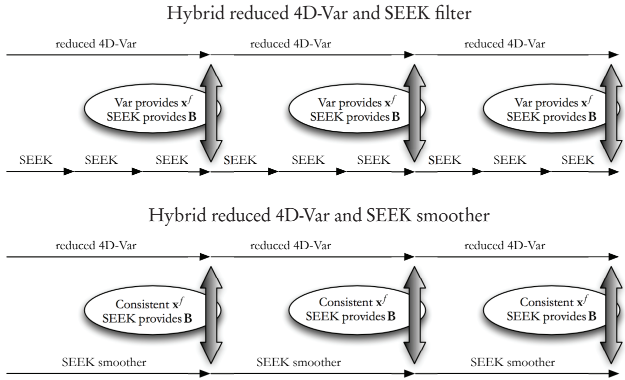


Figure 5.3. *Hybridization of the reduced 4D-Var and SEEK filter/smoother algorithms.*

5.4.2.1 ▪ Hybrid reduced/incremental 4D-Var

Reduced 4D-Var has been applied to a more realistic ocean experiment with real data in Hoteit and Köhl [2006] and Robert et al. [2006a]. Reduced 4D-Var was shown to have a tremendous impact on the convergence rate. However, it was unable to reach a good enough minimum, contrary to the classical unreduced incremental 4D-Var. This seems to be due mostly to the presence of a significant model error, which is more important when using only a few EOFs than with the full model. It was therefore proposed to hybridize both methods: first perform a few iterations of reduced 4D-Var, as a preprocessing tool, and then perform a few iterations of classical incremental 4D-Var. This method, called two-step 4D-Var by Robert et al. [2006a] and the hybrid reduced adjoint method by Hoteit and Köhl [2006], combined the advantages of both methods: good minimization and computational savings, so that the same final result was obtained with a computational time divided by two.

5.4.2.2 ▪ Hybrid reduced 4D-Var/SEEK filter

Hybrid methods coupling variational and filtering methods are numerous and will be treated in more detail in Chapter 7. This subsection is restricted to the hybridization of *reduced* variational and filtering methods.

A full comparison between the reduced 4D-Var and SEEK filter was done by Robert et al. [2006b] in an idealized, perfect model, twin experiment framework. The authors also proposed a new algorithm based on both methods. The idea is simply to perform in parallel a reduced 4D-Var assimilation and three cycles of the SEEK filter on the same assimilation window, starting from the same background information, as shown in Figure 5.3. At the end of the assimilation window,

1. the reduced 4D-Var analysis provides the new background state, and
2. the SEEK filter provides the new background error covariance matrix.

As the SEEK filter and the reduced 4D-Var did not provide strictly the same results, both were required to run fully on the same time window, resulting in doubling the computing time compared to using only one or the other. However, as both are reduced methods, the time savings are still significant.

An updated version of this algorithm was proposed with the SEEK smoother instead of the SEEK filter by Krysta et al. [2011], based on a idea of Veersé et al. [2000]—see Figure 5.3 for a schematic presentation. The SEEK smoother and the reduced 4D-Var methods provide consistent results at the end of the time window, i.e., the same analysis; covariance of the SEEK consistent with the reduced cost function, when using the same background state at the initial time; the same background error covariance matrix; and the same set of EOFs for the reduction. The coupling then works as follows:

1. run a reduced 4D-Var analysis and use it to compute the new background state, and

2. in parallel, perform a SEEK smoother analysis *only for the analysis error covariance matrix* on the same time window, and feed it to the reduced 4D-Var.

This method offers significantly improved results compared to the classical 4D-Var, due to the background error covariance matrix evolving cycle after cycle.

Chapter 6

The ensemble Kalman filter

In Chapter 3 our goal was to track the truth of the partially observed dynamical system,

$$\begin{cases} \mathbf{x}_{k+1} = \mathscr{M}_{k+1}(\mathbf{x}_k) + \boldsymbol{\epsilon}_k^{\mathrm{q}}, \\ \mathbf{y}_k = \mathscr{H}_k(\mathbf{x}_k) + \boldsymbol{\epsilon}_k^{\mathrm{o}}, \end{cases} \tag{6.1}$$

where $\mathbf{x}_k \in \mathbb{R}^n$ is the true state of the system at time t_k. The first equation is the forecast step. It simulates the evolution of the model from t_k to t_{k+1} using the model propagator $\mathscr{M}_{k+1}$. The term $\boldsymbol{\epsilon}_k^{\mathrm{q}}$ is the model error of $\mathscr{M}_{k+1}$ when compared to the true physical processes. It is assumed that this noise is unbiased, uncorrelated in time (white noise), and of model error covariance matrix $\mathbf{Q}_k \in \mathbb{R}^{n\times n}$, i.e.,

$$\mathrm{E}[\boldsymbol{\epsilon}_k^{\mathrm{q}}] = \mathbf{0} \qquad \text{and} \qquad \mathrm{E}[\boldsymbol{\epsilon}_k^{\mathrm{q}}(\boldsymbol{\epsilon}_l^{\mathrm{q}})^{\mathrm{T}}] = \mathbf{Q}_k \delta_{kl}.$$

The second equation in (6.1) is the observation equation, where $\mathscr{H}_k$ is the observation operator at time t_k. The term $\boldsymbol{\epsilon}_k^{\mathrm{o}} \in \mathbb{R}^p$ is the observation error, which is assumed to be unbiased, uncorrelated in time (white noise), and of error covariance matrix $\mathbf{R}_k \in \mathbb{R}^{p\times p}$, i.e.,

$$\mathrm{E}[\boldsymbol{\epsilon}_k^{\mathrm{o}}] = \mathbf{0} \qquad \text{and} \qquad \mathrm{E}\left[\boldsymbol{\epsilon}_k^{\mathrm{o}}(\boldsymbol{\epsilon}_l^{\mathrm{o}})^{\mathrm{T}}\right] = \mathbf{R}_k \delta_{kl}.$$

As an additional assumption, though quite a realistic one, we suppose that there is no correlation between model error and observation error. As a consequence,

$$\mathrm{E}\left[\boldsymbol{\epsilon}_k^{\mathrm{o}}(\boldsymbol{\epsilon}_l^{\mathrm{q}})^{\mathrm{T}}\right] = \mathbf{0}.$$

To track the truth, we used the extended Kalman filter (EKF) that was introduced in Section 3.6. For the clarity of this chapter, we begin by presenting the algorithm of the EKF in Algorithm 6.1.

There are two major drawbacks to the EKF. First of all, the EKF is numerically scarcely affordable in high-dimensional systems. One has to manipulate the error covariance matrix, which requires $n(n+1)/2$ scalars to be stored. Clearly this is impossible in high-dimensional systems for which the storage of a few state vectors is already challenging. Moreover, during the forecast step from t_k to t_{k+1}, one has to compute a forecast error covariance matrix (3.19),

$$\mathbf{P}_{k+1}^{\mathrm{f}} = \mathbf{M}_{k+1}\mathbf{P}_k^{\mathrm{a}}\mathbf{M}_{k+1}^{\mathrm{T}} + \mathbf{Q}_k,$$

Algorithm 6.1 Algorithm of the EKF

Require: For $k = 0,\ldots,K$: the observation error covariance matrices $\mathbf{R}_k$, the model error covariance matrices $\mathbf{Q}_k$, the observation models $\mathscr{H}_k$ and their tangent linear $\mathbf{H}_k$, the forward models $\mathscr{M}_k$ and their tangent linear $\mathbf{M}_k$.

1: Initialize system state $\mathbf{x}^{\mathrm{f}}_0$ and error covariance matrix $\mathbf{P}^{\mathrm{f}}_0$.
2: **for** $k = 0,\ldots,K$ **do**
3: Compute the gain: $\mathbf{K}_k = \mathbf{P}^{\mathrm{f}}_k\mathbf{H}^{\mathrm{T}}_k\left(\mathbf{H}_k\mathbf{P}^{\mathrm{f}}_k\mathbf{H}^{\mathrm{T}}_k + \mathbf{R}_k\right)^{-1}$.
4: Compute the state analysis: $\mathbf{x}^{\mathrm{a}}_k = \mathbf{x}^{\mathrm{f}}_k + \mathbf{K}_k\left(\mathbf{y}_k - \mathscr{H}_k[\mathbf{x}^{\mathrm{f}}_k]\right)$.
5: Compute the analysis error covariance matrix: $\mathbf{P}^{\mathrm{a}}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}^{\mathrm{f}}_k$.
6: Compute the forecast state: $\mathbf{x}^{\mathrm{f}}_{k+1} = \mathscr{M}_{k+1}[\mathbf{x}^{\mathrm{a}}_k]$.
7: Compute the forecast error covariance matrix: $\mathbf{P}^{\mathrm{f}}_{k+1} = \mathbf{M}_{k+1}\mathbf{P}^{\mathrm{a}}_k\mathbf{M}^{\mathrm{T}}_{k+1} + \mathbf{Q}_k$.
8: **end for**

where $\mathbf{M}_{k+1}$ is the tangent linear model (TLM) of $\mathscr{M}_{k+1}$. In addition to the derivation of the TLM, such a computation would require the use of the TLM $2n$ times (a left matrix multiplication by $\mathbf{M}_{k+1}$, followed by a right matrix multiplication by $\mathbf{M}^{\mathrm{T}}_{k+1}$). For a high-dimensional model, this is likely to be much too costly, even when resorting to parallel computing.

Second, the EKF is approximate for nonlinear systems, which is another major drawback. It relies on the tangent linear of the model to describe error propagation. When the tangent linear approximation is breached, for instance when the time step between two consecutive updates is long enough so that the nonlinearity of the model can develop, the forecast error covariance matrix may become underestimated, which is likely to lead to the divergence of the filter.

6.1 ▪ The reduced-rank square root filter

Following ideas put forward in Chapter 5, reduced-rank KFs [Verlaan and Heemink, 1997; Pham et al., 1998; Segers et al., 2000; Pham, 2001; Verlaan and Heemink, 2001] offer a solution to the major problem of the dimensionality. We shall focus on a specific variant, the reduced-rank square root filter, denoted RRSQRT. In this filter, the issue of the computation and propagation of the error covariance matrices is astutely circumvented. The error covariance matrices are represented by their principal axes (those with the largest eigenvalues), that is to say by a limited set of modes. The update and the forecast step will therefore apply to a limited number of modes, a collection of $m \ll n$ vectors for a state space of dimension n, rather than on huge matrices of size $n \times n$.

Suppose the initial system state is $\mathbf{x}^{\mathrm{f}}_0$, with an error covariance matrix $\mathbf{P}^{\mathrm{f}}_0$. We assume a decomposition in terms of the principal modes of $\mathbf{P}^{\mathrm{f}}_0 \in \mathbb{R}^{n\times n}$: $\mathbf{P}^{\mathrm{f}}_0 \simeq \mathbf{S}^{\mathrm{f}}_0(\mathbf{S}^{\mathrm{f}}_0)^{\mathrm{T}}$, where $\mathbf{S}^{\mathrm{f}}_0$ is a matrix of size $n \times m$ composed of m n-vector columns that coincide with the first m dominant modes of $\mathbf{P}^{\mathrm{f}}_0$. The representation of the background has radically changed: rather than thinking in terms of $\mathbf{P}^{\mathrm{f}}_0$, we now think about its dominant modes, $\mathbf{S}^{\mathrm{f}}_0$.

Let us focus on an analysis at a given time t_k. That is why the time index, k, is dropped in the following. We also consider the transformed matrix $\mathbf{Y}_{\mathrm{f}} = \mathbf{H}\mathbf{S}_{\mathrm{f}}$ of size $p \times m$ ranging in the observation space. $\mathbf{H}$ is either the observation operator when it is linear, or its tangent linear otherwise. These matrices appear when considering the

Kalman gain needed in the analysis step (3.16),

$$\begin{aligned}\mathbf{K} &= \mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}+\mathbf{R}\right)^{-1}\\ &= \mathbf{S}_{\mathrm{f}}\mathbf{S}_{\mathrm{f}}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{S}_{\mathrm{f}}\mathbf{S}_{\mathrm{f}}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}+\mathbf{R}\right)^{-1}\\ &= \mathbf{S}_{\mathrm{f}}(\mathbf{H}\mathbf{S}_{\mathrm{f}})^{\mathrm{T}}\left[(\mathbf{H}\mathbf{S}_{\mathrm{f}})(\mathbf{H}\mathbf{S}_{\mathrm{f}})^{\mathrm{T}}+\mathbf{R}\right]^{-1}.\end{aligned}$$

Hence, the Kalman gain, computed at the analysis step, is simply expressed with the help of the $\mathbf{S}_{\mathrm{f}}$ (or $\mathbf{Y}_{\mathrm{f}}$) matrix of size $p \times m$:

$$\mathbf{K} = \mathbf{S}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\left(\mathbf{Y}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}+\mathbf{R}\right)^{-1}.$$

The analysis estimate, $\mathbf{x}^{\mathrm{a}}$, can be obtained from this gain and the standard update formula (3.21).

Then, what happens to the formula for the analysis error covariance matrix $\mathbf{P}^{\mathrm{a}}$? We have (3.22)

$$\begin{aligned}\mathbf{P}^{\mathrm{a}} &= (\mathbf{I}_n-\mathbf{K}\mathbf{H})\mathbf{P}^{\mathrm{f}}\\ &= \left[\mathbf{I}_n-\mathbf{S}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\left(\mathbf{Y}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}+\mathbf{R}\right)^{-1}\mathbf{H}\right]\mathbf{S}_{\mathrm{f}}\mathbf{S}_{\mathrm{f}}^{\mathrm{T}}\\ &= \mathbf{S}_{\mathrm{f}}\left[\mathbf{I}_m-\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\left(\mathbf{Y}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}+\mathbf{R}\right)^{-1}\mathbf{Y}_{\mathrm{f}}\right]\mathbf{S}_{\mathrm{f}}^{\mathrm{T}},\end{aligned}$$

where $\mathbf{I}_n$ ($\mathbf{I}_m$) is the $n \times n$ ($m \times m$) identity matrix. We look for a *square root* matrix such that $\mathbf{S}_{\mathrm{a}}\mathbf{S}_{\mathrm{a}}^{\mathrm{T}} = \mathbf{P}^{\mathrm{a}}$. One such matrix of size $n \times m$ is

$$\mathbf{S}_{\mathrm{a}} = \mathbf{S}_{\mathrm{f}}\left[\mathbf{I}_m-\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}(\mathbf{Y}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}+\mathbf{R})^{-1}\mathbf{Y}_{\mathrm{f}}\right]^{\frac{1}{2}} \tag{6.2}$$

and represents a collection of m state vectors, that is to say a posterior ensemble. The square root matrix that we use in (6.2) is defined as follows. Given $\mathbf{A}$ a diagonalizable matrix, such that $\mathbf{A} = \boldsymbol{\Omega}\boldsymbol{\Lambda}\boldsymbol{\Omega}^{-1}$, where $\boldsymbol{\Lambda}$ is diagonal with nonnegative entries and $\boldsymbol{\Omega}$ is invertible, then the square root of $\mathbf{A}$ is defined by $\mathbf{A}^{\frac{1}{2}} = \boldsymbol{\Omega}\boldsymbol{\Lambda}^{\frac{1}{2}}\boldsymbol{\Omega}^{-1}$.

This factorization avoids a brutal calculation of the error covariance matrices. The computation of the square root matrix might look numerically costly. However, this is not the case since $\mathbf{Y}_{\mathrm{f}}$ is of reduced size and $\mathbf{I}_m-\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}(\mathbf{Y}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}+\mathbf{R})^{-1}\mathbf{Y}_{\mathrm{f}}$ is of dimension $m \times m$. In addition, the conditioning of the square root matrix is better than that of the initial error covariance matrix, as it becomes the square root of the original conditioning. This ensures finer numerical precision. This square root update in matrix form was first proposed by Andrews [1968].

After the analysis step, we seek to reduce the dimension of the system. We wish to shrink the number of modes from m to $m-q$. We first diagonalize $\mathbf{S}_{\mathrm{a}}^{\mathrm{T}}\mathbf{S}_{\mathrm{a}} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{\mathrm{T}}$, where $\mathbf{V}$ is an orthogonal matrix and $\boldsymbol{\Lambda}$ is a diagonal matrix of nonnegative entries. We consider the first $m-q$ eigenmodes with the largest eigenvalues. We keep the first $m-q$ eigenvectors that correspond to the first $m-q$ columns of $\mathbf{V}$ if the diagonal entries of $\boldsymbol{\Lambda}$ are stored in decreasing order. These $m-q$ vectors are stored in the $m \times (m-q)$ matrix $\widetilde{\mathbf{V}}$. Then $\mathbf{S}_{\mathrm{a}}$ is reduced to $\widetilde{\mathbf{S}}_{\mathrm{a}} \equiv \mathbf{S}_{\mathrm{a}}\widetilde{\mathbf{V}}$, which is an $n \times (m-q)$ matrix.

At the forecast step, the analysis is forecast by $\mathbf{x}_{k+1}^{\mathrm{f}} = \mathscr{M}_{k+1}(\mathbf{x}_k^{\mathrm{a}})$. The square root matrix $\mathbf{S}_k^{\mathrm{a}}$ is propagated with the TLM. More precisely, the matrix is enlarged by

Algorithm 6.2 Algorithm for RRSQRT

Require: For $k=0,\ldots,K$: the observation error covariances $\mathbf{R}_k$, the model error covariances $\mathbf{Q}_k$, the observation models $\mathscr{H}_k$ and their tangent linear $\mathbf{H}_k$, the forward models $\mathscr{M}_k$ and their tangent linear $\mathbf{M}_k$.

1: Initialize system state $\mathbf{x}_0^{\mathrm{f}}$ and error covariance matrix $\mathbf{P}_0^{\mathrm{f}}$.
Decomposition: $\mathbf{Q}_k=\mathbf{T}_k\mathbf{T}_k^{\mathrm{T}}$ and $\mathbf{P}_0^{\mathrm{f}}=\mathbf{S}_0^{\mathrm{f}}\left(\mathbf{S}_0^{\mathrm{f}}\right)^{\mathrm{T}}$.
2: **for** $k=0,\ldots,K$ **do**
3: Compute the gain: $\mathbf{Y}_k=\mathbf{H}_k\mathbf{S}_k^{\mathrm{f}}$, $\quad\mathbf{K}_k=\mathbf{S}_k^{\mathrm{f}}\mathbf{Y}_k^{\mathrm{T}}\left(\mathbf{Y}_k\mathbf{Y}_k^{\mathrm{T}}+\mathbf{R}_k\right)^{-1}$.
4: Compute the state analysis: $\mathbf{x}_k^{\mathrm{a}}=\mathbf{x}_k^{\mathrm{f}}+\mathbf{K}_k\left[\mathbf{y}_k-\mathscr{H}_k(\mathbf{x}_k^{\mathrm{f}})\right]$.
5: Compute the (square root) matrix of the modes:
$\mathbf{S}_k^{\mathrm{a}}=\mathbf{S}_k^{\mathrm{f}}\left(\mathbf{I}_m-\mathbf{Y}_k^{\mathrm{T}}(\mathbf{Y}_k\mathbf{Y}_k^{\mathrm{T}}+\mathbf{R}_k)^{-1}\mathbf{Y}_k\right)^{\frac{1}{2}}$, where $\mathbf{S}_k^{\mathrm{a}}$ has m modes.
6: Diagonalization: $\mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{\mathrm{T}}=\left(\mathbf{S}_k^{\mathrm{a}}\right)^{\mathrm{T}}\mathbf{S}_k^{\mathrm{a}}$.
7: Selection of the first $m-q$ modes of $\mathbf{V}\to\widetilde{\mathbf{V}}$.
8: Reduction of the ensemble: $\widetilde{\mathbf{S}}_k^{\mathrm{a}}=\mathbf{S}_k^{\mathrm{a}}\widetilde{\mathbf{V}}$. Then $\widetilde{\mathbf{S}}_k^{\mathrm{a}}$ has $m-q$ modes.
9: Compute the forecast state: $\mathbf{x}_{k+1}^{\mathrm{f}}=\mathscr{M}_{k+1}[\mathbf{x}_k^{\mathrm{a}}]$
10: Compute the forecast modes: $\mathbf{S}_{k+1}^{\mathrm{f}}=[\mathbf{M}_{k+1}\widetilde{\mathbf{S}}_k^{\mathrm{a}},\mathbf{T}_k]$, where $\mathbf{S}_{k+1}^{\mathrm{f}}$ has m modes.
11: **end for**

adding q modes that are meant to introduce model error variability. The matrix of these augmented modes is of the form

$$\mathbf{S}_{k+1}^{\mathrm{f}}=[\mathbf{M}_{k+1}\widetilde{\mathbf{S}}_k^{\mathrm{a}},\mathbf{T}_k],$$

where $\mathbf{T}_k$ introduces m n-vector perturbations. The matrix $\mathbf{S}_{k+1}^{\mathrm{f}}$ has m modes, so that the assimilation procedure can be cycled. The full RRSQRT scheme is summarized in Algorithm 6.2.

This type of filter has been proposed in hydrology and oceanography by Verlaan and Heemink [1997, 2001]. As described in Chapter 5, successful variants are known under the names of SEEK and SEIK [Pham et al., 1998; Pham, 2001]. It has been advocated for air quality forecasts, where the number of chemical species increases the state space dimension considerably [Segers, 2002; Hanea et al., 2004; Wu et al., 2008].

These filters clearly overcome the main drawback of the KF, provided the dynamics of the system can be represented with a limited ($m\ll n$) number of modes. However, by still making use of the TLM, the propagation of uncertainty remains approximate. In that respect, one can propose an even finer reduced KF, the ensemble Kalman filter (EnKF).

6.2 ▪ The EnKF: Principle and classification

The EnKF was proposed by G. Evensen in 1994 and later amended in 1998 [Evensen, 1994; Burgers et al., 1998; Houtekamer and Mitchell, 1998; Evensen, 2009]. Its semi-empirical justification could be disconcerting and initially not as obvious as that of the RRSQRT, in spite of a certain elegance. Nevertheless, the EnKF has proven very efficient on a large number of both academic and operational DA problems. It has become very popular over the past 20 years in its many forms.

The EnKF could be seen as a reduced-order KF, like the RRSQRT filter we described. Just as the EKF and the RRSQRT, it only handles error statistics up to second order (i.e., mean and covariances), which is loosely summarized by designating it as a Gaussian filter. Because of this truncation of statistics, it has been shown that in the limit of a large number of particles, the EnKF does not solve the Bayesian filtering problem, as opposed to the particle filter (see Chapter 3), except when the models are linear and when the initial error distribution is Gaussian and is spanned by the ensemble deviations from the mean [Le Gland et al., 2011; Mandel et al., 2011]. Nonetheless, this does not prevent the EnKF from often being a good approximate algorithm for the filtering problem.

Just as with the particle filter and the RRSQRT filter, the EnKF is based on the concept of particles, a collection of state vectors, which are called the members of the ensemble. Rather than propagating huge covariance matrices, the errors are emulated by scattered particles, a collection of state vectors whose variability is meant to represent the uncertainty of the system's state, which comes from the forecaster's ignorance. Just like the particle filter, but unlike the RRSQRT, the members are to be propagated by the model, without any linearization. Not only does this avoid the derivation of the TLM, but it also circumvents the approximate linearization. Finally, as opposed to the particle filter, the EnKF does not irremediably suffer from the curse of dimensionality (see Chapter 3).

Essentially two main flavors of EnKF have been proposed. The EnKFs of the first class are termed *stochastic* because some random sampling noise is generated in the analysis. The EnKFs of the second class are termed *deterministic* since they do not use random perturbations, which is achieved by the square root formalism that was first described with the RRSQRT.

6.3 ▪ The stochastic EnKF

The focus is first on the stochastic EnKF and its analysis step.

6.3.1 ▪ The analysis step

The EnKF seeks to mimic the analysis step of the KF but with an ensemble of limited size in place of the cumbersome covariance matrices. The goal is to perform for each member of the ensemble an analysis of the form

$$\mathbf{x}_i^{\mathrm{a}} = \mathbf{x}_i^{\mathrm{f}} + \mathbf{K}\left[\mathbf{y} - \mathcal{H}(\mathbf{x}_i^{\mathrm{f}})\right], \tag{6.3}$$

where $i = 1, \ldots, m$ is the member index in the ensemble and $\mathbf{x}_i^{\mathrm{f}}$ is the forecast state vector i, which represents a background state or prior at the analysis time. To mimic the KF, $\mathbf{K}$ must be identified with the Kalman gain,

$$\mathbf{K} = \mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}\right)^{-1}, \tag{6.4}$$

that we wish to estimate from the ensemble statistics. First of all, we can estimate the forecast error covariance matrix as

$$\mathbf{P}^{\mathrm{f}} = \frac{1}{m-1}\sum_{i=1}^{m}\left(\mathbf{x}_i^{\mathrm{f}} - \overline{\mathbf{x}}^{\mathrm{f}}\right)\left(\mathbf{x}_i^{\mathrm{f}} - \overline{\mathbf{x}}^{\mathrm{f}}\right)^{\mathrm{T}}, \quad \text{with} \quad \overline{\mathbf{x}}^{\mathrm{f}} = \frac{1}{m}\sum_{i=1}^{m}\mathbf{x}_i^{\mathrm{f}}.$$

This forecast error covariance matrix can be factorized into

$$\mathbf{P}^{\mathrm{f}} = \mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}},$$

where $\mathbf{X}_\mathrm{f}$ is an $n \times m$ matrix whose columns are the *normalized anomalies* or *normalized perturbations*, i.e., for $i = 1, \ldots, m$,

$$[\mathbf{X}_\mathrm{f}]_i = \frac{\mathbf{x}_i^\mathrm{f} - \overline{\mathbf{x}}^\mathrm{f}}{\sqrt{m-1}}.$$

Here $\mathbf{X}_\mathrm{f}$ plays the role of the matrix of the reduced modes $\mathbf{S}_\mathrm{f}$ of the RRSQRT filter (see Section 6.1).

Thanks to (6.3), we can obtain a posterior ensemble, $\left\{\mathbf{x}_i^\mathrm{a}\right\}_{i=1,\ldots,m}$, from which we can compute the posterior statistics. Hence, the posterior state and an ensemble of posterior perturbations can be computed from

$$\overline{\mathbf{x}}^\mathrm{a} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{x}_i^\mathrm{a}, \quad [\mathbf{X}_\mathrm{a}]_i = \frac{\mathbf{x}_i^\mathrm{a} - \overline{\mathbf{x}}^\mathrm{a}}{\sqrt{m-1}}.$$

The normalized anomalies, $\mathbf{X}_i^\mathrm{a} \equiv [\mathbf{X}_\mathrm{a}]_i$, i.e., the normalized deviations of the ensemble members from the mean, are obtained from (6.3) minus the mean update,

$$\mathbf{X}_i^\mathrm{a} = \mathbf{X}_i^\mathrm{f} + \mathbf{K}\left(\mathbf{0} - \mathbf{H}\mathbf{X}_i^\mathrm{f}\right) = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\,\mathbf{X}_i^\mathrm{f}, \tag{6.5}$$

where $\mathbf{X}_i^\mathrm{f} \equiv [\mathbf{X}_\mathrm{f}]_i$, which yields the analysis error covariance matrix

$$\mathbf{P}^\mathrm{a} = \mathbf{X}_\mathrm{a}\mathbf{X}_\mathrm{a}^\mathrm{T} = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{X}_\mathrm{f}\mathbf{X}_\mathrm{f}^\mathrm{T}(\mathbf{I}_n - \mathbf{K}\mathbf{H})^\mathrm{T} = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{P}^\mathrm{f}(\mathbf{I}_n - \mathbf{K}\mathbf{H})^\mathrm{T}. \tag{6.6}$$

However, theoretically, in order to mimic the BLUE analysis of the KF, we should have obtained instead (see Chapter 3)

$$\mathbf{P}^\mathrm{a} = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{P}^\mathrm{f}(\mathbf{I}_n - \mathbf{K}\mathbf{H})^\mathrm{T} + \mathbf{K}\mathbf{R}\mathbf{K}^\mathrm{T} = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{P}^\mathrm{f}. \tag{6.7}$$

Therefore, the error covariances of (6.6) are underestimated since the second positive term in (6.7), related to the observation errors, is ignored, which is likely to lead to the divergence of the EnKF when the scheme is cycled.

An elegant solution to this problem is to perturb the observation vector for each member: $\mathbf{y} \to \mathbf{y}_i \equiv \mathbf{y} + \mathbf{u}_i$, where $\mathbf{u}_i$ is drawn from the Gaussian distribution $\mathbf{u}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$. Let us define $\overline{\mathbf{u}}$, the mean of the sampled $\mathbf{u}_i$, and the innovation perturbations

$$[\mathbf{Y}_\mathrm{f}]_i = \frac{\mathbf{H}\mathbf{x}_i^\mathrm{f} - \mathbf{u}_i - \mathbf{H}\overline{\mathbf{x}}^\mathrm{f} + \overline{\mathbf{u}}}{\sqrt{m-1}}, \tag{6.8}$$

which shows that the forecast observations $\mathbf{H}\mathbf{x}_i^\mathrm{f}$ are the quantities that are genuinely perturbed. The posterior anomalies are modified accordingly:

$$\mathbf{X}_i^\mathrm{a} = \mathbf{X}_i^\mathrm{f} - \mathbf{K}\mathbf{Y}_i^\mathrm{f} = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{X}_i^\mathrm{f} + \frac{\mathbf{K}(\mathbf{u}_i - \overline{\mathbf{u}})}{\sqrt{m-1}}. \tag{6.9}$$

Denoting $\mathbf{E}_i = (\mathbf{u}_i - \overline{\mathbf{u}})/\sqrt{m-1}$ and $\mathbf{E} = [\mathbf{E}_1, \ldots, \mathbf{E}_m]$, we have $\mathbf{X}_\mathrm{a} = \mathbf{X}_\mathrm{f} - \mathbf{K}\mathbf{Y}_\mathrm{f} = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{X}_\mathrm{f} + \mathbf{K}\mathbf{E}$, which yields the analysis error covariance matrix

$$\mathbf{P}^\mathrm{a} = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{P}^\mathrm{f}(\mathbf{I}_n - \mathbf{K}\mathbf{H})^\mathrm{T} + \mathbf{K}\mathbf{E}\mathbf{E}^\mathrm{T}\mathbf{K}^\mathrm{T} + (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{X}_\mathrm{f}\mathbf{E}^\mathrm{T}\mathbf{K}^\mathrm{T} + \mathbf{K}\mathbf{E}\mathbf{X}_\mathrm{f}^\mathrm{T}(\mathbf{I}_n - \mathbf{K}\mathbf{H})^\mathrm{T},$$

whose expectation over the random noise gives the proper expected posterior covariances:

$$\begin{aligned} \mathrm{E}\left[\mathbf{P}^\mathrm{a}\right] &= (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{P}^\mathrm{f}(\mathbf{I}_n - \mathbf{K}\mathbf{H})^\mathrm{T} + \mathbf{K}\mathrm{E}\left[\mathbf{E}\mathbf{E}^\mathrm{T}\right]\mathbf{K}^\mathrm{T} \\ &= (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{P}^\mathrm{f}(\mathbf{I}_n - \mathbf{K}\mathbf{H})^\mathrm{T} + \mathbf{K}\mathbf{R}\mathbf{K} = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{P}^\mathrm{f}. \end{aligned}$$

Note that the gain can be formulated in terms of the anomaly matrices only,

$$\mathbf{K} = \mathbf{X}_\mathrm{f}\mathbf{Y}_\mathrm{f}^\mathrm{T}\left(\mathbf{Y}_\mathrm{f}\mathbf{Y}_\mathrm{f}^\mathrm{T}\right)^{-1}, \tag{6.10}$$

since $\mathbf{X}_\mathrm{f}\mathbf{Y}_\mathrm{f}^\mathrm{T}$ is a sample estimate for $\mathbf{P}^\mathrm{f}\mathbf{H}^\mathrm{T}$ and $\mathbf{Y}_\mathrm{f}\mathbf{Y}_\mathrm{f}^\mathrm{T}$ is a sample estimate for $\mathbf{H}\mathbf{P}^\mathrm{f}\mathbf{H}^\mathrm{T}+\mathbf{R}$. In this form, it is striking that the updated perturbations are linear combinations of the forecast perturbations. The new perturbations are sought within the *ensemble subspace* of the initial perturbations. Similarly, the state analysis is sought within the affine space $\overline{\mathbf{x}}^\mathrm{f} + \mathrm{Vec}\left(\mathbf{X}_1^\mathrm{f}, \mathbf{X}_2^\mathrm{f}, \ldots, \mathbf{X}_m^\mathrm{f}\right)$.

6.3.2 ▪ The forecast step

In the forecast step, the updated ensemble obtained at the analysis step is propagated by the model over a time step:

$$\text{for} \quad i = 1, \ldots, m, \quad \mathbf{x}_{i,k+1}^\mathrm{f} = \mathscr{M}_{k+1}(\mathbf{x}_{i,k}^\mathrm{a}).$$

A forecast can be computed from the mean of the forecast ensemble, while the forecast error covariances can be estimated from the forecast perturbations. Yet, these are only optional diagnostics in the scheme and they are not required in the cycling of the EnKF. It is important to observe that the use of the TLM operator is avoided. This makes a significant difference with the RRSQRT filter, especially in a significantly nonlinear regime. However, in this case, the EnKF is itself outdone by schemes known as the iterative EnKF (IEnKF) and the iterative ensemble Kalman smoother (IEnKS). These will be discussed in Chapter 7.

6.3.3 ▪ Avoiding the observation tangent linear operator

Like the particle filter or 3D-Var, the EnKF does not require the tangent linear of the forecast model. More interestingly, it can also avoid the computation of the adjoint of the observation operator. Let us consider the formula of the Kalman gain (6.10), which makes use of the adjoint of the observation operator via (6.8). The matrix products $\mathbf{P}^\mathrm{f}\mathbf{H}^\mathrm{T}$ and $\mathbf{H}\mathbf{P}^\mathrm{f}\mathbf{H}^\mathrm{T}$ both require the tangent linear of the observation operator. Instead, these can be estimated by an analogue of finite differences using the full observation model [Evensen, 1994; Houtekamer and Mitchell, 1998],

$$\begin{aligned}
\overline{\mathbf{y}}^\mathrm{f} &= \frac{1}{m}\sum_{i=1}^{m}\mathscr{H}\left(\mathbf{x}_i^\mathrm{f}\right), \\
\mathbf{P}^\mathrm{f}\mathbf{H}^\mathrm{T} &= \frac{1}{m-1}\sum_{i=1}^{m}\left(\mathbf{x}_i^\mathrm{f}-\overline{\mathbf{x}}^\mathrm{f}\right)\left[\mathbf{H}\left(\mathbf{x}_i^\mathrm{f}-\overline{\mathbf{x}}^\mathrm{f}\right)\right]^\mathrm{T} \\
&\simeq \frac{1}{m-1}\sum_{i=1}^{m}\left(\mathbf{x}_i^\mathrm{f}-\overline{\mathbf{x}}^\mathrm{f}\right)\left[\mathscr{H}\left(\mathbf{x}_i^\mathrm{f}\right)-\overline{\mathbf{y}}^\mathrm{f}\right]^\mathrm{T}, \\
\mathbf{H}\mathbf{P}^\mathrm{f}\mathbf{H}^\mathrm{T} &= \frac{1}{m-1}\left[\mathbf{H}\left(\mathbf{x}_i^\mathrm{f}-\overline{\mathbf{x}}^\mathrm{f}\right)\right]\left[\mathbf{H}\left(\mathbf{x}_i^\mathrm{f}-\overline{\mathbf{x}}^\mathrm{f}\right)\right]^\mathrm{T} \\
&\simeq \frac{1}{m-1}\sum_{i=1}^{m}\left[\mathscr{H}\left(\mathbf{x}_i^\mathrm{f}\right)-\overline{\mathbf{y}}^\mathrm{f}\right]\left[\mathscr{H}\left(\mathbf{x}_i^\mathrm{f}\right)-\overline{\mathbf{y}}^\mathrm{f}\right]^\mathrm{T},
\end{aligned} \tag{6.11}$$

where the key assumption in these derivations is the approximation

$$\mathbf{H}\left(\mathbf{x}_i^\mathrm{f}-\overline{\mathbf{x}}^\mathrm{f}\right) \simeq \mathscr{H}\left(\mathbf{x}_i^\mathrm{f}\right)-\overline{\mathbf{y}}^\mathrm{f},$$

Algorithm 6.3 Algorithm for the (stochastic) EnKF

Require: For $k=0,\dots,K$: the observation error covariance matrices $\mathbf{R}_k$, the observation models $\mathscr{H}_k$, the forward models $\mathscr{M}_k$.

Initialize the ensemble $\left\{\mathbf{x}^{\mathrm{f}}_{i,0}\right\}_{=1,\dots,m}$.

for $k=0,\dots,K$ **do**

  Draw a statistically consistent observation set:
  for $i=1,..,m$: $\mathbf{y}_{i,k}=\mathbf{y}_k+\mathbf{u}_i$, with $\mathbf{u}_i\sim\mathcal{N}(0,\mathbf{R}_k)$

  Compute the ensemble means
  $\overline{\mathbf{x}}^{\mathrm{f}}_k=\frac{1}{m}\sum_{i=1}^m\mathbf{x}^{\mathrm{f}}_{i,k}$, $\overline{\mathbf{u}}=\frac{1}{m}\sum_{i=1}^m\mathbf{u}_i$, $\overline{\mathbf{y}}^{\mathrm{f}}_k=\frac{1}{m}\sum_{i=1}^m\mathscr{H}_k\left(\mathbf{x}^{\mathrm{f}}_{i,k}\right)$
  and the normalized anomalies
  $[\mathbf{X}_{\mathrm{f}}]_{i,k}=\frac{\mathbf{x}^{\mathrm{f}}_{i,k}-\overline{\mathbf{x}}^{\mathrm{f}}_k}{\sqrt{m-1}}$, $[\mathbf{Y}_{\mathrm{f}}]_{i,k}=\frac{\mathscr{H}_k\left(\mathbf{x}^{\mathrm{f}}_{i,k}\right)-\mathbf{u}_i-\overline{\mathbf{y}}^{\mathrm{f}}_k+\overline{\mathbf{u}}}{\sqrt{m-1}}$.

  Compute the gain: $\mathbf{K}_k=\mathbf{X}^{\mathrm{f}}_k\left(\mathbf{Y}^{\mathrm{f}}_k\right)^{\mathrm{T}}\left\{\mathbf{Y}^{\mathrm{f}}_k\left(\mathbf{Y}^{\mathrm{f}}_k\right)^{\mathrm{T}}\right\}^{-1}$

  Update of the ensemble:
  for $i=1,..,m$: $\mathbf{x}^{\mathrm{a}}_{i,k}=\mathbf{x}^{\mathrm{f}}_{i,k}+\mathbf{K}_k\left(\mathbf{y}_{i,k}-\mathscr{H}_k(\mathbf{x}^{\mathrm{f}}_{i,k})\right)$,

  Compute the ensemble forecast:
  for $i=1,..,m$ $\mathbf{x}^{\mathrm{f}}_{i,k+1}=\mathscr{M}_{k+1}(\mathbf{x}^{\mathrm{a}}_{i,k})$.

end for

which can be seen as finite differences if the spread of the ensemble is small enough. This avoids the use of not only the adjoint model but also the tangent linear of the observation operator. This suggests modifying the definition of the perturbations as

$$[\mathbf{Y}_{\mathrm{f}}]_i=\frac{\mathscr{H}\left(\mathbf{x}^{\mathrm{f}}_i\right)-\mathbf{u}_i-\overline{\mathbf{y}}^{\mathrm{f}}+\overline{\mathbf{u}}}{\sqrt{m-1}}.$$

With the important exception of the Kalman gain's computation, all the operations on the ensemble members are independent. This implies that their parallelization can be carried out straightforwardly. This is one of the main reasons for the success and popularity of the EnKF. The algorithm of the stochastic EnKF is given in Algorithm 6.3.

The stochastic EnKF has one disadvantage over other deterministic variants of the EnKF (which we will investigate in Section 6.4): it introduces stochastic noise. This may impact the performance of the filter to an extent that depends on the precise DA setup; on the dynamics nonlinearity; and, above all, on the ensemble size.

6.3.4 ▪ Numerical illustration

The discrete model

$$x_0=0,\quad x_1=1,\quad \text{and for } 1\le k\le N,\quad x_{k+1}-2x_k+x_{k-1}=\omega^2 x_k-\lambda^2 x_k^3,\quad (6.12)$$

is a numerical implementation of the anharmonic oscillator of a material point,

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2}-\Omega^2x+\Lambda^2x^3=0,$$

where Ω^2 is proportional to ω^2 and Λ^2 is proportional to λ^2. The related potential energy is $V(x)=-\frac{1}{2}\Omega^2x^2+\frac{1}{4}\Lambda^2x^4$. The second term of $V(x)$ stabilizes the oscillator and plays the role of a spring force, whereas the first term destabilizes the point of origin, $x = 0$, leading to two potential wells. It is a second-order discrete equation, with a state vector that can be advantageously written as

$$\mathbf{u}_k = \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}. \tag{6.13}$$

From (6.12) and (6.13), the state-dependent transition matrix is

$$\mathcal{M}_{k+1} = \begin{bmatrix} 2+\omega^2-\lambda^2 x_k^2 & -1 \\ 1 & 0 \end{bmatrix},$$

so that $\mathbf{u}_{k+1} = \mathcal{M}_{k+1}(\mathbf{u}_k)$.

The EKF and the stochastic EnKF are tested with this nonlinear system in the absence of model error. We choose an ensemble of size $m = 10$, much greater than the state system dimension ($n = 2$). That is why the focus of this experiment is not on the impact of the reduction of dimensionality obtained by the limited size ensemble, but on the impact of the nonlinear dynamics on the filter's performance.

Figure 6.1 gives an example of synthetic (or twin) DA experiments. The true trajectory, *the truth*, is run as a reference, with $\omega = 3.5 \times 10^{-2}$ and $\lambda = 3 \times 10^{-4}$. Its trajectory is shown with a thick dashed black line. Obviously, the trajectory seems periodic. The truth is not meant to be directly accessible. However, it is observed through a set of observations that measure x_k, so that the observation operator is $\mathbf{H}_k = [1, 0]$. The observation equation is

$$y_k = \mathbf{H}_k \mathbf{u}_k + \epsilon_k.$$

Each noiseless observation, $\mathbf{H}_k \mathbf{u}_k$, is independently perturbed with an unbiased Gaussian noise, ϵ_k, of standard deviation $\sigma = 10$. The perturbed observations are indicated by circles. In the top panels, the system is observed every 25 time steps, whereas it is observed every 50 time steps in the bottom panels. Clearly, with 50 time steps between observations, the model nonlinearity has a stronger impact than with an interval of 25 time steps.

The assimilation is meant to track the truth using the perfect model, in particular with the same ω and λ, and the perturbed observations. However, the initial conditions are not known. The best estimate (i.e., through the analysis/forecast cycles) trajectory is plotted as a full line curve for the EKF and the EnKF. One can observe several abrupt corrections of the trajectory of the best estimate when an analysis takes place, especially when the innovation of the assimilation is stronger.

In the frequent-observation case (upper panels), both the EKF and the EnKF keep track of the truth. However, in the less frequent case (lower panels), the EKF loses track of the truth, while the EnKF still follows it. The EKF is said to have diverged (from the truth) even though the trajectory remains bounded. The divergence occurs because of the approximate propagation of the errors with the TLM, which could lead to an underestimation of the uncertainty. The EKF becomes increasingly but wrongly confident, which leads to the divergence. This behavior can be reproduced with other initial conditions, although the divergence of the EKF may happen at a different time. In the context of this experiment, the only significant difference between the two methods is the handling of nonlinearity, which is finer in the EnKF than in the EKF.

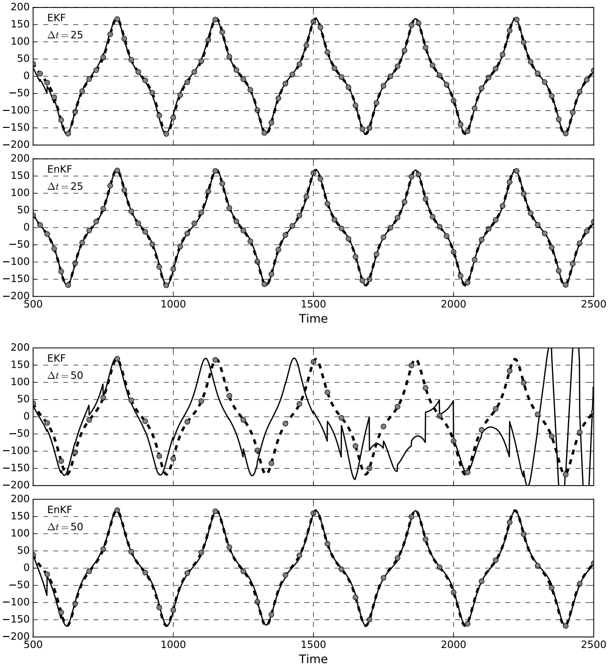


Figure 6.1. *Synthetic DA experiments with the anharmonic oscillator. The observations are represented by disks, the truth by a thick dashed line, and the EKF and EnKF best estimate trajectories by full line curves. The upper panels correspond to frequent observations (every* 25 *time steps), whereas the bottom panels correspond to half as frequent observations (every* 50 *time steps).*

6.4 ▪ The deterministic EnKF

The stochastic EnKF enables the individual tracking of each member of the ensemble, with the analysis step for the members to interact via the Kalman gain matrix. This nice property, central to stochastic filtering, comes with a price: we need to independently perturb the observation vector of each member. Although this seems elegant, it also introduces numerical noise when drawing the perturbations, $\mathbf{u}_i$. This can affect the performance of the stochastic EnKF.

An alternative idea for performing a statistically consistent EnKF analysis is to follow the square root approach implemented in the RRSQRT. This yields the *ensemble square root Kalman filter (EnSRKF)*. Since with this scheme the observations are not perturbed, it is sometimes called the *deterministic EnKF*. There are several variants of the EnSRKF [Tippett et al., 2003]. Be aware that they are, to a large extent, algebraically equivalent. Yet, their practical implementations may lead to discrepancies, especially when localization is applied (see Section 6.5). In this chapter, we shall present the EnSRKF where the analysis is performed in state space [Whitaker and Hamill, 2002], as well as the so-called ensemble-transform variant of the deterministic EnKF, usually abbreviated *ETKF* [Bishop et al., 2001; Hunt et al., 2007]. In the latter case, rather than performing the linear algebra in state or observation space, the algebra is mostly performed in the ensemble subspace of small dimension.

6.4.1 ▪ Spanning the ensemble subspace

Let us define the ensemble subspace. Following the stochastic EnKF, we can write the forecast error covariance matrix as

$$\mathbf{P}^{\mathrm{f}} = \frac{1}{m-1}\sum_{i=1}^{m}\left(\mathbf{x}_i^{\mathrm{f}} - \overline{\mathbf{x}}^{\mathrm{f}}\right)\left(\mathbf{x}_i^{\mathrm{f}} - \overline{\mathbf{x}}^{\mathrm{f}}\right)^{\mathrm{T}} = \mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}},$$

where $\mathbf{X}_{\mathrm{f}}$ is an $n \times m$ matrix whose columns are the *normalized* anomalies

$$[\mathbf{X}_{\mathrm{f}}]_i = \frac{\mathbf{x}_i^{\mathrm{f}} - \overline{\mathbf{x}}^{\mathrm{f}}}{\sqrt{m-1}}.$$

We shall assume that the analysis belongs to the affine subspace whose vectors $\mathbf{x}$ are of the form

$$\mathbf{x} = \overline{\mathbf{x}}^{\mathrm{f}} + \mathbf{X}_{\mathrm{f}}\mathbf{w},$$

where $\mathbf{w}$ is a vector of coefficients in the ensemble subspace $\mathbb{R}^m$. The restriction to the ensemble subspace is an assumption shared with the stochastic EnKF. However, here, the state vector is parameterized in the ensemble subspace by $\mathbf{w}$. Note that the decomposition of $\mathbf{x}$ is not unique because the vector of coefficients $\mathbf{w} + \lambda\mathbf{1}$ with $\mathbf{1} = (1,\dots,1)^{\mathrm{T}} \in \mathbb{R}^m$ yields the same state vector $\mathbf{x}$, since $\mathbf{X}_{\mathrm{f}}\mathbf{1} = \mathbf{0}$. Following the terminology used in physics, this degree of freedom is called a *gauge* degree of freedom.

Again we shall use the notation $\mathbf{Y}_{\mathrm{f}}$ to represent $\mathbf{H}\mathbf{X}_{\mathrm{f}}$ if the observation operator is linear (or linearized). If it is nonlinear, we consider $\mathbf{Y}_{\mathrm{f}}$ to be the matrix of the *observation anomalies* (see Section 6.3.3):

$$[\mathbf{Y}_{\mathrm{f}}]_i = \frac{\mathscr{H}(\mathbf{x}_i^{\mathrm{f}}) - \overline{\mathbf{y}}^{\mathrm{f}}}{\sqrt{m-1}} \quad \text{with} \quad \overline{\mathbf{y}}^{\mathrm{f}} = \frac{1}{m}\sum_{i=1}^{m}\mathscr{H}(\mathbf{x}_i^{\mathrm{f}}).$$

Note that no random perturbation was added to these anomalies, as opposed to the definition of $\mathbf{Y}_{\mathrm{f}}$ in the stochastic EnKF.

6.4.2 ▪ State analysis in state space and ensemble subspace

As opposed to the stochastic EnKF, we wish to perform a single analysis followed by the generation of a new set of perturbations centered on it, rather than performing an analysis for each member of the ensemble. For the state update, we can adapt the mean analysis of the stochastic EnKF

$$\mathbf{x}^{\mathrm{a}} = \overline{\mathbf{x}}^{\mathrm{f}} + \mathbf{K}\left[\mathbf{y} - \mathscr{H}(\overline{\mathbf{x}}^{\mathrm{f}})\right], \tag{6.14}$$

where we used $\mathbf{K} = \mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}\right)^{-1}$. Here, the Kalman gain is computed using $\mathbf{P}^{\mathrm{f}} = \mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}$ and a known predetermined $\mathbf{R}$.

This analysis can be reformulated in the ensemble subspace. This means finding the corresponding optimal coefficient vector $\mathbf{w}^{\mathrm{a}}$ that parameterizes $\mathbf{x}^{\mathrm{a}}$,

$$\mathbf{x}^{\mathrm{a}} = \overline{\mathbf{x}}^{\mathrm{f}} + \mathbf{X}_{\mathrm{f}}\mathbf{w}^{\mathrm{a}}.$$

Inserting this decomposition into (6.14), and denoting the innovation vector as $\boldsymbol{\delta} = \mathbf{y} - \mathscr{H}(\overline{\mathbf{x}}^{\mathrm{f}})$, we obtain

$$\overline{\mathbf{x}}^{\mathrm{f}} + \mathbf{X}_{\mathrm{f}}\mathbf{w}^{\mathrm{a}} = \overline{\mathbf{x}}^{\mathrm{f}} + \mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}\right)^{-1}\boldsymbol{\delta},$$

which suggests

$$\mathbf{w}^{\mathrm{a}} = \mathbf{X}_{\mathrm{f}}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}\right)^{-1}\boldsymbol{\delta} = \mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\left(\mathbf{Y}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}} + \mathbf{R}\right)^{-1}\boldsymbol{\delta}.$$

The gain, $\mathbf{K}$, is usually computed in the observation space. Using the Sherman–Morrison–Woodbury formula, it is possible to compute the gain in ensemble subspace. Remember that we learned in Section 2.4.3 that the gain, $\mathbf{K}$, can be written in state or observation space as

$$\mathbf{K} = \mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}\right)^{-1} = \left(\mathbf{P}^{\mathrm{f}^{-1}} + \mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}.$$

Using this identity and with the identifications $\mathbf{P}^{\mathrm{f}} \equiv \mathbf{I}_m$ and $\mathbf{H} \equiv \mathbf{Y}_{\mathrm{f}}$, we finally obtain

$$\mathbf{w}^{\mathrm{a}} = \left(\mathbf{I}_m + \mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{Y}_{\mathrm{f}}\right)^{-1}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\mathbf{R}^{-1}\boldsymbol{\delta}.$$

The gain is now computed in the ensemble subspace rather than in the observation or state space. This is numerically efficient provided the ensemble is smaller than the number of observations. We refer to Hunt et al. [2007] for a detailed numerical complexity analysis.

6.4.3 ▪ Generating the posterior ensemble (ensemble subspace)

Writing the state analysis in ensemble subspace is merely a reformulation of the stochastic EnKF. The genuine difference between the deterministic EnKF and the stochastic EnKF appears in the generation of the posterior ensemble.

To generate a posterior ensemble of perturbations that would be representative of the posterior uncertainty, we would like to factorize $\mathbf{P}^{\mathrm{a}} = \mathbf{X}_{\mathrm{a}}\mathbf{X}_{\mathrm{a}}^{\mathrm{T}}$. We can repeat the derivation of the RRSQRT, (6.2):

$$\begin{aligned}
\mathbf{P}^{\mathrm{a}} &= (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{P}^{\mathrm{f}} \\
&= \left(\mathbf{I}_n - \mathbf{X}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\left(\mathbf{Y}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}} + \mathbf{R}\right)^{-1}\mathbf{H}\right)\mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}} \\
&= \mathbf{X}_{\mathrm{f}}\left(\mathbf{I}_m - \mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\left(\mathbf{Y}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}} + \mathbf{R}\right)^{-1}\mathbf{Y}_{\mathrm{f}}\right)\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}.
\end{aligned}$$

That is why we choose

$$\mathbf{X}_{\mathrm{a}} = \mathbf{X}_{\mathrm{f}}\left(\mathbf{I}_m - \mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}(\mathbf{Y}_{\mathrm{f}}\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}} + \mathbf{R})^{-1}\mathbf{Y}_{\mathrm{f}}\right)^{\frac{1}{2}}\mathbf{U}, \tag{6.15}$$

with $\mathbf{U}$ an arbitrary orthogonal matrix. This expression can be simplified into

$$\begin{aligned}\mathbf{X}_\mathrm{a} &= \mathbf{X}_\mathrm{f}\left(\mathbf{I}_m - \mathbf{Y}_\mathrm{f}^\mathrm{T}(\mathbf{Y}_\mathrm{f}\mathbf{Y}_\mathrm{f}^\mathrm{T}+\mathbf{R})^{-1}\mathbf{Y}_\mathrm{f}\right)^{\frac{1}{2}}\mathbf{U}\\ &= \mathbf{X}_\mathrm{f}\left(\mathbf{I}_m - \left(\mathbf{I}_m+\mathbf{Y}_\mathrm{f}^\mathrm{T}\mathbf{R}^{-1}\mathbf{Y}_\mathrm{f}\right)^{-1}\mathbf{Y}_\mathrm{f}^\mathrm{T}\mathbf{R}^{-1}\mathbf{Y}_\mathrm{f}\right)^{\frac{1}{2}}\mathbf{U}\\ &= \mathbf{X}_\mathrm{f}\left[\left(\mathbf{I}_m+\mathbf{Y}_\mathrm{f}^\mathrm{T}\mathbf{R}^{-1}\mathbf{Y}_\mathrm{f}\right)^{-1}\left(\mathbf{I}_m+\mathbf{Y}_\mathrm{f}^\mathrm{T}\mathbf{R}^{-1}\mathbf{Y}_\mathrm{f}-\mathbf{Y}_\mathrm{f}^\mathrm{T}\mathbf{R}^{-1}\mathbf{Y}_\mathrm{f}\right)\right]^{\frac{1}{2}}\mathbf{U}\\ &= \mathbf{X}_\mathrm{f}\left(\mathbf{I}_m+\mathbf{Y}_\mathrm{f}^\mathrm{T}\mathbf{R}^{-1}\mathbf{Y}_\mathrm{f}\right)^{-\frac{1}{2}}\mathbf{U}, \end{aligned} \tag{6.16}$$

where we have used the Sherman–Morrison–Woodbury formula again between the first and the second lines. It is often called the *right transform* of the initial perturbations.

One critical property of the initial anomalies is that they are centered, which reads $\mathbf{X}_\mathrm{f}\mathbf{1} = \mathsf{0}$, where we recall that $\mathbf{1} = (1,\dots,1)^\mathrm{T}$. One would also want the updated perturbations to be centered on $\mathbf{x}^\mathrm{a}$. To ensure that $\mathbf{X}_\mathrm{a}\mathbf{1} = \mathsf{0}$, it is sufficient to satisfy $\mathbf{U1} = \mathbf{1}$, which can easily be checked by right-multiplying the members of (6.15) by $\mathbf{1}$. Not only is this intuitively more appealing but it was also proven to lead to a more precise algorithm [Wang et al., 2004; Sakov and Oke, 2008a; Livings et al., 2008].

Moreover, even though this linear analysis is independent of the choice of $\mathbf{U}$, this choice may have an impact on the nonlinear ensemble forecast. The physical balance of the members of the ensemble can be affected differently by distinct $\mathbf{U}$'s. The $\mathbf{U}$ that minimizes the displacement between the prior perturbation and the updated perturbations is $\mathbf{U} = \mathbf{I}_m$ [Ott et al., 2004]. This is likely to mitigate the imbalance generated by the approximate linear analysis when dealing with nonlinear models.

This posterior ensemble of anomalies is all that we need to cycle the deterministic EnKF. Defining $\mathbf{T} = \left(\mathbf{I}_m + \mathbf{Y}_\mathrm{f}^\mathrm{T}\mathbf{R}^{-1}\mathbf{Y}_\mathrm{f}\right)^{-1}$, we can build the posterior ensemble by setting, for all $i = 1,\dots,m$,

$$\mathbf{x}_i^\mathrm{a} = \mathbf{x}^\mathrm{a} + \sqrt{m-1}\mathbf{X}_\mathrm{f}\left[\mathbf{T}^{\frac{1}{2}}\mathbf{U}\right]_i = \overline{\mathbf{x}}^\mathrm{f} + \mathbf{X}_\mathrm{f}\left(\mathbf{w}^\mathrm{a} + \sqrt{m-1}\left[\mathbf{T}^{\frac{1}{2}}\mathbf{U}\right]_i\right). \tag{6.17}$$

The $\mathbf{T}$ matrix is acting on the right of the anomaly matrix $\mathbf{X}_\mathrm{f}$. Again, this shows that posterior perturbations are linear combinations of the initial perturbations.

The analysis update in ensemble subspace is followed by an ensemble forecast similar to the stochastic EnKF. The resulting algorithm of this ETKF is proposed in Algorithm 6.4 in the form of a pseudocode.

6.4.4 ▪ Generating the posterior ensemble (state space)

It can also be convenient, as we shall see later, to have an equivalent update formula but in state space, rather than algebraically obtained in the ensemble subspace. Hence, one seeks a linear transform acting on the left of the prior perturbations $\mathbf{X}_\mathrm{f}$. To find one, we will need a simple formula for matrices.

Lemma 6.1. *Assume that the scalar map $x \mapsto f(x)$ can be expanded as a formal power series: $f(x) = \sum_{k=0}^{\infty}\alpha_k x^k$. Then, we have, at a formal level, $\mathbf{A}f(\mathbf{BA}) = f(\mathbf{AB})\mathbf{A}$ for any two matrices $\mathbf{A}$ and $\mathbf{B}$ of compatible dimensions since*

$$\mathbf{A}f(\mathbf{BA}) = \mathbf{A}\sum_{k=0}^{\infty}\alpha_k(\mathbf{BA})^k = \sum_{k=0}^{\infty}\alpha_k\mathbf{A}(\mathbf{BA})^k = \sum_{k=0}^{\infty}\alpha_k(\mathbf{AB})^k\mathbf{A} = f(\mathbf{AB})\mathbf{A}.$$

Algorithm 6.4 Pseudocode for a complete cycle of the ETKF

Require: Observation operator $\mathcal{H}$ at current time; $\mathbf{E}$, the forecast ensemble at current time; $\mathbf{y}$, the observation at current time; $\mathbf{U}$, an orthogonal matrix in $\mathbb{R}^{m\times m}$ satisfying $\mathbf{U1}=\mathbf{1}$; $\mathcal{M}$, the model resolvent from current time to the next analysis time; $\mathbf{R}$, is the error covariance matrix.

$\overline{\mathbf{x}}=\mathbf{E1}/m$

$\mathbf{X}=\left(\mathbf{E}-\overline{\mathbf{x}}\mathbf{1}^{\mathrm{T}}\right)/\sqrt{m-1}$

$\mathbf{Z}=\mathcal{H}(\mathbf{E})$

$\overline{\mathbf{y}}=\mathbf{Z1}/m$

$\mathbf{S}=\mathbf{R}^{-\frac{1}{2}}\left(\mathbf{Z}-\overline{\mathbf{y}}\mathbf{1}^{\mathrm{T}}\right)/\sqrt{m-1}$

$\boldsymbol{\delta}=\mathbf{R}^{-\frac{1}{2}}\left(\mathbf{y}-\overline{\mathbf{y}}\right)$

$\mathbf{T}=\left(\mathbf{I}_m+\mathbf{S}^{\mathrm{T}}\mathbf{S}\right)^{-1}$

$\mathbf{w}=\mathbf{T}\mathbf{S}^{\mathrm{T}}\boldsymbol{\delta}$

$\mathbf{E}:=\overline{\mathbf{x}}\mathbf{1}^{\mathrm{T}}+\mathbf{X}\left(\mathbf{w}\mathbf{1}^{\mathrm{T}}+\sqrt{m-1}\,\mathbf{T}^{\frac{1}{2}}\mathbf{U}\right)$

$\mathbf{E}:=\mathcal{M}(\mathbf{E})$

We apply Lemma 6.1 (the so-called matrix shift lemma), $\mathbf{A}f(\mathbf{BA})=f(\mathbf{AB})\mathbf{A}$, to (6.16) with $\mathbf{A}=\mathbf{X}_{\mathrm{f}}$, $\mathbf{B}=\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}$ and $f(x)=(1+x)^{-\frac{1}{2}}$:

$$\begin{aligned}
\mathbf{X}_{\mathrm{a}}&=\mathbf{X}_{\mathrm{f}}\left(\mathbf{I}_m+\mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\mathbf{X}_{\mathrm{f}}\right)^{-\frac{1}{2}}\mathbf{U}\\
&=\left(\mathbf{I}_n+\mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-\frac{1}{2}}\mathbf{X}_{\mathrm{f}}\mathbf{U}\\
&=\left(\mathbf{I}_n+\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-\frac{1}{2}}\mathbf{X}_{\mathrm{f}}\mathbf{U}.
\end{aligned}\tag{6.18}$$

Hence, the transform matrix $\mathbf{T}=\left(\mathbf{I}_n+\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-\frac{1}{2}}$ acts on the left of the anomaly matrix, i.e., in state space. It is often called the *left transform*. This condensed form of the transform matrix was first suggested by Sakov and Bertino [2011].

This left transform can also be rewritten so that it looks like the update of the stochastic EnKF but with a modified Kalman gain. To do so, we need another matrix lemma.

Lemma 6.2. *Assume that the scalar map $x\mapsto f(x)$ can be expanded as a formal power series: $f(x)=\sum_{k=0}^{\infty}\alpha_k x^k$ with, in addition, $f(0)=1$. Define $g(x)=(f(x)-1)/x$. The formal power series of g is $g(x)=\sum_{k=0}^{\infty}\alpha_{k+1}x^k$. If $\mathbf{A}$ is a matrix of size $n\times p$ and if $\mathbf{B}$ is a matrix of size $p\times n$, then*

$$f(\mathbf{AB})=\sum_{k=0}^{\infty}\alpha_k(\mathbf{AB})^k=\mathbf{I}_n+\mathbf{A}\left(\sum_{k=0}^{\infty}\alpha_{k+1}(\mathbf{BA})^k\right)\mathbf{B}=\mathbf{I}_n+\mathbf{A}g(\mathbf{BA})\mathbf{B}.$$

In particular, if $f(x)=\frac{1}{\sqrt{1+x}}$, then $g(x)=-\frac{1}{1+x+\sqrt{1+x}}$. When applied to $\mathbf{AB}$, *one obtains [Bocquet.* 2016*]*

$$(\mathbf{I}_n+\mathbf{AB})^{-\frac{1}{2}}=\mathbf{I}_n-\mathbf{A}\left(\mathbf{I}_p+\mathbf{BA}+\sqrt{\mathbf{I}_p+\mathbf{BA}}\right)^{-1}\mathbf{B}.\tag{6.19}$$

Choosing $\mathbf{A}=\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}$ and $\mathbf{B}=\mathbf{R}^{-1}\mathbf{H}$ in (6.19), we find that

$$\mathbf{X}_{\mathrm{a}}=\left(\mathbf{I}_n-\widetilde{\mathbf{K}}\mathbf{H}\right)\mathbf{X}_{\mathrm{f}}\mathbf{U},\tag{6.20}$$

with

$$\widetilde{\mathbf{K}} = \mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{R} + \mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}\sqrt{\mathbf{I}_p + \mathbf{R}^{-1}\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}}\right)^{-1},$$

which was put forward in Whitaker and Hamill [2002] following Andrews [1968]. Note that this can also be written using the original Kalman gain,

$$\widetilde{\mathbf{K}} = \mathbf{K}\left(\mathbf{I}_p + \left[\mathbf{I}_p + \mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\right]^{-\frac{1}{2}}\right)^{-1}. \tag{6.21}$$

6.5 ▪ Localization and inflation

We have traded the EKF for a seemingly considerably cheaper filter meant to achieve similar performances. But this comes with some significant drawbacks. Fundamentally, one cannot hope to represent the full error covariance matrix of a complex high-dimensional system with only a few modes $m \ll n$, usually from a few dozen to a few hundred. This implies large *sampling errors*, meaning that the error covariance matrix is only sampled by a limited number of modes. This rank deficiency is accompanied by spurious correlations at long distances that strongly affect the filter performance. Even though the unstable degrees of freedom of dynamical systems that we wish to control with a filter are usually far fewer than the dimension of the system, they often still represent a nonnegligible fraction of the total degrees of freedom. Forecasting an ensemble of this size is usually not affordable.

The consequence of this issue is always the divergence of the filter. Hence, the EnKF is useful on condition that efficient fixes are applied. In other words, to make it a viable algorithm, one first needs to cope with the rank deficiency of the filter and with its manifestations, i.e., sampling errors.

Fortunately, there are clever tricks to overcome this major issue, known as *localization* and *inflation*, which explains, ultimately, the broad success of the EnKF in geosciences and engineering.

6.5.1 ▪ Localization

Localization relies on the idea that, for most geophysical systems, distant observables are weakly correlated. In other words, two distant parts of the system are almost independent, at least for short time scales. It is possible to exploit this relative independence and spatially localize the analysis [Houtekamer and Mitchell, 2001; Hamill et al., 2001; Evensen, 2003; Ott et al., 2004]. This has naturally been termed *localization*.

6.5.1.1 ▪ Domain localization

Broadly speaking, there are two main ways to implement this idea. The first one is called *domain localization*, or *local analysis*. Instead of performing a global analysis valid at any location in the domain, we perform a local analysis to update the local state variables using local observations. Typically, one would update a state variable at a location by assimilating only the observations within a fixed range of this point. If this range, the *localization length*, is too long, then the analysis becomes global, as before, and may fail because of the rank deficiency. On the other hand, if the localization length is too small, some short- to medium-range correlation might be neglected, resulting in a viable but less precise DA system.

One would then update all the state variables performing these local analyses. This may seem a formidable computational task, but all these analyses can be carried out in parallel (again we refer to Hunt et al. [2007] for an analysis of the numerical complexity). Besides, since the number of local observations is limited, the computation of the local gain is much faster than that of the global gain. All in all, such an approach is viable even on very large systems. Figure 6.2 is a schematic representation of a local update in the domain localization approach where only the surrounding observations of a grid cell to update are assimilated.

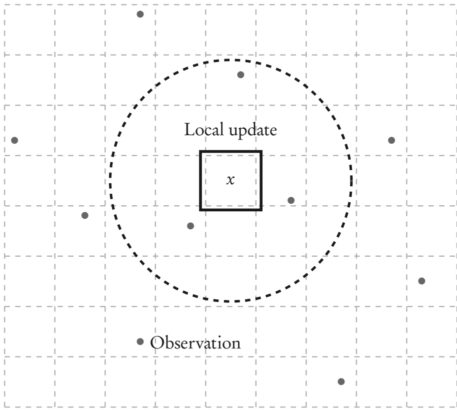


Figure 6.2. *Schematic representation of the local update of the variable defined over the framed grid cell with three surrounding observations (dots within the circle).*

The simplest strategy consists of selecting the observations within a disk around the center, $\mathbf{x}$, of the local analysis: the boxcar scheme. This is equivalent to making the variances and covariances of observations outside the domain go to infinity, i.e., replace, for each local analysis, $\mathbf{R}$ by a local $\mathbf{R}_\mathbf{x}$ restricted to the local observations.

Another analysis centered on a nearby point might share many observations but incorporate and exclude others, resulting in a discrete and not necessarily smooth transition between the two nearby analyses. To mitigate these transitions and reconstruction and obtain more consistent state and perturbation updates, it is possible to taper the inverse of the local observation error covariance matrix by a cutoff function that decreases with distance from $\mathbf{x}$. This way, the transition from one local domain to a closer one is smoother.

One possible cutoff function can be defined using the Gaspari–Cohn function [Gaspari and Cohn, 1999], which is a fifth-order piecewise rational function $G(r)$:

$$G(r) = \begin{cases} \text{if} \quad 0 \leq r < 1: & 1 - \frac{5}{3}r^2 + \frac{5}{8}r^3 + \frac{1}{2}r^4 - \frac{1}{4}r^5, \\ \text{if} \quad 1 \leq r < 2: & 4 - 5r + \frac{5}{3}r^2 + \frac{5}{8}r^3 - \frac{1}{2}r^4 + \frac{1}{12}r^5 - \frac{2}{3r}, \\ \text{if} \quad r \geq 2: & 0. \end{cases}$$

The cutoff function would be defined by $r \in \mathbb{R}^+ \mapsto G(r/c)$, where c is a length scale called the localization radius. It mimics a Gaussian distribution but vanishes beyond $r \geq 2c$, which is numerically efficient (G is compactly supported). This

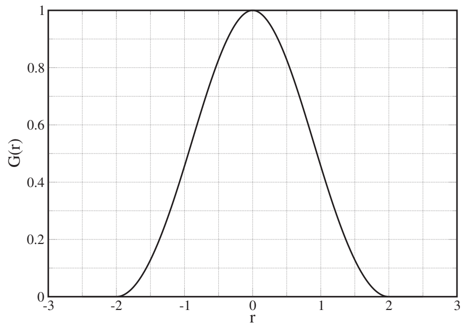


Figure 6.3. *Plot of the Gaspari–Cohn fifth-order piecewise rational function, which resembles a Gaussian distribution but has a compact support.*

approach is equivalent to tapering the ensemble members *and* the innovations by $r \mapsto \sqrt{G(r/c)}$, instead of modifying the local observation error covariances [Sakov and Bertino, 2011]. The Gaspari–Cohn cutoff function is represented in Figure 6.3.

6.5.1.2 ▪ Covariance localization

The second approach, called *covariance localization* or *Schur localization*, focuses on the forecast error covariance matrix. It is based on the remark that the forecast error covariance matrix $\mathbf{P}^{\mathrm{f}}$ is of low rank, at most $m-1$, and that this rank deficiency could be cured by filtering these empirical covariances. Physically, even though the empirical $\mathbf{P}^{\mathrm{f}}$ is probably a good approximation of the true error covariance matrix at short distances, the insufficient rank will induce long-range correlations. These correlations are *spurious*: they are likely to be nonphysical and are not present in the true forecast error covariance matrix. Hence, the idea is to regularize $\mathbf{P}^{\mathrm{f}}$ by smoothing out these long-range spurious correlations and, simultaneously, increasing the rank of $\mathbf{P}^{\mathrm{f}}$. To do so, one can regularize $\mathbf{P}^{\mathrm{f}}$ by multiplying it pointwise with a short-range predefined correlation matrix $\rho \in \mathbb{R}^{n\times n}$. The pointwise multiplication is called a Schur product and denoted by a $\circ$:

$$\left[\rho \circ \mathbf{P}^{\mathrm{f}}\right]_{i,j} = \left[\mathbf{P}^{\mathrm{f}}\right]_{i,j} [\rho]_{i,j}\,. \tag{6.22}$$

The Schur product theorem [Horn and Johnson, 2012] ensures that if ρ and $\mathbf{P}^{\mathrm{f}}$ are positive semidefinite (resp. positive definite), then the Schur product, $\rho\circ\mathbf{P}^{\mathrm{f}}$, is positive semidefinite (resp. positive definite). For instance, positive definite covariance matrices can be built with the Gaspari–Cohn function. This will ensure that the Schur product is positive semidefinite. As can be read from (6.22), the spurious correlations are leveled off by ρ if ρ is short ranged. This regularized $\rho \circ \mathbf{P}^{\mathrm{f}}$ will be used in the EnKF analysis as well as in the generation of the posterior ensemble of perturbations, as a replacement for $\mathbf{P}^{\mathrm{f}}$.

Figure 6.4 illustrates covariance localization. We consider a one-dimensional state space of $n = 200$ variables and a true error covariance matrix defined by $[\mathbf{P}]_{i,j} = e^{-|i-j|/L}$, where $i, j = 1,\dots,n$ and $L = 10$ (correlation length). An ensemble of $m = 20$

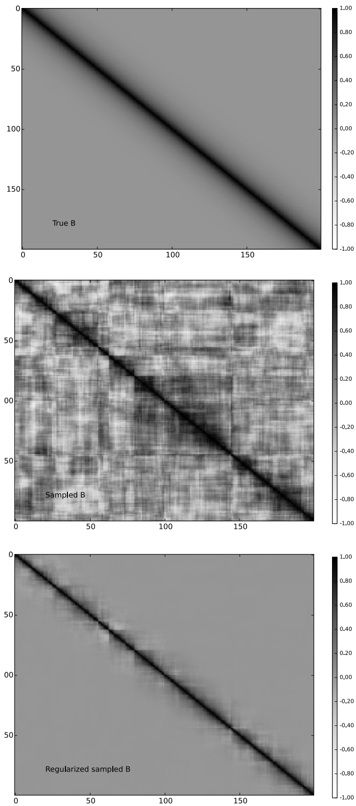


Figure 6.4. *Covariance localization. Upper panel: the true covariance matrix. Middle panel: the empirical covariance matrix from a* 20*-member ensemble. Bottom panel: the regularized empirical covariance matrix using Schur localization.*

members is generated from the exact Gaussian distribution of error covariance matrix $\mathbf{P}$. The empirical error covariance matrix is computed from this ensemble, and a Schur localization is applied to the empirical covariance matrix to form a regularized covariance matrix. The Gaspari–Cohn correlation function is used to generate the regularizing correlation function. For the sake of illustration, we take c of the Gaspari–Cohn function to be the correlation length of the true covariance matrix, which does not necessarily guarantee an optimal localization.

6.5.1.3 ▪ Implementation of localization

For the stochastic EnKF, the main task is to compute the Kalman gain. The regularization of the sample forecast error covariance matrix should happen at this stage. Because localization will transform the sample $\mathbf{P}^{\mathrm{f}}$ into a full-rank $\rho \circ \mathbf{P}^{\mathrm{f}}$, the computation of $(\rho \circ \mathbf{P}^{\mathrm{f}})\mathbf{H}^{\mathrm{T}}$, of the innovation statistics $\mathbf{R}+\mathbf{H}(\rho \circ \mathbf{P}^{\mathrm{f}})\mathbf{H}^{\mathrm{T}}$, and of its inverse in the Kalman gain may become computationally unfavorable for high-dimensional systems. One can take benefit from an often significantly smaller number of observations compared to the number of control variables to reduce the complexity of the gain computation. Furthermore, assume that the observations are in situ. At the simplest, assume that observation i is located within the grid cell with index $l(i)$. Then

$$\left[\left(\rho \circ \mathbf{P}^{\mathrm{f}}\right)\mathbf{H}^{\mathrm{T}}\right]_{k,i} = [\rho]_{k,l(i)}\left[\mathbf{P}^{\mathrm{f}}\right]_{k,l(i)}[\mathbf{H}]_{i,l(i)} = \left[\rho \circ \left(\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\right)\right]_{k,i},$$

which, in this case, explicitly extends the use of localization to observation space in direct correspondence with state space. Then, to estimate the Kalman gain, it often becomes much more efficient to compute instead

$$\left(\rho \circ \mathbf{P}^{\mathrm{f}}\right)\mathbf{H}^{\mathrm{T}} \longrightarrow \rho \circ (\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}), \qquad \mathbf{H}\left(\rho \circ \mathbf{P}^{\mathrm{f}}\right)\mathbf{H}^{\mathrm{T}} \longrightarrow \rho \circ (\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}).$$

Hence the Kalman gain could be estimated as [Houtekamer and Mitchell, 2001]

$$\mathbf{K} = \rho \circ (\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}})\left[\mathbf{R} + \rho \circ (\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}})\right]^{-1}.$$

The main drawback of localization is that it might also remove some physical and true long-distance correlations, thus inducing some physical imbalance in the analysis [Kepert, 2009; Greybush et al., 2011].

Covariance localization and localization by local analyses seem quite different. Indeed they are mathematically distinct, though they are both based on two sides of the same paradigm. However, it can be shown that they should yield very similar results if the innovation at each time step is not too strong [Sakov and Bertino, 2011], i.e., when the analysis remains close to the prior.

Because DA systems based on high-dimensional geophysical models very often yield inhomogeneous error fields, it would be useful to design localization schemes that are adaptive. They could depend on the local density of observations or on flow-dependent errors, etc. Preliminary statistical studies could be carried out to determine the optimal localization function for a given DA system in a given regime [Anderson and Lei, 2013]. The optimal localization function parameters could be determined on the fly from the ensemble only, using optimal linear filtering applied to the estimation of the regularized covariances [Ménétrier et al., 2015a,b].

The local ensemble transform Kalman filter (LETKF) has become an emblem of the success of the EnKF in conjunction with localization on significantly high-dimensional academic, but also operational, models. It was illustrated not only in several fields of the geosciences, such as atmosphere (including extraterrestrial atmospheres [Hoffman et al., 2010]), ocean, and solid earth sciences, but also in engineering, such as adaptive optics for future extra-large telescopes [Gray et al., 2014].

6.5.2 ▪ Inflation

Even when the analysis is made local, the error covariance matrices are still evaluated with an ensemble of limited size. This often leads to sampling errors and spurious correlations. With a proper localization scheme, they might be significantly reduced. No matter how small the residual errors are, they will accumulate and they will carry over to the next cycles of the sequential EnKF scheme. As a consequence, there is always a risk that the filter may ultimately diverge. One way around this is to inflate the error covariance matrix by a factor λ^2 slightly greater than 1, before or after the analysis [Pham et al., 1998; Anderson and Anderson, 1999]. For instance, after the analysis,

$$\mathbf{P}^{\mathrm{a}} \longrightarrow \lambda^2 \mathbf{P}^{\mathrm{a}}.$$

Another way to achieve this is to inflate the ensemble,

$$\mathbf{x}_i^{\mathrm{a}} \longrightarrow \overline{\mathbf{x}}^{\mathrm{a}} + \lambda\left(\mathbf{x}_i^{\mathrm{a}} - \overline{\mathbf{x}}^{\mathrm{a}}\right),$$

which can alternatively be enforced on the prior (forecast) ensemble. This type of inflation is called *multiplicative inflation*.

In a perfect but nonlinear model context, the multiplicative inflation is meant to compensate for sampling errors that are the indirect consequence of the nonlinearity in the ensemble forecast [Bocquet, 2011; Whitaker and Hamill, 2012; Bocquet et al., 2015]. In this case, it helps cure an intrinsic source of error of the EnKF scheme.

Yet, inflation can also compensate for the underestimation of the errors due to the presence of unaccounted-for model error, a source of error external to the EnKF scheme. In this context, an additive type of inflation can also be used, either by

$$\mathbf{P}^{\mathrm{f}} \longrightarrow \mathbf{P}^{\mathrm{f}} + \mathbf{Q}$$

or by adding noise to the ensemble members,

$$\mathbf{x}_i^{\mathrm{f}} \longrightarrow \mathbf{x}_i^{\mathrm{f}} + \epsilon_i \quad \text{with} \quad \mathrm{E}\left[\epsilon_i \epsilon_i^{\mathrm{T}}\right] = \mathbf{Q}.$$

The need and magnitude of a proper inflation is very dependent on the system, on its dynamics, on the observation network, or on the particular EnKF variant. It can also be made variable in space. For instance, in the absence of observation and of an update in a specific area, there is no need to locally update the ensemble, so that no inflation is required, in contrast to densely observed regions.

Perhaps even more than localization, inflation is a trick, even though conveniently simple. There has been development to estimate the proper inflation required by the EnKF, often depending on a ratio of information content between the observation and the prior. Because it can be diagnosed from its environment, inflation can be made adaptive (in time and/or in space). Some information on the adaptive inflation can even be carried over from one cycle to the next. Efficient adaptive inflation methods have been proposed by, e.g., Wang and Bishop [2003], Anderson [2007], Li et al. [2009], Zheng [2009], Brankart et al. [2010], Bocquet [2011], Miyoshi [2011], Bocquet and Sakov [2012], Liang et al. [2012], and Ying and Zhang [2015] to account for sampling errors and possibly extrinsic model error.

6.6 ▪ Numerical illustrations with the Lorenz-95 model

The Lorenz-95 low-order model is a one-dimensional toy model introduced by the famous meteorologist Edward Lorenz in 1995 (later published in 1996 in the ECMWF's seminar proceedings and in 1998 as Lorenz and Emmanuel [1998]). It represents a midlatitude zonal circle of the global atmosphere. It has $n = 40$ variables, $\{x_i\}_{i=1,\dots,n}$. Its dynamics are given by the following set of ODEs:

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = (x_{i+1} - x_{i-2})x_{i-1} - x_i + F$$

for $i = 1, \dots, n$. The domain on which the 40 variables are defined is circle-like. Hence, it is assumed periodic, and the following definitions are enforced: $x_0 = x_{40}$, $x_{-1} = x_{39}$, $x_{41} = x_1$. The term $(x_{i+1} - x_{i-2})x_{i-1}$ is a nonlinear convective term. It preserves the energy $\sum_{i=1}^{n} x_i^2$. The term $-x_i$ is a dampening term, while F is an energy injection/extraction term depending on the sign of the variables. Here F is chosen to be 8. This makes the dynamics of this model chaotic. With this choice, the model has 13 positive Lyapunov exponents. This implies that, out of the 40 degrees of freedom of this model, 13 directions lead to growing modes. Moreover, one Lyapunov exponent is zero, which corresponds to a neutral mode. In practice and for short-term forecasts, roughly 13 directions (which change with time) out of the 40 in model state space make a small perturbation grow under the action of the model. A time step of 0.05 is meant to represent a time interval of 6 hours in the real atmosphere. Here, the model is integrated using a fourth-order Runge–Kutta scheme with a time step of $\delta t = 0.05$ because the scheme is precise and one can afford the numerical cost in this low-order context.

Figure 6.5 displays a trajectory of the model state. The model is characterized by about eight nonlinear waves that interact (this number depends on F). As can be seen, it seems difficult to predict the behavior of these waves except that they have the tendency to drift eastward in spite of a phase velocity shifting the peaks and lows westward. These waves are meant to represent large-scale Rossby waves.

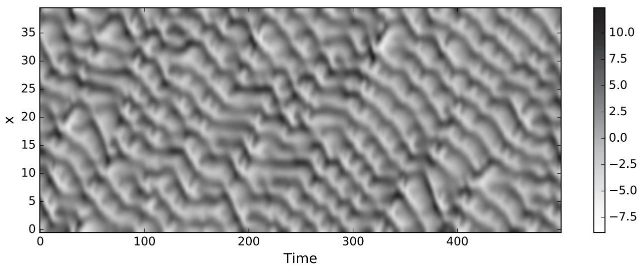


Figure 6.5. *Trajectory of a state of the Lorenz-95 model. The x coordinate represents time in units of $\Delta t = 0.05$. The y coordinate represents the 40 state variables.*

A twin experiment is conducted (see Section 6.3.4 for a definition). The truth is represented by a free model run, meant to be tracked by the DA system. The system is assumed to be fully observed ($p = 40$) every Δt, so that $\mathbf{H}_k \equiv \mathbf{I}_{40}$, with the observation error covariance matrix $\mathbf{R}_k \equiv \mathbf{I}_{40}$. The time interval between observational updates is set to $\Delta t = 0.05$, meant to be representative of a DA cycle of global meteorological models (6 hours), as we previously discussed. With such a value for Δt, the DA system is considered to be weakly nonlinear, yielding statistics of the errors weakly diverging from Gaussianity. The related synthetic observations are generated from the truth and perturbed by a Gaussian noise with the same distribution as the observation error prior. In this experiment, the performance of a scheme is measured by the temporal mean of a root mean square difference between a state estimate ($\mathbf{x}^{\mathrm{a}}$) and the truth ($\mathbf{x}^{\mathrm{t}}$). Typically, one averages over time the following analysis root mean square error (RMSE):

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i^{\mathrm{a}}-x_i^{\mathrm{t}}\right)^2}. \tag{6.23}$$

All DA runs will extend over 10^5 cycles, after a burn-in period of 5×10^3 cycles, which guarantees satisfying convergence of the error statistics, due to ergodicity.

We vary the ensemble size from $m=5$ to $m=50$ and compare the performance in terms of RMSE of

- the EnKF without inflation and without localization,
- the EnKF with optimally tuned inflation and without localization,
- the EnKF without inflation and with optimally tuned localization, and
- the EnKF with optimally tuned inflation and with optimally tuned localization.

Optimally tuned means that the selected parameterization corresponds to the best RMSE. The EnKF variant is chosen to be the ETKF. The average RMSEs of the analyses over the long run are displayed in Figure 6.6.

If one agrees that the application of the EnKF to this low-order model captures several of the difficulties of realistic DA, it becomes clear from these numerical results that localization and inflation are indeed mandatory ingredients of a satisfying implementation of the EnKF. In particular, localization cannot be avoided with an ensemble size smaller than about 15, which is about the size of the unstable and neutral model subspace (equal to the number of nonnegative Lyapunov exponents).

6.7 ▪ Other important flavors of the EnKF

6.7.1 ▪ Variants of the EnKF

6.7.1.1 ▪ Ensemble adjustment Kalman filter

The ensemble adjustment Kalman filter (EAKF) [Anderson, 2001] is an EnSRKF where the ensemble update is computed by left-multiplication of the prior perturbations. Because the EAKF was developed quite early, the update formula's expression is not as compact as in (6.18) or (6.20), but the crucial ingredients are in it.

6.7.1.2 ▪ Serial EnKF

The serial EnKF is an EnKF where the observations are assimilated serially, i.e., one by one. Consider the scalar observation y, of error variance r, and the related observation operator $\mathbf{h}:\mathbb{R}^n\mapsto\mathbb{R}$. Then the analysis update is $\mathbf{x}^{\mathrm{a}}=\mathbf{x}^{\mathrm{f}}+\mathbf{K}\left[y-h(\mathbf{x}^{\mathrm{f}})\right]$, where the simplified Kalman gain of size $n\times 1$ is a vector that reads

$$\mathbf{K}=\mathbf{P}^{\mathrm{f}}\mathbf{h}^{\mathrm{T}}/(r+\mathbf{h}\mathbf{P}^{\mathrm{f}}\mathbf{h}^{\mathrm{T}}),$$

where the tangent linear, $\mathbf{h}$, of the observation operator is a row vector in $\mathbb{R}^n$. The modified gain, $\widetilde{\mathbf{K}}$, of the ensemble square root approach simplifies to

$$\widetilde{\mathbf{K}}=\frac{1}{1+1/\sqrt{1+r^{-1}\mathbf{h}\mathbf{P}^{\mathrm{f}}\mathbf{h}^{\mathrm{T}}}}\mathbf{K}.$$

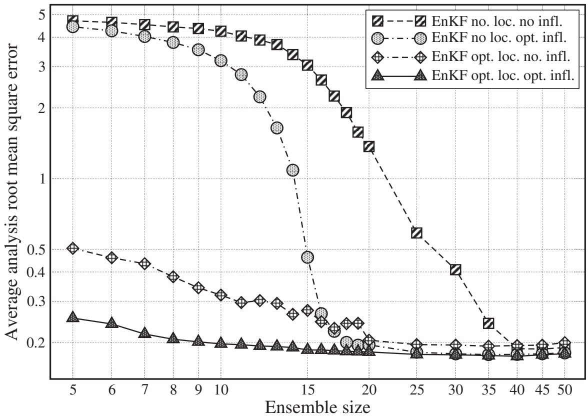


Figure 6.6. *Average analysis RMSE for a deterministic EnKF (ETKF) without localization and without inflation (boxes); without localization and with optimally tuned inflation (disks); with optimally tuned localization and no inflation (diamonds); with optimally tuned localization and with optimally tuned inflation (triangles).*

The correction due to the ensemble square root (Potter) scheme is conveniently reduced to a scalar.

If the ensemble is updated after each observation assimilation, then the serial EnKF should be mathematically equivalent to the *batch* EnKF, i.e., based on matrix update and the assimilation of observation vectors. Yet, this is only true for diagonal error covariance matrices.

Localization in the serial EnKF is elegantly simple. Focusing on scalar variables as the serial EnKF does, the covariance between the state variable x_i in grid cell i and the observation y, assumed to be local and located in grid cell j, is a scalar (a regression coefficient), which must be multiplied by the tapering correlation function $\rho_{i,j}$ for regularization. The Kalman gain of the stochastic EnKF is changed using

$$\mathbf{P}^{\mathrm{f}}\mathbf{h}^{\mathrm{T}} \longrightarrow \rho \circ \mathbf{P}^{\mathrm{f}}\mathbf{h}^{\mathrm{T}} \simeq \rho \circ \left(\mathbf{P}^{\mathrm{f}}\mathbf{h}^{\mathrm{T}}\right)$$

and using the approximation discussed in Section 6.5.1. Remarkably, $\mathbf{h}\mathbf{P}^{\mathrm{f}}\mathbf{h}^{\mathrm{T}}$ is unchanged, since $\rho_{i,i} = 1$. Moreover, the correction of the modified gain, $\widetilde{\mathbf{K}}$, of the ensemble square root approach is also unchanged for the same reason.

Finally, note that the equivalence between serial and batch assimilation in the EnKF is broken when localization is used.

6.7.1.3 ▪ DEnKF

We use the term *deterministic EnKF* to generally name the ensemble square root update scheme that avoids the need of stochastic sampling. Sakov and Oke [2008b] proposed a simplification of the EnSRKF, which they called the *deterministic EnKf or DEnKF*, because it astutely mimics the stochastic EnKF while being deterministic.

In the DEnKF, the state update is the same as in all previously discussed EnKF schemes. The perturbation update scheme, however, is an approximate but elegantly simple square root update scheme. Let us consider (6.21). Using the natural order on the positive definite matrices, we have

$$\mathbf{I}_p + \mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1} \geq \mathbf{I}_p,$$

so that

$$\left(\mathbf{I}_p + \left[\mathbf{I}_p + \mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\right]^{-\frac{1}{2}}\right)^{-1} \geq \frac{1}{2}\mathbf{I}_p. \tag{6.24}$$

The gain defined by $\widehat{\mathbf{K}} \equiv \frac{1}{2}\mathbf{K}$ could be used for the perturbation update as an approximation of the theoretical optimal gain $\mathbf{K}$. It is clear from (6.24) that the weaker this assimilation, i.e., $\mathbf{R} \gg \mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}$, the more precise the approximation. Hence, the update formula of the DEnKF is the square root–free

$$\mathbf{X}_{\mathrm{a}} \simeq \left(\mathbf{I}_n - \frac{1}{2}\mathbf{K}\mathbf{H}\right)\mathbf{X}_{\mathrm{f}}. \tag{6.25}$$

This scheme avoids the need to compute any square root, making it very similar to the original stochastic EnKF but with a very simple modification. It is computationally very competitive. Using the approximate gain, $\widehat{\mathbf{K}}$, instead of the deterministic gain, $\widetilde{\mathbf{K}}$, defined by (6.21) impacts the posterior error covariances, making the scheme a suboptimal but cautious approximation; it can be shown [Sakov and Oke, 2008b] that the expected posterior error covariances of the DEnKF are

$$\widehat{\mathbf{P}}_{\mathrm{a}} = (\mathbf{I}_n - \mathbf{K}\mathbf{H})\,\mathbf{P}^{\mathrm{f}} + \frac{1}{4}\mathbf{K}\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\mathbf{K}^{\mathrm{T}}.$$

The first term, $(\mathbf{I}_n - \mathbf{K}\mathbf{H})\,\mathbf{P}^{\mathrm{f}}$, is the EnKF optimal $\mathbf{P}^{\mathrm{a}}$. The extra term, which adds to the posterior covariances, is positive semidefinite. Hence $\widehat{\mathbf{P}}_{\mathrm{a}} \geq \mathbf{P}^{\mathrm{a}}$. We conclude that choosing $\widehat{\mathbf{K}}$ instead of $\widetilde{\mathbf{K}}$ impacts the posterior error covariances, making the scheme suboptimal but a safe approximation (the main risk being to underestimate the errors).

The state and perturbations scheme can even be reformulated as

$$\mathbf{x}_i^{\mathrm{a}} = \mathbf{x}_i^{\mathrm{f}} + \mathbf{K}\left[\mathbf{y} - \mathscr{H}\left(\frac{\mathbf{x}_i^{\mathrm{f}} + \overline{\mathbf{x}}^{\mathrm{f}}}{2}\right)\right],$$

to be compared to the stochastic EnKF update.

6.7.1.4 ▪ Mollified EnKF

One of the conceptual limitations of DA is the standpoint adopted earlier of the discretized representation of fields. This is obviously justified by the need for numerical implementation of the models and control variables. However, time continuous representations are often more satisfying from the conceptual standpoint, by ignoring implementation details. One interesting standpoint on the EnKF update step is offered by the mollified EnKF [Bergemann and Reich, 2010]. Instead of the algebraic update of the ensemble members as in the stochastic EnKF, or of the mean and the perturbations in the DEnKF, it is tempting to represent the update as the integration of ODEs (or possibly PDEs) from the prior to the posterior over a fictitious interval of time [Simon, 2006; Bergemann and Reich, 2010]. One would update the ensemble members via

$$\frac{\mathrm{d}\mathbf{x}_i(s)}{\mathrm{d}s} = \frac{1}{2}\mathbf{P}(s)\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\left[2\mathbf{y} - \mathbf{H}\left\{\mathbf{x}_i(s) + \overline{\mathbf{x}}(s)\right\}\right], \tag{6.26}$$

where $\mathbf{x}_i$ is integrated from $s = 0$ to $s = 1$, with the initial condition $\mathbf{x}_i(0) = \mathbf{x}_i^{\mathrm{f}}$. Then, it can be shown that $\mathbf{x}_i(1) = \mathbf{x}_i^{\mathrm{a}}$. We note the clear resemblance with the DEnKF. This shows that the DEnKF update represents a linearized integration of (6.26), i.e., maintaining the value of $\mathbf{P}(s)$ and $\overline{\mathbf{x}}(s)$ at $\mathbf{P}^{\mathrm{f}}$ and $\overline{\mathbf{x}}^{\mathrm{f}}$ over the interval.

6.7.1.5 ▪ Assimilation in the unstable subspace

Let us consider a minimalist DA problem (already encountered, in various forms, in the first three chapters) with a one-variable, perfect, linear model $x_{k+1} = \alpha x_k$, with k the time index. If $\alpha^2 > 1$, the model is unstable, and if $\alpha^2 < 1$ it is stable. Let us denote by b_k the forecast/prior error variance, r the static observation error variance, and a_k the error analysis variance. Sequential DA implies the following recursions for the variances:

$$a_k^{-1} = b_k^{-1} + r^{-1} \quad \text{and} \quad b_{k+1} = \alpha^2 a_k,$$

whose asymptotic solution ($a_k \to a_\infty$) is

$$a_\infty = 0 \text{ if } \alpha^2 < 1 \quad \text{and} \quad a_\infty = (1 - 1/\alpha^2)r \text{ if } \alpha^2 \geq 1.$$

Very roughly, it tells us that only the growing modes need to be controlled, i.e., that DA should be targeted at preventing errors to increase indefinitely in the space generated by the growing modes. This paradigm is called *assimilation in the unstable space* or *AUS* [see Palatella et al., 2013, and references therein]. It is tempting to identify the unstable subspace with the time-dependent space generated by the Lyapunov vectors with nonnegative exponents, which, strictly speaking, is the unstable and neutral subspace. Applied to the KF and possibly the EnKF, it is intuitively known that the error covariance matrix tends to collapse to this unstable and neutral subspace [Trevisan and Palatella, 2011]. This can be made rigorous in the linear model Gaussian statistics case. The generalization of the paradigm to nonlinear dynamical systems is more speculative, but Ng et al. [2011] and Palatella and Trevisan [2015] put forward some enlightening arguments about it. A connection between the AUS paradigm and a justification of multiplicative inflation was established in Bocquet et al. [2015].

6.7.2 ▪ Variational analysis

We recall one of the key elementary results of DA seen in Chapters 1 and 3, namely that when the observation model is linear, the BLUE analysis is equivalent to a 3D variational problem (3D-Var). That is to say, evaluating the matrix formula

$$\mathbf{x}^{\mathrm{a}} = \mathbf{x}^{\mathrm{f}} + \mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{R} + \mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\right)^{-1}\left(\mathbf{y} - \mathbf{H}\mathbf{x}^{\mathrm{f}}\right)$$

is equivalent to solving the minimization problem

$$\mathbf{x}^{\mathrm{a}} = \underset{\mathbf{x}}{\operatorname{argmin}}\, \mathscr{L}(\mathbf{x}) \quad \text{with} \quad \mathscr{L}(\mathbf{x}) = \frac{1}{2}\|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}}^2 + \frac{1}{2}\left\|\mathbf{x} - \overline{\mathbf{x}}^{\mathrm{f}}\right\|_{\mathbf{P}^{\mathrm{f}}}^2,$$

where $||\mathbf{x}||^2_{\mathbf{A}} = \mathbf{x}^{\mathrm{T}}\mathbf{A}^{-1}\mathbf{x}$ for any symmetric positive definite matrix $\mathbf{A}$. We assume momentarily that $\mathbf{P}^{\mathrm{f}}$ is full rank so that it is invertible and positive-definite. This equivalence can be fruitful with high-dimensional systems, where tools of numerical optimization can be used in place of linear algebra. This equivalence is also of theoretical interest because it enables an elegant generalization of the BLUE update to the case where the observation operator is nonlinear. Simply put, the cost function is now replaced with

$$\mathscr{L}_{\mathrm{NL}}(\mathbf{x}) = \frac{1}{2}||\mathbf{y} - \mathscr{H}(\mathbf{x})||^2_{\mathbf{R}} + \frac{1}{2}\left\|\mathbf{x} - \overline{\mathbf{x}}^{\mathrm{f}}\right\|^2_{\mathbf{P}^{\mathrm{f}}},$$

where $\mathscr{H}$ is nonlinear.

6.7.2.1 ▪ The maximum likelihood ensemble filter

This equivalence was put to use in the maximum likelihood ensemble filter (MLEF) introduced by Zupanski [2005]. To describe the MLEF in a framework we have already detailed, let us formulate it in terms of the ETKF (following Bocquet and Sakov [2013]). Hence, we write the analysis of the MLEF in ensemble subspace closely following Section 6.4. However, we must first write the corresponding cost function. Recall that the state vector is parameterized in terms of the vector of coefficients $\mathbf{w}$ in $\mathbb{R}^m$, $\mathbf{x} = \overline{\mathbf{x}}^{\mathrm{f}} + \mathbf{X}\mathbf{w}$. The reduced cost function is denoted by

$$\mathscr{J}_{\mathrm{NL}}(\mathbf{w}) = \mathscr{L}_{\mathrm{NL}}(\overline{\mathbf{x}}^{\mathrm{f}} + \mathbf{X}\mathbf{w}).$$

Its first term can easily be written in the reduced ensemble subspace:

$$\mathscr{J}^{\mathrm{o}}_{\mathrm{NL}}(\mathbf{w}) = \frac{1}{2}\left\|\mathbf{y} - \mathscr{H}(\overline{\mathbf{x}}^{\mathrm{f}} + \mathbf{X}\mathbf{w})\right\|^2_{\mathbf{R}}.$$

To proceed with the background term, $\mathscr{J}^{\mathrm{b}}(\mathbf{w})$, of the cost function, we first have to explain what the inverse of $\mathbf{P}^{\mathrm{f}} = \mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}$ of incomplete rank is whenever $m \leq n$. Because for most EnKFs, the analysis is entirely set in ensemble subspace as we repeatedly pointed out, the inverse, $\mathbf{P}^{\mathrm{f}} = \mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}$, must be the Moore–Penrose inverse [Golub and van Loan, 2013] of $\mathbf{P}^{\mathrm{f}}$, denoted by $\mathbf{P}^{\dagger}_{\mathrm{f}}$. Indeed, it is defined in the range of $\mathbf{P}^{\mathrm{f}}$. It is even more direct to introduce the SVD of the perturbation matrix $\mathbf{X} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathrm{T}}$, where $\boldsymbol{\Sigma} > 0$ is the diagonal matrix of the $m' \leq m$ positive singular values, $\mathbf{V}$ is of size $m \times m'$ such that $\mathbf{V}^{\mathrm{T}}\mathbf{V} = \mathbf{I}_{m'}$, and $\mathbf{U}$ is of size $n \times m'$ such that $\mathbf{U}^{\mathrm{T}}\mathbf{U} = \mathbf{I}_{m'}$. Note that $\mathbf{P}^{\dagger}_{\mathrm{f}} = \mathbf{U}\boldsymbol{\Sigma}^{-2}\mathbf{U}^{\mathrm{T}}$. Then we have

$$\begin{aligned}\mathscr{J}^{\mathrm{b}}(\mathbf{w}) &= \frac{1}{2}||\mathbf{X}_{\mathrm{f}}\mathbf{w}||^2_{\mathbf{P}^{\mathrm{f}}} = \frac{1}{2}\mathbf{w}^{\mathrm{T}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}\mathbf{P}^{\dagger}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}\mathbf{w} = \frac{1}{2}\mathbf{w}^{\mathrm{T}}\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^{\mathrm{T}}\left(\mathbf{U}\boldsymbol{\Sigma}^2\mathbf{U}^{\mathrm{T}}\right)^{\dagger}\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathrm{T}}\mathbf{w} \\ &= \frac{1}{2}\mathbf{w}^{\mathrm{T}}\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^{\mathrm{T}}\mathbf{U}\boldsymbol{\Sigma}^{-2}\mathbf{U}^{\mathrm{T}}\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathrm{T}}\mathbf{w} = \frac{1}{2}\mathbf{w}^{\mathrm{T}}\mathbf{V}\mathbf{V}^{\mathrm{T}}\mathbf{w}.\end{aligned}$$

As was mentioned earlier, there is a freedom in $\mathbf{w}$ that makes the solution of the minimization problem degenerate. Clearly $\mathscr{J}^{\mathrm{o}}_{\mathrm{NL}}(\mathbf{w})$ is unchanged if $\mathbf{w}$ is shifted by $\lambda\mathbf{1}$. So is $\mathscr{J}^{\mathrm{b}}(\mathbf{w})$ because $\mathbf{1}$ is in the null space of $\mathbf{V}$ since $\mathbf{X}\mathbf{1} = \mathbf{0}$. One way to lift the degeneracy of the variational problem is to add a *gauge-fixing* term that will constrain the solution in the null space of $\mathbf{X}$, or $\mathbf{V}$. In practice, one can add

$$\mathscr{J}^{\mathrm{g}}(\mathbf{w}) = \frac{1}{2}\mathbf{w}^{\mathrm{T}}(\mathbf{I}_m - \mathbf{V}\mathbf{V}^{\mathrm{T}})\mathbf{w}$$

to the cost function $\mathscr{J}_{\rm NL}$ to obtain the regularized cost function

$$\mathscr{J}(\mathbf{w}) = \frac{1}{2}\left\|\mathbf{y} - \mathscr{H}(\overline{\mathbf{x}}^{\rm f} + \mathbf{X}\mathbf{w})\right\|^2_{\mathbf{R}} + \frac{1}{2}\|\mathbf{w}\|^2,$$

where $\|\mathbf{w}\|^2 = \mathbf{w}^{\rm T}\mathbf{w}$. The cost function still has the same minimum but it is achieved at a non-degenerate $\mathbf{w}^\star$ such that $\left(\mathbf{I}_m - \mathbf{V}\mathbf{V}^{\rm T}\right)\mathbf{w}^\star = 0$. This is the cost function of the ETKF [Hunt et al., 2007] but with a nonlinear observation operator.

The update step of this EnKF can now be seen as a nonlinear variational problem, which can be solved using a variety of iterative methods, such as a Gauss–Newton method, a quasi-Newton method, of a Levenberg–Marquardt method [Nocedal and Wright, 2006]. For instance, with the Gauss–Newton method, we would define the iterate, the gradient, and an approximation of the Hessian as

$$\begin{aligned}
\mathbf{x}^{(j)} &= \overline{\mathbf{x}}^{\rm f} + \mathbf{X}_{\rm f}\mathbf{w}^{(j)},\\
\nabla\mathscr{J}_{(j)} &= -\mathbf{Y}_{(j)}^{\rm T}\mathbf{R}^{-1}\left(\mathbf{y} - \mathscr{H}(\mathbf{x}^{(j)})\right) + \mathbf{w}^{(j)},\\
\mathrm{H}_{(j)} &= \mathbf{I}_m + \mathbf{Y}_{(j)}^{\rm T}\mathbf{R}^{-1}\mathbf{Y}_{(j)},
\end{aligned}$$

respectively. The Gauss–Newton iterations are indexed by j and given by

$$\mathbf{w}^{(j+1)} = \mathbf{w}^{(j)} - \mathrm{H}_{(j)}^{-1}\nabla\mathscr{J}_{(j)}.$$

The scheme is iterated until a satisfying convergence is reached, for instance when the norm of $\left\|\mathbf{w}^{(j+1)} - \mathbf{w}^{(j)}\right\|$ crosses below a given threshold. The vector $\mathbf{Y}_{(j)}$ is defined as the image of the initial ensemble perturbations $\mathbf{X}_{\rm f}$ through the tangent linear model of the observation operator computed at $\mathbf{x}^{(j)}$ by $\mathbf{Y}_{(j)} = \mathbf{H}_{|\mathbf{x}^{(j)}}\mathbf{X}_{\rm f}$. Following Sakov et al. [2012], there are at least two ways to compute these sensitivities. One explicitly mimics the tangent linear by a downscaling of the perturbations by ε such that $0 < \varepsilon \ll 1$ before application of the full nonlinear operator $\mathscr{H}$ followed by an upscaling by ε^{-1}. The operation reads

$$\mathbf{Y}_{(j)} \approx \frac{1}{\varepsilon}\mathscr{H}\left(\mathbf{x}^{(j)}\mathbf{1}^{\rm T} + \varepsilon\mathbf{X}_{\rm f}\right)\left(\mathbf{I}_m - \frac{\mathbf{1}\mathbf{1}^{\rm T}}{m}\right).$$

Note that ε accounts for a normalization factor of $\sqrt{m-1}$. The second way consists of avoiding resizing the perturbations because this implies applying the observation operator to the ensemble an extra time. Instead of downscaling the perturbations, we can (i) generate transformed perturbations by applying the right-multiplication operator

$$\mathbf{T} = \left[\mathbf{I}_m + \mathbf{Y}_{(j)}^{\rm T}\mathbf{R}^{-1}\mathbf{Y}_{(j)}\right]^{-\frac{1}{2}},$$

(ii) build a new ensemble from these transformed perturbations around $\mathbf{x}^{(j)}$, (iii) apply $\mathscr{H}$ to this ensemble, and finally (iv) rotate back the new perturbations around $\mathbf{x}^{(j+1)}$ by applying $\mathbf{T}^{-1}$. Through the $\mathbf{T}$-transformation the second scheme also ensures a resizing of the perturbations where $\mathscr{H}$ is in a close-to-linear regime. However, as opposed to the first scheme, the last propagation of the perturbation can be used to directly estimate the final approximation of the Hessian and the final updated set of perturbations, which can be numerically efficient.

Algorithm 6.5 Pseudocode for a complete cycle of the MLEF, as a variant in ensemble subspace following Zupanski [2005], Carrassi et al. [2009], Sakov et al. [2012], and Bocquet and Sakov [2013]

Require: Observation operator $\mathcal{H}$ at current time; algorithm parameters: e, $j_{\max}$, ε; $\mathbf{E}$, the prior ensemble; $\mathbf{y}$, the observation at current time; $\mathbf{U}$, an orthogonal matrix in $\mathbb{R}^{m\times m}$ satisfying $\mathbf{U1}=\mathbf{1}$; $\mathcal{M}$, the model resolvent from current time to the next analysis time.

$\overline{\mathbf{x}}=\mathbf{E1}/m$
$\mathbf{X}=\left(\mathbf{E}-\overline{\mathbf{x}}\mathbf{1}^{\mathrm{T}}\right)/\sqrt{m-1}$
$\mathbf{T}=\mathbf{I}_m$
$j=0$, $\mathbf{w}=\mathbf{0}$
repeat
  $\mathbf{x}=\overline{\mathbf{x}}+\mathbf{Xw}$
  Bundle: $\mathbf{E}=\mathbf{x1}^{\mathrm{T}}+\varepsilon\mathbf{X}$
  Transform: $\mathbf{E}=\mathbf{x1}^{\mathrm{T}}+\sqrt{m-1}\mathbf{XT}$
  $\mathbf{Z}=\mathcal{H}(\mathbf{E})$
  $\overline{\mathbf{y}}=\mathbf{Z1}/m$
  Bundle: $\mathbf{Y}=\left(\mathbf{Z}-\overline{\mathbf{y}}\mathbf{1}^{\mathrm{T}}\right)/\varepsilon$
  Transform: $\mathbf{Y}=\left(\mathbf{Z}-\overline{\mathbf{y}}\mathbf{1}^{\mathrm{T}}\right)\mathbf{T}^{-1}/\sqrt{m-1}$
  $\nabla\mathcal{J}=\mathbf{w}-\mathbf{Y}^{\mathrm{T}}\mathbf{R}^{-1}(\mathbf{y}-\overline{\mathbf{y}})$
  $\mathrm{H}=\mathbf{I}_m+\mathbf{Y}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{Y}$
  Solve $\mathrm{H}\Delta\mathbf{w}=\nabla\mathcal{J}$
  $\mathbf{w}:=\mathbf{w}-\Delta\mathbf{w}$
  Transform: $\mathbf{T}=\mathrm{H}^{-\frac{1}{2}}$
  $j:=j+1$
until $||\Delta\mathbf{w}||\le e$ **or** $j\ge j_{\max}$
Bundle: $\mathbf{T}=\mathrm{H}^{-\frac{1}{2}}$
$\mathbf{E}=\mathbf{x1}^{\mathrm{T}}+\sqrt{m-1}\mathbf{XTU}$
$\mathbf{E}=\mathcal{M}(\mathbf{E})$

Note that each iteration amounts to solving an inner loop problem with the quadratic cost function

$$\mathcal{J}^{(j)}(\mathbf{w})=\frac{1}{2}\left\|\mathbf{y}-\mathcal{H}(\mathbf{x}^{(j)})-\mathbf{Y}_{(j)}(\mathbf{w}-\mathbf{w}^{(j)})\right\|^2_{\mathbf{R}}+\frac{1}{2}\left\|\mathbf{w}-\mathbf{w}^{(j)}\right\|^2.$$

The update of the perturbations follows that of the ETKF, i.e., equation (6.17). A full cycle of the algorithm is given in Algorithm 6.5 as a pseudocode. Either the bundle (finite differences with the ε-rescaling) scheme or the transform (using $\mathbf{T}$) scheme is needed to compute the sensitivities. Both are indicated in the algorithm. Inflation, possibly localization, should be added to the scheme to make it functional. In summary, the MLEF is an EnKF scheme that can use nonlinear observation operators in a consistent way using a variational analysis.

6.7.2.2 ▪ Numerical illustration

The (bundle) MLEF as implemented by Algorithm 6.5 is tested against the EnKF (ETKF implementation) using a setup similar to that of the Lorenz-95 model. To exhibit a difference of performance, the observation operator has been chosen to be nonlinear. Each of the 40 variables is observed with the nonlinear observation operator

$$\mathscr{H}(\mathbf{x}) = \frac{\mathbf{x}}{2}\left\{1 + \left(\frac{|\mathbf{x}|}{10}\right)^{\gamma-1}\right\}, \tag{6.27}$$

where $|\mathbf{x}|$ is the componentwise absolute value of $\mathbf{x}$. The second nonlinear term in the brackets is meant to be of the order of magnitude of the first linear term to avoid numerical overflow. Obviously, γ tunes the nonlinearity of the observation operator, with $\gamma = 1$ corresponding to the linear case $\mathscr{H}(\mathbf{x}) = \mathbf{x}$. The prior observation error is chosen to be $\mathbf{R} \equiv \mathbf{I}_p$. The ensemble size is $m = 20$, which, in this context, makes localization unnecessary. For both the ETKF and the MLEF, the need for inflation is addressed either by using the Bayesian hierarchical scheme for the EnKF, known as the finite-size EnKF or EnKF-N, which we shall describe later, or by optimally tuning a uniform inflation (which comes with a significant numerical cost).

The MLEF is expected to offer strong performance in the first cycles of the DA scheme when the spread of the ensemble is large enough, over a span where the tangent linear observation model is not a good approximation. To measure this performance, the length of the DA run is set to 10^2 cycles and these runs are repeated 10^3 times, over which a mean analysis RMSE is computed. The spread of the initial ensemble is chosen to be 3, i.e., roughly the climatological variability of a single Lorenz-95 variable.

The overall performances of the schemes are computed as a function of γ, i.e., the nonlinearity strength of the observation operator, and reported in Figure 6.7.

Since the model is fully observed, an efficient DA scheme should have an RMSE smaller than the prior observation error, i.e., 1, which is indeed the case for all RMSEs computed in these experiments. As γ departs from $\gamma = 1$, the performance of

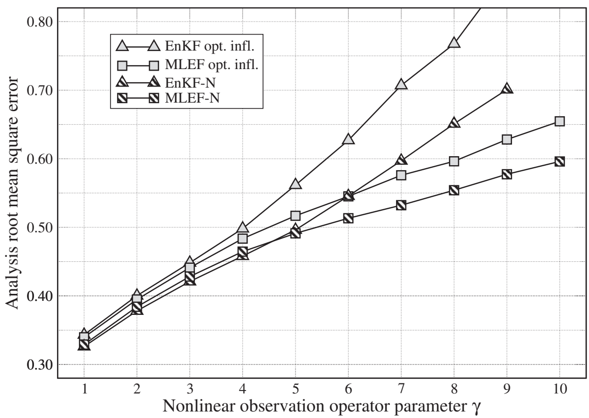


Figure 6.7. *Average analysis RMSE of a deterministic EnKF (ETKF) and of the MLEF with the Lorenz-95 model and the nonlinear observation operator equation (6.27), as a function of γ the nonlinearity strength in the observation operator. For each RMSE, 10^3 DA experiments are run over 10^2 cycles. The final RMSE is the mean over those 10^3 experiments. The EnKF-N and MLEF-N are hierarchical filter counterparts to the EnKF and MLEF that will be discussed in Section 6.7.3.*

both filters consistently degrades. Beyond $\gamma > 9$, the EnKF diverges from the truth. However, the MLEF better handles the model nonlinearity, especially beyond $\gamma > 4$ and the gap between the EnKF and the MLEF increases with γ. We also note that the EnKF-N and MLEF-N offer better performance than an optimal but uniformly tuned EnKF. This is explained by the fact that the finite-size scheme is adaptive and adjusts the inflation as the ensemble spread decreases.

For much longer runs, the RMSE gap between the EnKF and the MLEF decreases significantly. Indeed, in the permanent regime of these experiments, the spread of the ensemble is significantly smaller than 1 and the system is closer to linearity, a regime where both the EnKF and the MLEF perform equally well. The gap between the finite-size and the optimally tuned ensemble filters also decreases since the adaptation feature is not necessarily important in a permanent regime.

The MLEF will be generalized later by including not only the observation operator but also the forecast model in the analysis over a 4D DA window, leading to the IEnKF and IEnKS.

6.7.2.3 ▪ The α control variable trick

The analysis using covariance localization with a localization matrix $\boldsymbol{\rho}$ can be equivalently formulated in terms of ensemble coefficients similar to the $\mathbf{w}$ of the ETKF, but that have been made space dependent. Instead of a vector $\mathbf{w}$ of size m that parameterizes the state vector $\mathbf{x}$,

$$\mathbf{x} = \overline{\mathbf{x}} + \sum_{i=1}^{m} w_i \frac{\mathbf{x}_i - \overline{\mathbf{x}}}{\sqrt{m-1}},$$

we choose a much larger control vector $\boldsymbol{\alpha}$ of size mn that parameterizes the state vector $\mathbf{x}$ as

$$\mathbf{x} = \overline{\mathbf{x}} + \sum_{i=1}^{m} \boldsymbol{\alpha}_i \circ \frac{\mathbf{x}_i - \overline{\mathbf{x}}}{\sqrt{m-1}},$$

where $\boldsymbol{\alpha}_i$ is a subvector of size n. This change of control variable is known as the $\boldsymbol{\alpha}$ control variable [Lorenc, 2003; Buehner, 2005; Wang et al., 2007]. It is better formulated in a variational setting. Consider the cost function

$$\mathscr{L}(\mathbf{x}, \boldsymbol{\alpha}) = \frac{1}{2} \left\| \mathbf{y} - \mathscr{H}\left\{ \overline{\mathbf{x}} + \sum_{i=1}^{m} \boldsymbol{\alpha}_i \circ \frac{\mathbf{x}_i - \overline{\mathbf{x}}}{\sqrt{m-1}} \right\} \right\|_{\mathbf{R}}^2 + \frac{1}{2} \sum_{i=1}^{m} \|\boldsymbol{\alpha}_i\|_{\boldsymbol{\rho}}^2, \tag{6.28}$$

which is formally equivalent to the Lagrangian,

$$\mathscr{L}(\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{1}{2} \|\mathbf{y} - \mathscr{H}(\mathbf{x})\|_{\mathbf{R}}^2 + \frac{1}{2} \sum_{i=1}^{m} \|\boldsymbol{\alpha}_i\|_{\boldsymbol{\rho}}^2 + \boldsymbol{\beta}^{\mathrm{T}} \left\{ \mathbf{x} - \overline{\mathbf{x}} - \sum_{i=1}^{m} \boldsymbol{\alpha}_i \circ \frac{\mathbf{x}_i - \overline{\mathbf{x}}}{\sqrt{m-1}} \right\},$$

where $\boldsymbol{\beta}$ is a vector in $\mathbb{R}^n$ of Lagrange multipliers. Yet, as opposed to $\mathscr{L}(\mathbf{x}, \boldsymbol{\alpha})$, the Lagrangian $\mathscr{L}(\mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ is quadratic in $\boldsymbol{\alpha}$ and can be analytically minimized on $\boldsymbol{\alpha}$. The saddle point condition on $\boldsymbol{\alpha}_i$ is, denoting as before $\mathbf{X}_i = \frac{\mathbf{x}_i - \overline{\mathbf{x}}}{\sqrt{m-1}}$,

$$\boldsymbol{\alpha}_i = \boldsymbol{\rho} \boldsymbol{\beta} \circ \mathbf{X}_i.$$

Its substitution in $\mathscr{L}(\mathbf{x},\alpha,\beta)$ implies computing

$$\sum_{i=1}^{m}(\beta\circ\mathbf{X}_i)^{\mathrm{T}}\rho(\beta\circ\mathbf{X}_i)=\sum_{i=1}^{m}\sum_{k,l=1}^{n}\beta_k[\mathbf{X}_i]_k\rho_{k,l}\beta_l[\mathbf{X}_j]_l$$
$$=\sum_{k,l=1}^{n}\beta_k\left[\rho\circ\mathbf{X}\mathbf{X}^{\mathrm{T}}\right]_{k,l}\beta_l=\beta^{\mathrm{T}}\rho\circ\mathbf{P}\beta,$$

where $\mathbf{P}$ is the sample covariance matrix $\mathbf{P}=\frac{1}{m-1}\sum_{i=1}^{m}(\mathbf{x}_i-\overline{\mathbf{x}})(\mathbf{x}_i-\overline{\mathbf{x}})^{\mathrm{T}}$. This yields the new Lagrangian

$$\mathscr{L}(\mathbf{x},\beta)=\frac{1}{2}\|\mathbf{y}-\mathscr{H}(\mathbf{x})\|^2_{\mathbf{R}}+\frac{1}{2}\beta^{\mathrm{T}}(\rho\circ\mathbf{P})\beta+\beta^{\mathrm{T}}(\mathbf{x}-\overline{\mathbf{x}}).$$

This can be further optimized on β to yield the cost function

$$\mathscr{L}(\mathbf{x})=\frac{1}{2}\|\mathbf{y}-\mathscr{H}(\mathbf{x})\|^2_{\mathbf{R}}+\frac{1}{2}\|\mathbf{x}-\overline{\mathbf{x}}\|^2_{\rho\circ\mathbf{P}}. \tag{6.29}$$

Hence we have obtained a formal equivalence between (6.28) and the cost function (6.29) that governs the analysis of the EnKF with covariance localization. Both approaches are used in the literature.

6.7.3 ▪ Hierarchical EnKFs

Key assumptions of the EnKF are that the mean and the error covariance matrix are exactly given by the sampled moments of the ensemble. As we have seen, this is bound to fail without fixes. Indeed, the uncertainty is underestimated and may often lead to the divergence of the filter. As seen in Section 6.5, inflation and localization are fixes that address this issue in a satisfying manner.

The use of a *Bayesian statistical hierarchy* has been more recently explored as a distinct route toward solving this issue. If $\mathbf{x}$ is a state vector that depends on parameters θ and is observed by $\mathbf{y}$, Bayes' rule (see section 3.2) tells us that the probability of the variables and parameters conditioned on the observations is

$$p(\mathbf{x},\theta|\mathbf{y})\propto p(\mathbf{y}|\theta,\mathbf{x})p(\mathbf{x},\theta),$$

where the proportionality factor only depends on $\mathbf{y}$. But if θ is further assumed to be an uncertain parameter vector obeying a prior distribution, $p(\theta)$, and if the likelihood of $\mathbf{y}$ only depends on θ through $\mathbf{x}$, one can further decompose this posterior distribution into

$$p(\mathbf{x},\theta|\mathbf{y})\propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}|\theta)p(\theta).$$

We have created a hierarchy of random variables: $\mathbf{x}$ at the first level and θ at a second level. In this context, $p(\mathbf{x}|\theta)$ is still called a prior, while $p(\theta)$ is termed a *hyperprior* to emphasize that it operates at a second level of the Bayesian hierarchy [Gelman et al., 2014]. Applied to the EnKF, a Bayesian hierarchy could be enforced by seeing the moments of the true error distribution as multivariate random variables, rather than coinciding with the sampled moments of the ensemble.

One of the simplest ways to enforce this idea is to marginalize $p(\mathbf{x},\theta|\mathbf{y})$, and hence $p(\mathbf{x}|\theta)$, over all potential θ. In the following, θ will be the ensemble sampled moments.

Specifically, Bocquet [2011] recognized that the ensemble mean $\overline{\mathbf{x}}$ and ensemble error covariance matrix $\mathbf{P}$ used in the EnKF may be different from the unknown first- and second-order moments of the true error distribution, $\mathbf{x}_b$ and $\mathbf{B}$, where $\mathbf{B}$ is a positive definite matrix. The mismatch is due to the finite size of the ensemble, which leads to sampling errors. It is claimed in Bocquet et al. [2015] that these errors are mainly induced by the nonlinear ensemble propagation in the forecast step.

Let us account for the uncertainty in $\mathbf{x}_b$ and in $\mathbf{B}$. As in Section 6.4.1, we denote by $\mathbf{E} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m]$ the ensemble of size m formatted as an $n \times m$ matrix; $\overline{\mathbf{x}} = \mathbf{E}\mathbf{1}/m$ the ensemble mean, where $\mathbf{1} = (1, \ldots, 1)^{\mathrm{T}}$; and $\mathbf{X} = (\mathbf{E} - \overline{\mathbf{x}}\mathbf{1}^{\mathrm{T}})/\sqrt{m-1}$ the normalized perturbation matrix. Hence, $\mathbf{P} = \mathbf{X}\mathbf{X}^{\mathrm{T}}$ is the empirical covariance matrix of the ensemble. Marginalizing over all potential $\mathbf{x}_b$ and $\mathbf{B}$, the prior of $\mathbf{x}$ reads

$$p(\mathbf{x}|\mathbf{E}) = \int \mathrm{d}\mathbf{x}_b \mathrm{d}\mathbf{B}\, p(\mathbf{x}|\mathbf{E}, \mathbf{x}_b, \mathbf{B}) p(\mathbf{x}_b, \mathbf{B}|\mathbf{E}).$$

The symbol $\mathrm{d}\mathbf{B}$ corresponds to the Lebesgue measure on all independent entries $\prod_{i \le j}^{n} \mathrm{d}[\mathbf{B}]_{ij}$, but the integration is restricted to the cone of positive definite matrices. Since $p(\mathbf{x}|\mathbf{E}, \mathbf{x}_b, \mathbf{B})$ is conditioned on the knowledge of the true prior statistics, it does not depend on $\mathbf{E}$, so that

$$p(\mathbf{x}|\mathbf{E}) = \int \mathrm{d}\mathbf{x}_b \mathrm{d}\mathbf{B}\, p(\mathbf{x}|\mathbf{x}_b, \mathbf{B}) p(\mathbf{x}_b, \mathbf{B}|\mathbf{E}).$$

Bayes' rule can be applied to $p(\mathbf{x}_b, \mathbf{B}|\mathbf{E})$, yielding

$$p(\mathbf{x}|\mathbf{E}) = \frac{1}{p(\mathbf{E})} \int \mathrm{d}\mathbf{x}_b \mathrm{d}\mathbf{B}\, p(\mathbf{x}|\mathbf{x}_b, \mathbf{B}) p(\mathbf{E}|\mathbf{x}_b, \mathbf{B}) p(\mathbf{x}_b, \mathbf{B}). \tag{6.30}$$

Assuming independence of the samples, the likelihood of the ensemble $\mathbf{E}$ can be written

$$p(\mathbf{E}|\mathbf{x}_b, \mathbf{B}) = \prod_{i=1}^{m} p(\mathbf{x}_i|\mathbf{x}_b, \mathbf{B}).$$

The last factor in (6.30), $p(\mathbf{x}_b, \mathbf{B})$, is the hyperprior. The distribution represents our beliefs about the forecast filter statistics, $\mathbf{x}_b$ and $\mathbf{B}$, prior to actually running any filter. We recall that this distribution is termed *hyperprior* because it represents a prior for the background information in the first stage of a Bayesian hierarchy.

Assuming one subscribes to this view of the EnKF, it shows that more information is actually required in the EnKF, in addition to the observations and the prior ensemble, which are potentially insufficient for an inference.

A simple choice was made in Bocquet [2011] for the hyperprior: the Jeffreys' prior is an analytically tractable and uninformative hyperprior of the form

$$p_{\mathrm{J}}(\mathbf{x}_b, \mathbf{B}) \propto |\mathbf{B}|^{-\frac{n+1}{2}}, \tag{6.31}$$

where $|\mathbf{B}|$ is the determinant of the background error covariance matrix $\mathbf{B}$ of dimension $n \times n$. A more sophisticated hyperprior meant to hold static information, the normal-inverse-Wishart distribution, was proposed in Bocquet et al. [2015].

With a given hyperprior, the marginalization over $\mathbf{x}_b$ and $\mathbf{B}$, (6.30), can in principle be carried out to obtain $p(\mathbf{x}|\mathbf{E})$. We choose to call it a *predictive prior* to comply with the traditional view that sees it as prior before assimilating the observations. Note, however, that statisticians would rather call it a *predictive posterior* distribution as the outcome of a first-stage inference of a Bayesian hierarchy, where $\mathbf{E}$ is the data.

Using Jeffreys' hyperprior, Bocquet [2011] showed that the integral can be obtained analytically and that the predictive prior is a multivariate t-distribution,

$$p(\mathbf{x}|\mathbf{E}) \propto \left|\frac{(\mathbf{x}-\overline{\mathbf{x}})(\mathbf{x}-\overline{\mathbf{x}})^{\mathrm{T}}}{m-1} + \varepsilon_m \mathbf{P}\right|^{-\frac{m}{2}}, \tag{6.32}$$

where $|\cdot|$ denotes the determinant and $\varepsilon_m = 1 + 1/m$. The determinant is computed in the ensemble subspace $\xi = \overline{\mathbf{x}} + \mathrm{Vec}(\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m)$, i.e., the affine space spanned by the perturbations of the ensemble so that it is not singular. Moreover, we impose $p(\mathbf{x}|\mathbf{E}) = 0$ if $\mathbf{x}$ is not in ξ. This distribution has fat tails, thus accounting for the uncertainty in $\mathbf{B}$. The factor ε_m is a result of the uncertainty in $\mathbf{x}_b$; if $\mathbf{x}_b$ were known to coincide with the ensemble mean $\overline{\mathbf{x}}$, then ε_m would be 1 instead. For a Gaussian process, $\varepsilon_m \mathbf{P}$ is an unbiased estimator of the squared error of the ensemble mean $\overline{\mathbf{x}}$ [Sacher and Bartello, 2008], where ε_m stems from the uncertain $\mathbf{x}_b$, which does not coincide with $\overline{\mathbf{x}}$. In the derivation of Bocquet [2011], the $\varepsilon_m \mathbf{P}$ correction comes from integrating out on $\mathbf{x}_b$. Therefore, ε_m can be seen as an inflation factor on the prior covariance matrix that should actually apply to any type of EnKF.

This non-Gaussian prior distribution can be seen as a mixture of Gaussian distributions weighted according to the hyperprior. It can be shown that (6.32) can be rearranged as

$$p(\mathbf{x}|\mathbf{E}) \propto \left\{1 + \frac{(\mathbf{x}-\overline{\mathbf{x}})^{\mathrm{T}} (\varepsilon_m \mathbf{P})^{\dagger} (\mathbf{x}-\overline{\mathbf{x}})}{m-1}\right\}^{-\frac{m}{2}} \tag{6.33}$$

for $\mathbf{x} \in \xi$ and $p(\mathbf{x}|\mathbf{E}) = 0$ if $\mathbf{x} \notin \xi$; $\mathbf{P}^{\dagger}$ is the Moore–Penrose inverse of $\mathbf{P}$.

In comparison, the traditional EnKF implicitly assumes that the hyperprior is $\delta(\mathbf{B}-\mathbf{P})\delta(\mathbf{x}_b-\overline{\mathbf{x}})$, where δ is a Dirac multidimensional distribution. In other words, the background statistics generated from the ensemble coincide with the true background statistics. As a result, one obtains in this case the Gaussian prior

$$p(\mathbf{x}|\mathbf{E}) \propto \exp\left\{-\frac{1}{2}(\mathbf{x}-\overline{\mathbf{x}})^{\mathrm{T}} \mathbf{P}^{\dagger} (\mathbf{x}-\overline{\mathbf{x}})\right\} \tag{6.34}$$

for $\mathbf{x} \in \xi$ and $p(\mathbf{x}|\mathbf{E}) = 0$ if $\mathbf{x} \notin \xi$.

From these predictive priors given in state space, it is possible to derive a formally simple prior in ensemble subspace, i.e., in terms of the coefficients $\mathbf{w}$ that we used for the ETKF and MLEF ($\mathbf{x} = \overline{\mathbf{x}} + \mathbf{X}\mathbf{w}$). In turn, this prior leads to an effective cost function in the ensemble subspace for the analysis [Bocquet et al., 2015]

$$\mathscr{J}(\mathbf{w}) = \frac{1}{2}||\mathbf{y} - \mathscr{H}(\overline{\mathbf{x}} + \mathbf{X}\mathbf{w})||^2_{\mathbf{R}} + \frac{m+1}{2}\ln\left(\varepsilon_m + \frac{||\mathbf{w}||^2}{m-1}\right), \tag{6.35}$$

which should be used in place of the related cost function of the ETKF, which reads [Hunt et al., 2007]

$$\mathscr{J}(\mathbf{w}) = \frac{1}{2}||\mathbf{y} - \mathscr{H}(\overline{\mathbf{x}} + \mathbf{X}\mathbf{w})||^2_{\mathbf{R}} + \frac{1}{2}||\mathbf{w}||^2. \tag{6.36}$$

The EnKF that results from the effective cost function (6.35) has been called the finite-size EnKF because it sees the ensemble in the asymptotic limit, but as a finite set. It is denoted *EnKF-N*, where the N indicates an explicit dependence on the size of the ensemble.

It was further shown in Bocquet and Sakov [2012] that it is enlightening to separate the angular degrees of freedom of $\mathbf{w}$, i.e., $\mathbf{w}/|\mathbf{w}|$, from its radial one $|\mathbf{w}|$ in the cost function. This amounts to defining a Lagrangian of the form

$$\mathscr{L}(\mathbf{w},\rho,\zeta) = \frac{1}{2}\|\mathbf{y}-\mathscr{H}(\overline{\mathbf{x}}+\mathbf{X}\mathbf{w})\|^2_{\mathbf{R}} + \frac{\zeta}{2}\left(\frac{\|\mathbf{w}\|^2}{m-1}-\rho\right) + \frac{m+1}{2}\ln(\varepsilon_m+\rho),$$

where ζ is a Lagrange parameter used to enforce the decoupling. When the observation operator is linear or linearized, this Lagrangian turns out to be equivalent to a dual cost function of the ζ parameter, which is

$$\mathscr{D}(\zeta) = \frac{1}{2}\boldsymbol{\delta}^{\mathrm{T}}\left(\mathbf{R}+\frac{m-1}{\zeta}\mathbf{Y}\mathbf{Y}^{\mathrm{T}}\right)^{-1}\boldsymbol{\delta} + \frac{\varepsilon_m\zeta}{2} + \frac{m+1}{2}\ln\frac{m+1}{\zeta} - \frac{m+1}{2}, \tag{6.37}$$

where $\boldsymbol{\delta} = \mathbf{y}-\mathscr{H}(\overline{\mathbf{x}})$ is the innovation vector. The dual cost function is defined over the interval $]0,(m+1)/\varepsilon_m]$. Although it is not necessarily convex, its global minimum can easily be found numerically because it is a one-dimensional optimization problem. To perform an EnKF-N analysis using this dual cost function, one would first minimize (6.37) to obtain the optimal ζ_{a}. The analysis is then

$$\mathbf{w}_{\mathrm{a}} = \left(\mathbf{Y}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{Y}+\frac{\zeta_{\mathrm{a}}}{m-1}\mathbf{I}_m\right)^{-1}\mathbf{Y}^{\mathrm{T}}\mathbf{R}^{-1}\boldsymbol{\delta} = \mathbf{Y}^{\mathrm{T}}\left(\frac{\zeta_{\mathrm{a}}}{m-1}\mathbf{R}+\mathbf{Y}\mathbf{Y}^{\mathrm{T}}\right)^{-1}\boldsymbol{\delta}. \tag{6.38}$$

Based on the effective cost function (6.35), an updated set of perturbations can be obtained:

$$\mathbf{X}_{\mathrm{a}} = \mathbf{X}[\mathrm{H}_{\mathrm{a}}]^{-\frac{1}{2}}\mathbf{U} \quad \text{with} \quad \mathrm{H}_{\mathrm{a}} = \mathbf{Y}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{Y}+\frac{\zeta_{\mathrm{a}}}{m-1}\mathbf{I}_m - \frac{2}{m+1}\left(\frac{\zeta_{\mathrm{a}}}{m-1}\right)^2\mathbf{w}_{\mathrm{a}}\mathbf{w}_{\mathrm{a}}^{\mathrm{T}}. \tag{6.39}$$

The last term of the Hessian, $-\frac{2}{m+1}\left(\frac{\zeta_{\mathrm{a}}}{m-1}\right)^2\mathbf{w}_{\mathrm{a}}\mathbf{w}_{\mathrm{a}}^{\mathrm{T}}$, which is related to the covariances of the angular and radial degrees of freedom of $\mathbf{w}$, can very often be neglected. If so, the update equations are equivalent to those of the ETKF but with an inflation of the prior covariance matrix by a factor $(m-1)/\zeta_{\mathrm{a}}$. Hence the EnKF-N implicitly determines an adaptive optimal inflation.

If an SVD of $\mathbf{Y}$ is available, the minimization of $\mathscr{D}(\zeta)$ is immediate, using for instance a dichotomous search. Such a decomposition is often already available because it was meant to be used to compute (6.38) and (6.39).

Practically, it was found in several low-order models and in perfect model conditions that the EnKF-N does not require any inflation and that its performance is close to that of an equivalent ETKF, where a uniform inflation would have been optimally tuned to obtain the best performance of the filter. In a preliminary study by Bocquet et al. [2015], the use of a more informative hyperprior, such as the normal-inverse-Wishart distribution, was proposed to avoid the need for localization while still avoiding the need for inflation.

6.7.3.1 ▪ Numerical illustrations

The three-variable Lorenz model [Lorenz, 1963] (Lorenz-63 hereafter) is the emblem of chaotic systems. It is defined by the ODEs

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \sigma(y - x),$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = \rho x - y - xz,$$
$$\frac{\mathrm{d}z}{\mathrm{d}t} = xy - \beta z,$$

where $\sigma = 10$, $\rho = 28$, and $\beta = 8/3$. This model is chaotic, with $(0.91, 0, -14.57)$ as its Lyapunov exponents. The model doubling time is 0.78 time units. Its attractor has the famous butterfly shape with two distinct wings, or lobes, which were illustrated in Section 2.5.1. It was used by Edward Lorenz to explain the finite horizon of predictability in meteorology.

To demonstrate the relevance of the EnKF-N with this model, a numerical twin experiment similar to that used with the Lorenz-95 model is designed. The system is assumed fully observed so that $\mathbf{H}_k \equiv \mathbf{I}_3$, with the observation error covariance matrix $\mathbf{R}_k \equiv 4\mathbf{I}_3$. The time interval between observational updates is varied from $\Delta t = 0.10$ to $\Delta t = 0.50$. The larger Δt is, the stronger the impact of model nonlinearity on the state estimation, and the stronger the need for an inflation correction. The performance of the DA schemes is measured by the analysis RMSE (6.23) averaged over a very long run. We test a standard ETKF where the uniform inflation is optimally tuned to minimize the RMSE (about 20 values are tested). We compare it to the EnKF-N, which does not require inflation. In both cases, the ensemble size is set to $m = 3$.

The skills of both filters are shown in Figure 6.8. It is remarkable that the EnKF-N achieves an even better performance than the optimally tuned ETKF, without any tuning. As Δt increases, the ETKF requires a significantly stronger inflation. This is mostly needed at the transition between the two lobes of the Lorenz-63 attractor. Within the lobes, the DA system is effectively much more linear and requires little inflation. By contrast, the EnKF-N, which is adaptive, applies a strong inflation only when needed, i.e., at the transition between lobes.

An analogous experiment can be carried out, but with the Lorenz-95 model. The setup of Section 6.6 is used. The performance of both filters is shown in the left panel of Figure 6.9 when the ensemble size is varied and when $\Delta t = 0.05$. The EnKF-N achieves the same performance but without any tuning, and hence with numerical efficiency. A similar experiment is performed but varying Δt, and hence system nonlinearity, and setting $m = 20$. The results are reported in the right panel of Figure 6.9. Again, it can be seen that the same performance can be reached without any tuning with the EnKF-N.

The adaptation of the EnKF-N is also illustrated by Figure 6.7, where the MLEF and ETKF were compared in the first stages of a DA experiment with the Lorenz-95 model and a nonlinear observation operator. In this context, the finite-size MLEF and ETKF were shown to outperform the standard MLEF and ETKF with optimally tuned uniform inflation.

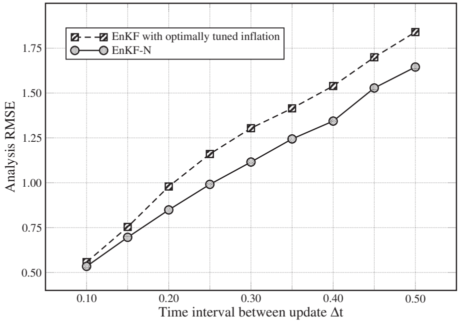


Figure 6.8. *Average analysis RMSE for a deterministic EnKF (ETKF) with optimally tuned inflation and for the EnKF-N. The model is Lorenz-*63.

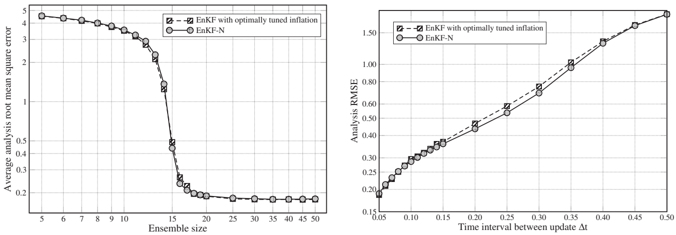


Figure 6.9. *Average analysis RMSE for a deterministic EnKF (ETKF) with optimally tuned inflation and for the EnKF-N. Left panel: the ensemble size is varied from $m = 5$ to $m = 50$ and $\Delta t = 0.05$. Right panel: Δt is varied from* 0.05 *to* 0.50 *and the ensemble size is set to $m = 20$. The model is Lorenz-*95.

6.7.3.2 ▪ Passing on hierarchical statistical information

The EnKF-N was built with the idea to remain algorithmically as close as possible to the EnKF. To avoid relying on any additional input, the hierarchy of information that was established on $\mathbf{x}_b$ and $\mathbf{B}$ was closed by marginalizing over all potential priors leading to effective cost functions.

However, it is actually possible to propagate the information that the filter carries about $\mathbf{x}_b$ and $\mathbf{B}$ from one cycle to the next. This idea was formalized in Myrseth and Omre [2010]. It relies on the natural conjugacy of the normal-inverse-Wishart distribution with the multivariate normal distribution. The ensemble can be seen as an observation set for the estimation of the moments of the true distribution, $\mathbf{x}_b$ and $\mathbf{B}$, which obeys a multivariate normal distribution. If $\mathbf{x}_b$ and $\mathbf{B}$ are supposed to follow a normal-inverse-Wishart distribution, then the posterior distribution will also follow a normal-inverse-Wishart distribution with parameters that are easily updated using the data, i.e., the ensemble members, thanks to the natural conjugacy. This defines a level-2 update scheme for the mean and the error covariances. The mean and error covariances used on the level-1 update scheme, i.e., the EnKF, are subsequently drawn from this updated distribution. This scheme is quite different from a traditional EnKF and truly accounts for the uncertainty in the moments. Another and rather similar attempt was documented in Tsyrulnikov and Rakitkoa [2016].

6.8 ▪ The ensemble Kalman smoother

Filtering consists of assimilating observations as they become available, making the best estimate at the present time. If $\mathbf{y}_{K:1} = \mathbf{y}_K, \mathbf{y}_{K-1}, \dots, \mathbf{y}_1$ is the collection of observations from t_1 to t_K, filtering aims at estimating the PDF $p(\mathbf{x}_K|\mathbf{y}_{K:1})$ at t_K. Only past and present observations are accounted for, which would necessarily be the case for real-time nowcasting and forecasting.

Smoothing, on the other hand, aims at estimating the state of the system (or a trajectory of it), using past, present, and possibly future observations. Indeed, assuming again t_K is the present time, one could also be interested in the distribution $p(\mathbf{x}_k|\mathbf{y}_{K:1})$, where $1 \leq k \leq K$. More generally, one would be interested in estimating $p(\mathbf{x}_{K:1}|\mathbf{y}_{K:1})$, where $\mathbf{x}_{K:1} = \mathbf{x}_K, \mathbf{x}_{K-1}, \dots, \mathbf{x}_1$ is the collection of state vectors from t_1 to t_K, i.e., a trajectory. Note that $p(\mathbf{x}_k|\mathbf{y}_{K:1})$ is a marginal distribution of $p(\mathbf{x}_{K:1}|\mathbf{y}_{K:1})$ obtained by integrating out $\mathbf{x}_l$, with $l = 1, \dots, k-1, k+1, \dots, K$. This is especially useful for hindcasting and reanalysis problems that aim at retrospectively obtaining the best estimate for the model state using all available information.

The KF can be extended to address the smoothing problem, leading to a variety of Kalman smoother algorithms [Anderson and Moore, 1979]. It has been introduced in the geosciences and studied in this context by Cohn et al. [1994].

The generalization of the Kalman smoother to the family of *ensemble* Kalman filters followed the development of the EnKF. Even for linear systems, where we theoretically expect equivalent schemes, there are several implementations of the smoother, even for a given flavor of EnKF [Cosme et al., 2012]. Its implementation may also depend on whether one wishes to estimate a state vector or a trajectory, or on how long the backward analysis can go. In the following, we shall focus on the fixed-lag ensemble Kalman smoother (EnKS) that was introduced and used in Evensen and van Leeuwen [2000], Zhu et al. [2003], Khare et al. [2008], Cosme et al. [2010], and Nerger et al. [2014]. If Δt is the time interval between updates, fixed-lag means that the analysis goes backward by $L\Delta t$ in time. The variable L measures the length of the time window in units of Δt. We will implement it following the ensemble transform framework as used in Bocquet and Sakov [2013, 2014].

The EnKS is built on the EnKF, which will be its backbone. There are two steps. The first step consists of running the EnKF from t_{k-1} to t_k. The second step consists of updating the state vectors at t_{k-1}, t_{k-2} back to t_{k-L}.

Let us make L the lag of the EnKS and define t_L as the present time. Hence, we wish to retrospectively estimate $\mathbf{x}_{L:0} = \mathbf{x}_L, \mathbf{x}_{L-1}, \dots, \mathbf{x}_0$. An EnKF has been run from the starting time to t_L. The collection of all posterior ensembles $\mathbf{E}_L, \mathbf{E}_{L-1}, \dots, \mathbf{E}_0$ is assumed to be stored, which is a significant technical constraint. For each $\mathbf{E}_k$ there is a corresponding mean state $\overline{\mathbf{x}}_k$ and a normalized perturbation matrix $\mathbf{X}_k$. This need for memory (at least $L \times m \times n$ scalars) is the main requirement of the EnKS.

The EnKF provides an approximation for the PDF $p(\mathbf{x}_L|\mathbf{y}_{L:})$, where $\mathbf{y}_{L:}$ represents the collection of observation vectors from the beginning of the DA experiment (earlier than t_0) to t_L. For the EnKF, this PDF can be approximated as a Gaussian distribution. Now, let us describe the backward pass, starting from the latest dates. Let us first derive a retrospective update for $\mathbf{x}$ one time step backward. Hence, we wish to compute a

Gaussian approximation for $p(\mathbf{x}_{L-1}|\mathbf{y}_{L:})$. From Bayes' rule, we obtain

$$p(\mathbf{x}_{L-1}|\mathbf{y}_{L:}) \propto p(\mathbf{y}_L|\mathbf{x}_{L-1},\mathbf{y}_{L-1:})p(\mathbf{x}_{L-1}|\mathbf{y}_{L-1:}),$$

which relates the smoothing PDF to the current observation likelihood and to the filtering distribution at t_{L-1}. From the EnKF's standpoint, $p(\mathbf{x}_{L-1}|\mathbf{y}_{L-1:})$ is approximately Gaussian in the affine subspace centered at $\mathbf{x}^{\mathrm{a}}_{L-1}$ and spanned by the columns of $\mathbf{X}^{\mathrm{a}}_{L-1}$. The affine subspace can be parameterized by $\mathbf{x} = \mathbf{x}^{\mathrm{a}}_{L-1} + \mathbf{X}^{\mathrm{a}}_{L-1}\mathbf{w}$. Using the variational expression of the ETKF or the MLEF as defined in ensemble subspace, $p(\mathbf{x}_{L-1}|\mathbf{y}_{L-1:})$ is proportional to $\exp(-\frac{1}{2}\|\mathbf{w}\|^2)$ when written in terms of the coordinates, $\mathbf{w}$, of the ensemble subspace (see Section 6.4.2). Then $p(\mathbf{y}_L|\mathbf{x}_{L-1},\mathbf{y}_{L-1:})$ is the likelihood and reads, in ensemble subspace,

$$p(\mathbf{y}_L|\mathbf{x}_{L-1},\mathbf{y}_{L-1:}) \propto \exp\left\{-\frac{1}{2}\left\|\mathbf{y}_L - \mathscr{H}_L \circ \mathscr{M}_{L:L-1}(\mathbf{x}^{\mathrm{a}}_{L-1} + \mathbf{X}^{\mathrm{a}}_{L-1}\mathbf{w})\right\|^2_{\mathbf{R}_L}\right\}.$$

Hence the complete cost function for the retrospective analysis on t_{L-1} is

$$\mathscr{L}(\mathbf{w}) = \frac{1}{2}\left\|\mathbf{y}_L - \mathscr{H}_L \circ \mathscr{M}_{L:L-1}(\mathbf{x}^{\mathrm{a}}_{L-1} + \mathbf{X}^{\mathrm{a}}_{L-1}\mathbf{w})\right\|^2_{\mathbf{R}_L} + \frac{1}{2}\|\mathbf{w}\|^2.$$

This type of potentially nonquadratic cost function will be at the heart of the IEnKF/IEnKS (see Chapter 7). Here, the update is expanded around $\mathbf{x}^{\mathrm{a}}_{L-1}$ using the TLM to make the cost function quadratic:

$$\begin{aligned}
\mathscr{L}(\mathbf{w}) &= \frac{1}{2}\left\|\mathbf{y}_L - \mathscr{H}_L \circ \mathscr{M}_{L:L-1}(\mathbf{x}^{\mathrm{a}}_{L-1} + \mathbf{X}^{\mathrm{a}}_{L-1}\mathbf{w})\right\|^2_{\mathbf{R}_L} + \frac{1}{2}\|\mathbf{w}\|^2 \\
&\simeq \frac{1}{2}\left\|\mathbf{y}_L - \mathscr{H}_L \circ \mathscr{M}_{L:L-1}(\mathbf{x}^{\mathrm{a}}_{L-1}) - \mathbf{H}_L\mathbf{M}_{L:L-1}\mathbf{X}^{\mathrm{a}}_{L-1}\mathbf{w}\right\|^2_{\mathbf{R}_L} + \frac{1}{2}\|\mathbf{w}\|^2 \\
&= \frac{1}{2}\left\|\mathbf{y}_L - \mathscr{H}_L(\mathbf{x}^{\mathrm{f}}_L) - \mathbf{H}_L\mathbf{X}^{\mathrm{f}}_L\mathbf{w}\right\|^2_{\mathbf{R}_L} + \frac{1}{2}\|\mathbf{w}\|^2 \\
&= \frac{1}{2}\left\|\boldsymbol{\delta}_L - \mathbf{Y}^{\mathrm{f}}_L\mathbf{w}\right\|^2_{\mathbf{R}_L} + \frac{1}{2}\|\mathbf{w}\|^2,
\end{aligned}$$

where $\boldsymbol{\delta}_L = \mathbf{y}_L - \mathscr{H}_L(\mathbf{x}^{\mathrm{f}}_L)$ and, as before, $\mathbf{Y}^{\mathrm{f}}_k = \mathbf{H}_k\mathbf{X}^{\mathrm{f}}_k$. From this cost function, the derivation of an ETKF-like analysis is immediate. We obtain

$$\mathbf{X}^{\mathrm{a},1}_{L-1} = \mathbf{X}^{\mathrm{a}}_{L-1}\sqrt{\boldsymbol{\Omega}_\star} \qquad \text{and} \qquad \mathbf{x}^{\mathrm{a},1}_{L-1} = \mathbf{x}^{\mathrm{a}}_{L-1} + \mathbf{X}^{\mathrm{a}}_{L-1}\mathbf{w}_\star, \tag{6.40}$$

with

$$\boldsymbol{\Omega}_\star = \left[\mathbf{I}_m + (\mathbf{Y}^{\mathrm{f}}_L)^{\mathrm{T}}\mathbf{R}_L^{-1}\mathbf{Y}^{\mathrm{f}}_L\right]^{-1} \qquad \text{and} \qquad \mathbf{w}_\star = \boldsymbol{\Omega}_\star\mathbf{Y}^{\mathrm{f}}_L\mathbf{R}_L^{-1}\boldsymbol{\delta}_L. \tag{6.41}$$

The superscript 1 indicates that the estimate of $\mathbf{x}_{L-1}$ now accounts for observation one time step ahead.

We can proceed backward and consider an updated estimation for $\mathbf{x}_{L-2}$. The EnKF had yielded $\mathbf{x}^{\mathrm{a}}_{L-2}$, two time steps earlier. The backward pass of the EnKS run, one time step earlier, must have updated $\mathbf{x}_{L-2}$ using the same formula that we derived for $\mathbf{x}_{L-1}$ a few lines above, yielding the estimate $\mathbf{x}^{\mathrm{a},1}_{L-2}$ and the updated ensemble $\mathbf{X}^{\mathrm{a},1}_{L-2}$. With $\mathbf{x}^{\mathrm{a},1}_{L-2}$, we have accounted for $\mathbf{y}_{L-1}$, but not $\mathbf{y}_L$ yet. Hence, using Bayesian estimation as a guide, we need to estimate

$$p(\mathbf{x}_{L-2}|\mathbf{y}_{L:}) \propto p(\mathbf{y}_L|\mathbf{x}_{L-2},\mathbf{x}_{L-1:})p(\mathbf{x}_{L-2}|\mathbf{y}_{L-1:}).$$

As above, we can derive from these distributions an approximate quadratic cost function for the retrospective analysis on $\mathbf{x}_{L-2}$. We formally obtain the cost function

$$\mathscr{L}(\mathbf{w}) = \frac{1}{2}\left\|\mathbf{y}_L - \mathscr{H}_L \circ \mathscr{M}_{L:L-2}(\mathbf{x}^{a,1}_{L-2} + \mathbf{X}^{a,1}_{L-2}\mathbf{w})\right\|^2_{\mathbf{R}_L} + \frac{1}{2}\|\mathbf{w}\|^2,$$

where $\mathbf{x}_{L-2}$ is parameterized as $\mathbf{x}_{L-2} = \mathbf{x}^{a,1}_{L-2} + \mathbf{X}^{a,1}_{L-2}\mathbf{w}$. The outcome is formally the same,

$$\mathscr{L}(\mathbf{w}) = \frac{1}{2}\left\|\boldsymbol{\delta}_L - \mathbf{Y}^{\mathrm{f}}_L\mathbf{w}\right\|^2_{\mathbf{R}_L} + \frac{1}{2}\|\mathbf{w}\|^2,$$

with $\boldsymbol{\delta}_L = \mathbf{y}_L - \mathscr{H}_L(\mathbf{x}^{\mathrm{f}}_L)$ and $\mathbf{Y}^{\mathrm{f}}_L = \mathbf{H}_L\mathbf{X}^{\mathrm{f}}_L$, but where $\mathbf{w}$ is a vector of coefficients that applies to a different ensemble of perturbations ($\mathbf{X}^{a,1}_{L-2}$ instead of $\mathbf{X}^{a}_{L-1}$). From this cost function, an ETKF-like analysis is immediate. We obtain

$$\mathbf{x}^{a,2}_{L-2} = \mathbf{x}^{a,1}_{L-2} + \mathbf{X}^{a,1}_{L-2}\mathbf{w}_\star \qquad \text{and} \qquad \mathbf{X}^{a,2}_{L-2} = \mathbf{X}^{a,1}_{L-2}\sqrt{\boldsymbol{\Omega}_\star},$$

with

$$\boldsymbol{\Omega}_\star = \left[\mathbf{I}_m + (\mathbf{Y}^{\mathrm{f}}_L)^{\mathrm{T}}\mathbf{R}_L^{-1}\mathbf{Y}^{\mathrm{f}}_L\right]^{-1} \qquad \text{and} \qquad \mathbf{w}_\star = \boldsymbol{\Omega}_\star\mathbf{Y}^{\mathrm{f}}_L\mathbf{R}_L^{-1}\boldsymbol{\delta}_L.$$

The state vectors have been updated retrospectively down to two time steps in the past (fixed-lag EnKS of $L = 2$). The scheme could be extended backward for longer lags following the same rationale. The schematic Figure 6.10 of the EnKS depicts the EnKF step followed by a backward sweep for smoothing for a time window length of $L = 4$. Hence, provided one can afford to store the ensembles, the backward pass is rather cheap as it only performs the linear algebra of the analysis. The forecast model is only used in the EnKF pass. The algorithm of the EnKS in ensemble subspace is given in Algorithm 6.6 as a pseudocode.

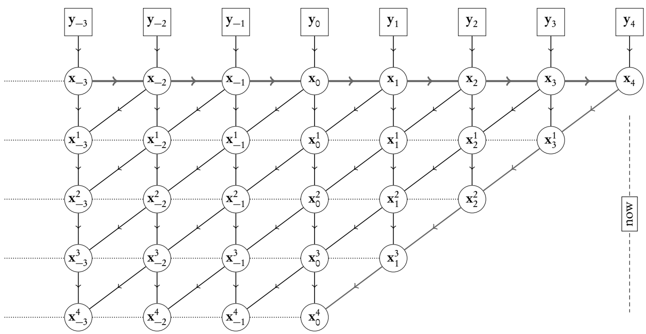


Figure 6.10. *Schematic of the EnKS. The far right of the figure (t_4) corresponds to the present time. The upper level corresponds to the EnKF pass, from left to right. Lower levels indicate updates that account for more recent observations. The backward smoothing pass follows a diagonal moving down and backward in time. The arrows indicate the fluxes of information.*

Algorithm 6.6 Pseudocode for a complete cycle of the EnKS in ensemble subspace.

Require: Observation operator $\mathscr{H}$ at current time; for $l = 0, L-1$: $\mathbf{E}^l$, the previous L analysis ensembles; $\mathbf{E}^L$, the forecast ensemble at current time; $\mathbf{y}$, the observation at current time; $\mathbf{U}$, an orthogonal matrix in $\mathbb{R}^{m\times m}$ satisfying $\mathbf{U1} = \mathbf{1}$; $\mathscr{M}$, the model resolvent from current time to the next analysis time.

$\mathbf{E} = \mathbf{E}^L$
$\overline{\mathbf{x}} = \mathbf{E1}/m$
$\mathbf{X} = \left(\mathbf{E} - \overline{\mathbf{x}}\mathbf{1}^{\mathrm{T}}\right)/\sqrt{m-1}$
$\mathbf{Z} = \mathscr{H}(\mathbf{E})$
$\overline{\mathbf{y}} = \mathbf{Z1}/m$
$\mathbf{S} = \mathbf{R}^{-\frac{1}{2}}\left(\mathbf{Z} - \overline{\mathbf{y}}\mathbf{1}^{\mathrm{T}}\right)/\sqrt{m-1}$
$\boldsymbol{\delta} = \mathbf{R}^{-\frac{1}{2}}(\mathbf{y} - \overline{\mathbf{y}})$
$\mathbf{T} = \left(\mathbf{I}_m + \mathbf{S}^{\mathrm{T}}\mathbf{S}\right)^{-1}$
$\mathbf{w} = \mathbf{TS}^{\mathrm{T}}\boldsymbol{\delta}$
$\mathbf{E}^L = \overline{\mathbf{x}}\mathbf{1}^{\mathrm{T}} + \mathbf{X}\left(\mathbf{w}\mathbf{1}^{\mathrm{T}} + \sqrt{m-1}\,\mathbf{T}^{\frac{1}{2}}\mathbf{U}\right)$
for $l = 0, \ldots, L-1$ **do**
    $\overline{\mathbf{x}} = \mathbf{E}^l\mathbf{1}/m$
    $\mathbf{X} = \left(\mathbf{E}^l - \overline{\mathbf{x}}\mathbf{1}^{\mathrm{T}}\right)/\sqrt{m-1}$
    $\mathbf{E}^l = \overline{\mathbf{x}}\mathbf{1}^{\mathrm{T}} + \mathbf{X}\left(\mathbf{w}\mathbf{1}^{\mathrm{T}} + \sqrt{m-1}\,\mathbf{T}^{\frac{1}{2}}\mathbf{U}\right)$
end for
for $l = 0, \ldots, L-1$ **do**
    $\mathbf{E}^l := \mathbf{E}^{l+1}$
end for
$\mathbf{E}^L = \mathscr{M}(\mathbf{E}^{L-1})$

Using the same setup as in Section 6.6 and $m = 20$, we test the EnKS with the Lorenz-95 model. The lag is fixed to $L = 50$. The results are reported in Figure 6.11. The EnKS provides not only a filtering estimation but also a retrospective estimation of the state $l\Delta t$ in the past with $0 \le l \le L$. The analysis time-average RMSEs are computed as a function of $0 \le l \le L$. As expected, for $l = 0$, one recovers the filtering performance, while for $l = 50$, one recovers the maximal smoothing performance. In accordance with a few experiments in the literature on several models, a performance saturation is observed when the lag, L, is increased. This is due to the truncation of second-order statistics (the Gaussian assumption) so that information is necessarily lost with time.

One common mistake when implementing the EnKS is to apply inflation to account for sampling errors in both the EnKF and the backward updating pass. It must only be used in the EnKF, not the backward updating, because sampling errors are already accounted for in the EnKF and the updated ensembles once and for all. Hence, applying inflation in the backward pass would result in a suboptimal estimation, and a degradation of smoothing as L increases, or within the time window for fixed L.

Finally, let us mention that the ideas of the EnKS are quite useful in assimilating *asynchronous* observations within an EnKF DA system [Sakov et al., 2010]. Some DA systems, for instance the operational DA systems, continuously receive the observations they are meant to assimilate. Most of them have been obtained within the time window of the DA cycle, typically 3 to 6 hours for numerical weather forecasting systems. They are asynchronous with the update. Those observations, collected within

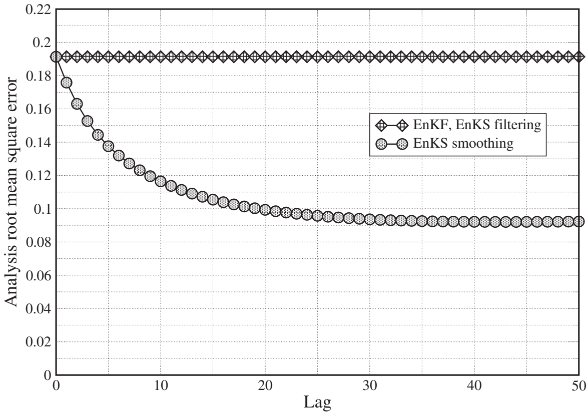


Figure 6.11. *Analysis RMSE of the EnKS with* $L = 50$ *using the same setup as in Section* 6.6 *and* $m = 20$, *applied to the Lorenz-*95 *model. The diamonds indicate the RMSE for the EnKF, which is also the filtering RMSE of the EnKS. The circles indicate the smoothing RMSE of the EnKS, i.e., the mean RMSE of the reanalysis of the state vector,* $L\Delta t$, *in the past.*

the time window $[t_k, t_{k+1}]$, can be used to update the ensemble at time t_k of the most recent scheduled update of an EnKF DA system. This can be achieved provided that an ensemble forecast throughout $[t_k, t_{k+1}]$ has been computed. Then, applying the smoothing equations of the EnKS, for instance (6.40) and (6.41), leads to the formulation of a consistent update at t_k. This, however, relies on linearity assumptions, which are indeed taken for granted in the EnKS.

6.9 ▪ A widespread and popular DA method

The EnKF, its variants, and precursors of the method have been successfully developed and implemented in meteorology and oceanography, including in operations. Because the method is simple to implement, it has been used in a huge number of studies in these fields. But it has spread out to other geoscience disciplines and beyond. For instance, to quote just a very few references, it has been applied in greenhouse gas inverse modeling [Kang et al., 2012], air quality forecasting [Wu et al., 2008], extraterrestrial atmosphere forecasting [Hoffman et al., 2010], detection and attribution in climate sciences [Hannart et al., 2016], geomagnetism reanalysis [Fournier et al., 2013], and ice-sheet parameter estimation and forecasting [Bonan et al., 2014]. It has also been used in petroleum reservoir estimation and in adaptive optics for extra-large telescopes [Gray et al., 2014].

Chapter 7

Ensemble variational methods

In this chapter, we look into the many ways to combine the benefits of variational and ensemble Kalman methods. The focus will mainly be on these developments in the domain of numerical weather prediction (NWP). We believe, however, that their applicability is much broader.

Combining the advantages of both classes of methods, one also wishes to avoid their drawbacks. From a theoretical standpoint, the EnKF propagates the error estimates and has a dynamical, flow-dependent representation of them. It also does not require the tangent linear and adjoint models of the observation operator, as seen from (6.11) of Chapter 6. On the other hand, 4D-Var by definition operates on a DA time window over which asynchronous observations can consistently, i.e., modelwise, be assimilated. Moreover, 4D-Var can perform a full nonlinear analysis within its DA window (abbreviated DAW in this chapter) thanks to numerical optimization techniques. On the downside, the EnKF requires the use of regularization techniques, inflation and localization, to filter out sampling errors and address the rank deficiency issue of the ensemble. 4D-Var requires the use of the tangent linear and adjoint models (evolution and observation), which are very time-consuming to derive and maintain. Other advantages and drawbacks, some of them pertaining to the practical implementation of the methods in high-dimensional NWP systems, are documented in Lorenc [2003] and Kalnay et al. [2007].

The investigations of NWP centers on the performance of both classes of methods have not shown very significant differences for synoptical meteorology. Indeed, the current operational DA systems for synoptic scale meteorology are rather Gaussian because of the dense and frequent observations of the atmosphere. Hence, it may be expected that well-tuned EnKF and 4D-Var should perform similarly in this context. However, this does not hold in general and in particular for significantly non-Gaussian DA systems.

The performance of the EnKF and 4D-Var can be compared on the Lorenz-95 model (see Section 6.7.3.1) in various regimes of nonlinearity. For instance, Figure 7.1 compares the performance of the EnKF (and the smoother EnKS described in Section 6.8) with 4D-Var as a function of the length of the DAW. For short windows, the EnKF and EnKS outperform 4D-Var because they both propagate the error statistics between two analyses using the perturbations simulated by the ensemble, whereas 4D-Var only propagates a mean state estimator. For longer DAWs, 4D-Var outperforms the EnKF

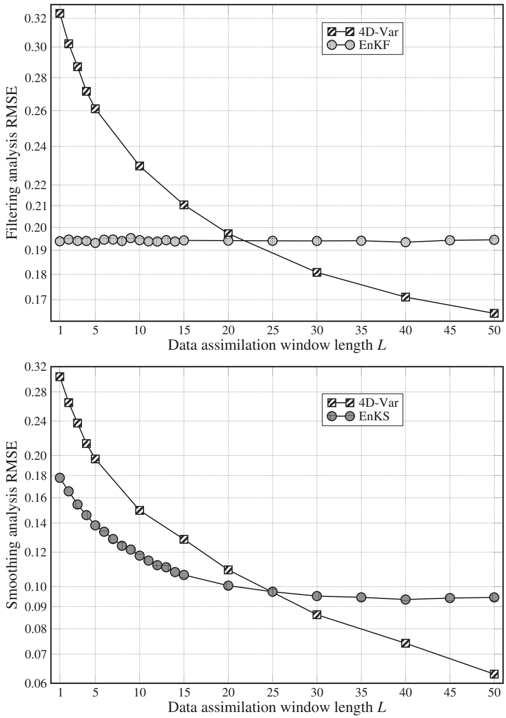


Figure 7.1. *Synthetic DA experiments with the Lorenz-*95 *model, which emphasizes the distinct performance of the EnKF and* 4*D-Var. The upper panel shows the filtering analysis RMSE of optimally tuned EnKF and* 4*D-Var DA experiments as a function of the length of the* 4*D-Var DAW. The lower panel shows the smoothing analysis RMSE of optimally tuned EnKS and* 4*D-Var as a function of the length of their DAW. The optimal RMSE is chosen within the window for* 4*D-Var. The EnKF and the EnKS use an ensemble of* $m = 20$, *which avoids the need for localization but requires inflation. The length of the DAW is* $L \times \Delta t$, *where* $\Delta t = 0.05$.

and the EnKS because of its full nonlinear analysis within the DAW. Hence, there must be room for an improved approach by properly combining the two methods—see Figure 1.5.

In this chapter, we will discuss four main classes of emerging systems that combine ensemble and variational methods. All these methods can generically be called *ensemble variational* or *EnVar* methods, as recommended by A.C. Lorenc and a WMO THORPEX working group.[32] These methods share ideas, principles, and many

[32]`http://www.wcrp-climate.org/WGNE/BlueBook/2013/individual-articles/01_Lorenc_Andrew_EnVar_nomenclature.pdf.`

techniques, but they originate from quite distinct objectives. We will first discuss the *hybrid* methods that consider two DA systems running concurrently (building on already existing systems), but that additionally exchange information in hopes of improving the global performance. Second, we will discuss systems based on an *ensemble of DA*, or *EDA*. It is usually based on an already existing variational system such as a 4D-Var. An ensemble of such 4D-Var is generated with the goal of introducing dynamical errors into the background error covariance matrix of 4D-Var. Third, 4DEnVar systems stem from the need to avoid the development and maintenance of complex adjoint operators for NWP centers that already operate a 4D-Var. The idea is to perform the 4D variational analysis within the reduced space of a 4D ensemble of model trajectories. Finally, we will discuss the IEnKS, which uses a 4D ensemble of trajectories to perform a nonlinear variational analysis and to represent flow-dependent errors. Its initial goal was to endow the EnKF with the ability to perform nonlinear analysis and generate a consistent updated ensemble, which requires iterating as in the numerical optimization of nonlinear objective functions, and with the ultimate goal of systematically outperforming the EnKF and 4D-Var, especially in strongly nonlinear regimes.

These new DA approaches significantly blur the lines described in the concluding remarks by Kalnay et al. [2007] regarding the future of DA techniques. For challenging DA problems, pure EnKF or pure 4D-Var state-of-the-art systems don't exist any more. These methods have cross-fertilized.

Because this topic of DA is relatively new and rapidly evolving, one must keep in mind that this chapter is a snapshot of the state of the art in early 2016.

7.1 ▪ The hybrid methods

Hybrid could refer to two different concepts that are nevertheless intertwined. Hybrid DA defines a system where two DA methods, usually already implemented, are run concurrently and exchange information about the errors and the estimated model state to get an even better estimation of these. In practice, however, a hybrid DA system combines static and predetermined prior error statistics, such as in OI, 3D-Var, or 4D-Var, with flow-dependent error statistics such as those produced by an EnKF. Consequently, it is actually the error covariances that are hybridized, not necessarily the DA systems. Nonetheless, these two types of error covariances were historically produced by two distinct DA systems (3D-Var and EnKF, typically), so that the second definition is actually often connected to the first.

From this consideration on terminology, one can guess that the goal of this hybridization is to have systems based on OI, 3D-Var, and 4D-Var benefit from a dynamical estimation of the errors and, reciprocally, to have an EnKF system that has a dynamical but degenerate representation of the errors benefit from a full-rank prior representation of all errors. This echoes our initial discussion on the flaws and merits of 4D-Var and of the EnKF.

7.1.1 ▪ Hybridizing the EnKF and 3D-Var

Here, we shall mostly focus on the hybridization of an EnKF system with a 3D-Var (possibly OI) system. The 3D-Var system relies on a full-rank static (and/or predetermined) error covariance matrix, $\mathbf{C}$. The EnKF estimates the flow-dependent errors through the perturbations $\mathbf{P}^{\mathrm{f}} = \frac{1}{m-1}\sum_{i=1}^{m}(\mathbf{x}_i - \overline{\mathbf{x}})(\mathbf{x}_i - \overline{\mathbf{x}})^{\mathrm{T}} = \mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}$, where $\mathbf{X}_{\mathrm{f}}$ is the normalized anomaly matrix (see Section 6.4). Whatever the type of EnKF, the simplest idea is to perform a state analysis using a linear combination of these error covariances,

$$\mathbf{B} = \gamma \mathbf{C} + (1-\gamma)\mathbf{P}^{\mathrm{f}},$$

where $\gamma \in [0,1]$ is a scalar parameter that controls the blending of the covariances: using $\mathbf{B}$ with $\gamma = 1$ corresponds to a 3D-Var state analysis, while using $\mathbf{B}$ with $\gamma = 0$ corresponds to the pure EnKF state analysis.

If the EnKF is stochastic, then one can update each member, $i = 1, \dots, m$, of the ensemble using a state analysis with $\mathbf{B}$, which essentially solves the variational problem with the following cost function:

$$\mathscr{L}_i(\mathbf{x}) = \frac{1}{2}\|\mathbf{y} + \boldsymbol{\epsilon}_i - \mathscr{H}(\mathbf{x})\|^2_{\mathbf{R}} + \frac{1}{2}\|\mathbf{x} - \mathbf{x}_i\|^2_{\mathbf{B}}, \tag{7.1}$$

where $\mathbf{x}_i$ is the first guess of member i and $\boldsymbol{\epsilon}_i$ is a member-dependent perturbation drawn from the observation error prior (see Section 6.3). Recall that $\|\mathbf{z}\|^2_{\mathbf{B}} = \mathbf{z}^{\mathrm{T}}\mathbf{B}^{-1}\mathbf{z}$. The elegance of the scheme lies in the fact that it yields a statistically consistent update of the ensemble thanks to the stochastic representation of errors. This consistency is exact if $\mathscr{H}$ is linear and approximate otherwise. It was first proposed by Hamill and Snyder [2000]. This EnKF–3D-Var scheme was shown to improve the performance of DA over the EnKF when the ensemble is small, and over 3D-Var when the observation network is not dense enough. Of course, this requires a proper tuning of γ.

If the EnKF is deterministic, the construction of the hybrid is not as straightforward, especially concerning the update of the perturbation ensemble. First, one extracts from the ensemble the mean state and perturbations, $\{\mathbf{x}_i\}_{i=1,\dots,m} \longrightarrow \overline{\mathbf{x}}, \mathbf{X}_{\mathrm{f}}$. Second, one can perform an analysis on the mean state, i.e., solve the following variational problem:

$$\mathbf{x}^{\mathrm{a}} = \underset{\mathbf{x}}{\operatorname{argmin}} \mathscr{L}(\mathbf{x}) \quad \text{where} \quad \mathscr{L}(\mathbf{x}) = \frac{1}{2}\|\mathbf{y} - \mathscr{H}(\mathbf{x})\|^2_{\mathbf{R}} + \frac{1}{2}\|\mathbf{x} - \overline{\mathbf{x}}\|^2_{\mathbf{B}}. \tag{7.2}$$

This mean state analysis is equivalent to that of the hybrid stochastic EnKF–3D-Var (7.1) if the perturbed observations are centered and if $\mathscr{H}$ is close to being linear.

However, the deterministic update of the perturbations is not as simple as in the stochastic hybrid case. To make it simple, and in the absence of localization of the perturbations, Wang et al. [2007, 2008] advocated using the right-transform perturbations update of the ETKF scheme (see (6.16) of Section 6.4.3):

$$\mathbf{X}_{\mathrm{a}} = \mathbf{X}_{\mathrm{f}}\left(\mathbf{I}_m + \mathbf{Y}_{\mathrm{f}}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{Y}_{\mathrm{f}}\right)^{-\frac{1}{2}}\mathbf{U}, \quad \text{where} \quad \mathbf{Y}_{\mathrm{f}} = \mathbf{H}\mathbf{X}_{\mathrm{f}}.$$

This is simple but only updates the perturbations based on the knowledge of the dynamical errors, neglecting the static background errors. It is as if γ were set to 0 in the perturbations update step. This may lead to a misestimation of the errors and could require inflation to account for a possible underestimation. These updated perturbations obtained from $\mathbf{X}_{\mathrm{a}}$ are then added to the state analysis, $\mathbf{x}^{\mathrm{a}}$, to obtain the updated ensemble.

However, there are ways to improve these perturbation updates. We saw in Section 6.4.4 that a full-rank left-transform update is also theoretically possible:

$$\mathbf{X}_{\mathrm{a}} = \left(\mathbf{I}_n + \mathbf{B}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-\frac{1}{2}}\mathbf{X}_{\mathrm{f}}\mathbf{U}, \tag{7.3}$$

with $\mathbf{B} = \gamma\mathbf{C} + (1-\gamma)\mathbf{P}^{\mathrm{f}}$, which is meant to be full rank. With high-dimensional systems, this inverse square root is not numerically affordable. But it could be estimated using iterative Lanczos-based methods (see Golub and van Loan [2013]) or randomized SVD techniques (see Halko et al. [2011]), because the full-rank left transform acts on the small-dimensional vector subspace of the perturbations, $\mathbf{X}_{\mathrm{f}}$.

The perturbations could actually be computed by extracting the dominant modes of the Hessian of the cost function (7.2). This could be achieved using iterative Krylov-based methods [Golub and van Loan, 2013]. Extracting the Lanczos vectors was advocated by T. Auligné and implemented in the EVIL (Ensemble Variational Integrated Lanczos) DA system.

Bocquet et al. [2015] proposed generalizing the left-transform update formula (7.3) of Sakov and Bertino [2011]. This formula is meant to extract some residual information from the static background error statistics that is not accounted for in (7.3) if used in an hybrid system. It reads

$$\mathbf{X}_{\mathrm{a}} = \left[\mathbf{I}_n + \left(\gamma\mathbf{C} + (1-\gamma)\mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}\right)\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right]^{-\frac{1}{2}} \left[(1-\gamma)\mathbf{I}_n + \gamma\mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\dagger}\mathbf{C}\left(\mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}\right)^{\dagger}\right]^{\frac{1}{2}} \mathbf{X}_{\mathrm{f}}\mathbf{U},$$

where † denotes the Moore–Penrose inverse of a matrix.

Most of the hybrid methods focus on a linear combination of covariance matrices. However, Penny [2014] proposed using a linear combination of Kalman gains that depends in a nontrivial way on the gain that would be obtained from the two DA methods that are part of the hybrid. Penny [2014] defines a hybrid/mean-LETKF: it runs an LETKF to obtain a mean update $\overline{\mathbf{x}}_{\mathrm{a}}$, which is used in the subsequent 3D-Var analysis defined by the cost function

$$\mathscr{L}(\mathbf{x}) = \frac{1}{2}\,\|\mathbf{y} - \mathscr{H}(\mathbf{x})\|_{\mathbf{R}}^2 + \frac{1}{2}\,\|\mathbf{x} - \overline{\mathbf{x}}_{\mathrm{a}}\|_{\mathbf{C}}^2 .$$

The updated ensemble of the LETKF is then recentered on $\mathbf{x}^{\mathrm{a}}_{\mathrm{hybrid}} = \underset{\mathbf{x}}{\operatorname{argmin}}\,\mathscr{L}(\mathbf{x})$. This was shown to be equivalent to the update

$$\mathbf{x}^{\mathrm{a}}_{\mathrm{hybrid}} = \overline{\mathbf{x}} + \widehat{\mathbf{K}}(\mathbf{y} - \mathscr{H}(\overline{\mathbf{x}})), \quad \text{with} \quad \widehat{\mathbf{K}} = \mathbf{K} + \gamma\mathbf{K}^{\mathrm{C}}\left(\mathbf{I}_p - \mathbf{H}\mathbf{K}\right),$$

where $\overline{\mathbf{x}}$ is the ensemble mean,

$$\mathbf{K} = \mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}\right)^{-1}$$

is the flow-dependent gain and

$$\mathbf{K}^{\mathrm{C}} = \mathbf{C}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{C}\mathbf{H}^{\mathrm{T}} + \mathbf{R}\right)^{-1}$$

is the static gain.

7.1.2 ▪ Augmented state formalism

The analysis process that results from the blending of several covariances as many sources of statistical information can be represented by the straightforward linear combinations of the covariances. It can also be represented through an augmented state formalism where the analysis increments coming from each part of the combined covariance matrix seem independent. This has been suggested by Lorenc [2003] and Buehner [2005].

The two representations are actually equivalent [Wang et al., 2007]. Let us give a minimal demonstration of this equivalence by only considering the prior of a hybrid analysis where $\mathbf{B}_1$ and $\mathbf{B}_2$ are the two background error covariance matrices that are combined. For the sake of simplicity, they are both considered full-rank matrices, but the generalization to the case where one of them is degenerate does not pose any difficulty. Here, the mixing coefficients (such as γ) are hidden in the definition of $\mathbf{B}_1$ and $\mathbf{B}_2$. The prior term of the cost function reads

$$\mathcal{L}_{\mathrm{b}}(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}-\overline{\mathbf{x}}\|^2_{\mathbf{B}_1+\mathbf{B}_2}.$$

Introducing the Lagrangian,

$$\mathcal{L}_{\mathrm{b}}(\mathbf{x},\boldsymbol{\eta}) = \boldsymbol{\eta}^{\mathrm{T}}(\mathbf{x}-\overline{\mathbf{x}}) - \frac{1}{2}\|\boldsymbol{\eta}\|^2_{(\mathbf{B}_1+\mathbf{B}_2)^{-1}} = \boldsymbol{\eta}^{\mathrm{T}}(\mathbf{x}-\overline{\mathbf{x}}) - \frac{1}{2}\|\boldsymbol{\eta}\|^2_{\mathbf{B}_1^{-1}} - \frac{1}{2}\|\boldsymbol{\eta}\|^2_{\mathbf{B}_2^{-1}},$$

which is a function of $\mathbf{x}$ and a vector of Lagrange multipliers $\boldsymbol{\eta}$, we have

$$\max_{\boldsymbol{\eta}} \mathcal{L}_{\mathrm{b}}(\mathbf{x},\boldsymbol{\eta}) = \mathcal{L}_{\mathrm{b}}(\mathbf{x}),$$

by simple maximization of a quadratic cost function.

The same trick can be enforced once again but on the two last quadratic terms separately using ancillary variables $\delta\mathbf{x}_1$ and $\delta\mathbf{x}_2$. Denoting

$$\mathcal{L}_{\mathrm{b}}(\mathbf{x},\boldsymbol{\eta},\delta\mathbf{x}_1,\delta\mathbf{x}_2) = \boldsymbol{\eta}^{\mathrm{T}}(\mathbf{x}-\overline{\mathbf{x}}) - \boldsymbol{\eta}^{\mathrm{T}}\delta\mathbf{x}_1 + \frac{1}{2}\|\delta\mathbf{x}_1\|^2_{\mathbf{B}_1} - \boldsymbol{\eta}^{\mathrm{T}}\delta\mathbf{x}_2 + \frac{1}{2}\|\delta\mathbf{x}_2\|^2_{\mathbf{B}_2},$$

we have

$$\min_{\delta\mathbf{x}_1,\delta\mathbf{x}_2} \mathcal{L}_{\mathrm{b}}(\mathbf{x},\boldsymbol{\eta},\delta\mathbf{x}_1,\delta\mathbf{x}_2) = \mathcal{L}_{\mathrm{b}}(\mathbf{x},\boldsymbol{\eta}).$$

Factorizing on $\boldsymbol{\eta}$, we obtain the Lagrangian

$$\mathcal{L}_{\mathrm{b}}(\mathbf{x},\boldsymbol{\eta},\delta\mathbf{x}_1,\delta\mathbf{x}_2) = \boldsymbol{\eta}^{\mathrm{T}}(\mathbf{x}-\overline{\mathbf{x}}-\delta\mathbf{x}_1-\delta\mathbf{x}_2) + \frac{1}{2}\|\delta\mathbf{x}_1\|^2_{\mathbf{B}_1} + \frac{1}{2}\|\delta\mathbf{x}_2\|^2_{\mathbf{B}_2},$$

where $\boldsymbol{\eta}$ clearly plays the role of a vector of Lagrange multipliers, which enforces the constraint $\mathbf{x} = \overline{\mathbf{x}} + \delta\mathbf{x}_1 + \delta\mathbf{x}_2$, where $\delta\mathbf{x}_1$ and $\delta\mathbf{x}_2$ are two a priori independent increments.

When applied to the full analysis cost function, this yields an optimization problem over $\delta\mathbf{x}_1$ and $\delta\mathbf{x}_2$ in place of $\mathbf{x}$ but under the constraint $\mathbf{x} = \overline{\mathbf{x}} + \delta\mathbf{x}_1 + \delta\mathbf{x}_2$. Indeed, we have

$$\begin{aligned}
\mathcal{L} &= \min_{\mathbf{x}} \mathcal{L}(\mathbf{x}) \\
&= \min_{\mathbf{x}} \left\{ \frac{1}{2}\|\mathbf{y}-\mathcal{H}(\mathbf{x})\|^2_{\mathbf{R}} + \mathcal{L}_{\mathrm{b}}(\mathbf{x}) \right\} \\
&= \min_{\mathbf{x}} \left\{ \frac{1}{2}\|\mathbf{y}-\mathcal{H}(\mathbf{x})\|^2_{\mathbf{R}} + \max_{\boldsymbol{\eta}} \min_{\delta\mathbf{x}_1,\delta\mathbf{x}_2} \mathcal{L}_{\mathrm{b}}(\mathbf{x},\boldsymbol{\eta},\delta\mathbf{x}_1,\delta\mathbf{x}_2) \right\} \\
&= \min_{\mathbf{x},\delta\mathbf{x}_1,\delta\mathbf{x}_2 : \mathbf{x}=\overline{\mathbf{x}}+\delta\mathbf{x}_1+\delta\mathbf{x}_2} \left\{ \frac{1}{2}\|\mathbf{y}-\mathcal{H}(\mathbf{x})\|^2_{\mathbf{R}} + \frac{1}{2}\|\delta\mathbf{x}_1\|^2_{\mathbf{B}_1} + \frac{1}{2}\|\delta\mathbf{x}_2\|^2_{\mathbf{B}_2} \right\} \\
&= \min_{\delta\mathbf{x}_1,\delta\mathbf{x}_2} \left\{ \frac{1}{2}\|\mathbf{y}-\mathcal{H}(\overline{\mathbf{x}}+\delta\mathbf{x}_1+\delta\mathbf{x}_2)\|^2_{\mathbf{R}} + \frac{1}{2}\|\delta\mathbf{x}_1\|^2_{\mathbf{B}_1} + \frac{1}{2}\|\delta\mathbf{x}_2\|^2_{\mathbf{B}_2} \right\} \\
&= \min_{\delta\mathbf{x}_1,\delta\mathbf{x}_2} \mathcal{L}(\delta\mathbf{x}_1,\delta\mathbf{x}_2).
\end{aligned} \tag{7.4}$$

This proves the equivalence between minimizing $\mathcal{L}(\mathbf{x})$ and minimizing $\mathcal{L}(\delta\mathbf{x}_1, \delta\mathbf{x}_2)$, which is defined by the last equality of (7.4).

One advantage of this augmented state representation is that a preconditioned variational problem for $\mathcal{L}(\delta\mathbf{x}_1, \delta\mathbf{x}_2)$ can be obtained more easily. For instance, if we use $\mathbf{B} = \gamma\mathbf{C} + (1-\gamma)\mathbf{X}_\mathrm{f}\mathbf{X}_\mathrm{f}^\mathrm{T}$, and assuming we know a square root of $\mathbf{C}$, we can obtain the equivalent augmented state cost function

$$\mathcal{L}(\delta\mathbf{x}_\mathrm{c}, \mathbf{w}) = \frac{1}{2}\left\|\mathbf{y} - \mathcal{H}\left(\overline{\mathbf{x}} + \sqrt{\gamma}\mathbf{C}^{\frac{1}{2}}\delta\mathbf{x}_\mathrm{c} + \sqrt{1-\gamma}\mathbf{X}_\mathrm{f}\mathbf{w}\right)\right\|^2_\mathbf{R} + \frac{1}{2}\|\delta\mathbf{x}_\mathrm{c}\|^2 + \frac{1}{2}\|\mathbf{w}\|^2,$$

where $\|\mathbf{z}\|^2 = \mathbf{z}^\mathrm{T}\mathbf{z}$. This cost function may be much more efficiently minimized than

$$\mathcal{L}(\mathbf{x}) = \frac{1}{2}\|\mathbf{y} - \mathcal{H}(\mathbf{x})\|^2_\mathbf{R} + \frac{1}{2}\|\mathbf{x} - \overline{\mathbf{x}}\|^2_{\gamma\mathbf{C}+(1-\gamma)\mathbf{X}_\mathrm{f}\mathbf{X}_\mathrm{f}^\mathrm{T}}$$

because of the preconditioning of $\delta\mathbf{x}_\mathrm{c}$ and $\mathbf{w}$.

7.1.3 ▪ Localization of the perturbations

As can be seen from $\mathbf{B} = \gamma\mathbf{C} + (1-\gamma)\mathbf{X}_\mathrm{f}\mathbf{X}_\mathrm{f}^\mathrm{T}$, the effective background error covariance matrix is full rank as long as the static prior remains nondegenerate, even if the ensemble size is small. This actually performs a kind of localization called *shrinkage* in statistics. That is why an EnKF can be made robust by this hybridization even if the ensemble is small and even in the absence of the main localization techniques that we discussed in Chapter 6 [Hamill and Snyder, 2000; Wang et al., 2007]. However, in theory and in practice, shrinkage does not seem as efficient as localization for geophysical models. It is therefore useful to further localize the perturbations in the combined covariance matrix $\mathbf{B}$ with

$$\mathbf{B} = \gamma\mathbf{C} + (1-\gamma)\boldsymbol{\rho}\circ(\mathbf{X}_\mathrm{f}\mathbf{X}_\mathrm{f}^\mathrm{T}),$$

using covariance localization by a Schur product with a short-range correlation matrix $\boldsymbol{\rho}$ (see Section 6.5.1.2). The variational analysis is then based on the cost function

$$\mathcal{L}(\mathbf{x}) = \frac{1}{2}\|\mathbf{y} - \mathcal{H}(\mathbf{x})\|^2_\mathbf{R} + \frac{1}{2}\|\mathbf{x} - \overline{\mathbf{x}}\|^2_{\gamma\mathbf{C}+(1-\gamma)\boldsymbol{\rho}\circ(\mathbf{X}_\mathrm{f}\mathbf{X}_\mathrm{f}^\mathrm{T})}. \tag{7.5}$$

The augmented-state trick can be used to separate the increments that project onto the static background error covariance matrix and onto the localized sampled error covariance matrix, as follows:

$$\begin{aligned}\mathcal{L}(\delta\mathbf{x}_\mathrm{c}, \delta\mathbf{x}_\mathrm{p}) =& \frac{1}{2}\left\|\mathbf{y} - \mathcal{H}(\overline{\mathbf{x}} + \delta\mathbf{x}_\mathrm{c} + \delta\mathbf{x}_\mathrm{p})\right\|^2_\mathbf{R} \\ &+ \frac{1}{2}\|\delta\mathbf{x}_\mathrm{c}\|^2_{\gamma\mathbf{C}} + \frac{1}{2}\left\|\delta\mathbf{x}_\mathrm{p}\right\|^2_{(1-\gamma)\boldsymbol{\rho}\circ(\mathbf{X}_\mathrm{f}\mathbf{X}_\mathrm{f}^\mathrm{T})}.\end{aligned}$$

Then, we can reuse the $\boldsymbol{\alpha}$ control variable trick of Section 6.7.2.3 on the flow-dependent part of the background error covariances. We note that this $\boldsymbol{\alpha}$ control variable trick can actually be seen as an augmented spacewise state representation. We obtain

$$\begin{aligned}\mathcal{L}(\delta\mathbf{x}_\mathrm{c}, \boldsymbol{\alpha}) =& \frac{1}{2}\left\|\mathbf{y} - \mathcal{H}\left(\overline{\mathbf{x}} + \delta\mathbf{x}_\mathrm{c} + \sum_{i=1}^m \boldsymbol{\alpha}_i \circ [\mathbf{X}_\mathrm{f}]_i\right)\right\|^2_\mathbf{R} \\ &+ \frac{1}{2}\|\delta\mathbf{x}_\mathrm{c}\|^2_{\gamma\mathbf{C}} + \frac{1}{2}\sum_{i=1}^m \|\boldsymbol{\alpha}_i\|^2_{(1-\gamma)\boldsymbol{\rho}}.\end{aligned}$$

Recall that $\alpha = [\alpha_1, \ldots, \alpha_m]$ can be viewed as a matrix of size $n \times m$, and that $[\mathbf{X}_\mathrm{f}]_i$ is the ith perturbation of the normalized perturbation matrix $\mathbf{X}_\mathrm{f}$. With preconditioning, one finally obtains

$$\mathscr{L}(\delta\mathbf{x}_\mathrm{c}, \alpha) = \frac{1}{2}\left\|\mathbf{y} - \mathscr{H}\left(\overline{\mathbf{x}} + \sqrt{\gamma}\mathbf{C}^{\frac{1}{2}}\delta\mathbf{x}_\mathrm{c} + \sqrt{1-\gamma}\rho^{\frac{1}{2}}\sum_{i=1}^{m}\alpha_i \circ [\mathbf{X}_\mathrm{f}]_i\right)\right\|_\mathbf{R}^2 + \frac{1}{2}\|\delta\mathbf{x}_\mathrm{c}\|^2 + \frac{1}{2}\sum_{i=1}^{m}\|\alpha_i\|^2.$$

This representation, which was first advocated by Lorenc [2003] and Buehner [2005], implies that the solution is parameterized as

$$\mathbf{x} = \overline{\mathbf{x}} + \sqrt{\gamma}\mathbf{C}^{\frac{1}{2}}\delta\mathbf{x}_\mathrm{c} + \sqrt{1-\gamma}\rho^{\frac{1}{2}}\sum_{i=1}^{m}\alpha_i \circ [\mathbf{X}_\mathrm{f}]_i.$$

7.2 ▪ Ensemble of variational data assimilations (EDA)

Another class of hybrid systems stems from NWP centers where 4D-Var is in use. It consists of processing an *ensemble of data assimilations* or *EDA*. More generally, an EDA system denotes a DA system that processes several variational analyses in parallel. The goal is to introduce some flow dependence in the 4D-Var operational schemes that were initially only based on the static background error covariance matrix. This scheme actually closely mimics the stochastic EnKF and, even more to the point, the hybrid EnKF–3D-Var scheme introduced by Hamill and Snyder [2000] and described earlier.

The main idea is to maintain an ensemble of Var, which will be assumed to be a 4D-Var in the following. This is usually numerically costly for 4D-Var and may require degrading model resolution. Each analysis uses the same static background error covariance matrix $\mathbf{B}$, which may have been obtained from the sampled covariances whose variances have been properly filtered and whose correlations have been regularized, as well as the static background covariances [Raynaud et al., 2009; Berre and Desroziers, 2010]. Just as in the stochastic EnKF, it is necessary for each analysis to have perturbed observations to maintain statistical consistency (Section 6.3). The strong-constraint 4D-Var cost function for each analysis, $i = 1, \ldots, m$, has the form

$$\mathscr{L}_i(\mathbf{x}_0) = \frac{1}{2}\sum_{k=0}^{K}\left\|\mathbf{y}_k + \epsilon_k^i - \mathscr{H}_k \circ \mathscr{M}_{k:0}(\mathbf{x}_0)\right\|_{\mathbf{R}_k}^2 + \frac{1}{2}\left\|\mathbf{x}_0 - \mathbf{x}_0^i\right\|_\mathbf{B}^2,$$

where $\mathscr{M}_{k:0}$ is the resolvent of the forecast model from t_0 to t_k, $\mathscr{H}_k$ is the observation operator at t_k, and ϵ_k^i is the random noise added to observation $\mathbf{y}_k$ and is related to the ith member analysis. The symbol $\circ$ stands for the composition operator. This generates an ensemble of updates, one for each $\mathbf{x}_0^i$, similarly to the stochastic EnKF. It is also possible to perturb each member of the ensemble in the forecast step to account for (parametric or not) model error. Hence, the ensemble will be instrumental in accounting for flow dependence and model error.

This methodology has been implemented at Météo-France [Raynaud et al., 2009; Berre and Desroziers, 2010; Berre et al., 2015] as well as at the ECMWF [Bonavita et al., 2011, 2012] to enhance their 4D-Var system and to introduce flow dependence of the errors. From a mathematical standpoint, their operational 4D-Var is now quite close to a hybrid DA system based on a stochastic EnKF. The regularization techniques parallel those used in the EnKF: inflation and localization. The localization is, however, enforced via a regularization based on wavelets [Berre et al., 2015], a technique that has been long perfected in operational 4D-Var at the ECMWF and Météo-France.

As a rather consistent stochastic sampling method, an ensemble of 4D-Vars following the EDA scheme is likely to approximately sample from the Bayesian conditional distribution of the DA problem over the 4D-Var DAW. M. Jardak and O. Talagrand have numerically shown that a single analysis of such an EDA approach leads to a reliable updated ensemble and a quasi-flat rank diagram histogram, which supports the seemingly Bayesian character of the EDA sampling (personal communication).

7.3 ▪ 4DEnVar

The primary goal of NWP centers that already use 4D-Var (the ECMWF, the UK MetOffice, Météo-France, Environment Canada, the Japan Meteorological Agency) has been to circumvent the derivation and maintenance of the tangent linear and adjoint forecast models. A secondary goal has been to incorporate flow dependence into 4D-Var. As we have seen, this second objective can be achieved using an EDA system that essentially mimics the stochastic EnKF.

7.3.1 ▪ A 4D-Var without the forecast model adjoint

A solution for the first problem draws its inspiration from the elegant way the EnKF handles the tangent linear and adjoint of the observation operator. We saw in Chapter 6 that any occurrence of the error statistics or any regression coefficient that involves the tangent linear of the observation operator can be estimated thanks to the ensemble of perturbations. For instance, recalling (6.11) from Section 6.3.3,

$$
\begin{aligned}
\overline{\mathbf{y}}^{\mathrm{f}} &= \frac{1}{m}\sum_{i=1}^{m}\mathcal{H}\left(\mathbf{x}_i^{\mathrm{f}}\right),\\
\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}} &\simeq \frac{1}{m-1}\sum_{i=1}^{m}\left(\mathbf{x}_i^{\mathrm{f}}-\overline{\mathbf{x}}^{\mathrm{f}}\right)\left[\mathcal{H}\left(\mathbf{x}_i^{\mathrm{f}}\right)-\overline{\mathbf{y}}^{\mathrm{f}}\right]^{\mathrm{T}},\\
\mathbf{H}\mathbf{P}^{\mathrm{f}}\mathbf{H}^{\mathrm{T}} &\simeq \frac{1}{m-1}\sum_{i=1}^{m}\left[\mathcal{H}\left(\mathbf{x}_i^{\mathrm{f}}\right)-\overline{\mathbf{y}}^{\mathrm{f}}\right]\left[\mathcal{H}\left(\mathbf{x}_i^{\mathrm{f}}\right)-\overline{\mathbf{y}}^{\mathrm{f}}\right]^{\mathrm{T}}.
\end{aligned}
$$

In a 4D-Var, we are also concerned with the sensitivities to observations within the DAW to the initial state vector. These are typically related to the asynchronous and synchronous statistics $\mathbf{P}_0^{\mathrm{f}}(\mathbf{H}_k\mathbf{M}_{k:0})^{\mathrm{T}}$ and $(\mathbf{H}_k\mathbf{M}_{k:0})\mathbf{P}_0^{\mathrm{f}}(\mathbf{H}_k\mathbf{M}_{k:0})^{\mathrm{T}}$, respectively, where $\mathbf{M}_{k:0}$ is the tangent linear of the forecast model that relates an error at the beginning of the DAW, i.e., at t_0, to an error within the DAW, at t_k, whereas $\mathbf{H}_k\mathbf{M}_{k:0}$ relates an error, t_0, to an observation error within the DAW.

Using a similar trick as in the EnKF, Liu et al. [2008] proposed using the ensemble to estimate these sensitivities within the DAW, which, in the absence of the ensemble, would require the adjoint models. The ensemble is forecast within the DAW, with $\mathcal{H}_k \circ \mathcal{M}_{k:0}$ playing the role of $\mathcal{H}_k$ in the EnKF. In essence, the adjoint can be computed in this scheme because it is estimated from the tangent linear by an ansatz of the finite differences, and because the action of the model within the ensemble subspace can easily be represented by a matrix whose adjoint is simply the matrix transpose operator.

Therefore, to avoid the use of the adjoint in a 4D-Var, one could perform the variational optimization in the reduced basis of the ensemble perturbations. This idea clearly borrows from the reduced-order 4D-Vars developed in ocean DA [Robert et al., 2005, 2006a], which avoid the use of the adjoint model and offer an efficient preconditioning for the minimization (see Section 5.4.2).

More precisely, let us start with the 4D-Var cost function over the DAW $[t_0, t_K]$,

$$\mathscr{L}(\mathbf{x}_0) = \frac{1}{2}\sum_{k=0}^{K} ||\mathbf{y}_k - \mathscr{F}_{k:0}(\mathbf{x}_0)||^2_{\mathbf{R}_k} + \frac{1}{2}||\mathbf{x}_0 - \overline{\mathbf{x}}_0||^2_{\mathbf{P}^{\mathrm{f}}},$$

where observations at $t_0, t_1, \dots, t_K$ are assimilated. We have defined $\mathscr{F}_{k:0} = \mathscr{H}_k \circ \mathscr{M}_{k:0}$, where $\circ$ stands for the composition of operators. Suppose that the background error covariance matrix is sampled from an ensemble of centered perturbations, $\mathbf{P}^{\mathrm{f}} = \mathbf{X}_{\mathrm{f}}\mathbf{X}_{\mathrm{f}}^{\mathrm{T}}$. Following the ETKF construction (see Chapter 6), we look for a solution in the affine subspace generated by $\overline{\mathbf{x}}_0$ and the perturbations $\mathbf{X}_{\mathrm{f}}$,

$$\mathbf{x}_0 = \overline{\mathbf{x}}_0 + \mathbf{X}_{\mathrm{f}}\mathbf{w}.$$

The goal is to minimize the reduced-order cost function

$$\mathscr{L}(\mathbf{w}) = \frac{1}{2}\sum_{k=0}^{K} ||\mathbf{y}_k - \mathscr{F}_{k:0}(\overline{\mathbf{x}}_0 + \mathbf{X}_{\mathrm{f}}\mathbf{w})||^2_{\mathbf{R}_k} + \frac{1}{2}||\mathbf{w}||^2,$$

whose quadratic approximation obtained by linearizing around the first guess is

$$\mathscr{L}(\mathbf{w}) \simeq \frac{1}{2}\sum_{k=0}^{K} ||\mathbf{y}_k - \mathscr{F}_{k:0}(\overline{\mathbf{x}}_0) + \mathbf{H}_k\mathbf{M}_{k:0}\mathbf{X}_{\mathrm{f}}\mathbf{w}||^2_{\mathbf{R}_k} + \frac{1}{2}||\mathbf{w}||^2.$$

The argument of the minimum of this cost function is

$$\mathbf{w}^{\mathrm{a}} = \left[\mathbf{I}_m + \sum_{k=0}^{K}(\mathbf{H}_k\mathbf{M}_{k:0}\mathbf{X}_{\mathrm{f}})^{\mathrm{T}}\mathbf{R}_k^{-1}(\mathbf{H}_k\mathbf{M}_{k:0}\mathbf{X}_{\mathrm{f}})\right]^{-1} \sum_{k=0}^{K}(\mathbf{H}_k\mathbf{M}_{k:0}\mathbf{X}_{\mathrm{f}})^{\mathrm{T}}\mathbf{R}_k^{-1}(\mathbf{y}_k - \mathscr{F}_{k:0}(\overline{\mathbf{x}}_0)).$$

The key point is that $\mathbf{H}_k\mathbf{M}_{k:0}\mathbf{X}_{\mathrm{f}}$, as well as its transpose matrix, can be estimated with a single ensemble forecast through the DAW,

$$\mathbf{H}_k\mathbf{M}_{k:0}\mathbf{X}_{\mathrm{f}} \approx \frac{1}{\varepsilon}\mathscr{F}_{k:0}\left(\overline{\mathbf{x}}_0\mathbf{1}^{\mathrm{T}} + \varepsilon\mathbf{X}_{\mathrm{f}}\right)\left(\mathbf{I}_m - \frac{\mathbf{1}\mathbf{1}^{\mathrm{T}}}{m}\right), \tag{7.6}$$

where $0 < \varepsilon \ll 1$ is a small scaling parameter needed to compute the finite differences. These can also be estimated by choosing $\varepsilon = 1$ if the ensemble spread is small enough.

This stands as the analysis step of the method that is now commonly called *4DEnVar*, where $\varepsilon = 1$ is chosen to minimize the number of ensemble forecasts. It can hardly be seen as a new DA method as it is based on or revisits already-known techniques such as reduced-order 4D-Var. But it does stand as a class of implementation techniques meant to circumvent the use of the adjoint model in the 4D-Var, and possibly as in EDA systems to incorporate flow-dependent errors.

7.3.2 ▪ Variants

Another ingredient needed to fully specify a 4DEnVar scheme is the choice of the perturbations. In Liu et al. [2009] and Buehner et al. [2010a], the perturbations were obtained from a stochastic EnKF similarly to Hamill and Snyder [2000]. This introduced the flow dependence of the errors. Quite often, the 4DEnVar performs variational analysis using the perturbations generated by an independent EnKF implementation. If the updated perturbations are obtained deterministically from an ETKF-like update formula derived from the main variational analysis, rather than from an independent EnKF running concurrently, then this DA scheme actually mathematically coincides with the so-called 4D-LETKF [Hunt et al., 2004; Fertig et al., 2007].

If an adjoint model used in the original 4D-Var system is available and if the focus is more on building an EnVar to represent the flow dependence of the errors, then the method is usually referred to as a *4D-Var-Ben* [Buehner et al., 2010a,b], emphasizing that the method still resembles a 4D-Var where the background error covariance matrix has been obtained from an ensemble.

In Section 7.1, we mostly focused on the hybridization of an EnKF with a 3D-Var. A similar approach can be implemented but with a 4D-Var [Zhang and Zhang, 2012; Clayton et al., 2013]. Again the scheme mainly relies on an EnKF. A 4D variational optimization is performed for state estimation using the combination of a static background covariance matrix and the flow-dependent sampled covariances of the EnKF. The latter can be localized using, for instance, a 4D generalization of the α control variable trick. This yields a state analysis. The updated ensemble is built on the updated ensemble of the EnKF but centered on the variational state analysis. Similar to 4D-Var-Ben, this EnVar approach benefits from the available adjoint of the 4D-Var.

To generate updated perturbations in an EDA system where a single state analysis is obtained from a 4DEnVar, one would have to consider an ensemble of 4DEnVar, an option that is considered at the UK MetOffice [Lorenc et al., 2015].

7.3.3 ▪ Localization and weak-constraint formalism

The main drawback of performing the four-dimensional variational analysis within the ensemble subspace is that, as for any ensemble-based method, it is likely to be rank deficient with realistic high-dimensional models. That is why localization is mandatory. Unfortunately, spatial localization in such a 4D context is not as simple as in a 3D context, where covariance localization only requires the definition of a proper localization matrix, usually the direct product of horizontal and vertical correlation functions. With the inclusion of the time component, one needs to be able to localize covariances such as $\mathbf{X}_0\mathbf{Y}_k^{\mathrm{T}}$ between state variables at t_0 and observations at t_k within the DAW, as well as $\mathbf{Y}_k\mathbf{Y}_k^{\mathrm{T}}$ between observations at t_k within the DAW. It is sound to assume that one can localize the background error covariance matrix at t_0, $\mathbf{X}_0\mathbf{X}_0^{\mathrm{T}} \mapsto \rho \circ \left(\mathbf{X}_0\mathbf{X}_0^{\mathrm{T}}\right)$, using the localization matrix ρ defined at the beginning of the DAW. But the regularization of covariances such as $\mathbf{X}_0\mathbf{Y}_k^{\mathrm{T}}$, $\mathbf{Y}_k\mathbf{Y}_k^{\mathrm{T}}$ is not as straightforward. Indeed, the model flow and a static localization operator (here a Schur product) do not commute in general, i.e.,

$$\mathbf{M}_{k:0}(\rho \circ \mathbf{P}_0)\mathbf{M}_{k:0}^{\mathrm{T}} \neq \rho \circ \left(\mathbf{M}_{k:0}\mathbf{P}_0\mathbf{M}_{k:0}^{\mathrm{T}}\right), \tag{7.7}$$

as emphasized in Fairbairn et al. [2014] and Bocquet and Sakov [2014].

Assuming that the model is perfect, and that a spatial localization operator is defined at the beginning of the DAW at t_0, it is formally possible to derive the equation that governs the dynamics of the localization operator [Bocquet, 2016]. Let us consider a *linear* localization operator that applies to covariances, $\mathbb{L}:\mathbf{P}\mapsto\mathbb{L}\cdot\mathbf{P}$. Schur localization is an example of such a generic linear operator. The transport of the covariances by the TLM from t to $t+\delta t$ is formalized with the linear operator

$$\mathscr{T}_{t+\delta t:t}:\mathbf{P}_t\mapsto\mathscr{T}_{t+\delta t:t}\cdot\mathbf{P}_t=\mathbf{M}_{t+\delta t:t}\mathbf{P}_t\mathbf{M}_{t+\delta t:t}^{\mathrm{T}}.$$

Assuming that the localization operator evolves in time, we wish to impose that localization commutes with the TLM, which can be written as

$$\mathscr{T}_{t+\delta t:t}\cdot\mathbb{L}_t\cdot\mathbf{P}_t=\mathbb{L}_{t+\delta t}\cdot\mathscr{T}_{t+\delta t:t}\cdot\mathbf{P}_t. \tag{7.8}$$

Ideally, this dynamical constraint should apply to any $t,\delta t$, and $\mathbf{P}_t$. In the limit where δt goes to zero, the transport operator can be expanded as

$$\mathscr{T}_{t+\delta t:t}=\mathbf{I}\otimes\mathbf{I}+\mathscr{K}_t\delta t+o(\delta t),$$

where $\mathscr{K}_t$ is a linear operator that can be written $\mathscr{K}_t=\mathbf{M}_t\otimes\mathbf{I}+\mathbf{I}\otimes\mathbf{M}_t$ and $\mathbf{I}\otimes\mathbf{I}$ is the identity operator in the space of the covariance matrices. As a consequence, (7.8) can in turn be expanded in the $\delta t\to 0$ limit to yield

$$\frac{\partial\mathbb{L}_t}{\partial t}=[\mathscr{K}_t,\mathbb{L}_t], \tag{7.9}$$

where $[A,B]=AB-BA$ is the commutator of operators. Formally, $\mathbb{L}_t$ actually acts on vec($\mathbf{P}_t$), i.e., the vector formed by the stacked columns of $\mathbf{P}_t$. This (quantum) *Liouville equation* prescribes the dynamics of a consistent localization operator for synchronous covariances of the form $\mathbf{Y}_k\mathbf{Y}_k^{\mathrm{T}}$. A consistent asynchronous localization operator that can be used to regularize covariances such as $\mathbf{X}_0\mathbf{Y}_k^{\mathrm{T}}$ is also needed. In this case, the same Liouville equation still applies but with an asymmetrical transport $\mathscr{K}_t=\mathbf{I}\otimes\mathbf{M}_t$ (or $\mathscr{K}_t=\mathbf{M}_t\otimes\mathbf{I}$ for $\mathbf{Y}_k\mathbf{X}_0^{\mathrm{T}}$).

An exact solution to this equation can only be obtained in restrictive cases, for instance when the dynamics of the system are governed by a hyperbolic PDE, which can be solved using characteristic curves. This is not a realistic assumption but it is hoped to be a reliable approximation. In any case, localization should be *covariant* with the flow as much as possible.

This view does suggest a way to localize the perturbations within the DAW more consistently than using a static localization operator. The ensemble should be forecast exactly within the DAW, while the localization operator should be propagated using a surrogate model of advection [Bocquet, 2016; Desroziers et al., 2016]. The cost of the surrogate model of advection is small since an *embarrassingly* parallel Lagrangian transport scheme can be implemented. If the main model represents to some extent the transport of, for instance, potential vorticity, the advection field of the surrogate model could be identified with the wind field of the main model, or a filtered output of this wind field. This approximate covariant localization scheme within the DAW of a 4DEnVar has a counterpart when using domain localization that involves pulling back the locations of the in situ observations within the DAW to t_0 using a simple Lagrangian model of advection [Bocquet, 2016].

The advantage of hybrid ensemble–4DVar or 4D-Var-Ben systems that do maintain an adjoint model in comparison to 4DEnVar systems is that covariance localization within the DAW is much easier. Indeed, the minimization can be performed using the full-rank localized background error covariance matrix thanks to the adjoint model [Fairbairn et al., 2014; Poterjoy and Zhang, 2015; Bocquet, 2016].

A weak-constraint formalism for 4DEnVar has not been investigated thoroughly so far. However, a formal basis was established by Desroziers et al. [2014]. In particular, a clear connection was established between a weak-constraint 4DEnVar and a weak-constraint 4DVar, which opens the way to parallelization in time of 4DEnVar implementations [Fisher et al., 2011].

7.3.4 ▪ A replacement for 4DVar?

The 4DEnVar systems have been evaluated and sometimes validated by meteorologists in NWP centers. Quite often, their first goal, which was to achieve similar performance to 4D-Var, has been reached [Buehner et al., 2013; Gustafsson et al., 2014; Lorenc et al., 2015; Kleist and Ide, 2015; Buehner et al., 2015]. This often applies to synoptical scale meteorology, which is rather well observed so that the DA system has error statistics that are quite close to Gaussian. In particular, the EnVar and hybrid methods discussed so far did not rely on an outer loop. However, as suggested in the introduction of this chapter, EnVar methods have more generally the potential to systematically outperform both 4D-Var and EnKF systems. This will be discussed in the next section.

7.4 ▪ The iterative ensemble Kalman smoother (IEnKS)

The hybrid and EnVar methods described so far have not been derived from first principles. In the following, we shall describe another EnVar method, the iterative ensemble Kalman smoother (IEnKS), which can be heuristically derived from Bayes' rule. As a consequence, all approximations and sources of suboptimality in the scheme are known and accounted for. Moreover, because of this derivation, the scheme is heuristically guaranteed to outperform 4D-Var and the EnKF in terms of precision, regardless of the numerical cost. It can either estimate the tangent linear and adjoint models, like 4DEnVar, or it can make use of already available tangent and adjoint models, like En-4DVar and 4D-Var-Ben. We shall mostly suppose here that the tangent linear and adjoint models are not available.

The IEnKS was introduced and justified through a Bayesian derivation in Bocquet and Sakov [2014]. Here, we shall focus on the description of the algorithm. The IEnKS extends to a 4D-variational analysis the so-called iterative ensemble Kalman filter (IEnKF) of Sakov et al. [2012]. The IEnKS was named after the paper by Bell [1994], who first described such an algorithm in the Kalman filter context. The IEnKS is meant not only for smoothing but also for filtering as well as for forecasting.

In the following, we first precisely describe a simple variant of the algorithm. Here, we shall not be concerned with hybrid covariances. We will only consider flow-dependent errors estimated by the perturbations of an ensemble. Moreover, we shall assume that the model is perfect, as in most implementations of 4D-Var, although weak-constraint generalizations of the IEnKS are possible.

7.4.1 ▪ The algorithm

As in Section 7.2, we assume that batches of observations, $\mathbf{y}_k \in \mathbb{R}^{p_k}$, are collected every Δt, at times t_k. The observations are related to the state vector through a possibly nonlinear, possibly time-dependent observation operator $\mathscr{H}_k$. The observation errors are assumed to be Gaussian-distributed, unbiased, and uncorrelated in time, with observation error covariance matrices $\mathbf{R}_k$.

7.4.1.1 ▪ Analysis step

Let us first consider the analysis step of the scheme. It is performed over a window of L intervals of length Δt ($L\Delta t$ in time units). By convention, the time at the end of the DAW will be called t_L, while the time at the beginning of the DAW will be called t_0. At t_0 (i.e., $L\Delta t$ in the past), the prior distribution is estimated from an ensemble of m state vectors of $\mathbb{R}^n$: $\mathbf{x}_{0,[1]},\ldots,\mathbf{x}_{0,[i]},\ldots,\mathbf{x}_{0,[m]}$. Index 0 refers to time, while $[i]$ refers to the ensemble member index. They can be gathered into the ensemble matrix $\mathbf{E}_0 = [\mathbf{x}_{0,[1]},\ldots,\mathbf{x}_{0,[m]}]$ of size $n\times m$. The ensemble can equivalently be given in terms of its mean, $\overline{\mathbf{x}}_0 = \frac{1}{m}\sum_{i=1}^m \mathbf{x}_{0,[i]}$, and its (normalized) anomaly matrix, $\mathbf{X}_0 = \frac{1}{\sqrt{m-1}}[\mathbf{x}_{0,[1]} - \overline{\mathbf{x}}_0,\ldots,\mathbf{x}_{0,[m]} - \overline{\mathbf{x}}_0]$.

As in the EnKF, this prior is modeled as a Gaussian distribution of mean $\overline{\mathbf{x}}_0$ and covariance matrix $\mathbf{X}_0\mathbf{X}_0^{\mathrm{T}}$, the first- and second-order sampled moments of the ensemble. The background is rarely full rank since the anomalies of the ensemble span a vector space of dimension smaller than or equal to $m-1$ and in a realistic context $m \ll n$. Hence, in the absence of localization, one seeks an analysis state vector $\mathbf{x}_0$ in the ensemble subspace $\overline{\mathbf{x}}_0 + \mathrm{Vec}\left\{\mathbf{x}_{[1]}-\overline{\mathbf{x}}_0,\ldots,\mathbf{x}_{[m]}-\overline{\mathbf{x}}_0\right\}$ that can be written $\mathbf{x}_0 = \overline{\mathbf{x}}_0 + \mathbf{X}_0\mathbf{w}$, where $\mathbf{w}$ is a vector of coefficients in $\mathbb{R}^m$. The analysis will be defined as the most likely deterministic state of a cost function in the reduced subspace of the ensemble and will later be identified as the mean of the analysis ensemble.

As in Section 7.3, $\mathscr{F}_{k:0}$ stands for the composition of $\mathscr{H}_k$ and the resolvent $\mathscr{M}_{k:0}$. The variational analysis of the IEnKS over $[t_0, t_L]$ stems from a cost function. The restriction of this cost function to the ensemble subspace reads

$$\mathscr{J}(\mathbf{w}) = \sum_{k=L-S+1}^{L} \frac{1}{2}\|\mathbf{y}_k - \mathscr{F}_{k:0}(\overline{\mathbf{x}}_0 + \mathbf{X}_0\mathbf{w})\|^2_{\mathbf{R}_k} + \frac{1}{2}\|\mathbf{w}\|^2. \tag{7.10}$$

The second term represents the prior in ensemble subspace [Hunt et al., 2007]. Here S batches of observations are assimilated in this analysis. The case $L=0$, $S=1$, where $\mathscr{M}_{0:0}$ is defined as the identity, corresponds to the MLEF that we described in Section 6.7.2.1; the case $L=1$, $S=1$ corresponds to the IEnKF [Sakov et al., 2012]. Otherwise, one has $1\le S\le L+1$ and a 4D variational analysis.

This cost function is iteratively minimized in the reduced subspace of the ensemble following a Gauss–Newton algorithm,

$$\mathbf{w}^{(j+1)} = \mathbf{w}^{(j)} - \mathrm{H}_{(j)}^{-1}\nabla\mathscr{J}_{(j)}(\mathbf{w}^{(j)}), \tag{7.11}$$

using the gradient $\nabla\mathscr{J}$ and an approximation, H, of the Hessian in reduced space,

$$\begin{aligned}
\mathbf{x}_0^{(j)} &= \overline{\mathbf{x}}_0 + \mathbf{X}_0\mathbf{w}^{(j)},\\
\nabla\mathscr{J}_{(j)} &= \mathbf{w}^{(j)} - \sum_{k=L-S+1}^{L} \mathbf{Y}_{k,(j)}^{\mathrm{T}}\mathbf{R}_k^{-1}\left[\mathbf{y}_k - \mathscr{F}_{k:0}(\mathbf{x}_0^{(j)})\right],\\
\mathrm{H}_{(j)} &= \mathbf{I}_m + \sum_{k=L-S+1}^{L} \mathbf{Y}_{k,(j)}^{\mathrm{T}}\mathbf{R}_k^{-1}\mathbf{Y}_{k,(j)}.
\end{aligned}$$

The notation (j) refers to the iteration index of the minimization. At the first iteration, one sets $\mathbf{w}^{(0)} = \mathbf{0}$ so that $\mathbf{x}_0^{(0)} = \overline{\mathbf{x}}_0$. Recall that $\mathbf{I}_m$ is the identity matrix in $\mathbb{R}^m$. Then $\mathbf{Y}_{k,(j)} = [\mathscr{F}_{k:0}]'_{|\mathbf{x}_0^{(j)}} \mathbf{X}_0$ is the tangent linear of the operator from $\mathbb{R}^m$ to the observation space. Each iteration solves the inner loop quadratic minimization problem with cost function

$$\mathscr{J}_{(j)}(\mathbf{w}) = \sum_{k=L-S+1}^{L} \frac{1}{2} \left\| \mathbf{y}_k - \mathscr{F}_{k:0}(\mathbf{x}^{(j)}) - \mathbf{Y}_{k,(j)}(\mathbf{w} - \mathbf{w}^{(j)}) \right\|^2_{\mathbf{R}_k} + \frac{1}{2} \left\| \mathbf{w} - \mathbf{w}^{(j)} \right\|^2.$$

Here, this inner loop problem is directly dealt with by solving the associated linear system. An elegant idea is to solve it instead using the EnKS [Mandel et al., 2015].

Note that other iterative minimization schemes could be used in place of Gauss–Newton, such as quasi-Newton, Levenberg–Marquardt, or trust-region methods [Bocquet and Sakov, 2012; Mandel et al., 2015; Nino Ruiz and Sandu, 2016], with distinct convergence properties.

The estimation of these sensitivities using the ensemble avoids the use of the model adjoint. These sensitivities can be computed using finite differences (the so-called bundle IEnKS variant). The ensemble is rescaled closer to the mean trajectory by a factor ε. It is then propagated through the model and the observation operator ($\mathscr{F}_{k:0}$), after which it is rescaled back by the inverse factor ε^{-1}. Following (7.6), this can be written

$$\mathbf{Y}_{k,(j)} \approx \frac{1}{\varepsilon} \mathscr{F}_{k:0}\left(\mathbf{x}_0^{(j)} \mathbf{1}^\mathrm{T} + \varepsilon \mathbf{X}_0\right)\left(\mathbf{I}_m - \frac{\mathbf{1}\mathbf{1}^\mathrm{T}}{m}\right), \tag{7.12}$$

where $\mathbf{1} = (1, \ldots, 1)^\mathrm{T} \in \mathbb{R}^m$. Note that the $\sqrt{m-1}$ factor needed to scale the anomalies has been incorporated in ε. It is also possible to compute these sensitivities without resorting to finite differences, choosing $\varepsilon = 1$, with very similar IEnKS performance, which is the so-called transform variant [Sakov et al., 2012; Bocquet and Sakov, 2014]. In this alternative, the sensitivities are computed similarly to a 4DEnVar. But this goes further, since $\varepsilon = 1$ can be made consistent within the nonlinear variational minimization. Let us finally mention that these sensitivities can obviously be computed with the adjoint model if it is available.

The iterative minimization is stopped when $\|\mathbf{w}^{(j)} - \mathbf{w}^{(j-1)}\|$ reaches a predetermined threshold, e. Let us denote by $\mathbf{w}_0^\star$ the solution of the cost function minimization. The $\star$ symbol will be used with any quantity obtained at the minimum. Subsequently, a posterior ensemble can be generated at t_0 by

$$\mathbf{E}_0^\star = \mathbf{x}_0^\star \mathbf{1}^\mathrm{T} + \sqrt{m-1}\mathbf{X}_0 \mathbf{W}\mathbf{U},$$

where

$$\mathbf{W} = \left[\mathbf{I}_m + \sum_{k=L-S+1}^{L} \left(\mathbf{Y}_k^\star\right)^\mathrm{T} \mathbf{R}_k^{-1} \mathbf{Y}_k^\star\right]^{-\frac{1}{2}}, \tag{7.13}$$

$\mathbf{U}$ is an arbitrary orthogonal matrix that satisfies $\mathbf{U}\mathbf{1} = \mathbf{1}$ meant to keep the posterior ensemble centered on the analysis, and $\mathbf{x}_0^\star = \overline{\mathbf{x}}_0 + \mathbf{X}_0 \mathbf{w}_0^\star$. The update of the anomalies, $\mathbf{X}_0^\star = \mathbf{X}_0 \mathbf{W}$, is a right transform of the initial anomalies, $\mathbf{X}_0$. A left transform more suitable for hybrid IEnKS and covariance localization could be used instead [Bocquet, 2016]. If filtering is considered, it additionally requires an independent model forecast throughout the DAW and possibly beyond the present time for forecasting.

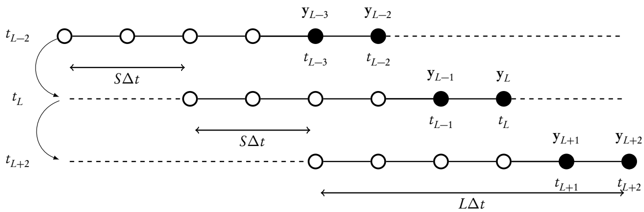


Figure 7.2. *Cycling of the SDA IEnKS, where $L = 5$ and $S = 2$, in units of Δt, the time interval between two updates. The method performs a smoothing update throughout the window but only assimilates the newest observation vectors (that have not been already assimilated), marked by black dots. Note that the time index of the dates and the observations is absolute for this schematic, not relative.*

7.4.1.2 ▪ Forecast step

During the forecast step of the scheme cycle, the ensemble is propagated for $S\Delta t$, with S an integer introduced via (7.10):

$$\mathbf{E}_S^\star = \mathscr{M}_{S:0}(\mathbf{E}_0^\star).$$

This forecast ensemble at t_S will form the prior for the next analysis. A typical cycling of the analysis and forecast steps is schematically displayed in Figure 7.2.

A pseudocode of the IEnKS is proposed in Algorithm 7.1. It describes a full cycle of the (single data assimilation or SDA) IEnKS that includes the analysis and forecast steps, with the exception of the model forecast within the DAW required to complete the analysis for all times in the DAW, which is independent from the main IEnKS cycle.

Because it combines the propagation of the errors via the ensemble with a nonlinear 4D variational analysis, the IEnKS has been shown to outperform 4D-Var, the EnKF, and the EnKS on several low-order models (Lorenz-63, Lorenz-95, Lorenz-95 coupled to a tracer model or a chemistry model, and a 2D barotropic model). This good performance persisted in all tested regimes of nonlinearity, applied to smoothing, filtering, or forecasting. It was also shown to be useful for parameter estimation because it combines the straightforward state augmentation technique of the EnKF with a variational analysis where the adjoint sensitivity to the parameters is estimated within the scheme in ensemble subspace [Bocquet and Sakov, 2013; Haussaire and Bocquet, 2016]. The only fundamental approximations of the IEnKS scheme are the Gaussian assumption made in the generation of the posterior ensemble and the second-order statistics assumption made when estimating the prior from the ensemble. When $S = L$, and using only the first iteration of the minimization, the single DA transform variant IEnKS then becomes formally equivalent to the so-called 4D-ETKF [Hunt et al., 2004]. As such, it can also be understood as a rigorously derived generalization of the 4DEnVar methods. In the case where $S = 1$ and $L = 0$, it also coincides with an ensemble-transform variant of the MLEF [Zupanski, 2005].

The performance of the single DA IEnKS has been compared to that of 4D-Var and the EnKF/EnKS with the Lorenz-95 model as a function of the DAW length. The results are shown in Figure 7.3. This follows up on the initial illustration of this chapter that compared 4D-Var with the EnKF. The shift, S, is chosen to be 1, which makes the

Algorithm 7.1 A cycle of the lag-L/shift-S/SDA/bundle/Gauss-Newton IEnKS.

Require: t_L is present time. Transition model $\mathcal{M}_{k+1:k}$, observation operators $\mathcal{H}_k$ at t_k. Algorithm parameters: ε, e, $j_{\max}$. $\mathbf{E}_0$, the ensemble at t_0, $\mathbf{y}_k$ the observation at t_k. $\mathbf{U}$ is an orthogonal matrix in $\mathbb{R}^{m\times m}$ satisfying $\mathbf{U1}=\mathbf{1}$.

$j=0$, $\mathbf{w}=\mathbf{0}$, $P=L-S+1$
$\mathbf{x}_0^{(0)}=\mathbf{E}_0\mathbf{1}/m$
$\mathbf{X}_0=\left(\mathbf{E}_0-\mathbf{x}_0^{(0)}\mathbf{1}^{\mathrm{T}}\right)/\sqrt{m-1}$
repeat
  $\mathbf{x}_0=\mathbf{x}_0^{(0)}+\mathbf{X}_0\mathbf{w}$
  $\mathbf{E}_0=\mathbf{x}_0\mathbf{1}^{\mathrm{T}}+\varepsilon\mathbf{X}_0$
  $\mathbf{E}_P=\mathcal{M}_{P:0}(\mathbf{E}_0)$
  $\overline{\mathbf{y}}_P=\mathcal{H}_P(\mathbf{E}_P)\mathbf{1}/m$
  $\mathbf{Y}_P=(\mathcal{H}_P(\mathbf{E}_P)-\overline{\mathbf{y}}_P\mathbf{1}^{\mathrm{T}})/\varepsilon$
  for $k=P+1,\dots,L$ **do**
    $\mathbf{E}_k=\mathcal{M}_{k:k-1}(\mathbf{E}_{k-1})$
    $\overline{\mathbf{y}}_k=\mathcal{H}_k(\mathbf{E}_k)\mathbf{1}/m$
    $\mathbf{Y}_k=(\mathcal{H}_k(\mathbf{E}_k)-\overline{\mathbf{y}}_k\mathbf{1}^{\mathrm{T}})/\varepsilon$
  end for
  $\nabla\mathcal{J}=\mathbf{w}-\sum_{k=P}^{L}\mathbf{Y}_k^{\mathrm{T}}\mathbf{R}_k^{-1}(\mathbf{y}_k-\overline{\mathbf{y}}_k)$
  $\mathrm{H}=\mathbf{I}_m+\sum_{k=P}^{L}\mathbf{Y}_k^{\mathrm{T}}\mathbf{R}_k^{-1}\mathbf{Y}_k$
  Solve $\mathrm{H}\Delta\mathbf{w}=\nabla\mathcal{J}$
  $\mathbf{w}:=\mathbf{w}-\Delta\mathbf{w}$
  $j:=j+1$
until $\|\Delta\mathbf{w}\|\le e$ **or** $j\ge j_{\max}$
$\mathbf{E}_0=\mathbf{x}_0\mathbf{1}^{\mathrm{T}}+\sqrt{m-1}\mathbf{X}_0\mathrm{H}^{-\frac{1}{2}}\mathbf{U}$
$\mathbf{E}_S=\mathcal{M}_{S:0}(\mathbf{E}_0)$

system close to *quasi-static*: for large L the DAW is slightly shifted at each iteration. We can expect that the underlying cost function is only slightly modified from one analysis to the next, so that the global minimum can more easily be tracked (following the ideas initially developed by Pires et al. [1996]) . The SDA IEnKS performs as theoretically expected up to $L=15$ for filtering and up to $L=30$ for smoothing. Beyond these lengths the performance deteriorates. In the regime where the IEnKS is reliable, it clearly outperforms 4D-Var and the EnKS. Again, this performance is due to (i) the flow dependence of the error, (ii) the fully nonlinear analysis within the DAW, and, to a lesser extent, (iii) the quasi-static $S=1$.

7.4.2 ▪ Single and multiple assimilation of observations

An important idea of the IEnKS is to assimilate observations far ahead in time from the updated state at t_0. This is meant to stabilize the scheme in the directions of the unstable modes of the chaotic model. This has been shown to be very efficient but ultimately fails for DAWs that are too long. An idea to stabilize the system for long DAWs is to assimilate more observations within the DAW. Depending on the precise choice of S and L, this may lead to assimilating observations several times, which is statistically inconsistent. Let us see how such a design can actually be justified [Bocquet and Sakov, 2014].

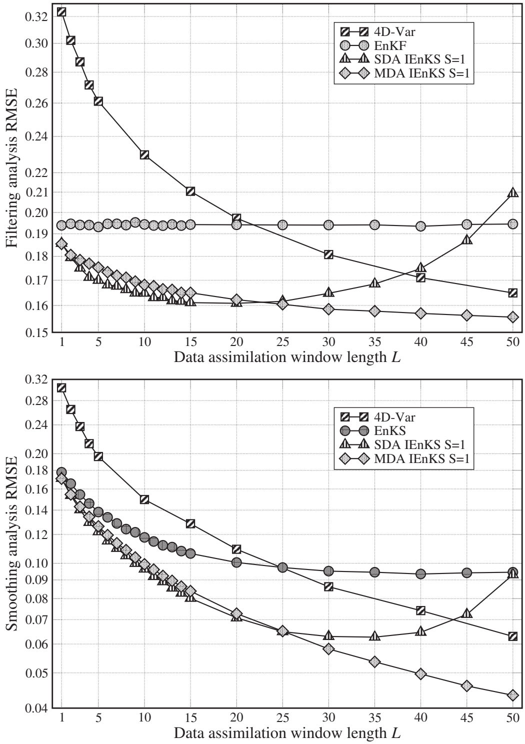


Figure 7.3. *Synthetic data assimilation experiments with the Lorenz-*95 *model, which show the theoretical superiority of the IEnKS regardless of its numerical cost. The upper panel shows the filtering analysis RMSE of optimally tuned EnKF,* 4*D-Var, IEnKS assimilation experiments, as a function of the length of the DAW. The lower panel shows the smoothing analysis RMSE of optimally tuned EnKS,* 4*D-Var, and IEnKS as a function of the length of their DAW. The optimal RMSE is chosen within the window for* 4*D-Var and it is taken at the beginning of the window for the IEnKS. The EnKF, EnKS, and IEnKS use an ensemble of* $m = 20$ *which avoids the need for localization but requires inflation. The length of the DAW is* $L \times \Delta t$, *where* $\Delta t = 0.05$. *Both RMSEs for the SDA and MDA IEnKS are shown.*

The IEnKS cost function can be generalized with the goal of assimilating any observation within the DAW,

$$\mathscr{J}(\mathbf{w}) = \sum_{k=0}^{L} \frac{\beta_k}{2} \|\mathbf{y}_k - \mathscr{F}_{k:0}(\overline{\mathbf{x}}_0 + \mathbf{X}_0\mathbf{w})\|^2_{\mathbf{R}_k} + \frac{1}{2}\|\mathbf{w}\|^2, \tag{7.14}$$

which introduces a set of coefficients, $0 \le \beta_k \le 1$, that weight the innovation terms.

There are some degrees of freedom in the choice of L, S, and the $\{\beta_k\}_{0\le k\le L}$. Let us just mention a few legitimate choices for them. First, for any choice of L and S such that $1 \le S \le L+1$, the most natural choice for the $\{\beta_k\}_{0\le k\le L}$ is to set $\beta_k = 1$ for $k = L-S+1,\dots,L$, and $\beta_k = 0$ otherwise. That way, the observations are assimilated once and only once. This leads back to the original formulation of the IEnKS (7.10) and explains why it was called the single data assimilation scheme (SDA IEnKS). It is simple, and the optional analysis of the update step is merely a forecast of the analyzed state at t_0, or possibly a forecast of the full ensemble from t_0. When $S \ge L$, the DAWs do not overlap, while they do so if $S < L$.

For very long DAWs, the use of *multiple data assimilation* (or *splitting*) of observations, denoted MDA in the following, can be proven to be numerically efficient. An observation vector $\mathbf{y}$ is said to be assimilated with weight β ($0 \le \beta \le 1$) if the following Gaussian observation likelihood is used in the analysis:

$$p(\mathbf{y}^\beta|\mathbf{x}) = \frac{e^{-\frac{\beta}{2}\|\mathbf{y}-\mathcal{H}(\mathbf{x})\|^2_{\mathbf{R}}}}{\sqrt{(2\pi/\beta)^p|\mathbf{R}|}}, \tag{7.15}$$

where $|\mathbf{R}|$ is the determinant of $\mathbf{R}$. The upper index of $\mathbf{y}^\beta$ refers to its partial assimilation with weight β. The prior errors attached to the several occurrences of one observation are *chosen* to be independent. In that light, the $\{\beta_k\}_{0\le k\le L}$ are merely the weights of the observation vectors $\{\mathbf{y}_k\}_{0\le k\le L}$ within the DAW. Statistical consistency imposes that a unique observation vector be assimilated so that the sum of all its weights in the DA experiment is one. For instance, if $1 = S \le L+1$, one requires $\sum_{k=0}^{L}\beta_k = 1$. In the more general case where the observation vectors have the same number of nonzero weights, then L is a multiple of S: $L = QS$, where Q is an integer. As a result, consistency requires $\sum_{q=0}^{Q-1}\beta_{Sq+l} = 1$, with $l = 0,\dots,S$.

In the MDA case, unlike the SDA subcase, the optional estimation of the $\mathbf{x}_k$ states, besides $\mathbf{x}_0$, within the DAW is more complex. It requires reweighting the observations within the DAW and performing a second analysis within the DAW with these balanced weights. More details can be found in Bocquet and Sakov [2014] about this so-called balancing step.

Note that when the constraint $\sum_{k=0}^{L}\beta_k = 1$ is not satisfied, the underlying smoothing PDF will not be the targeted one but, with well-chosen $\{\beta_k\}_{0\le k\le L}$, could be a power of it, which can be put to use.

The chaining of the DA cycles in the MDA case is displayed in Figure 7.4. A pseudocode of the MDA IEnKS is proposed in Algorithm 7.2. It describes a full cycle of the (MDA) IEnKS that includes the analysis at the beginning of the DAW and the forecast step, with the exception of the balancing step, which is required to complete the analysis for all times in the DAW, but which is independent from the main IEnKS cycle.

These MDA approaches are mathematically consistent in the sense that they are demonstrated to be correct in the linear model Gaussian statistics case. In Bocquet and Sakov [2014], a heuristic argument based on Bayesian ideas justifies the use of the method in the nonlinear case.

The accuracy of the SDA and MDA IEnKS, as well as 4D-Var and the EnKF, is compared in Figure 7.3. The results emphasize the accuracy of the IEnKS compared to 4D-Var and the EnKF. Moreover, the SDA IEnKS is shown to be more accurate than the MDA IEnKS for short DAWs, since the latter is not as consistent with nonlinear models. Yet, the MDA IEnKS is much more stable for longer DAWs.

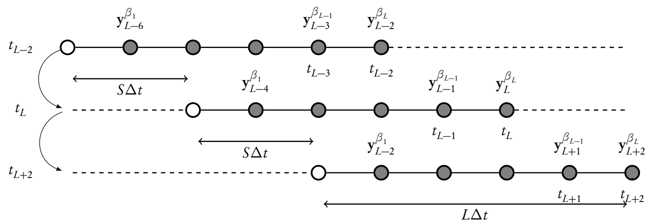


Figure 7.4. *Chaining of the MDA IEnKS cycles. The schematic illustrates the case $L = 5$ and $S = 2$. The method performs a smoothing update throughout the window, potentially using all observations within the window (marked by gray dots), except for the first observation vector, assumed to be already entirely assimilated. Note that the time index for the dates and the observations is absolute for this schematic, not relative.*

Algorithm 7.2 A cycle of the lag-L/shift-S/MDA/bundle/Gauss-Newton IEnKS.

Require: t_L is present time. Transition model $\mathcal{M}_{k+1:k}$, observation operators $\mathcal{H}_k$ at t_k. Algorithm parameters: ε, e, $j_{\max}$. $\mathbf{E}_0$, the ensemble at t_0, $\mathbf{y}_k$ the observation at t_k. $\mathbf{U}$ is an orthogonal matrix in $\mathbb{R}^{m\times m}$ satisfying $\mathbf{U1} = \mathbf{1}$. β_k for $k = 0, \ldots, L$ are the weights (see main text).

$j = 0$, $\mathbf{w} = \mathbf{0}$
$\mathbf{x}_0^{(0)} = \mathbf{E}_0 \mathbf{1}/m$
$\mathbf{X}_0 = \left(\mathbf{E}_0 - \mathbf{x}_0^{(0)}\mathbf{1}^{\mathrm{T}}\right)/\sqrt{m-1}$
repeat
    $\mathbf{x}_0 = \mathbf{x}_0^{(0)} + \mathbf{X}_0\mathbf{w}$
    $\mathbf{E}_0 = \mathbf{x}_0\mathbf{1}^{\mathrm{T}} + \varepsilon\mathbf{X}_0$
    $\overline{\mathbf{y}}_0 = \mathcal{H}_0(\mathbf{E}_0)\mathbf{1}/m$
    $\mathbf{Y}_0 = (\mathcal{H}_0(\mathbf{E}_0) - \overline{\mathbf{y}}_0\mathbf{1}^{\mathrm{T}})/\varepsilon$
    for $k = 1, \ldots, L$ **do**
        $\mathbf{E}_k = \mathcal{M}_{k:k-1}(\mathbf{E}_{k-1})$
        $\overline{\mathbf{y}}_k = \mathcal{H}_k(\mathbf{E}_k)\mathbf{1}/m$
        $\mathbf{Y}_k = (\mathcal{H}_k(\mathbf{E}_k) - \overline{\mathbf{y}}_k\mathbf{1}^{\mathrm{T}})/\varepsilon$
    end for
    $\nabla\mathcal{J} = \mathbf{w} - \sum_{k=0}^{L}\beta_k\mathbf{Y}_k^{\mathrm{T}}\mathbf{R}_k^{-1}(\mathbf{y}_k - \overline{\mathbf{y}}_k)$
    $\mathrm{H} = \mathbf{I}_m + \sum_{k=0}^{L}\beta_k\mathbf{Y}_k^{\mathrm{T}}\mathbf{R}_k^{-1}\mathbf{Y}_k$
    Solve $\mathrm{H}\Delta\mathbf{w} = \nabla\mathcal{J}$
    $\mathbf{w} := \mathbf{w} - \Delta\mathbf{w}$
    $j := j + 1$
until $||\Delta\mathbf{w}|| \leq e$ **or** $j \geq j_{\max}$
$\mathbf{E}_0 = \mathbf{x}_0\mathbf{1}^{\mathrm{T}} + \sqrt{m-1}\mathbf{X}_0\mathrm{H}^{-\frac{1}{2}}\mathbf{U}$
$\mathbf{E}_S = \mathcal{M}_{S:0}(\mathbf{E}_0)$

Greater stability can be achieved with the SDA IEnKS when using a quasi-static variational assimilation within each analysis, a procedure that may have a significant numerical cost depending on the implementation (A. Fillion, M. Bocquet, and S. Gratton, personal communication).

7.4.3 ▪ Localization

Like any EnVar approach, the IEnKS requires localization for a suitable use with high-dimensional models. Localization of the IEnKS is as difficult as with 4DEnVar. It can be even more critical, since localization may partially destroy the consistency between consecutive iterations in the nonlinear variational analysis. As for 4DEnVar, an approximate solution consists of propagating the ensemble within the DAW, but also the localization operator using a surrogate advection model. This also enables the estimation of the updated ensemble. However, if an adjoint of the model is available, it becomes possible, as in En-4DVar or 4D-Var-Ben, to exactly (albeit implicitly) propagate localization within the DAW. Going further than En-4DVar or 4D-Var-Ben, the adjoint also enables an exact construction of the updated ensemble. More details on the theory and numerical results are given in Bocquet [2016].

Figure 7.5 shows an example of the IEnKS with the Lorenz-95 model used in conjunction with localization. The ensemble is set to $m = 10$, lower than the dimension of the unstable and neutral subspace of the model (14), which means that localization is mandatory. It appears from this experiment that, at least in this idealized context, a covariant (i.e., evolving with the model flow) localization achieves a performance as good as that of an IEnKS with localization where the model adjoint is available and where the localized covariances are implicitly propagated.

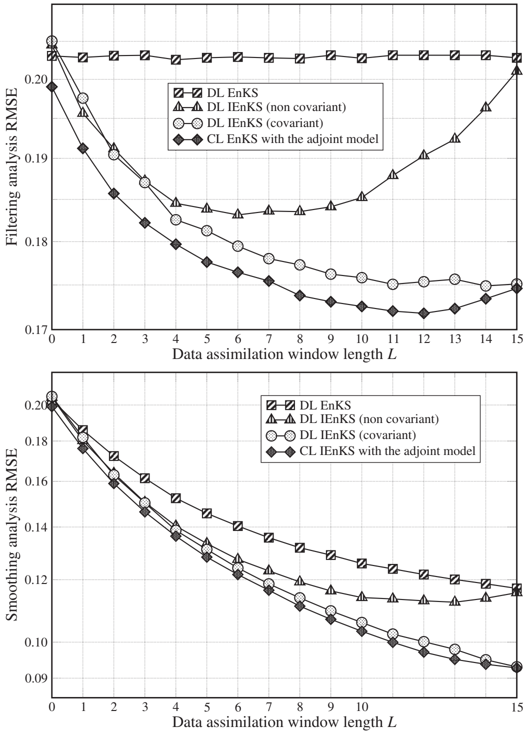


Figure 7.5. *Synthetic DA experiments with the Lorenz-*95 *model to compare several localization strategies when* $m = 10$. *The inflation is optimally tuned. The upper panel shows the filtering analysis RMSE as a function of the length of the DAW. The lower panel shows the smoothing analysis RMSE as a function of the length of the DAW. Four local methods are compared: a DL (domain localization) EnKF/EnKS, the IEnKS with a static domain (i.e., noncovariant) localization, the IEnKS where the local domains are covariantly advected eastward with the velocity* $v = 6$ *grid cells per time unit, and a flow-dependent covariance localization IEnKS where the adjoint is supposed to be available and used. Covariance localization and domain localization use a Gaspari–Cohn localization function of radius* 12 *grid cells (see Section* 6.5*). The length of the DAW is* $L \times \Delta t$, *where* $\Delta t = 0.05$. *Adapted from Bocquet [*2016*].*

Part III

Applications and case studies

Given the generality of the statistical and variational algorithmic approaches presented in the preceding two parts of this book, we can now proceed to the applications. We will see that their field of applicability is vast, ranging from environmental to medical, engineering, and even financial case studies. In each chapter, we provide easily accessible examples and case studies that refer extensively to Parts I and II. We include, in each chapter, pointers and references to numerous realistic and complex cases.

Chapter 8

Applications in environmental sciences

8.1 ▪ Physical oceanography

8.1.1 ▪ Presentation of the domain

After numerical weather forecasting, oceanography is the second most active application domain for DA. It started in the late 1980s, following the atmosphere community. Specific difficulties caused a 10- to 20-year delay between atmosphere and ocean assimilation, in particular: cost and complexity of data acquisition (especially at depth), presence of complex boundaries, flux modeling at the air–sea boundary, interactions with chemistry and biology, model errors, and strong turbulence. The arrival of satellites produced a rapid increase in data quantity and quality and drastically improved both models and assimilation systems for the ocean. The dynamism of the field is such that an extensive review is out of reach—we therefore propose a selection of significant questions and contributions.

8.1.1.1 ▪ Global assimilation and operational forecasting

In the late 1990s, the Global Ocean Data Assimilation Experiment (GODAE) was established. The idea was to use state-of-the art ocean models and combine them with similarly advanced DA methods to demonstrate the feasibility of global ocean DA. This 10-year experiment had two objectives. First, provide short-term ocean forecasts (or initial conditions for climate models). Second, provide global (re)analyses to improve the oceanographic knowledge base and develop tools to improve the observing system.

Associated with this experiment, national operational forecasting agencies were created, aiming to provide real-time and accurate ocean forecasts for coastal and offshore industry, fishing, and other marine activities.

Difficulties for real-time or global ocean DA are similar to the weather forecasting problem, in particular the trade-off between accuracy and computing power.

8.1.1.2 ▪ Coastal Lagrangian search and rescue

Because of human activities and presence along the coast, coastal assimilation (including forecast for search and rescue) is of course of great interest. It is related to Lagrangian assimilation, which aims to retrieve the ocean state (currents in particular) from drifting buoys' location information, which is a convenient way to obtain additional local data at a reasonable cost.

8.1.1.3 ▪ Ocean and atmosphere or biogeochemistry coupling

Another hot topic is DA for coupled models. Coupling may vary depending on the application. For climate (or tropical storm), ocean–atmosphere models are considered, whereas for marine biology, models coupling the physical processes with chemistry and/or (phyto)plankton are developed. The latter are related to ocean color images, which give information about chlorophyll content.

8.1.1.4 ▪ Error and bias management

As for the atmosphere, and even more so because of the inaccessible ocean depths, managing model biases as well as correlated observation errors is an extremely difficult task, which is kept quiet in most of the DA literature. However, as DA systems improve, it is becoming more and more difficult to ignore.

8.1.2 ▪ Examples of DA problems in this context

8.1.2.1 ▪ Global assimilation and operational forecasting

The GODAE [Smith, 2000; Bell et al., 2009], as well as other initiatives, such as ECCO (Estimating the Circulation and Climate of the Oceans) and WOCE (World Ocean Circulation Experiment), instigated many global DA experiments. Among the significant developments, MIT developed a global ocean DA model based on automatic differentiation [Wunsch and Heimbach, 2007], for example, in Stammer [2004] it provided improved estimates of air–sea fluxes for climate models.

On the operational side, we can cite two representative works, which take advantage of two other ocean models: Brasseur et al. [2005] describe the French operational forecasting project MERCATOR, based on the European ocean model NEMO; Chao et al. [2009] rely on the Regional Ocean Modeling System (ROMS) to design a forecasting system for the coast of central California.

8.1.2.2 ▪ Coastal Lagrangian search and rescue

Another interesting avenue of oceanographic applications is the assimilation of Lagrangian data (drifting float position information); see, e.g., Nodet [2006], Kuznetsov et al. [2003], Fan et al. [2004], Castellari et al. [2001]. These observations are not straightforward to use, because the observation operator involved is nonlinear and quite sensitive to both initial perturbations and small scale phenomena, but they provide "cheap" (compared to satellite data) observations. They are sometimes used for search and rescue operations [Davidson et al., 2009] and/or along the coast, as well as for oil spill monitoring [Jordi et al., 2006]. There are also a few attempts at coastal morphodynamics forecasting [Smith et al., 2009], which is obviously extremely difficult even from a direct modeling point of view.

8.1.2.3 ▪ Ocean and atmosphere or biogeochemistry coupling

Model coupling is a hot topic in geoscience modeling. Indeed, as computing power and modeling progress, more realism is gained by correctly representing external forcing and possible interactions between a system and connected systems, and model coupling is an accurate way to do so. For example, for El Niño forecasting, it is crucial to correctly represent the ocean–atmosphere interaction [Behringer et al., 1998; Kleeman et al., 1995]. DA in this framework can be complicated, as technical implementations of coupling can complicate the DA algorithms (e.g., adjoint methods need to be able to differentiate the coupling interaction and communication). Other than ocean–atmosphere coupling, coupled models are used in marine biogeochemistry [Brasseur et al., 2009; Carmillet et al., 2001; Gregg, 2008; Natvik and Evensen, 2003]. DA in these models usually relies on ocean color observations, as the ocean color is strongly related to its chlorophyll and phytoplankton content.

8.1.2.4 ▪ Error and bias management

DA with errors (systematic bias in the model or correlated observation errors) is quite common. Regarding systematic error and model bias correction, we can mention Bell et al. [2004] and Huddleston et al. [2004]. Regarding observation errors, the usual assumption implies that the observation error covariance matrix $\mathbf{R}$ is diagonal; hence the error is uncorrelated. This causes most satellite images to be thinned before use, so that in the end only a small percentage of the data is actually used. There have been some attempts at proposing nondiagonal $\mathbf{R}$ matrices; e.g., Chabot et al. [2015] use a change of variable into wavelet space to put correlation information back into $\mathbf{R}$.

8.1.3 ▪ Focus: Operational oceanography

Operational oceanography aims to provide monthly and seasonal forecasts of the ocean state. These ocean forecasts can have multiple uses: for coastal industries, off-shore activities, fisheries, and sailing races for example. They can also be used as forcings for monthly and seasonal operational weather forecasting. Indeed, at this range, the ocean impact on the weather must be precisely taken into account. This is the object of this focus. The work of Vidard et al. [2009] deals with the operational ocean system analysis of ECMWF, which provides initial ocean states for the weather forecasts. This work presents the first implementation of the assimilation of altimetric sea level satellite data into the ocean analysis system.

In this work, the assimilation scheme is quite simple—we use OI. OI is a simplified version of the BLUE algorithm, where only relevant (local) observations are taken into account for the computation of the analysis at a given point.

Figure 8.1 presents correlation maps between currents computed by the ECMWF ocean analysis system and the ones produced by the OSCAR project (Ocean Surface Current Analysis), which have been proven to be of good quality, in particular in the tropical ocean. The closer to one the correlation is, the better the results. As we can see, the combination of both in situ and satellite assimilation yields the best correlation improvement.

8.2 ▪ Glaciology

8.2.1 ▪ Presentation of the domain

Glaciology is a multifaceted domain whose various subfields are crucial for the understanding and forecasting of current climate change. We can cite many reasons why this is so.

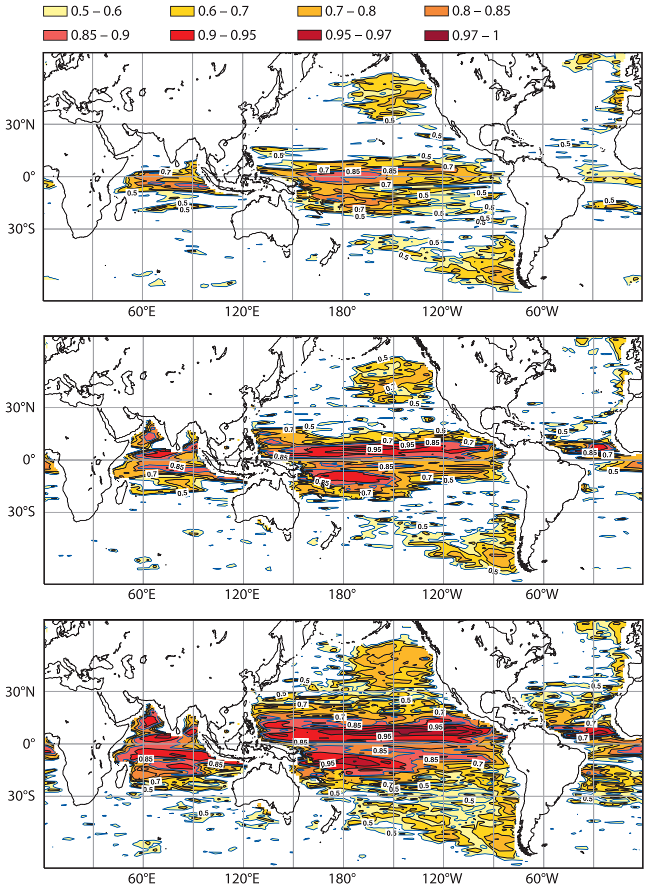


Figure 8.1. *Correlation map between currents produced by ECMWF ocean analysis and the OSCAR surface currents. Top panel: control run (without assimilation). Middle panel: assimilation of in situ temperature and salinity observations only. Bottom panel: combined assimilation of in situ (temperature and salinity) and altimetric (sea level height) observations. The darker the color, the closer to 1 the correlation is, and the better the performance. Reprinted with permission from American Meteorological Society [Vidard et al., 2009].*

8.2.1.1 ▪ Paleoclimatology

First, glaciology, through ice core modeling, gives plenty of information about the climate of the past. Indeed, when ice forms, air bubbles are confined inside and act as a snapshot of the current atmosphere content. Deep ice cores then carry information about past atmosphere gas composition, and isotopic analysis allows us to some extent to reconstruct temperature history. Ice cores also contain ash, pollen, and dust, and all of these combined with the gas bubbles and the ice volume give information about many past phenomena (such as volcanic eruptions, the sun's activity, ocean volume, precipitation, atmospheric chemistry, fires, and plants).

8.2.1.2 ▪ Climate 1: Albedo

Second, the ice on the planet (ice sheets, ice caps, glaciers, sea ice) is a crucial component of climate models because of its link to the earth albedo and the associated positive feedback loop. Indeed, the ice albedo, though variable, is generally high (meaning that a lot of sunlight energy is reflected away). If the ice starts to melt, then the albedo in that area will decrease, leading to more heat being absorbed, which in turn will accelerate the ice melt and the albedo decrease. The reverse is also true (at least locally in time and space, under some temperature and humidity assumptions), as a snowfall would increase the albedo and lower the temperature, allowing more snow to fall, etc.

8.2.1.3 ▪ Climate 2: Sea level rise

Third, terrestrial ice (everything but sea ice) makes a significant contribution to sea level change and is also associated with interactions with the ocean (and the climate in general) in this regard. Global warming affects the ice sheets, ice caps, and glaciers and causes them to contribute to half of the current total sea level rise (which is about 3 mm/year). More dramatically, the melting of the two ice sheets of the planet (Antarctica and Greenland) would cause a sea level rise of 61.1 and 7.2 meters, respectively. Such a massive melt would of course take time, but a positive feedback loop also exists here, called the "small ice sheet instability," caused by the link between altitude and temperature: if the ice melts so that the ice sheet elevation goes beyond a given altitude, then its surface temperature will reach melting temperature, which in turn will increase the melt and decrease the altitude again, accelerating the phenomenon and causing the ice sheet to disappear completely. Due to the massive impact of sea level rise on human population, forecasting the contribution of Antarctica and Greenland is therefore a very important topic in glaciology.

8.2.2 ▪ Examples of DA problems in this context

Glaciology's specific difficulties are numerous: coexistence of very small and very large scales in both time and space; strong nonlinearities in the fluid behavior (non-Newtonian viscosity); complex thermomechanical behavior of the ice structure and ice material properties; difficulty of observation acquisition (inside and below the ice); etc. Still, the progress of both modeling and observing systems allows the use of DA methods to answer a couple of questions.

8.2.2.1 ▪ Paleoclimatology

Providing a good reconstruction of past temperatures is a key ingredient to understanding climate variations. Many proxies (indirect observations) give information about it: atmospheric composition from ice cores, oxygen isotopes from ocean sediment cores, pollen from soil cores, and so on. DA aims to combine these indirect observations with ice sheet dynamics models to reconstruct paleotemperatures.

For example, Bintanja et al. [2004] used the nudging method combined with observations of sea level into a large-scale 3D model of ice dynamics to reconstruct northern hemisphere temperature from 120 000 years ago until now. More recently, Bonan et al. [2012] used the adjoint method on an academic test case to study the assimilation of ice volume data to recover past temperature.

Another example is the dating of ice cores, which aims to combine ice and gas stratigraphic observations and glaciological modeling to reconstruct the chronology of the ice. Lemieux-Dudon et al. [2008] and Lemieux-Dudon et al. [2010] developed a variational assimilation scheme to consistently date multiple Antarctic ice cores at once and provide confidence intervals on the inferred dating.

8.2.2.2 ▪ Sea level rise forecast

The Intergovernmental Panel on Climate Change (IPCC) deemed it urgent in its fourth assessment report in 2007 to develop robust forecasting methods to predict Antarctica's and Greenland's contribution to sea level rise, in particular the part due to ice calving (ice discharge to the ocean, happening at the boundary of the ice sheets). This is a complex problem, as it depends on the ice velocity in ice streams (or outlet glaciers) and ice shelves, which in turn is highly sensitive to poorly observed parameters, such as the elevation of the bedrock and the basal boundary conditions (which can vary greatly between two extremes: grounded frozen ice and melting fast sliding ice).

Various DA methods were implemented to infer these parameters from surface observations, for example Arthern and Gudmundsson [2010], Arthern and Hindmarsh [2006], Arthern [2003], and Gillet-Chaulet et al. [2012] (approximate adjoint method); Heimbach and Bugnion [2009] (automatic differentiation adjoint); and Bonan et al. [2014] (ETKF).

8.2.2.3 ▪ Local glacier and ice stream monitoring

On smaller scales the problem of inferring basal properties of the ice from satellite surface data to forecast the evolution of glaciers and ice shelves is also of strong interest for the glaciological community.

Many studies used DA methods to do so, e.g., Jay-Allemand et al. [2011], Larour [2005], MacAyeal [1992], Morlighem et al. [2010], Petra et al. [2012], and Vieli et al. [2007], who used diverse variations of the adjoint method to study ice streams, glaciers, and ice shelves for various applications: study of a glacier surge, reconstruction of the ice shelf rheology, reconstruction of the basal condition of an ice stream, etc.

8.2.3 ▪ Focus: Sea level rise

The first implementation of DA methods to provide a forecast of Greenland's contribution to sea level rise was done in Gillet-Chaulet et al. [2012]. The main issue was to estimate the basal parameter in the friction law at the bedrock interface. The ice discharge (volume loss) is indeed highly sensitive to this parameter, as it governs the sliding velocity, which can therefore vary strongly with this parameter. Then the

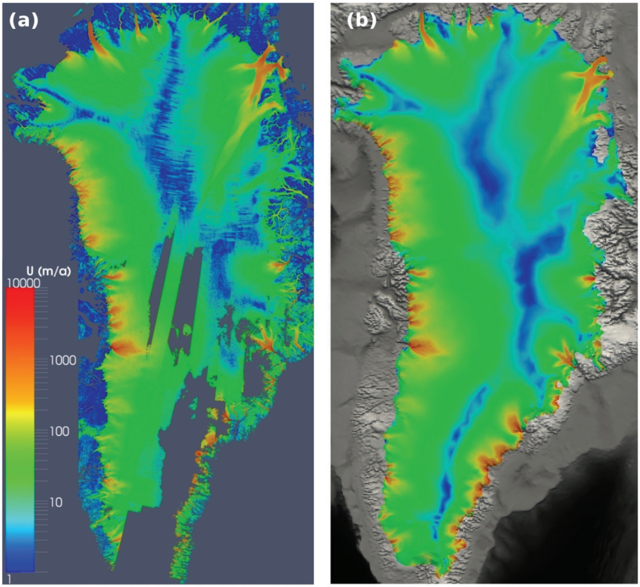


Figure 8.2. *Greenland surface velocities, on the left* (a) *observed velocities; on the right* (b) *velocities after reconstruction of the basal coefficient field by an approximate adjoint method. Reprinted with permission from European Geosciences Union [Gillet-Chaulet et al.,* 2012*].*

sliding velocity in turn influences the discharge of ice toward the sea. As this parameter (which is not a single number but a high-dimensional variable, with one value for every grid point on all the bedrock) concerns basal boundary conditions, it is very difficult to estimate directly because measurements are simply impossible.

DA methods allow us to use surface observations to recover this parameter field. An approximate adjoint method was used, in other words a variational assimilation method in which the adjoint was simplified. The simplification was necessary at that point because the ice sheet model that was used is a state-of-the-art finite element model with unstructured and adaptive mesh (using classical error estimates), for which no full adjoint is yet available.

For this experiment, the surface observations were the surface height of the ice and the surface velocity. In this framework, with real data, exact validation is impossible as the true state of the unknown parameter is unreachable. Therefore, the validation was made by comparing the simulated surface velocities after assimilation and the observed velocities; see Figure 8.2. As we can see, the main features of the flow are well reproduced: low velocities in the central areas, fast-flowing ice streams individualized and well localized, and good rendering of the largest outlet glaciers and their watersheds, all of which are crucial for sea level rise estimation.

8.3 ▪ Fluid–biology coupling; marine biology

8.3.1 ▪ Presentation of the domain

In this section we consider applications in biology and ecology coupled with fluid dynamics (be it the ocean or a tank of water). This includes the study of marine ecosystems as well as the engineering use of algae or bacteria for ecological purposes. Both these questions are of significant importance for the future and are strongly related to climate change and human activities.

8.3.1.1 ▪ Marine ecosystems

One such problem is the evolution of fish populations. Because millions of people depend on fisheries, this question is clearly important. The fish population evolution depends on multiple factors: fishing industry catches, predator health/number/location, food availability (other smaller fishes, plankton, or phytoplankton), ocean temperature/currents/chemistry, climate change, etc. This is a typical ecology problem, and the complete answer would require the modeling of the whole food chain and a complete ecosystem. Similarly, the study of the response of marine ecosystems (not only fish populations) and food chain equilibriums to global change is of ecological interest.

8.3.1.2 ▪ Microbiology and ecological remediation

Another problem related to fluid–biology coupling is the use of microbiology for ecological engineering. For example, knowing that a given bacterium is known to consume a given synthetic molecule contributing to pollution, how could we monitor, forecast, and control the evolution of the bacteria population so that we achieve bioremediation/pollution control? In the same area, people are also interested in using algae to produce biofuels, and the same questions about monitoring and controlling the total biomass hold.

8.3.2 ▪ Examples of DA problems in this context

8.3.2.1 ▪ Fishing industry

Due to the arrival of factory ships (e.g., purse super seiners, factory ships that can fish hundreds of tons in one catch), overfishing has been a growing concern. In this framework, modeling and forecasting of the fish population evolution can be a huge help for decision-makers. The fish population depends not only on fishing catch but also on various geophysical parameters (climate, currents, food availability, etc.), so that coupled and complex marine ecosystem models are required to provide accurate trends. Increased computational power also allows an increase of complexity and improved modeling capacity. However, and this is a common problem in ecology, it also causes the multiplication of unknown parameters in the models.

DA, more specifically the adjoint method, has been used to calibrate model parameters in this context, e.g., Lawson et al. [1995] for an academic predator–prey model, McGillicuddy et al. [1998] for a coupled plankton–ocean model, and Dueri et al. [2012a] and Faugeras and Maury [2005] for skipjack tuna population evolution in the Indian ocean.

8.3.2.2 ▪ Marine ecology

Closely linked to fish population ecology, there has been an increase of interest in marine ecology. In particular, models were developed for the coupling between physical oceanography, biogeochemistry, and plankton biology, as key ingredients for fishing ecology studies. As mentioned above, despite the upgraded complexity, these models still rely on strong parameterization of unresolved scales and phenomena, involving many unknown parameters. Brasseur et al. [2009] propose a short review of the domain, and more comprehensive studies can be found in Dowd [2007] (Bayesian statistical assimilation) and Faugeras et al. [2004, 2003] (variational assimilation). The review Luo et al. [2011] offers a broader view of DA of ecosystems modeling, not only in the marine framework.

8.3.2.3 ▪ Microbiology

Biological remediation is a tool to achieve depollution of large water bodies, such as reservoirs, lakes, and rivers. It relies on the simple principle that putting (well-chosen) bacteria into polluted water will lead the bacteria to consume the contaminant, grow, and produce biomass (dead bacteria containing the contaminant). Extracting the biomass while renewing the polluted water ensures that the bacteria still live and gradually clean the water. This process can be done in tanks that pump the contaminated water, remove the biomass, and pump clean water back into the water body. The question is then to optimize the flow output into the tank to achieve maximal depollution in minimal time.

This problem is modeled by coupling the biology (bacteria/contaminant population evolution) and fluid dynamics (contaminant transport versus clean water diffusion into the water body). Control methods were successfully applied, both with simple ODE-based models [Gajardo et al., 2011] and more complex spatialized PDE-based models [Barbier et al., 2016].

8.3.3 ▪ Focus: Tuna population ecology

A model of tuna dynamics has been developed and presented in Dueri et al. [2012a] and Faugeras and Maury [2005]. This model is structured in 3D space and fish size, and couples the fish population, the marine ecology, and the physical oceanography. It is based on PDEs representing the evolution of the density of the fish at every given size. Of course, this kind of model is highly complex and depends on many unknown parameters that must be finely tuned to reproduce the real tuna dynamics well. Dueri et al. [2012a] proposed computing a cost function measuring the negative loglikelihood of various observations (fish catches, size frequencies) and minimizing it to calibrate the model parameter. The minimization required the gradient of the cost function, which was obtained using an automatically differentiated TLM.

After assimilation, the model proves to be satisfactory; e.g., when comparing the vertically integrated exploitable biomass to the catch distribution in Figure 8.3, we can see that it is able to properly represent the main features of the tuna distribution at different periods of the year and under different environmental conditions.

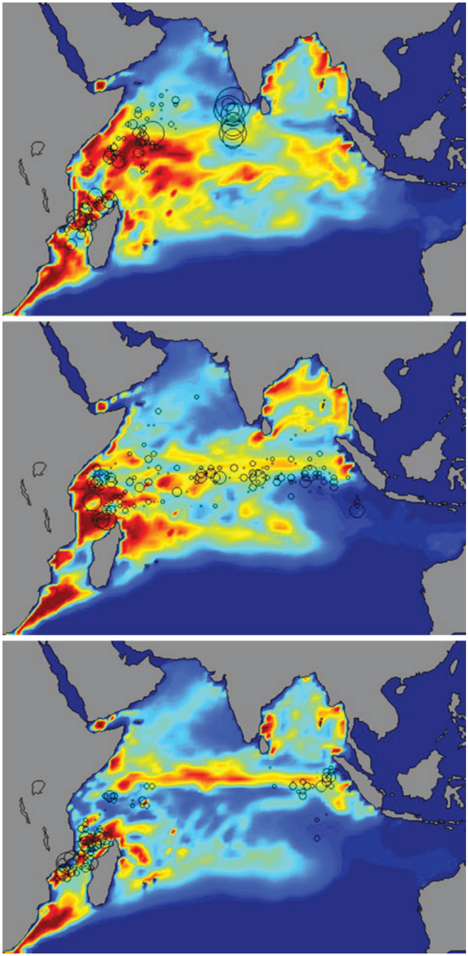


Figure 8.3. *Exploitable population of skipjack tuna computed by the model versus observations (circles) in the Indian Ocean. Top: April* 1993, *middle: February* 1998, *and bottom: April* 1998. *Reprinted with permission from Elsevier [Dueri et al.,* 2012b*].*

8.4 ▪ Land surface modeling and agroecology

8.4.1 ▪ Presentation of the domain

This section groups applications covering plant, water, and carbon cycles, e.g., agronomy, agriculture, forestry, soil sciences, carbon pools, and associated climate retroactions. This vast group of domains is equally crucial for the current and future well-being of human populations and the planet in general.

8.4.1.1 ▪ Agroecology for water quality and availability

Issues related to water and agriculture have an immediate impact on human lives. Supply of both water and plants (in terms of quantity as well as quality) is changing fast because of human activities and climate change. The understanding of their processes, as well as their monitoring and forecasting, is therefore of utmost importance. In this area, scientific issues are numerous; we can cite a few: evolution and monitoring of subsurface water reservoirs due to agricultural irrigation and precipitation changes, pollution of freshwater by pesticides and bioremediation using trees or buffer zones, and crop modeling and plant growth modeling for water saving optimization. It is clear that each of these questions directly impacts the quantity and quality of the water supply of the local population. They are closely related to agroecology and agriculture questions (use of pesticides, irrigation, crop management) and therefore food security (also in terms of both quantity and quality).

8.4.1.2 ▪ Vegetation, continental water, and carbon cycle modeling for climate

Agriculture, forestry, and surface and subsurface hydrology, as well as the carbon cycle, have many links and feedback loops with climate change, which are and will indirectly affect human, animal, and plant populations as well. A few of the questions of interest here are contribution of the forests/soil/plants to the carbon cycle and modeling of ecosystems; modeling of the water exchange and humidity on continental surfaces, and feedback with the atmosphere and climate; modeling of the ground cover distribution and evolution, link with the albedo and humidity modeling, and feedback into the climate models; and so on. These questions are important for our future. We will see that DA provides tools to look for answers and further our knowledge and understanding of them.

8.4.2 ▪ Examples of DA problems in this context

Land surface and agroecology modeling are complex for many reasons: scarcity of observations below the surface of the earth, strong spatial heterogeneity, multiple scales, reactive chemical species, nonlinearities, threshold phenomena, etc. Up until recently, most models were limited to conceptual models (in distributed hydrology, for example) or to empirical modeling (in crop modeling, for example). However, the increase of computer power and the availability of high-resolution satellite observations allow both the spatialization of models and the implementation of DA methods.

8.4.2.1 ▪ Agroecology

For example, Wu et al. [2012] implemented the adjoint method to optimize the water supply for sunflower growth. They identified that plant growth is sensitive to the supply frequency and total during each development phase of the plant and were able to propose an optimal strategy for fruit filling.

Another example is crop modeling, which provides useful tools for testing of agronomical strategies and decision-making to both minimize environmental consequences and maximize crop production. The review by Dorigo et al. [2007] studies the combined use of remote sensing observations and DA methods developed to improve agroecosystem modeling. For example, as has been said before, such models rely on poorly known soil parameters. Lauvernet et al. [2002] developed a variational method to assimilate vegetation properties from remote sensing images into a crop model, to estimate soil and crop characteristics, and to better forecast the production. Varella et al. [2010] have implemented an inverse model (importance sampling method that consists of one analysis step of the particle filter) to infer these parameters from real observations.

8.4.2.2 ▪ Vegetation, water, and climate

Land surface modeling is a key component of climate models, as it impacts albedo, humidity, carbon cycle, and more. Because of vegetation and evapotranspiration, feedback loops exist between (agricultural) land use and local climate and water resources. To study this interaction, Courault et al. [2014] developed a coupled model (microclimate/land surface), for which they calibrated the (unknown) soil parameters using variational DA of remote sensing observations. Other applications were performed on the same principle: use satellite data to retrieve poorly known vegetation/land surface biophysical parameters. For example, Pellenq and Boulet [2004] implemented the EnKF for a vegetation model, and Lauvernet et al. [2002, 2008] developed a variational method on a radiative transfer model to estimate vegetation properties at the top of the canopy from temporal series of upper atmosphere data. Compared to classical inversion methods used in radiative transfer, the adjoint model combined with temporal and spatial constraints from remote sensing images drastically improved the results on twin experiments.

8.4.2.3 ▪ Carbon cycle dynamics

Modeling of the carbon cycle on the land surface is a difficult task because it requires accurate modeling of a complex ecosystem (called the terrestrial carbon ecosystem) with many interactions and uncertainties. The terrestrial carbon ecosystem is a carbon sink (i.e., the land biosphere carbon absorption exceeds its losses). Compared to the other components of climate models, it presents the largest uncertainties, up to the point that predictions for the future remain largely unknown. There have therefore been recent attempts to constrain these models using DA; see e.g., Bloom and Williams [2015] (focus on the carbon pool ecosystem) and Delahaies et al. [2013] (focus on the variational assimilation system). At a smaller scale, Williams et al. [2005] studied forest carbon dynamics using the EnKF, and Ribbens et al. [1994] calibrated a model of tree seedling dispersion using an inverse method (direct likelihood maximization).

8.4.3 ▪ Focus: Crop modeling

In crop modeling, most of the input parameters are either empirical (since the functioning of vegetation is not a priori described by exact equations) or difficult to estimate, due to their large variability in time and space. LAI (leaf area index) is a key canopy variable that directly quantifies green vegetation biomass. Though it is the most observable canopy parameter (using remote sensing), assimilation of LAI is usually performed without capitalizing on the information from temporal and spatial dependencies of the vegetation. Lauvernet et al. [2014] performed such a study and assumed that the model parameters are governed by spatial dependencies.

Figure 8.4 (top) depicts observations of LAI (generated by their Bonsai model) on a whole-wheat growth cycle, from sowing to harvest, on 100 different pixels of the landscape: x-axis is time in days; y-axis is the pixels; and z-axis is the LAI value on each pixel, at each time step. One distinguishes the spatial levels considered: the two big groups of LAI represent two different varieties of wheat (from pixel 1 to 50, then 51 to 100), where the varietal parameters are equal in the model. For each variety, there are 5 plots with similar plot parameters (pixel 1 to 10, 11 to 20, etc., ...), and 10 pixels per plot, with independent parameters.

In Figure 8.4 (middle), LAI was assimilated without using spatial constraints, i.e., on each pixel independently. Results are quite satisfying in this example due to a very large set of observations of LAI (one image each day). If the number of observations decreases (e.g., one image every 20 days, like a satellite), the estimation quality decreases drastically. However, using spatial constraints (as has been done on Figure 8.4 (bottom)) allows us to get better estimates with a large dataset, but also to keep stability when the observation frequency decreases.

8.5 ▪ Natural hazards

8.5.1 ▪ Presentation of the domain

Climate change is most likely increasing, and will continue to do so for at least the next couple of decades, according to the probability of extreme event occurrence. This section deals with the modeling and forecasting of such events to mitigate the risk for human populations and activities.

8.5.1.1 ▪ Floods

As population increases in already dense areas, a certain disregard for the risk of rare events (e.g., floods with long return periods such as 25, 50, or 100 years) is not uncommon. Therefore, floods caused by rivers or sea storms impacting housing or commercial buildings are regular natural hazards. In parallel to the population increase, there is also a development of the observation network for rivers (e.g., satellite remote sensing or in situ monitoring systems), which allows for operational flood forecasting. Still, some scientific issues remain: hydrological modeling of extreme events, data sparsity, or, on the contrary super high resolution data versus large-scale models, etc.

8.5.1.2 ▪ Wildfires

As was said above, climate change will most likely increase the risk of natural hazards. Wildfires are also affected, as changes in precipitation and temperature will probably cause severe drought in some areas and increase the risk of large and destructive wildfires. Therefore, real-time forecasting of wildfire propagation is a crucial issue for both wildfire hazard prevention and emergency management. This is a complex problem, in terms of both modeling and DA, because of numerous difficulties: nonlinear and complex front propagation, strong dependency on atmospheric conditions, need of real-time observations and predictions, feedback loops between fire and atmosphere, etc.

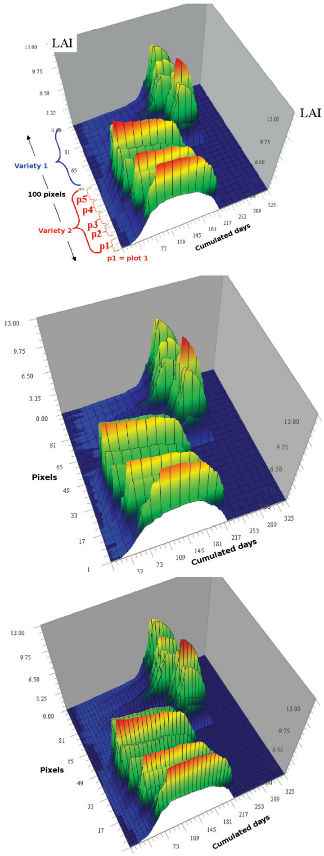


Figure 8.4. *LAI maps. Top: LAI observed in time and space (represented here without noise). Middle: LAI estimated after classical DA of noisy LAI (RMSE* = 0.39623*). Bottom: LAI estimated after DA of noisy LAI using spatial constraints (RMSE* = 0.36573*). Reprinted with permission from RNTI [Lauvernet et al.,* 2014*].*

8.5.1.3 ▪ Hurricanes

Hurricanes or tropical storms are extreme storms that develop over warm ocean water and lose intensity over land, so that they cause huge damage over coastal tropical areas. According to IPCC [Field, 2012], "attribution of single extreme events to anthropogenic climate change is challenging." Experts still disagree on the question, and the evolution of losses due to tropical storms is also unclear, because of two conflicting factors: population and development increase in coastal and sensitive areas, versus forecasting and prevention progress. From a scientific point of view, hurricane track forecasting has much improved over the last decades, but hurricane intensity forecasting is still lacking. Indeed, intensity is closely related to the inner core structure of the storm, which is both insufficiently modeled (as it requires very high resolution and/or fine parameterization of small-scale physics) and insufficiently observed.

8.5.2 ▪ Examples of DA problems in this context

8.5.2.1 ▪ Hydrology and floods

Many recent studies have tackled the problem of forecasting flash flood extreme events. Some of them focus on river hydraulics (using PDE models such as shallow-water models), while others study complete river catchments (using distributed models of independent grid cells connected by transfer functions).

For PDE-based models, real-time flash flood modeling requires accurate input variables. For example, Ricci et al. [2011] implemented the KF to retrieve the upstream inflow from river water level observations, while Habert et al. [2014] estimated the lateral inflow using the EKF.

On the other hand, Harader et al. [2012] studied the retrieval of the rainfall input in a rainfall-runoff model of a river catchment, which is a complex task because of numerous uncertainties and heterogeneities in the model. On this subject, the difficulties have been studied and highlighted by Coustau et al. [2013], who implemented the BLUE to estimate the peak discharge on the same catchment for real-time flood forecasting.

8.5.2.2 ▪ Wildfires

Wildfire modeling has recently improved, with complex and precise models for solving the flame structure (combustion coupled with computational fluid dynamics (CFD)). However, these models are too computationally expensive to be used for real-time forecasting, and simplifications need to be made for DA experiments.

Mandel et al. [2008] proposed a coupled PDE model of energy (temperature) and fuel balance and successfully performed synthetic experiments with the EnKF to provide temperature corrections.

However, as even this type of model is deemed too expensive to be used for real-time forecasting at the regional scale, recent approaches have developed simplified models based on a front-tracking solver in which the fire is considered as an interface (i.e., a front) propagating between the burning area and the unburnt area. Errors in the model inputs and equations translate into errors in the simulated characteristics of the fire, and thus should be reduced. For example, Rochoux et al. [2015, 2014] proposed a two-part comprehensive DA study to first retrieve biomass fuel and wind parameters and second provide real-time forecasts of the fire front.

8.5.2.3 ▪ Hurricanes

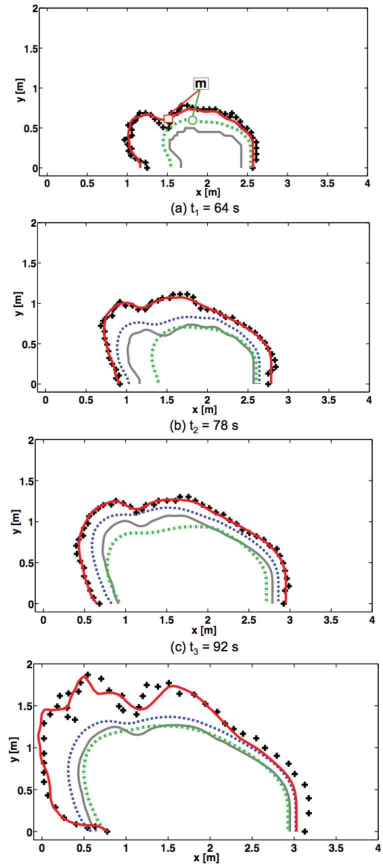
Real-time forecasting of tropical cyclones has of course been of major interest for decades. Earlier DA experiments used the nudging method; see e.g., Anthes [1974]. With the development of NWP, more sophisticated methods were implemented. The first implementations of variational methods gave poor results because fixed covariance information did not correspond well to the hurricanes' strong variability at their core. To overcome this problem, Xiao et al. [2000] developed a two-step variational method: the first step is dedicated to the correct positioning of the hurricane, and the second step is for the classical assimilation of the other variables. More recently, Torn and Hakim [2009] implemented the EnKF to make the most of its handling of error covariances. As the EnKF naturally evolves and adapts forecast error covariances, the results were indeed improved.

8.5.3 ▪ Focus: Wildfire modeling and real-time forecasting

In the context of wildfire modeling, Rochoux et al. [2014] performed DA on a front-tracking model to retrieve some poorly known model parameters and state. The biomass fuel properties and the near-surface wind as well as the position of the front are sequentially corrected as new observations become available. In Rochoux et al. [2015] they used the calibrated model to provide a real-time forecast of the fire front.

The DA ingredients are as follows: the control variable is either the input parameters of the front-tracking model [Rochoux et al., 2014] or the positions of markers along the fire front [Rochoux et al., 2015]; observations of this front are available (as geotracking tools now make it possible to obtain such data from mid-infrared imaging), and the EnKF is used to combine the data with the front-tracking model.

Figure 8.5 presents the assimilation results of a controlled grassland fire experiment with four EnKF assimilation cycles from initial time 50 s to final time 106 s. In this figure the benefit of the DA can be clearly seen, as it really provides a precise tracking of the fire front. This ability of DA to provide real-time analysis of the fire front over a simple real-life (albeit controlled) grassland fire shows the maturity of both the model and the method.



(a) $t_1 = 64$ s

(b) $t_2 = 78$ s

(c) $t_3 = 92$ s

(d) $t_4 = 106$ s

Figure 8.5. *Controlled grassland fire experiment (data from King's College London) with multiple assimilation cycles from* $t_0 = 50$ s *to* $t_4 = 106$ s *with fire front position estimation. Black crosses correspond to observations, the gray solid line corresponds to the initial condition of the assimilation cycle, the green dashed-dotted line corresponds to the free run (without DA), the blue dashed line corresponds to the mean forecast estimate (without DA for the first observation time or with DA at the previous observation time), and the red solid line corresponds to the mean analysis estimate (with a DA update at the current observation time). Assimilation cycles* (a) *[50; 64 s];* (b) *[64; 78 s];* (c) *[78; 92 s]; and* (d) *[92; 106 s]. Reprinted with permission from European Geoscience Union [Rochoux et al., 2015].*

Chapter 9

Applications in atmospheric sciences

9.1 ▪ Numerical weather prediction

9.1.1 ▪ Presentation of the domain

Obviously, weather forecasts have always been of paramount importance. They have often been the epitome of risk assessment in agriculture; ship navigation and sailing; air navigation; battlefield tactics; extreme events such as storms, tornadoes, and floods; and more recently for renewable energy, such as wind energy and solar energy.

At the end of the nineteenth century the Norwegian school of meteorology, led by Wilhelm Bjerknes, built the foundations of modern meteorology. Modern communications, e.g., telegrams, allowed meteorologists to almost instantaneously transfer observation results to draw charts of pressure, temperature, humidity, and wind speed. Weather forecasting emerged from the art of the forecaster to interpreting these charts and their extrapolation in space and time.

After the Second World War, scientific computation and new measurement devices, such as radar, fostered the ambition of weather forecasting. Solving at least approximately the fundamental equations that govern the atmospheric motions became possible, thus enabling comparison between observation charts and the numerical output of mathematical models. NWP was born.

With the increase in the number of observations and in the accuracy of the numerical models, enabled by the increase in computational power, the quality of meteorological forecasts has steadily increased over the years.

9.1.2 ▪ Examples of DA problems in this context

NWP is extremely challenging. Solving the primitive equations at very high resolution and incorporating finer physics, such as cloud microphysics and constituent chemistry and physics, is a first major hardship. The second hardship is fundamental and is due to the chaotic atmosphere dynamics, as first explained by the famous meteorologist Edward Lorenz. The main consequence is that any small perturbation in one of the meteorological fields increases exponentially with time. Hence, a fine estimation of the meteorological fields, whose deviations from the true ones can be seen as an error, is bound to diverge exponentially with time from the truth. NWP at the mid-latitudes has a finite horizon of predictability of about 10 days [Lorenz, 1965, 1982; Dalcher and Kalnay, 1987].

A solution to overcome this second hardship is to correct the estimation of the fields by correcting the initial condition of a forecast using information coming from measurements. That is why DA is critical in NWP. It was first introduced in the form of computationally cheap interpolation methods, Cressman interpolation, and later statistical interpolation. Using background climatological information and a variational analysis, this led to the introduction of 3D-Var in the 1990s. At the beginning of the 2000s, optimal control over a time window was used to obtain a model-consistent trajectory and led to the introduction of 4D-Var in the field [Rabier et al., 2000]. A few years later, the EnKF was also experimented with and put into operation [Houtekamer and Mitchell, 2005]. Nowadays, operational centers implement hybrid and 4DEnVar methods (see Chapter 7) to combine some of the benefits of the filtering and variational methods while trying to avoid their drawbacks.

As opposed to many components of the earth system, the atmosphere is a very well observed part. The improvements in NWP skills have come from the increase in the model resolution and from the ability of DA to make good use of observations, but also from the fast-increasing flux of observations, most of them stemming today from space platforms.

The exponential increase in the number of observations, mostly due to satellite measurements, has benefited NWP in the southern hemisphere, where the synoptic observation system (i.e., the global-scale traditional observation network) is much sparser than in the northern hemisphere due to the oceans. These observations, after thinning, are fed into the DA systems.

The increase in the model resolution and the improvement in the parameterizations have decreased representativeness errors and more generally model error to a point that errors have become of the same magnitude as the initial condition error growth induced by the dynamics, at least for the standard way of measuring NWP performance. In turn, this implies that the determination of the initial condition remains of paramount importance, as well as the role played by DA. DA is also meant to play a significant role in model error mitigation by, for instance, state parameter estimation.

9.1.3 ▪ Focus: Evaluating the skill of NWP

The improvement in NWP is objectively measured by standard skill tests routinely issued by the operational centers. The increase in performance has recently been qualified as "The quiet revolution of numerical weather prediction" in Bauer et al. [2015]. This is first justified by the achievements of the NWP community, which have not been publicized as much as others in other sciences, such as fundamental physics. Second, the progress has been due to many factors from both the scientific and technical sides, some of them already mentioned. Third, this progress has been steady over three decades, with no clearly publicized breakthrough. Figure 9.1 shows this steady improvement over the years using a standard skill score of NWP applied to an observable: the geopotential height at 500 hPa, from the ECMWF. As hinted at in the previous section, the gap between predictability over the northern and southern hemispheres has reduced considerably because of the uniform coverage of satellite observations.

An empirical model for the error growth in NWP is [Lorenz, 1982; Dalcher and Kalnay, 1987]

$$\frac{\mathrm{d}E}{\mathrm{d}t} = (\alpha E + \beta)\left(1 - \frac{E}{E_\infty}\right), \tag{9.1}$$

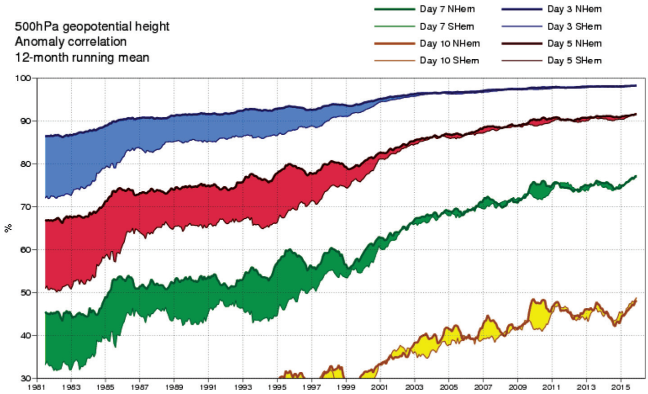


Figure 9.1. *Anomaly (difference between the actual value and the climatological value) correlation coefficient of the* 500 *hPa height forecasts for the extratropical northern hemisphere (upper curves) and southern hemisphere (lower curves), and for a forecast horizon of* 3, 5, 7, *and* 10 *days.* ©*2016 ECMWF, reprinted with permission.*

where α is the dynamical error growth due the chaotic nature of the model, β is the model error source term, and E_∞ is the asymptotic error level at which error saturates because of the finite volume of phase space. A typical error growth curve that follows this model is plotted in Figure 9.2. In the short term, error growth is dominated by model error, which exceeds the uncertainty on the initial condition. The exponential

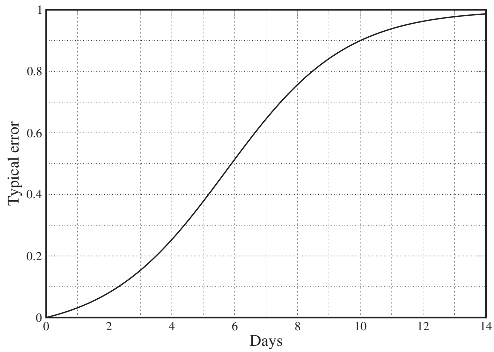


Figure 9.2. *Typical error growth following the empirical model* (9.1)*. The asymptotic error is* $E_\infty = 1$, $\alpha = 0.5$, *and* $\beta = 0.025$.

growth due to the uncertainty on the initial condition finally takes over but will ultimately saturate.

This empirical model allows us to roughly attribute the improvement in the forecast to either the estimation of the initial condition or the mitigation of model error, using a simple fit to the skill scores [Simmons and Hollingsworth, 2002; Magnusson and Källén, 2013]. Such an analysis confirms that the improvement mostly comes from both the reduction of the error in the initial condition (thanks to DA) and the reduction in model error. Both turn out to be equally important. The initial condition can be addressed by DA, whereas model error can only be partly mitigated by DA. The increase in the number of observations has been a less important factor in the most recent trend analysis [Magnusson and Källén, 2013], typically since the 2000s.

9.2 ▪ Atmospheric constituents

The primary goal of DA in meteorology or oceanography can be identified with the inverse problem that consists of estimating the initial condition of the state vector. This is justified by the strong sensitivity of the errors to the initial condition due to the chaotic dynamics of the primitive equations. In addition to the fluid itself, the primitive equations account for the transport and evolution of the water content of the atmosphere but do not describe all the other trace constituents of the atmosphere.

9.2.1 ▪ Presentation of the domain

These trace constituents are critical [Seinfeld and Pandis, 2016]. They can form air pollution with huge sanitary impacts, such as ozone or fine particulate matter in the boundary layer. They strongly affect the radiative budget and climate: the ozone in the stratospheric ozone layer to protect us from ultraviolet radiation and carbon dioxide as the main global contributor to global warming of the atmosphere. The atmosphere can hold harmful pollutants such as radionuclides in the wake of a nuclear accident, or ashes after a volcano eruption, with radiative forcing impact and with annoyance for airplanes.

To simulate these additional species, one should add to the primitive equations the transport equations of the species, the reactions among them, and the emission and loss processes in the atmosphere. There are hundreds of species of interest in various forms: gaseous, aqueous, particulate matter, and aerosols, to name a few. Their chemical and physical dynamics can be strongly nonlinear. However, their dynamics are barely chaotic.

9.2.2 ▪ Examples of DA problems in this context

As a consequence, the main issue in controlling and tracking these species may not be in determining the initial content of the species in the atmosphere. An accurate quantitative estimation of the concentrations usually requires the estimation of all the influential factors, such as the emissions or the sinks, the chemical kinetic rates, and the microphysical thermodynamical coefficients of the many parameterizations required to simulate the species. Moreover, the estimation of these forcings is often interesting per se, not only for nowcasting or predicting the species concentrations.

In particular, the estimation of the emissions is highly relevant. It is critical in the definition of abating policies for air quality or for the earth system's response, leading to global warming. When estimating these factors rather than the concentrations, DA is actually used to solve an inverse problem. There is indeed a very strong connection between the methods of DA and the mathematical techniques developed long ago in inverse problems, as already heavily emphasized in this book [see also Bocquet, 2012b, and references therein]. For instance, the concept of background in DA is tantamount to the concept of regularization in mathematical inverse problems, which is pivotal in the understanding of emission inverse problems.

The application of DA to the estimation of the fluxes of pollutants began in the 1990s with the goal of estimating greenhouse gas fluxes, such as carbon dioxide and methane [Chevallier, 2012, and references therein], but also to estimate primary pollutants or precursors of secondary pollutants, such as nitrogen oxides, carbon monoxide, sulfate dioxide, or other trace but harmful species, such as heavy metals [Elbern et al., 2012, and references therein].

With this objective in mind, filtering approaches are not the most efficient techniques. Indeed, to estimate an emission flux, one would like to later use concentration observations that are causal consequences of these emissions. Hence, smoothing techniques are usually preferred, such as 4D-Var, BLUE equations but applied over a time-window, or an EnKS. That said, the EnKF has also been successfully used for forecasting and parameter estimation in this context.

9.2.3 ▪ Focus: Inverse modeling of the radionuclide emission from the Fukushima-Daiichi nuclear power plant accident

In the case of an accidental release of pollutant in the atmosphere, it is crucial to forecast the plume and the contaminated zones, for instance to implement safety protocols to protect the population. Numerical transport models are used to simulate the dispersion of the pollutant. Figure 9.3 shows the large-scale dispersion of cesium-137 that came from the nuclear accident of the Fukushima-Daiichi nuclear power plant (FDNPP). Cesium-137 is a harmful radionuclide with a 30-year half-life that is carried over large distances in the form of particulate matter.

These models requires several input fields. In an accident context with a pointwise source, the critical inputs are the meteorological fields obtained from meteorological models, wind and precipitation, and, above all, the pollutant source term. An incomplete and erroneous knowledge of these inputs has dramatic consequences on the reliability of the plume forecast. Given that accidental releases are, fortunately, exceptional, and because it is dangerous to perform in situ measurements, the source term is very difficult to estimate.

The inverse modeling problem consists of locating the source or, more often, when the location is known, estimating the time rates of the release. This is called source term estimation.

Since in this context the model can often be assumed linear in the source term, several inference methods can be used to estimate the source term, such as 4D-Var, BLUE over the accident time window, an ensemble smoother, stochastic optimization, the simplex method, or Monte Carlo Markov chains. However, the main difficulty often comes from the paucity of the observations. Moreover, the knowledge of the prior statistics of the errors is rather poor. Hence, it is tempting to use the observations to infer part of these statistics, in a fashion similar to the estimation of the regularization parameters in inverse problems [Michalak et al., 2005; Davoine and Bocquet, 2007].

In the wake of earlier estimation of the Chernobyl source term, several estimations of the FDNPP source term using DA techniques have been published [Winiarek et al., 2012; Stohl et al., 2012; Saunier et al., 2013; Winiarek et al., 2014]. Figure 9.4 shows

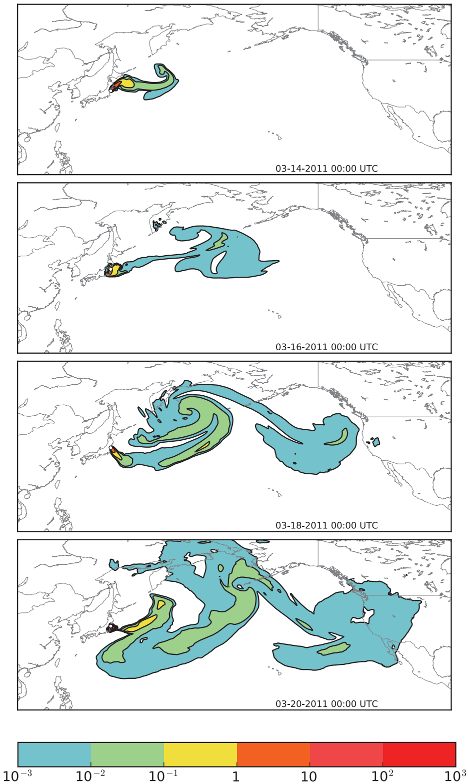


Figure 9.3. *Cesium-137 radioactive plume at ground level (activity concentrations in becquerel per cubic meter) emitted from the FDNPP in March 2011, as simulated by the* POLYPHEMUS/POLAIR3D *chemical transport numerical model [using the setup from Winiarek et al., 2014].*

an estimation of the cesium-137 source term from the FDNPP. It has been obtained using DA methods, where the error statistics are estimated while accounting for the non-Gaussianity of the errors induced by the positiveness of the emission rates. Once the source term is estimated, the numerical model can be used to simulate the contaminated zones, in the air on short notice and in the soil in the longer term. Figure 9.5 compares the observations of cesium-137 deposited near the FDNPP to the simulation deposition field obtained using the estimated source term.

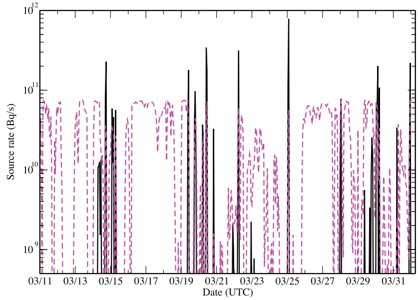


Figure 9.4. *Cesium-*137 *source term as inferred by inverse modeling (from Winiarek et al. [*2014*]). Estimated total:* 12×10^{15} *becquerel (Chernobyl was about* 85×10^{15} *becquerel). The pink curve represents the diagnosed uncertainty related to the inversion. Reprinted with permission from Elsevier [Winiarek et al.,* 2014*].*

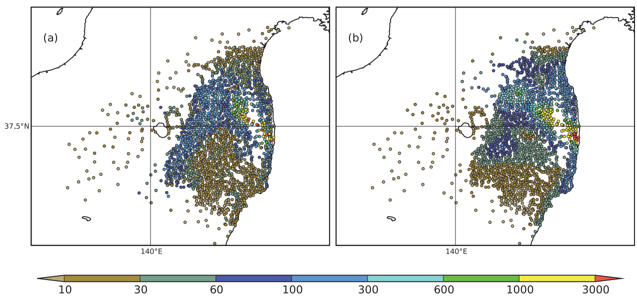


Figure 9.5. *Deposited cesium-*137 *(in kilobecquerel per square meter) measured* (a) *and hindcast* (b) *near the FDNPP. Reprinted with permission from Elsevier [Winiarek et al.,* 2014*].*

Chapter 10

Applications in geosciences

10.1 ▪ Seismology and exploration geophysics

10.1.1 ▪ Presentation of the domain

In geophysics in general, and seismology[33] in particular, the objective is to obtain information about the earth's internal structure. The unknown earth structure is represented by a model, $\mathbf{m}$; the synthetic data by $\mathbf{u}(\mathbf{m})$; and the measurements by $\mathbf{u}^{\mathrm{obs}}$ (see Section 2.3). The signals are generated either by natural sources, i.e., earthquakes and other seismic activity, or by artificial sources that usually take the form of explosives, pneumatic hammers, or boomers (used in underwater seismic exploration). The reflected signals are measured by seismic sensors known as geophones (hydrophones in underwater acoustics—see below). Recently, there have been interesting attempts to use the earth's background seismic noise as the input source—see Shapiro et al. [2005], Larose et al. [2008], Verbeke et al. [2012], Daskalakis et al. [2016], and Zigone et al. [2015]. The model is some form of wave equation [Aki and Richards, 2002; Courant and Hilbert, 1989b; Brekhovskikh, 1980] that describes the propagation of acoustic, elastic, or even electromagnetic waves through the earth's crust.

The inverse problem can be solved by either deterministic or statistical/probabilistic approaches. The statistical approach was formalized by Tarantola [2005]. The deterministic approach, based on adjoint methods, is described in the excellent book of Fichtner [2011]. Other, more classical approaches that are still employed in the petroleum industry are based on a range of diverse stacking and migration methods—interested readers should consult Claerbout [1976, 1985] and Aki and Richards [2002].

To quote J. Claerbout: "There is not one theory of inversion of seismic data, but many—maybe more theories than theoreticians." (see his website at `http://sep.stanford.edu/sep/prof/`, where all his monographs are freely available).

The geophysical images that are computed by inversion techniques are then used to visualize petroleum and mineral resource deposits, subsurface water, contaminant transport for environmental pollution studies, archeology, etc.

[33] The study of the propagation of elastic waves in and through the earth's crust—often in relation to earthquakes.

10.1.2 ▪ Examples of DA problems in this context

10.1.2.1 ▪ Petroleum or seismic prospecting

Oil and gas exploration remains one of the major areas for inversion techniques. Often this is coupled with reservoir modeling—see Wikipedia [2014]. The expenditure on seismic data acquisition campaigns is estimated at billions of dollars per year [Bret-Rouzaut and Favennec, 2011]. The subsequent data processing costs alone, can amount to $1 million for a single seismic survey. There is clearly a need for reliable and efficient inversion methods and algorithms to process all this data and to generate meaningful images.

There is a lot of information to be found on the websites of the numerous commercial companies, either oil and gas or those that provide services to oil and gas exploration. Among these are Schlumberger, Total, CGG, Statoil, and Ikon.

Recent research on this subject has addressed more exotic aspects, such as complex geological structures [Liu et al., 2006], joint inversions [Gyulai, 2013], combination of Bayesian inversion with rock physics [Grana and Della Rossa, 2010], stochastic (Monte Carlo Markov chain) methods [Martin et al., 2012], high-dimensional seismic inverse problems [Zhang et al., 2016], and full waveform inversion (FWI; see below).

10.1.2.2 ▪ Geological prospecting

For geological and mineral prospecting, magnetic and electromagnetic approaches are most often employed. In time domain electromagnetics (TDEM), electric impulses are used to induce transient electric and magnetic fields. Recent references on this approach are Newman and Commer [2005] and Haber et al. [2007].

10.1.2.3 ▪ Earthquake detection and analysis

Detection and reconstruction (by simulation) of earthquakes and other seismic events is a vital domain for today's research and for decision-making by public authorities. In particular, there has been recent progress toward the goal of early detection and thus prediction of earthquakes (see below for the case of volcanic eruptions). In fact, a lot of hope has recently been placed in the ability of background seismic noise monitoring and inversion as a promising avenue. The first major paper on this subject was Shapiro et al. [2005], even though the statistical inversion technique was inspired by the earlier acoustics work of Lobkis and Weaver [2001]. These results were subsequently justified mathematically by Bardos et al. [2008] and have been applied in many concrete cases—see Rivet et al. [2015], Frank and Shapiro [2014], Campillo et al. [2011], Yao et al. [2011], Stehly et al. [2009], and Lin et al. [2008].

Note that seismic detection is also widely used for monitoring nuclear tests. The recent explosions in North Korea are a good example [Wikipedia, 2016a].

10.1.2.4 ▪ Volcanic eruption detection and analysis

The Whisper project[34] was a trailblazer in the utilization of ambient seismic noise to monitor property changes in the solid earth. They were the first to succeed in using seismic ambient noise recordings to extract deterministic signals that led to imaging of the earth's interior at high resolution. Their main goal was to apply the proposed noise-based inversion methods to study the transient processes related to volcanic and

[34] `https://whisper.obs.ujf-grenoble.fr`.

tectonic activity through continuous measuring of mechanical changes in deep parts of the earth. One of the main research targets was active volcanic regions. For studies of volcanoes, they processed data from the La Réunion vulcanological observatory and from other volcanic areas in Japan, the U.S. Pacific Northwest, etc.

Numerous striking results have been obtained. We can cite Mordret et al. [2015], Droznin et al. [2015], Obermann et al. [2013], Brenguier et al. [2008, 2007], and many more references that can be found on the project's website.

10.1.3 ▪ Focus: An adjoint approach for FWI

As explained in Fichtner [2011], FWI has recently been applied in seismic tomography. The method is characterized by the numerical simulation of the elastoacoustic vector wave equation [Aki and Richards, 2002; Landau and Lifschitz, 1975],

$$\rho \boldsymbol{u}_{tt} - (\lambda + \mu)\nabla(\nabla \cdot \boldsymbol{u}) - \mu \nabla^2 \boldsymbol{u} = \boldsymbol{f}, \tag{10.1}$$

where $\boldsymbol{u} = (u_1, u_2, u_3)^T$ is the displacement in the x-, y- and z-directions; λ and μ are the Lamé coefficients; ρ is the medium density; and $\boldsymbol{f}$ is an initial impulse that represents the acoustic source. The relations between these coefficients and the wave speeds are

$$c_p^2 = \frac{\lambda + 2\mu}{\rho} \qquad \text{and} \qquad c_s^2 = \frac{\mu}{\rho},$$

where c_p is the pressure (or primary) wave speed and c_s is the shear (or secondary) wave speed. The physical domain of interest is subdivided into layers that are either geological (rock, sand, etc.) or water. Thus, an acoustic layer is obtained in the model by simply setting $\mu = 0$ locally.

The advantage of the system (10.1) is that it intrinsically models all the different types of waves that can arise in layered media—compressional and shear waves in the bulk and Love, Stoneley, and Rayleigh waves along the interfaces. These equations must be completed with physically relevant boundary and initial conditions. On the surface we usually specify a zero pressure condition. Between layers, continuity conditions on the normal components of u and the stresses, which are related to the gradient of u by Hooke's law, must be satisfied. At the bottom-most level we give a suitable absorbing condition. On the lateral boundaries, suitable absorbing/radiating conditions need to be specified [Komatitsch and Martin, 2007; Xie et al., 2014]. The initial condition is usually a Ricker wavelet (`http://subsurfwiki.org/wiki/Ricker_wavelet`), with the desired frequency content, located at the source position—other initial conditions are possible. The geophones (or other measurement devices) are simulated by simply recording the solution at those points that correspond to their locations.

Numerical solutions of the above system enable accurate modeling of seismic wave propagation through heterogeneous, realistic, geophysical earth models. When these direct solutions are combined with adjoint methods, exactly as presented in Chapter 2, we obtain excellent tomographic resolution of the subsurface structure. Numerous real-world applications can be found in the last three chapters of Fichtner [2011] and also in Tape et al. [2009], Virieux and Operto [2009], Peter et al. [2011], Monteiller et al. [2015], and Fichtner and Villasenor [2015].

10.2 ▪ Geomagnetism

10.2.1 ▪ Presentation of the domain

Dynamo theory [Wikipedia, 2015b] describes the generation of a magnetic field by a rotating body. The earth's dynamo (also called the geodynamo) is a geomagnetic source that gives rise to long-term variability—this is known as geomagnetic reversal, or "flip." On shorter time scales, of about one year, there are changes in declination. The understanding of the dynamical causes of this variability, based on available data, is a long-standing problem in geophysics with numerous fundamental and practical implications. DA has been applied to this problem thanks to the increasing quality and quantity of geomagnetic observations obtainable from satellites, coupled with our ability to produce accurate numerical models of earth-core dynamics. First attempts were based on the optimal interpolation approach [Kuang et al., 2008]. Reviews can be found in Fournier et al. [2010], Kuang and Tangborn [2011], and most recently Hulot et al. [2015].

Depending on whether we seek to understand fast or slow variability, we can resort to two-dimensional (quasi-geostrophic) or three-dimensional numerical models to assimilate geomagnetic observations. The former are well suited for describing the fast variability and are less demanding in resources. The latter are more complete and able to correctly represent the variability on longer time scales, but the computational cost is higher. More details can be found in Canet et al. [2009] and Aubert and Fournier [2011].

10.2.2 ▪ Focus: An EnKF for time-dependent analysis of the geomagnetic field

In Fournier et al. [2013], an ensemble Kalman filter (EnKF; see Chapter 6) is used to assimilate time-dependent observations based on a three-dimensional dynamo model. Twin experiments led to the choice of an ensemble size equal to 480 members. Forecasting capabilities of the assimilation were promising but required a full observation error covariance matrix to ensure good results in some cases. Note that the state vector in the calculations had a dimension of almost 10^6.

10.3 ▪ Geodynamics

10.3.1 ▪ Presentation of the domain

In earth sciences the knowledge of the state of the earth mantle and its evolution in time is of interest in numerous domains, such as internal dynamics, geological records, postglacial rebound, sea level change, ore deposit, tectonics, and geomagnetic reversals. Convection theory is the key to understanding and reconstructing the present and past state of the mantle. For the past 40 years, considerable efforts have been made to improve the quality of numerical models of mantle convection. However, these models are still sparsely used to estimate the convective history of the solid earth, especially when compared to ocean or atmospheric models for weather and climate prediction. The main shortcoming is their inability to successfully produce earth-like seafloor spreading and continental drift in a self-consistent way.

The ultimate goal is to reconstruct the deep earth from geological data and thus be able to image the deep earth from ancient times up until today. This is termed *hindcasting* and is simply forecasting in the direction of decreasing time; i.e., we are "predicting" the past.

10.3.2 ▪ Examples of DA problems in this context

In Coltice et al. [2012], convection models were used to predict the processes of seafloor spreading and continental drift. The results obtained prove that the combination of high-level DA methodologies and convection models together with advanced tectonic datasets can retrieve the earth's mantle history.

There is now hope to understand the causes of seismic anomalies in the deep earth. Another application is the understanding of geomagnetic phenomena—in particular the evolution of the earth's magnetic field (see above). In fact, knowledge of the earth's paleogeography has potentially wide-ranging applications in water resource research, mineral resource research, and even paleontology.

10.3.3 ▪ Focus: Sequential DA for joint reconstruction of mantle convection and surface tectonics

An extended Kalman filter (EKF) was employed in Bocher et al. [2016] and was able to recover the temperature field of a convective system with plate-like tectonics at its surface over several hundred megayears. The only observations used were surface heat fluxes and surface velocities.

Chapter 11

Applications in medicine, biology, chemistry, and physical sciences

11.1 ▪ Medicine

11.1.1 ▪ Presentation of the domain

The type and number of observations that are available in medicine today are impressive (see also Section 11.4). In cardiology alone, we can cite electrocardiograms, MRI, CT scans, flow measurements with ultrasound, pressure measurement with catheters, and myocardium thickness measurement with piezoelectric sensors. The grand challenge is to use and combine these sources to solve inverse problems that will then lead to personalized medicine. DA in this field is novel, and many promising avenues remain to explore.

Wherever we encounter reaction-diffusion equations, there exists the possibility to use adjoint and DA approaches. This has been the case for studying tumor growth [Bresch et al., 2010; Colin et al., 2014] and cardiac electrophysiology (see below). This was also seen above (see Section 8.5) for the modeling of forest fires and in Chapter 2 in a theoretical context. Hemodynamics is another field for potential DA where we can either propose simple models of blood flow (see Section 11.3 on fluid dynamics) or also take into account the mechanics of arterial walls in a fluid–structure interaction model—see for example Perego et al. [2011], Bertoglio et al. [2012, 2013, 2014], Pant et al. [2014], and the website of the euHeart project (`http://www.euheart.eu`), where a lot of groundbreaking work was done.

11.1.2 ▪ Examples of DA problems in this context

11.1.2.1 ▪ Tumor growth

The mathematical modeling of cancer is aimed at understanding and then being able to predict the growth of a tumor. Just as important is the capacity of these models to thus simulate and reproduce the effects of anticancer therapies, be they chemotherapy or radiotherapy or any combination of these, by including their effects in the model. With the ever-increasing availability of medical imaging, a new era is opening up for better parameterization of these models in light of these images. This is precisely where inversion methods and DA can begin to provide hope for personalized therapeutic protocols.

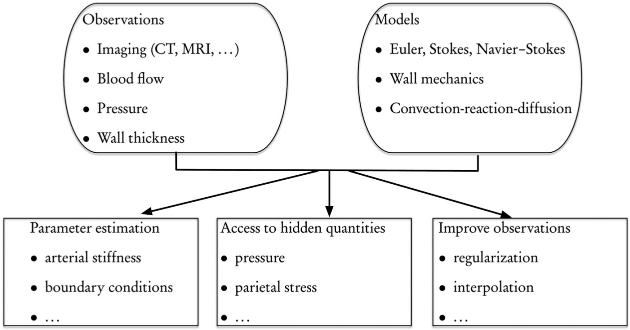


Figure 11.1. *Assimilation of medical data for the cardiovascular system (adapted from a presentation of J.-F. Gerbeau).*

The complete modeling of oncogenesis is considered to be out of reach, but there is no doubt that the coupling of simpler, PDE-based models with image sequences can provide helpful insight into the progression of the disease. In Lombardi et al. [2012], the case of lung metastases is studied.

Modeling of tumor growth has been described in Cristini and Lowengrub [2010]. In Lombardi et al. [2012] and Colin et al. [2014], inverse problems are formulated and solved for the identification of difficult to access parameters that lead to remarkably accurate predictions of tumor growth.

There are numerous perspectives in this domain, e.g., employing functional imaging (PET, flux, and diffusion MRI), use of biomarkers, and studies of placebo effects.

11.1.2.2 ▪ Hemodynamics

Understanding and modeling blood flow is a basic necessity for studying pathologies and estimating the effects of treatments in cardiology, e.g., stents, valves, etc. The physics of blood flow is in fact coupled with arterial wall movements, thus giving rise to fluid–structure interaction (FSI) problems. However, there is a hierarchy of models that can be applied to the problem, ranging from response functions up to full 3D PDEs [Formaggia et al., 2009]. In all cases, the need for estimating unknown parameters arises. With the deluge of measured data that is now available (see Figure 11.1), we are clearly in a position to formulate and solve DA and inverse problems for personalized medicine. In Bertoglio et al. [2014], the stiffness of the artery wall was estimated with the aid of a reduced-order unscented KF coupled with an FSI model.

Functional magnetic resonance imaging (fMRI) is based on studying the vascular response in the brain to neuronal activity and can be used to study mental activity. The ability to accurately model the evoked hemodynamic response to a neural event plays an important role in the analysis of fMRI data [Lindquist et al., 2008]. It is most commonly performed using blood oxygenation level–dependent (BOLD) contrast to study local changes in deoxyhemoglobin concentration in the brain. The primary goal of fMRI research is to use information provided by the BOLD signal to draw conclusions about the underlying, unobserved neuronal activity. This response can be modeled at different levels, starting from relatively simple linear time-independent (LTI) response functions and moving up to full fluid dynamic modeling (see the following section). Once the models are defined, parameter estimation inverse problems can be formulated and solved by comparing with measured data (as seen above), usually from imaging—see for example Liu and Hu [2012].

11.1.2.3 ▪ Cardiac electrophysiology

The function of the heart is to pump blood through the circulatory system. To achieve this, the chambers (atria, ventricles) contract during the cardiac cycle. This contraction is triggered by an electrical impulse that takes the form of a wave propagating across the heart and depolarizing the cardiac muscular cells. An ECG (electrocardiogram) provides a noninvasive measurement of this electrophysiology. The Dutch doctor Willem Einthoven was awarded the Nobel Prize in Medicine in 1924 for the invention of the ECG.

The progression of the depolarization wave can be modeled by a reaction-diffusion equation, and the resulting numerical simulations can produce remarkably accurate synthetic ECGs [Boulakia et al., 2010].

The next step is electromechanical coupling—see Focus in Section 11.1.3—where the electrophysiology is coupled with a mechanical model of the heart's muscular contraction by the myocardium. Based on this, interesting electromechanical inverse problems can be formulated.

11.1.3 ▪ Focus: Electrocardiography inverse problem

This inverse problem involves the reconstruction of the heart's electrical activity from measurements of the body surface potential (as obtained by an ECG). A multiphysics approach is proposed in Corrado et al. [2015]. This approach combines a nudging or Luenberger observer for the mechanical state estimation with a reduced-order, unscented KF for the identification of the electrophysiological parameters—it is thus a *hybrid* DA method based on a joint gain filter. In fact, the mechanical measurements are shown to improve the identifiability of the electrical problem and produce an improved reconstruction of the electrical state. The computational challenge is huge: different meshes are used for the mechanics (18 000 elements), the electrics (540 000 elements), and the complete thorax (1.25 million elements); a master–slave coupling is used to transfer information between the electrical and mechanical solvers; uncertainties are taken into account; both mechanical and electrical measurements are used.

This model can now be coupled with real patient data, coming from ECG, tagged MRI, anatomy, and physiology. A patient workflow can be defined and thus provide a complete and personalized diagnostic, prevention, or pre- or post-operation planning and surveillance tool.

11.2 ▪ Systems biology

11.2.1 ▪ Presentation of the domain

Systems biology is based on the premise that it is not enough to understand how cellular components function to study whole cells or organisms. To predict a complete system's behavior, we need to consider the interactions between system components. This can be achieved, in the biological context (as well as in others—see for example

Section 11.3 below on fluid dynamics), by combining experimental techniques with computational methods and thus developing predictive models. Of course, with the arrival of the "-omics," in particular genomics and metabolomics, there is now an enormous amount of available observation data for this.

11.2.2 ▪ Examples of DA problems in this context

In the excellent review paper of Engl et al. [2009], they define two major classes of inverse problems for systems biology: (1) parameter identification based on the vast amount of available data, and (2) qualitative inverse modeling, which aims at reaching targeted dynamics. An example of the first class is to seek parameter values that stop growth of certain cell types and thus inhibit malignant growth. Another example that corresponds to class (2) is how to adjust circadian rhythms so that they respond more rapidly and smoothly to pacemaker calibration to counteract effects of jet lag.

Traditionally, chemical reaction networks are modeled by large systems of coupled ODEs that originate from Michaelis–Menten reaction kinetics [Wikipedia, 2016c]—see also Section 11.6. The dimension of these systems can be several thousands of equations and parameters. Parameter identification problems for these systems are solved by the classical data-mismatch approach using local (gradient-based) or global (stochastic) methods. The ill-posedness of the inverse problem is usually treated by the use of regularization techniques (see Chapters 1 and 2 and Engl et al. [1996]). The qualitative inverse problems are based on bifurcation diagrams that capture the direct relation between the cell physiology and the qualitative dynamics of the system under study. This is also used in chemical and electrical engineering problems and is related to what are known as "robust system" design techniques. The algorithms use gradient-based methods to place bifurcation points at desired locations.

11.3 ▪ Fluid dynamics

11.3.1 ▪ Presentation of the domain

Fluid dynamics covers a vast field of applications. These range from aerodynamics to meteorology, passing through experimental fluid mechanics and dynamics, going down to the level of blood-flow monitoring within the human body.

The application of DA methods to experimental fluid dynamics (EFD) has recently begun to be explored and can hopefully bridge gaps between EFD and computational fluid dynamics (CFD). Adjoint methods have been used for some time already to solve the inverse problem of aerodynamics for optimal shape design of airfoils and even complete airplanes. Another recent area of research combines fluid dynamics with image processing and is being applied to oceanographic and meteorological contexts.

11.3.2 ▪ Examples of DA problems in this context

11.3.2.1 ▪ Aerodynamics

The use of adjoint methods for optimal design in aerodynamics was initiated by Pironneau [1974]. The approach follows exactly the presentation in Chapter 2.

Since a wing is an apparatus for controlling the flow around an airplane, one can apply the theory of optimal control of PDEs to their design [Pironneau, 1974; Jameson, 1988; Giles and Duta, 2003]. This is based on the Euler equations for compressible flows and the Navier–Stokes equations for turbulent, incompressible flows from CFD

[Courant and Friedrichs, 1976; Batchelor, 2000; Kundu and Cohen, 2002]. A quadratic cost function is defined as

$$J = \frac{1}{2}\int_{\mathscr{B}} (p - p_t)^2 \, \mathrm{d}\mathscr{B},$$

where $\mathscr{B}$ is the surface shape of the airfoil, p is the fluid pressure field, and p_t is the target pressure distribution. The surface shape, after being suitably parameterized by a set of geometric design variables, is treated as the control function, which is varied to minimize J. A variation in the shape causes a variation in the pressure field and a subsequent variation in the cost function. Then, using the classical adjoint approach, we can formulate a gradient algorithm for optimizing the shape, exactly as was done in Chapter 2. To avoid a solution with a nonsmooth shape, Jameson [2006] proposed a weighted Sobolev gradient that enables the generation of a sequence of smooth shapes. This approach has been successfully applied to a large number of practical design problems, ranging from classical NACA airfoils to Boeing 747 wing fuselages [Vassberg and Jameson, 2002; Leoviriyakit and Jameson, 2004].

11.3.2.2 ▪ Experimental fluid dynamics

In a recent special issue, Suzuki [2015], the application of DA methods to EFD is explored. The motivation is to study the coupling of EFD with CFD. This coupling has, until now, been developed chiefly in the fields of geophysics, oceanography, and meteorology (as we have amply seen in previous chapters of this book). The idea is to introduce the variational and sequential methods that have been developed in these fields into the domain of fluid dynamics. This could help to overcome mutual weaknesses in EFD and CFD.

Measurements, and measurement devices, can characterize real-world flows, but they are often inadequate due to their limitation in time and space, their inaccuracy, and the fact that a measurement instrument can actually perturb the flow that we want to measure. These shortcomings have traditionally been dealt with by various interpolation and approximation techniques. But thanks to recent developments in CFD and the increasing capacity of HPC (high-performance computing), we are now able to simulate and reproduce real flows and correctly represent complex geometries, fine-scale flow structures, boundary and initial conditions, and even turbulent flows. Whereas the first two are well within the reach of CFD, the last two could greatly benefit from coupling with EFD.

In Hayase [2015], the different measurement methods and techniques are reviewed, interpolation methods are listed, and the integration of numerical simulation with flow measurements is discussed. This leads to the notion of a "hybrid wind tunnel," where simulation and measurement can be efficiently combined.

11.3.2.3 ▪ Coupled fluid dynamics with image analysis

The coupling of image sequences with fluid dynamics models by DA methods has been applied to geophysical flows and in EFD (see above). This has been particularly interesting for the understanding and the prediction of turbulent flows.

Sequences of satellite images furnish the possibility to follow scalar quantities, such as temperature, pressure, water vapor density, or phytoplankton density, that are transported by the flow. But the transport phenomena can be well described by PDEs that link the spatial and temporal variations of the field image to the unknown distribution

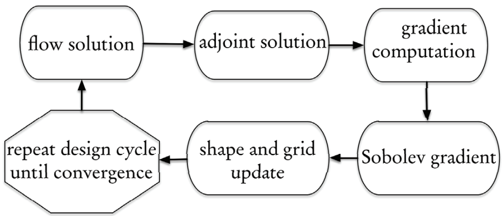


Figure 11.2. *Design cycle for aerodynamic shape optimization (adapted from Jameson [2006]).*

of flow velocities. We can thus constrain the observations by the model equations and proceed to assimilate the data.

A sequence of images inherently contains a large quantity of information, both spatial and temporal. The coupling with the Navier–Stokes equations can then be implemented by a variety of DA schemes and approaches. In Beyou et al. [2013], a weighted ETKF was used to assimilate ocean surface currents from images (observations) of sea surface temperatures. More recently, Yang et al. [2015] employed a hybrid ensemble-based 4D-Var scheme (4DEnVar—see Chapter 7) to assimilate a shallow-water model with simulated and real image sequences.

11.3.3 ▪ Focus: Optimal shape design in aerodynamics

Shape optimization [Delfour and Zolésio, 2001; Mohammadi and Pironneau, 2009] is a class of optimization problems where the modeling, optimization, or control variable is no longer a set of parameters or functions, but the shape or the structure of a geometric object. This is, in fact, an inverse problem. Adjoint methods are intensively used to compute gradients, and automatic differentiation (see Section 2.3.9) has often been employed. These gradient-based approaches are the only ones that make industrial problems tractable, and their use in aerodynamic design is now becoming widespread—see Figure 11.2.

In Jameson [2006], a continuous adjoint approach is applied to five concrete case studies: a transonic airfoil, a Boeing 747 planform, a super Boeing 747, a P51 racer, and a transonic executive jet. Results are obtained based on the Euler and Navier–Stokes equations. For the improvement of transonic performance, relatively small changes are needed in the wing cross-sections. To achieve truly optimal designs, the whole wing planform needs to be considered. The cost function must then include terms on both aerodynamics properties and overall structural weight. The functional takes the form

$$J = \alpha_1 C_D + \alpha_2 \frac{1}{2} \int_{\mathscr{B}} (p - p_t)^2 \,\mathrm{d}\mathscr{B} + \alpha_3 C_W,$$

where C_D is the drag coefficient and C_W is a dimensionless parameter of the wing weight. The three coefficients α_i permit the calculation of Pareto fronts for designs that have minimal weight for a given drag, or minimal drag for a given weight. The use of a smooth gradient—see also Schmidt et al. [2008], Schmidt [2010]—guarantees that the redesigned surface is always achievable.

11.4 ▪ Imaging and acoustics

11.4.1 ▪ Presentation of the domain

Imaging, in general, attempts to solve an inverse problem to characterize the medium (human body, machine part, etc.) from measurements taken on its surface. Usually a wave is emitted by a device, propagates through the medium, and is measured by a suitable sensor array. The waves can be elastic, acoustic, electromagnetic, or optic. Depending on the physical nature of the wave (basically its frequency and energy content), it will be able to penetrate different media at different resolutions.

11.4.2 ▪ Examples of DA problems in this context

11.4.2.1 ▪ Medical imaging

One of the most important successes in the field of inverse problems for imaging was the invention of an inversion algorithm for computed tomography (CT) by Cormack in 1963 [Wikipedia, 2015a] and its experimental demonstration by Hounsfield in 1973. These two received the Nobel Prize in Physiology or Medicine in 1979. Both CT and MRI are based on the theory of inverse problems. Recent work has addressed multimodal imaging, where two or more imaging techniques are combined to improve the accuracy and resolution of the inversion process [Ammari et al., 2011, 2012; Pulkkinen et al., 2014; Ammari et al., 2015].

11.4.2.2 ▪ Underwater acoustics

Similarly, in underwater acoustics [Jensen et al., 2011], there is an important emphasis on inverse problems. These inverse problems aim to elucidate properties of the sea water and the underlying sediment layers in the seabed from acoustic signals that are emitted by special acoustic arrays and measured on others, called hydrophones. Recently, adjoint methods have begun to be employed to this end [Asch et al., 2002; Hermand et al., 2006; Meyer et al., 2006]. Bayesian approaches have also been used [Dosso and Dettmer, 2011], but usually in a stationary setting—in fact the computations are generally performed in the frequency domain.

11.4.2.3 ▪ Geophysics and seismology

Of course, geophysical and seismic prospecting (see Chapter 10) is another context that has a lot in common with imaging and acoustics.

11.4.2.4 ▪ Image processing

Finally, a number of books are dedicated to inverse problems for image processing—see Colton and Kress [1998], Vogel [2002], and Kaipio and Somersalo [2005].

11.4.3 ▪ Focus: Inversion in underwater geoacoustics

Numerous inverse problems can be formulated in the field of underwater geoacoustics—see Figure 11.3. Among these we can single out the following:

1. ocean acoustic tomography (OAT), where we seek to reconstruct the sound-speed profile in the water layer, $c(z)$;

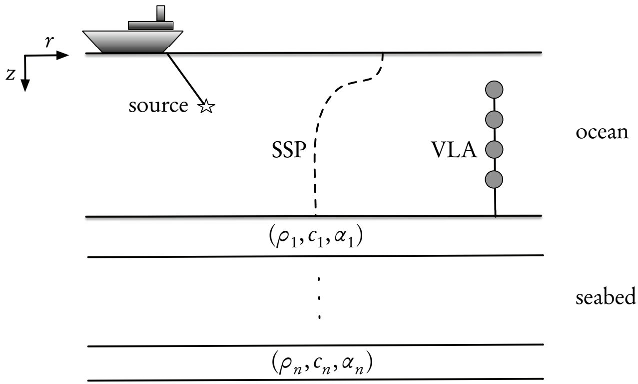


Figure 11.3. *Physical setup for a geoacoustics inverse problem. The ship tows an acoustic source, whose signal is measured on the vertical linear array (VLA). The acoustic signal propagates through the ocean, characterized by its sound-speed profile (SSP), $c(z)$, and penetrates into the n layers of sediments, characterized by their density ρ_i, sound speed c_i, and attenuation α_i. Note that the layer thicknesses can also be defined as parameters of the problem that are generally unknown.*

2. geoacoustic inversion (GI), where we seek to identify the sediment layer properties, $(\rho, c, \alpha)_m$, of layer m;

3. source or object detection;

4. buried object detection;

5. ocean impulse response, or Green's function reconstruction; and

6. tracking problems: sedimentation, currents, salinity, temperature, and other gradients.

According to D.M.F. Chapman, in Taroudakis and Makrakis [2001], "The goal of all geoacoustic inversions is to estimate environmental characteristics from measured acoustic field values, with the aid of a physically realistic computational acoustics model." A seismoacoustics model will clearly provide even better information for the inversions. In addition, ocean acoustics usually involves the measurement of a signal that propagates in a noisy environment. The array processing must take this into account. This gives rise to matched-field processing when the propagation medium is accounted for in the processing. In practice, although the acoustic measurements are obviously performed in the time domain, the processing is done in the frequency domain, thus requiring sophisticated Fourier transform techniques to have accurate spectral estimates. The use of spectral finite elements (SPECFEM) [Cristini and Komatitsch, 2012] will enable us to restore a full time domain signal processing capacity, thus leading (hopefully) to a much closer physical match with the recorded signals.

11.4.4 ▪ Focus: Photoacoustic tomography

The photoacoustic effect refers to the generation of acoustic waves by the absorption of optical energy [Fisher et al., 2007]. In photoacoustic imaging, energy absorption causes thermoelastic expansion of the tissue, which in turn leads to propagation of a pressure wave. This signal is measured by transducers distributed on the boundary of the organ, which is in turn used for imaging optical properties of the organ. Mathematically, the pressure, p, satisfies the wave equation

$$\frac{\partial^2 p}{\partial t^2}(x,t) - c^2 \Delta p(x,t) = 0, \quad x \in \Omega, \quad t \in]0,T[,$$

with the boundary condition

$$p(x,t) = 0 \quad \text{on } \partial\Omega \times]0,T[,$$

and the initial conditions

$$p(x,t)|_{t=0} = a\delta_{x=z} \quad \text{in } \Omega,$$

and

$$\partial_t p(x,t)|_{t=0} = 0 \quad \text{in } \Omega,$$

where a is the absorbed energy.

In Ammari et al. [2011], different imaging algorithms are used to reconstruct point obstacles. When only limited-view measurements are available, the coupling with an adjoint-based optimal control problem enables good reconstruction. It is shown that if one can accurately construct the geometric control, then one can perform imaging with the same resolution using limited-view as using full-view data, as can be seen in Figure 11.4.

11.5 ▪ Mechanics

11.5.1 ▪ Presentation of the domain

Mechanics is a field where a very large number of inverse problems exist. We can mention optimal design of mechanical structures (beams, trusses, plates), crack detection in materials, and parameter identification. Even the simplest mechanical system, made up of a mass, a spring, and a damper, gives rise to a number of inverse problems. The configuration of Figure 11.5 can be described by the linear dynamic state equation

$$M\ddot{\mathbf{u}} + C\dot{\mathbf{u}} + K\mathbf{u} = F,$$

with initial conditions

$$\mathbf{u}(0) = u_0, \quad \dot{\mathbf{u}}(0) = u_1.$$

Two classes of inverse problems can be defined:

1. Structural model updating, where for known $\mathbf{u}$ and F we seek to identify the system parameters, M, C, and K. This is also known as structural optimization.
2. Structural load updating, where for given $\mathbf{u}$, M, C, and K we seek the load, F.

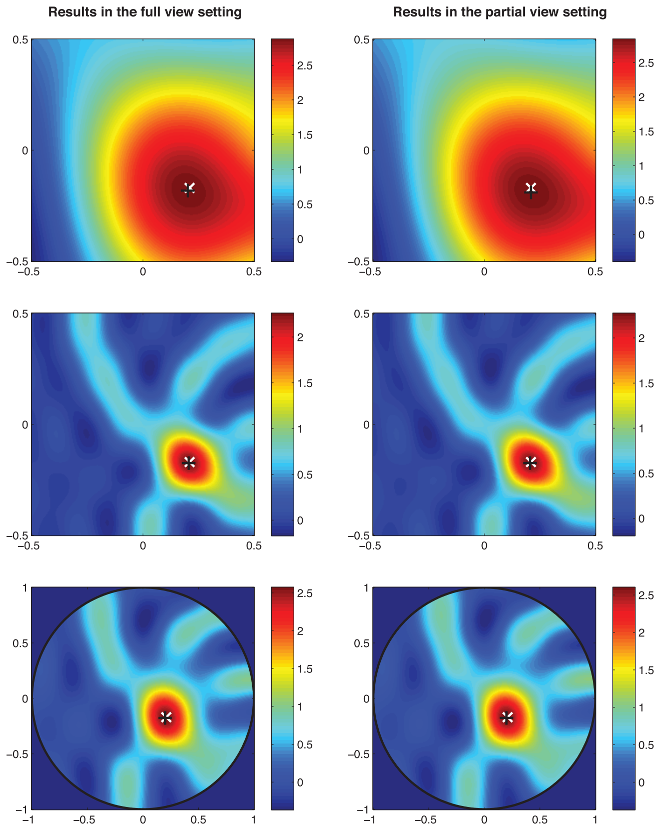


Figure 11.4. *Kirchhoff imaging algorithm results for source localization in square and circular geometries. The (black/white) x denotes the (numerical/theoretical) center of the source. Full- and limited-view results are essentially equivalent in their precision [Ammari et al., 2011].*

The theoretical background for these problems has already been treated in Chapter 2, Section 2.3.

A more complete setting can be found in Moireau et al. [2008] and Chapelle et al. [2009]. They describe PDE-based models where an extended Kalman filter (EKF) is coupled with a parameter estimation method to solve inverse problems in continuum mechanics. This approach was subsequently applied to cardiac biomechanics problems—see Section 11.1 above.

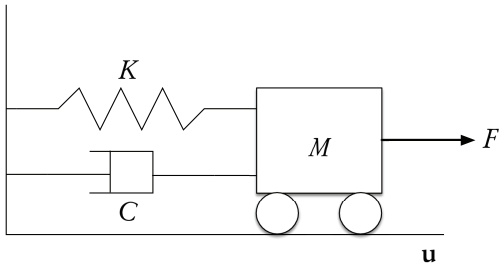


Figure 11.5. *A simple mechanical system, composed of a spring with stiffness K, a dashpot with coefficient C, and a mass M subject to a force F that produces a movement (displacement and velocity)* $\mathbf{u}$.

11.5.2 ▪ Examples of DA problems in this context

The applications in mechanics are numerous. They range from heat-conduction problems in steel manufacturing [Engl and Kugler, 2005] to crack identification [Bonnet and Constantinescu, 2005] and biomechanics [Imperiale et al., 2013].

Many other mechanics-related applications can be found in the other sections of this part of the book, in particular, fluid mechanics (see Section 11.3), medicine (see Section 11.1), and geomechanics (see Section 10.1).

11.6 ▪ Chemistry and chemical processes

11.6.1 ▪ Presentation of the domain

Chemistry and chemical processes have already been encountered in Section 9.2 on Atmospheric Constituents and Section 11.2 on Systems Biology. Here we will describe a class of applications that are based on differential algebraic equations (DAEs) or PDEs.

Adjoint methods are widely used in the control and sensitivity analysis of chemical processes. This is the case for systems described by all three classes of equations: ODEs, DAEs, and PDEs.

11.6.2 ▪ Examples of DA problems in this context

11.6.2.1 ▪ Fermentation processes

In Merger et al. [2016], an adjoint approach is used to control a wine-fermentation process (see below) that is modeled by a system of parabolic PDEs. Beer fermentation has also been studied, but based on more classical reaction kinetics described by ODEs—see Ramirez and Maciejowski [2007].

11.6.2.2 ▪ Petroleum products

In Biegler [2007], a case study of low-density polyethylene (LDPE), used in the manufacture of plastic bags, squeeze bottles, and plastic films, is analyzed. The system is composed of 130 ODEs and 500 DAEs. Gradients of the objective function with respect to the control coefficients and the model parameters are calculated either directly from the DAE sensitivity equations or by integrating the adjoint equations. This enables real-time optimization that can then be used as a decision-making aid. In the monograph of Biegler [2010], a lot of supplementary background material as well as additional applications can be found.

11.6.3 ▪ Focus: Wine fermentation process

Modeling of fermentation is important in food, chemical, and pharmaceutical production. For wine fermentation, if X is the yeast concentration and N, O, S, and E are the nitrogen, oxygen, sugar, and ethanol concentrations, respectively, then the reactions are given by

$$\begin{aligned} a_2 N + a_3 O + a_4 S &\xrightarrow{\mu_1} a_1 X, \\ a_5 S &\xrightarrow{\mu_2} a_6 E + a_7 T, \end{aligned}$$

where μ_1 and μ_2 are the reaction rates and $a_1, \dots, a_7$ are the yield coefficients. The models that are usually used to describe this reaction system are systems of ODEs derived from classical reaction kinetics.

If space-dependent concentrations and temperature variations are taken into account, we obtain a system of six PDEs of reaction-diffusion type. For example, the equation for X (see Merger et al. [2016] for full details of the system) is

$$\frac{\partial X}{\partial t} - \sigma_1 \Delta X = +a_1 \mu_1(X, N, O, S, T) - \Phi(E) X,$$

where Δ is the Laplacian operator and $\Phi(E)$ is a term that models the dying of the yeast population at the end of the fermentation process due to a toxic concentration of ethanol. Neumann boundary conditions are applied and initial concentrations are imposed at time zero. Diffusion-type PDEs have already been encountered numerous times in this book—see for example Sections 2.3 and 11.1.

To optimize the fermentation process, a cost function is defined in terms of an unknown control function, u, the applied temperature of the cooling/heating cycle. The gradient of the cost functional with respect to u is obtained from the adjoint equation, a backward reaction-diffusion equation, and the optimal control is then computed by a quasi-Newton BFGS (Broyden–Fletcher–Goldfarb–Shanno) algorithm [Press et al., 2007; Nocedal and Wright, 2006]. Numerical experiments have demonstrated the effectiveness of the proposed optimal control.

Chapter 12

Applications in human and social sciences

Over the last few years, there has been an explosion in the availability of continuous and spatially linked datasets, both formal (for example, economic time series) and informal (for example, crowd-sourced, geolocated twitter feeds). Social science is now in a position to better constrain model errors using dynamic DA. Indeed, this is essential if socioeconomic modeling is to fulfill its considerable potential. However, social scientists have largely restricted their research to static modeling traditions.

In this chapter, we will present some of the pioneering work in this vast domain. The first application is economics and finance, where there is obviously both a lot of dynamic data available and strong motivations to improve analysis and predictions. The second domain is that of traffic control, a subject with deep consequences for energy efficiency, pollution, global warming, and public health and safety. Finally, urban planning and management is an excellent use-case for exploiting the availability of linked geospatial and socioeconomic data.

12.1 ▪ Economics and finance

12.1.1 ▪ Presentation of the domain

One of the major challenges in finance today is the estimation of the volatility, σ. Adjoint methods can be, and are being, used to find the volatility for given stock prices. In macroeconomics, there is wider use of particle filter and sequential Monte Carlo methods for simulation and estimation of parameters.

12.1.2 ▪ Examples of DA problems in this context

12.1.2.1 ▪ Volatility estimation and option pricing

The evolution of stock market prices can be reasonably well described by Brownian motion [Oksendal, 2003] models of the form

$$\mathrm{d}S/S = \sigma\,\mathrm{d}W + \mu\,\mathrm{d}t,$$

which is an SDE, where $S(t)$ is the stock price, σ is the volatility, $\mathrm{d}W$ is the increment of a standard Brownian motion, μ is related to the interest rate, and t is time. Under suitable assumptions, and applying Itô calculus, this model can be transformed into a

PDE of parabolic type, known as the Black–Scholes equation [Wikipedia, 2016b], for the option price, $V(S,t)$. The equation is

$$\frac{\partial V}{\partial t}+\frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2}=rV-rS\frac{\partial V}{\partial S},$$

where r is the return, or interest rate. We can thus consider the volatility as a quantification of the uncertainty of the future price of an asset. But the volatility is in general unknown and needs to be estimated so that the observed market prices are well reproduced by the equation. This in an inverse problem for σ given the observations of $V(S,t)$. Adjoint methods provide a very efficient way of solving this problem and can produce estimates of the so-called Greeks that are the sensitivities (or gradient components) of V with respect to S, t, r, and σ.

In the comprehensive monograph of Achdou and Pironneau [2005], numerical algorithms and methods are described for adjoint-based solution of the inverse problem for finding the volatility of different options. Recently, these and other methods, in particular particle filters, have been applied to the domain of high-frequency trading [Platania and Rogers, 2004; Duembgen and Rogers, 2014].

12.1.2.2 ▪ DA in economics

In Yong and Wu [2013], a number of examples in macroeconomics and international finance are studied with linear state space models, where parameter estimation plays a central role. In economics, Pitt and Shephard [1999] and Kim et al. [1998] have pioneered the application of particle filters in financial econometrics. More recently, Fernandez-Villaverde and Rubio-Ramirez [2007] have shown how particle filtering facilitates likelihood-based inference in dynamic macroeconomic models.

12.2 ▪ Traffic control

12.2.1 ▪ Presentation of the domain

The objective of traffic control is to improve the performance of a traffic system or to mitigate, as far as possible, the negative side effects of traffic. It strongly relies on our ability to monitor and forecast current and future traffic flow.

12.2.2 ▪ Examples of DA problems in this context

Key ingredients of traffic monitoring and forecasting are a relevant traffic flow model and a large enough observation set, with reasonable accuracy and good estimates on error statistics. State-of-the-art traffic flow models are written as a nonlinear conservation law for the vehicle density; see, e.g., the model by Work et al. [2010], based on the Lighthill–Whitham–Richards PDE. In this model, the flux function depends on the velocity of the vehicles, which in turn depends empirically on the vehicle density. Regarding observations, two kinds are considered in the literature: fixed sensors, e.g., on a given sensitive area of a highway, and GPS-enabled mobile devices.

12.2.2.1 ▪ Real-time monitoring and forecasting using stationary sensors

This experiment was conducted in Grenoble by Canudas De Wit et al. [2015]. They equipped Grenoble south ring (highly subject to morning and evening congestion) with stationary sensors at regular intervals, which measured data such as vehicle flow, mean velocity or individual vehicle velocity, and occupancy. The Grenoble Traffic Lab platform then produces real-time traffic monitoring and forecasting, which has been shown by the authors to be promising. They strongly rely on DA methods. First, they use a least-squares type method to calibrate the fluid dynamics model parameters (using the data). Then, they use a KF approach to provide traffic forecasts, using the data and calibrated model.

12.2.2.2 ▪ Highway traffic estimation using GPS mobile devices

Another source of data for traffic comes from GPS-equipped mobile devices (smartphones). As the penetration rate of such devices is ever increasing, it is relevant to use the data for traffic estimation. Of course the implied privacy issues are quite sensitive and should be considered while designing a system. The Mobile Century experiment by a Berkeley team, Herrera et al. [2010], circumvented the privacy issue by asking a team of students (with Nokia phones) to patrol the highway on designated itineraries. Without engorging the roads, they attained a 2% to 5% penetration rate and proved their system to be efficient; see Work et al. [2010]. This experiment used an EnKF described by Work et al. [2008], thanks to data collected on given lines on the highway. They used other data to perform cross-validation and showed a good agreement between the model and the observations.

12.3 ▪ Urban planning

12.3.1 ▪ Presentation of the domain

Town and urban areas are ever changing: old buildings razed to the ground, new buildings built, parks or green areas created, neighborhoods restructured, roads and transportation system changed, and so on. This continuous evolution affects both the environment and the human population: access to green areas, air quality, local climate, hydrology, ecology, extreme events such as floods, and more generally people's quality of life. Urban planning, from a DA point of view, aims to provide tools to urban managers and policy makers to monitor, forecast, and plan for a safe environment and preserve or improve quality of life. Typical questions that can be answered using land-use evolution modeling and DA are as follows: What variation on the urban environment will ensue from a given societal change? Which decision should be made to improve population access to green areas? What is the impact of an isolated decision on future urban hydrology?

12.3.2 ▪ Examples of DA problems in this context

12.3.2.1 ▪ Particle filtering for uncertainty reduction in land-use modeling

As can be observed in many application domains, the need for DA methods arises when modeling has reached a certain maturity and simple methods of model calibrating have reached their limit. Urban planning is no exception, and DA has recently been introduced to improve land-use change models and in particular to take into account uncertainties.

A Belgian initiative is interested in urban planning modeling using remote-sensing data. In the MAMUD project, Van de Voorde et al. [2013] designed and calibrated a land-use model using satellite data to assess the impact of city growth on the urban and suburban hydrology. DA methods were required when it became obvious that the uncertainties in the model as well as the data were proving to be a bottleneck for forecast accuracy. In the follow-up ASIMUD project, van der Kwast et al. [2011] proposed a particle filter algorithm to calibrate model parameters, while taking into account both model and data uncertainty and providing improved confidence intervals on the forecasts.

Another example of particle filtering for land-use change modeling is found in Verstegen et al. [2016]. In this study, the authors wondered whether systemic change could be detected using DA. In a case study of sugar cane expansion in Brazil, the particle filter combined with satellite data highlighted that the model structure and parameterization were nonstationary. They realized that a complex societal change was not captured by the stationary model. Their particle filter allowed systemic change in the model and notably improved the projections' confidence intervals.

12.3.2.2 ▪ Calibration of land-use and transportation integrated models

Land-use and transportation integrated (LUTI) models consider the coupling of land-use models and models of transportation of goods and people. Because of the coupling, they are more realistic for the assessment of the impact of urban planning policies; however, they are also more complex, and their calibration is a difficult task. In the project CITiES, DA is used for such a calibration. Capelle et al. [2015] first proposed calibrating a LUTI model using variational-type assimilation: they formulated the calibration as parameter estimation through the Levenberg–Marquardt optimization of a cost function. Gilquin et al. [2016] then proceeded further by introducing sensitivity analysis into the calibration procedure: in a first step, global stochastic sensitivity analysis ascertains to which parameters the model is the most sensitive; in a second step, optimization is done on these important parameters using the data. Compared to the classically used trial-and-error calibration procedures for LUTI models, these two studies showed a significant improvement on both the calibration quality and the computing time.

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